

Thus, correct timing of the write operation at the inputs of the DRAM circuits in the stack may be ensured.

[0091] For example, a buffer chip (e.g. see the buffer chip 202 of FIGS. 2A-E, etc.) may have a cumulative latency of 2, in which case, the AMB may send the write data 2 clock cycles earlier to the buffered stack. It should be noted that this scheme may not be possible in the case of registered DIMMs since the memory controller sends the write data directly to the buffered stacks. As an option, a memory controller may be designed to send write data earlier so that write operations have the correct timing at the input of the DRAM circuits in the stack without requiring the buffer chip to delay the write address and operation.

[0092] FIG. 8 shows address bus conflicts 800 caused by delayed write operations, in accordance with yet another embodiment. As mentioned earlier, the delaying of the write addresses and operations may be performed by a buffer chip, or optionally a register, AMB, etc., in a manner that is completely transparent to the memory controller of a host system. However, since the memory controller is unaware of this delay, it may schedule subsequent operations, such as for example activate or precharge operations, which may collide with the delayed writes on the address bus from the buffer chip to the DRAM circuits in the stack. As shown, an activate operation 802 may interfere with a write operation 804 that has been delayed. Thus, a delay of activate operations may be employed, as will be described in further detail with respect to FIG. 9.

[0093] FIGS. 9A-B show variable delays 900 and 950 of operations through a buffer chip, in accordance with another embodiment. As an option, the variable delays 900 and 950 may be implemented in the context of the architecture and environment of FIGS. 1-8. Of course, however, the variable delays 900 and 950 may be carried out in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0094] In order to prevent conflicts on an address bus between the buffer chip and its associated stack(s), either the write operation or the precharge/activate operation may be delayed. As shown, a buffer chip (e.g. see the buffer chip 202 of FIGS. 2A-E, etc.) may delay the precharge/activate operations 952A-C/902A-C. In particular, the buffer chip may make the buffered stack appear as one or more larger capacity DRAM circuits that have longer tRCD (RAS to CAS delay) and tRP (i.e. precharge time) parameters.

[0095] For example, if the cumulative latency through a buffer chip is 2 clock cycles while the native read CAS latency of the DRAM circuits is 4 clock cycles, then in order to hide the delay of the address/control signals and the data signals through the buffer chip, the buffered stack may appear as one or more larger capacity DRAM circuits with a read CAS latency of 6 clock cycles to the memory controller. In addition, if the tRCD and tRP of the DRAM circuits is 4 clock cycles each, the buffered stack may appear as one or more larger capacity DRAM circuits with tRCD of 6 clock cycles and tRP of 6 clock cycles in order to allow a buffer chip (e.g., see the buffer chip 202 of FIGS. 2A-E, etc.) to delay the activate and precharge operations in a manner that is transparent to the memory controller. Specifically, a buffered stack that uses 4-4-4 DRAM circuits (i.e. CAS latency=4, tRCD=4, tRP=4) may appear as one or at least one larger capacity DRAM circuits with 6-6-6 timing (i.e. CAS latency=6, tRCD=6, tRP=6).

[0096] Since the buffered stack appears to the memory controller as having a tRCD of 6 clock cycles, the memory controller may schedule a column operation to a bank 6 clock cycles after an activate (e.g. row) operation to the same bank. However, the DRAM circuits in the stack may actually have a tRCD of 4 clock cycles. Thus, the buffer chip may have the ability to delay the activate operation by up to 2 clock cycles in order to avoid any conflicts on the address bus between the buffer chip and the DRAM circuits in the stack while still ensuring correct read and write timing on the channel between the memory controller and the buffered stack.

[0097] As shown, the buffer chip may issue the activate operation to the DRAM circuits one, two, or three clock cycles after it receives the activate operation from the memory controller, register, or AMB. The actual delay of the activate operation through the buffer chip may depend on the presence or absence of other DRAM operations that may conflict with the activate operation, and may optionally change from one activate operation to another.

[0098] Similarly, since the buffered stack may appear to the memory controller as at least one larger capacity DRAM circuit with a tRP of 6 clock cycles, the memory controller may schedule a subsequent activate (e.g. row) operation to a bank a minimum of 6 clock cycles after issuing a precharge operation to that bank. However, since the DRAM circuits in the stack actually have a tRP of 4 clock cycles, the buffer chip may have the ability to delay issuing the precharge operation to the DRAM circuits in the stack by up to 2 clock cycles in order to avoid any conflicts on the address bus between the buffer chip and the DRAM circuits in the stack. In addition, even if there are no conflicts on the address bus, the buffer chip may still delay issuing a precharge operation in order to satisfy the tRAS requirement of the DRAM circuits.

[0099] In particular, if the activate operation to a bank was delayed to avoid an address bus conflict, then the precharge operation to the same bank may be delayed by the buffer chip to satisfy the tRAS requirement of the DRAM circuits. The buffer chip may issue the precharge operation to the DRAM circuits one, two, or three clock cycles after it receives the precharge operation from the memory controller, register, or AMB. The actual delay of the precharge operation through the buffer chip may depend on the presence or absence of address bus conflicts or tRAS violations, and may change from one precharge operation to another.

[0100] FIG. 10 shows a buffered stack 1000 of four 512 Mb DRAM circuits mapped to a single 2 Gb DRAM circuit, in accordance with yet another embodiment. As an option, the buffered stack 1000 may be implemented in the context of the architecture and environment of FIGS. 1-9. Of course, however, the buffered stack 1000 may be carried out in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0101] The multiple DRAM circuits 1002A-D buffered in the stack by the buffer chip 1004 may appear as at least one larger capacity DRAM circuit to the memory controller. However, the combined power dissipation of such DRAM circuits 1002A-D may be much higher than the power dissipation of a monolithic DRAM of the same capacity. For example, the buffered stack may consist of four 512 Mb DDR2 SDRAM circuits that appear to the memory controller as a single 2 Gb DDR2 SDRAM circuit.

[0102] The power dissipation of all four DRAM circuits 1002A-D in the stack may be much higher than the power dissipation of a monolithic 2 Gb DDR2 SDRAM. As a result, a DIMM containing multiple buffered stacks may dissipate much more power than a standard DIMM built using monolithic DRAM circuits. This increased power dissipation may limit the widespread adoption of DIMMs that use buffered stacks.

[0103] Thus, a power management technique that reduces the power dissipation of DIMMs that contain buffered stacks of DRAM circuits may be utilized. Specifically, the DRAM circuits 1002A-D may be opportunistically placed in a precharge power down mode using the clock enable (CKE) pin of the DRAM circuits 1002A-D. For example, a single rank registered DIMM (R-DIMM) may contain a plurality of buffered stacks of DRAM circuits 1002A-D, where each stack consists of four $\times 4$ 512 Mb DDR2 SDRAM circuits 1002A-D and appears as a single $\times 4$ 2 Gb DDR2 SDRAM circuit to the memory controller. A 2 Gb DDR2 SDRAM may generally have eight banks as specified by JEDEC. Therefore, the buffer chip 1004 may map each 512 Mb DRAM circuit in the stack to two banks of the equivalent 2 Gb DRAM, as shown.

[0104] The memory controller of the host system may open and close pages in the banks of the DRAM circuits 1002A-D based on the memory requests it receives from the rest of the system. In various embodiments, no more than one page may be able to be open in a bank at any given time. For example, with respect to FIG. 10, since each DRAM circuit 1002A-D in the stack is mapped to two banks of the equivalent larger DRAM, at any given time a DRAM circuit 1002A-D may have two open pages, one open page, or no open pages. When a DRAM circuit 1002A-D has no open pages, the power management scheme may place that DRAM circuit 1002A-D in the precharge power down mode by de-asserting its CKE input.

[0105] The CKE inputs of the DRAM circuits 1002A-D in a stack may be controlled by the buffer chip 1004, by a chip on an R-DIMM, by an AMB on a FB-DIMM, or by the memory controller in order to implement the power management scheme described hereinabove. In one embodiment, this power management scheme may be particularly efficient when the memory controller implements a closed page policy.

[0106] Another optional power management scheme may include mapping a plurality of DRAM circuits to a single bank of the larger capacity DRAM seen by the memory controller. For example, a buffered stack of sixteen $\times 4$ 256 Mb DDR2 SDRAM circuits may appear to the memory controller as a single $\times 4$ 4 Gb DDR2 SDRAM circuit. Since a 4 Gb DDR2 SDRAM circuit is specified by JEDEC to have eight banks, each bank of the 4 Gb DDR2 SDRAM circuit may be 512 Mb. Thus, two of the 256 Mb DDR2 SDRAM circuits may be mapped by the buffer chip 1004 to a single bank of the equivalent 4 Gb DDR2 SDRAM circuit seen by the memory controller.

[0107] In this way, bank 0 of the 4 Gb DDR2 SDRAM circuit may be mapped by the buffer chip to two 256 Mb DDR2 SDRAM circuits (e.g. DRAM A and DRAM B) in the stack. However, since only one page can be open in a bank at any given time, only one of DRAM A or DRAM B may be in the active state at any given time. If the memory controller opens a page in DRAM A, then DRAM B may be placed in the precharge power down mode by de-asserting

its CKE input. As another option, if the memory controller opens a page in DRAM B, DRAM A may be placed in the precharge power down mode by de-asserting its CKE input. This technique may ensure that if p DRAM circuits are mapped to a bank of the larger capacity DRAM circuit seen by the memory controller, then p-1 of the p DRAM circuits may continuously (e.g. always, etc.) be subjected to a power saving operation. The power saving operation may, for example, comprise operating in precharge power down mode except when refresh is required. Of course, power-savings may also occur in other embodiments without such continuity.

[0108] FIG. 11 illustrates a method 1100 for refreshing a plurality of memory circuits, in accordance with still yet another embodiment. As an option, the method 1100 may be implemented in the context of the architecture and environment of any one or more of FIGS. 1-10. For example, the method 1100 may be carried out by the interface circuit 102 of FIG. 1. Of course, however, the method 1100 may be carried out in any desired environment. It should also be noted that the aforementioned definitions may apply during the present description.

[0109] As shown, a refresh control signal is received in operation 1102. In one optional embodiment, such refresh control signal may, for example, be received from a memory controller, where such memory controller intends to refresh a simulated memory circuit(s).

[0110] In response to the receipt of such refresh control signal, a plurality of refresh control signals are sent to a plurality of the memory circuits (e.g. see the memory circuits 104A, 104B, 104N of FIG. 1, etc.), at different times. See operation 1104. Such refresh control signals may or may not each include the refresh control signal of operation 1102 or an instantiation/copy thereof. Of course, in other embodiments, the refresh control signals may each include refresh control signals that are different in at least one aspect (e.g. format, content, etc.).

[0111] During use of still additional embodiments, at least one first refresh control signal may be sent to a first subset (e.g. of one or more) of the memory circuits at a first time and at least one second refresh control signal may be sent to a second subset (e.g. of one or more) of the memory circuits at a second time. Thus, in some embodiments, a single refresh control signal may be sent to a plurality of the memory circuits (e.g. a group of memory circuits, etc.). Further, a plurality of the refresh control signals may be sent to a plurality of the memory circuits. To this end, refresh control signals may be sent individually or to groups of memory circuits, as desired.

[0112] Thus, in still yet additional embodiments, the refresh control signals may be sent after a delay in accordance with a particular timing. In one embodiment, for example, the timing in which the refresh control signals are sent to the memory circuits may be selected to minimize a current draw. This may be accomplished in various embodiments by staggering a plurality of refresh control signals. In still other embodiments, the timing in which the refresh control signals are sent to the memory circuits may be selected to comply with a tRFC parameter associated with each of the memory circuits.

[0113] To this end, in the context of an example involving a plurality of DRAM circuits (e.g. see the embodiments of FIGS. 1-2E, etc.), DRAM circuits of any desired size may receive periodic refresh operations to maintain the integrity

of data therein. A memory controller may initiate refresh operations by issuing refresh control signals to the DRAM circuits with sufficient frequency to prevent any loss of data in the DRAM circuits. After a refresh control signal is issued to a DRAM circuit, a minimum time (e.g. denoted by tRFC) may be required to elapse before another control signal may be issued to that DRAM circuit. The tRFC parameter may therefore increase as the size of the DRAM circuit increases.

[0114] When the buffer chip receives a refresh control signal from the memory controller, it may refresh the smaller DRAM circuits within the span of time specified by the tRFC associated with the emulated DRAM circuit. Since the tRFC of the emulated DRAM circuits is larger than that of the smaller DRAM circuits, it may not be necessary to issue refresh control signals to all of the smaller DRAM circuits simultaneously. Refresh control signals may be issued separately to individual DRAM circuits or may be issued to groups of DRAM circuits, provided that the tRFC requirement of the smaller DRAM circuits is satisfied by the time the tRFC of the emulated DRAM circuits has elapsed. In use, the refreshes may be spaced to minimize the peak current draw of the combination buffer chip and DRAM circuit set during a refresh operation.

[0115] While various embodiments have been described above, it should be understood that they have been presented by way of example only, and not limitation. For example, any of the network elements may employ any of the desired functionality set forth hereinabove. Thus, the breadth and scope of a preferred embodiment should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

What is claimed is:

1. A sub-system, comprising:
an interface circuit in communication with a plurality of memory circuits and a system, the interface circuit operable to interface the memory circuits and the system for simulating at least one memory circuit with at least one aspect that is different from at least one aspect of at least one of the plurality of memory circuits;
wherein the at least one aspect is selected from the group consisting of a signal, a capacity, a timing, and a logical interface.
2. The sub-system of claim 1, wherein the at least one aspect includes the signal.
3. The sub-system of claim 2, wherein the signal is selected from the group consisting of an address signal, a control signal, and a data signal.
4. The sub-system of claim 1, wherein the at least one aspect includes the capacity.
5. The sub-system of claim 4, wherein the capacity includes a memory capacity.
6. The sub-system of claim 5, wherein memory capacity relates to a number of the memory circuits.
7. The sub-system of claim 5, wherein the interface circuit is operable to interface the memory circuits and the system

for simulating at least one memory circuit with a first memory capacity that is greater than a second memory capacity of at least one of the plurality of memory circuits.

8. The sub-system of claim 1, wherein the at least one aspect includes the timing.

9. The sub-system of claim 8, wherein the timing relates to a latency.

10. The sub-system of claim 9, wherein the latency includes a column address strobe (CAS) latency.

11. The sub-system of claim 10, wherein the interface circuit is operable to interface the memory circuits and the system for simulating at least one memory circuit with a first CAS latency that is longer than a second CAS latency of at least one of the plurality of memory circuits.

12. The sub-system of claim 9, wherein the latency includes a row address strobe (RAS) to CAS latency (tRCD).

13. The sub-system of claim 12, wherein the interface circuit is operable to interface the memory circuits and the system for simulating at least one memory circuit with a first tRCD that is longer than a second tRCD of at least one of the plurality of memory circuits.

14. The sub-system of claim 9, wherein the latency includes a row precharge latency (tRP).

15. The sub-system of claim 14, wherein the interface circuit is operable to interface the memory circuits and the system for simulating at least one memory circuit with a first tRP that is longer than a second tRP of at least one of the plurality of memory circuits.

16. The sub-system of claim 9, wherein the latency includes an activate to precharge latency (tRAS).

17. The sub-system of claim 1, wherein the at least one aspect includes the logical interface.

18. The sub-system of claim 1, wherein the memory circuits each include dynamic random access memory (DRAM).

19. A method, comprising:

interfacing a plurality of memory circuits and a system;
and

simulating at least one memory circuit with at least one aspect that is different from at least one aspect of at least one of the plurality of memory circuits;

wherein the at least one aspect is selected from the group consisting of a signal, a capacity, a timing, and a logical interface.

20. A sub-system, comprising:

an interface circuit in communication with a plurality of memory circuits and a system, the interface circuit operable to interface the memory circuits and the system for simulating at least one memory circuit with at least one aspect that is different from at least one aspect of at least one of the plurality of memory circuits;

wherein the interface circuit interfaces a majority of address or control signals of the memory circuits.

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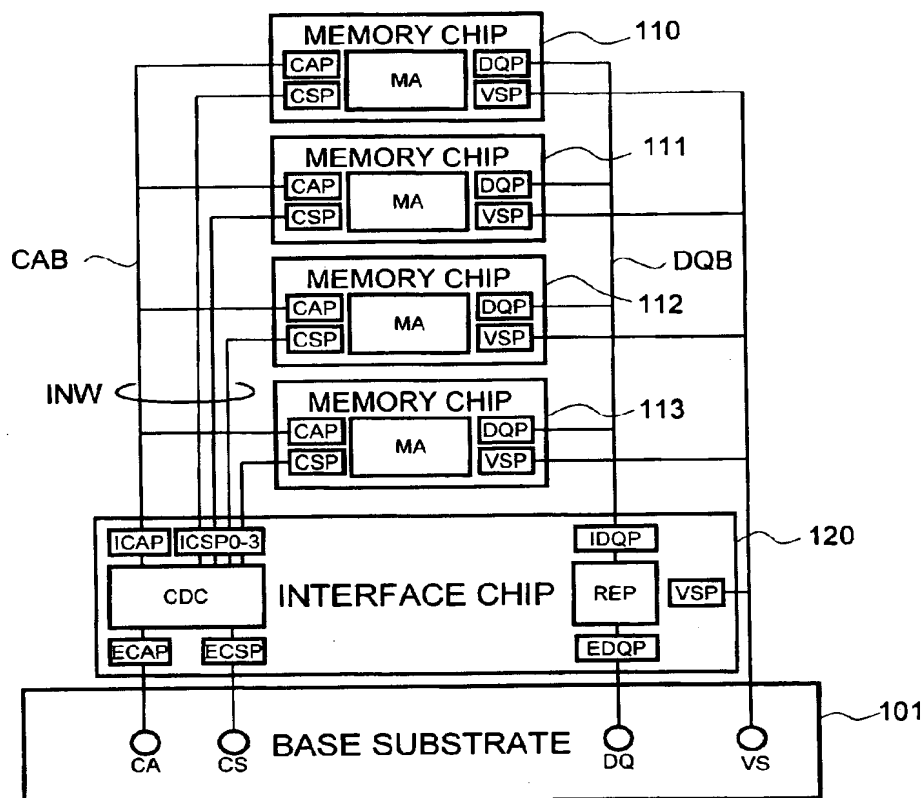
(19) **United States**(12) **Patent Application Publication**
Sekiguchi et al.(10) **Pub. No.: US 2006/0233012 A1**(43) **Pub. Date: Oct. 19, 2006**(54) **SEMICONDUCTOR STORAGE DEVICE
HAVING A PLURALITY OF STACKED
MEMORY CHIPS**(30) **Foreign Application Priority Data**

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Tokyo (JP); Osamu Nagashima, Tokyo
(JP); Mitsunaki Katagiri, Tokyo (JP);
Ichiro Anjo, Tokyo (JP)(51) **Int. Cl.**
GI1C 5/02 (2006.01)(52) **U.S. Cl.** 365/51(57) **ABSTRACT**

A semiconductor storage employs a base substrate (101) having a command/address external terminal group (CA), a data input/output external terminal group (DQ), and a single chip select external terminal (CS), and also comprises a plurality of memory chips (110) to (113) mounted on a base substrate (101), each of which can individually carry out read and write operations. The terminals (CA), (DQ), and (CS) are connected to an interface chip (120). The interface chip (120) has a chip select signal generation circuit that can individually activate a plurality of memory chips (110) to (113) on the basis of an address signal fed by way of the terminal (CA) and on the basis of a chip select signal fed by way of the terminal (CS).

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GROUP(21) **Appl. No.: 11/392,805**(22) **Filed: Mar. 30, 2006****100**

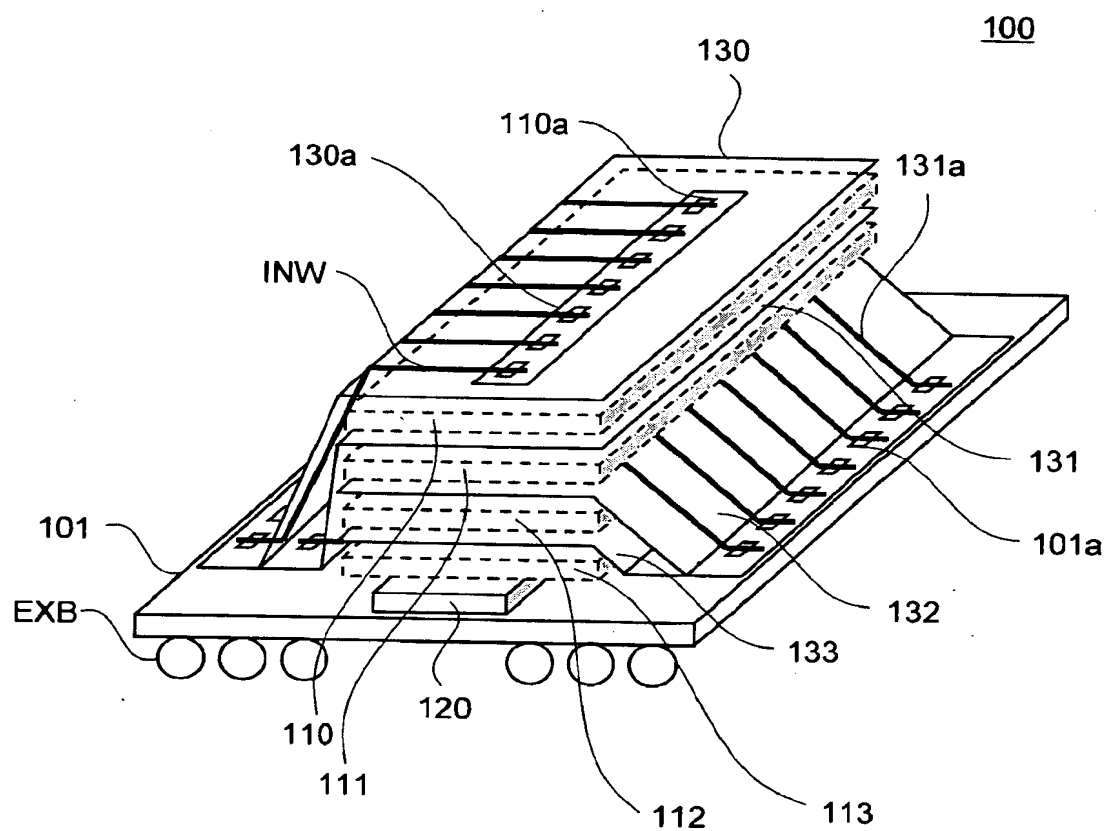


FIG. 1

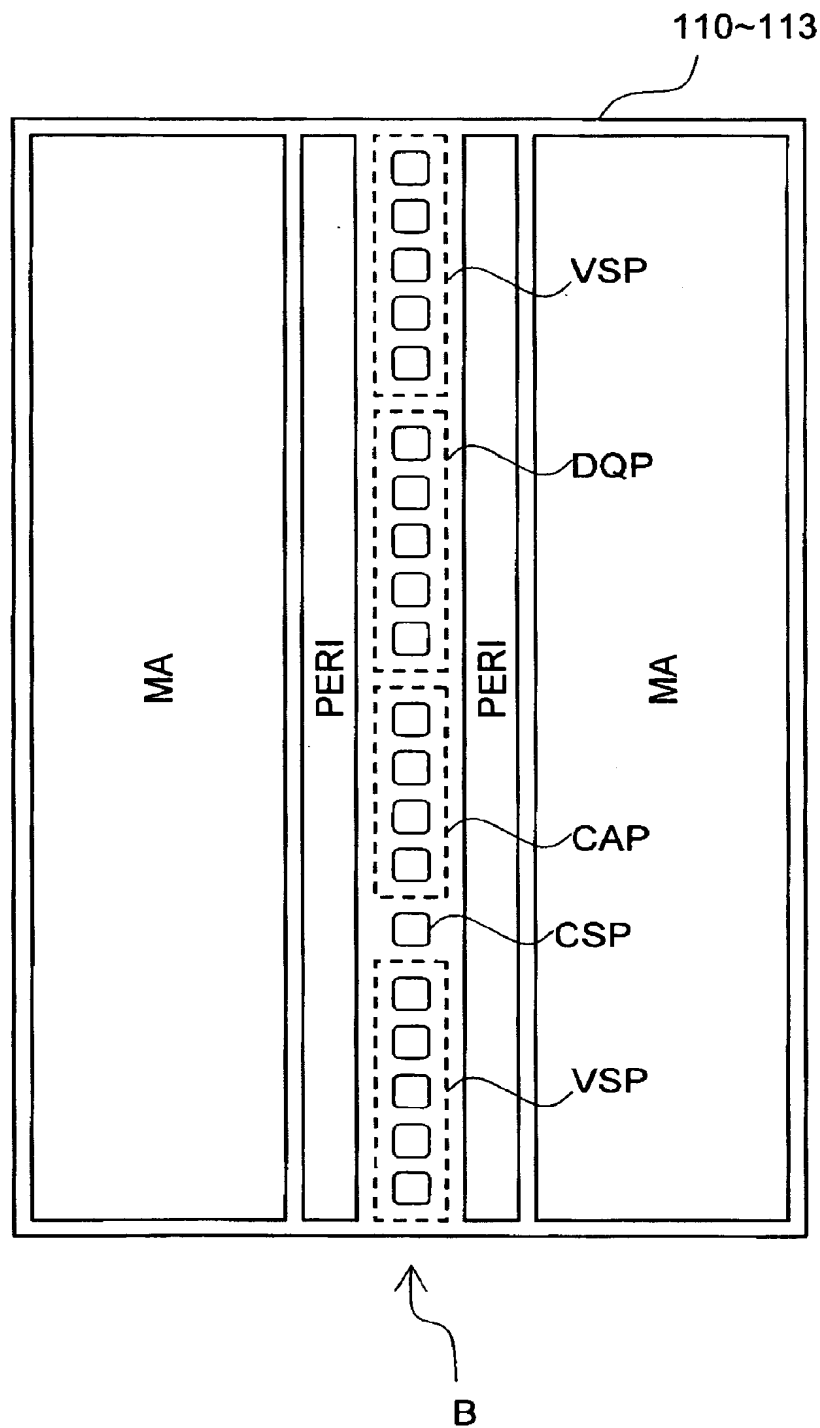


FIG.2

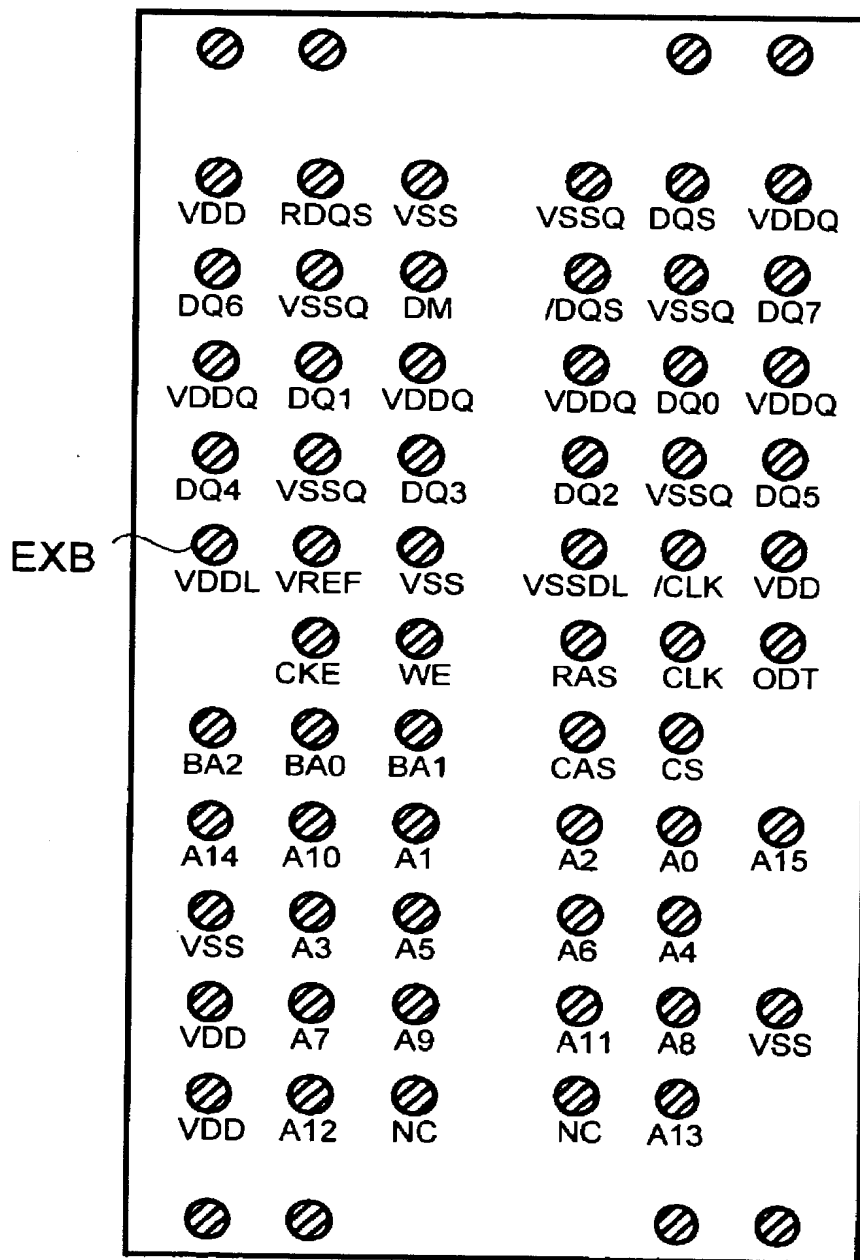


FIG.3

100

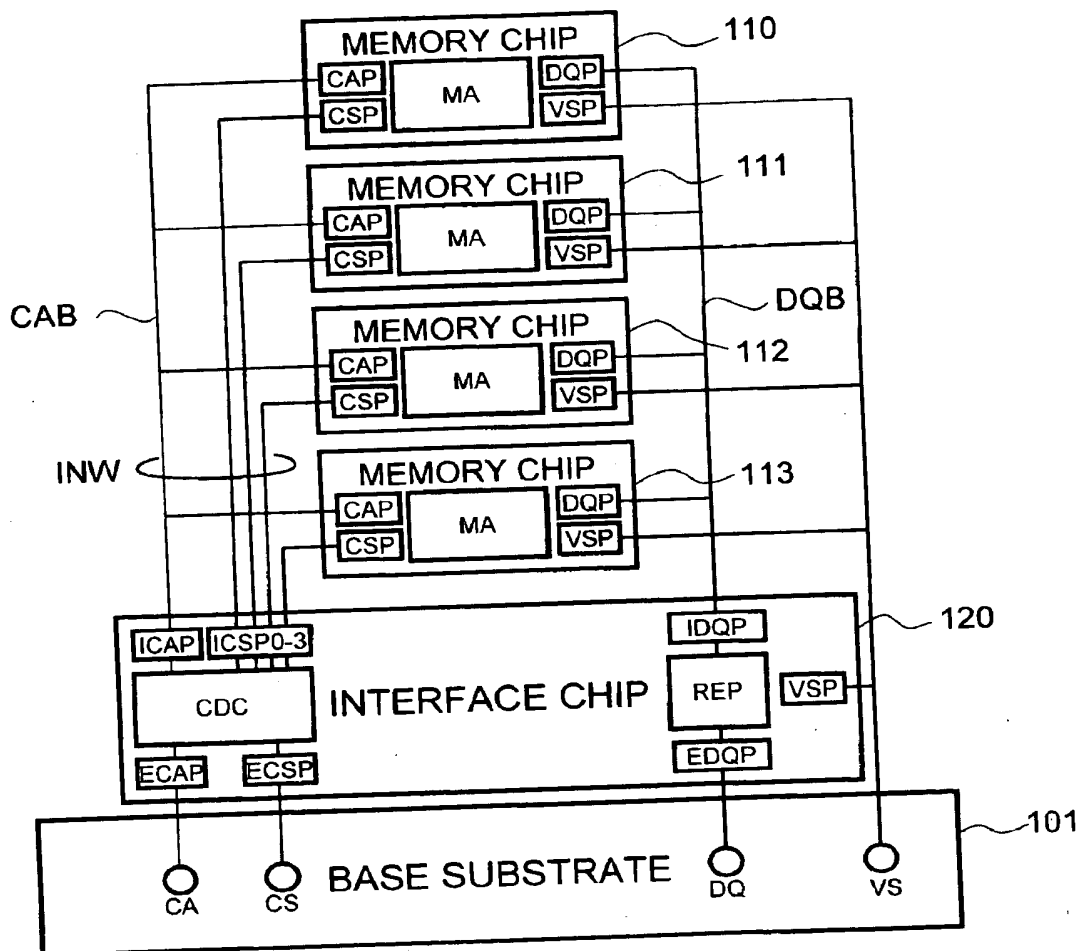


FIG.4

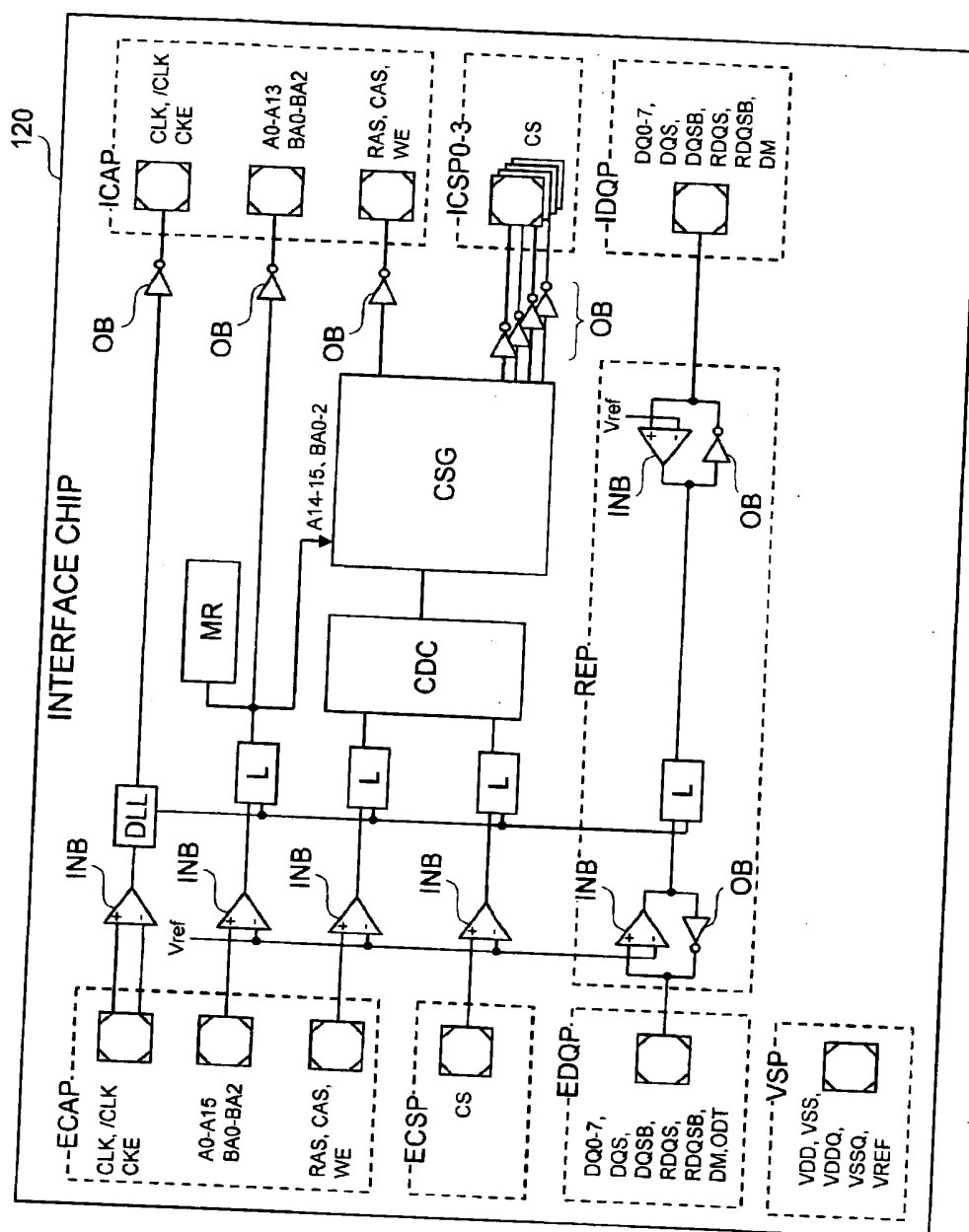


FIG.5

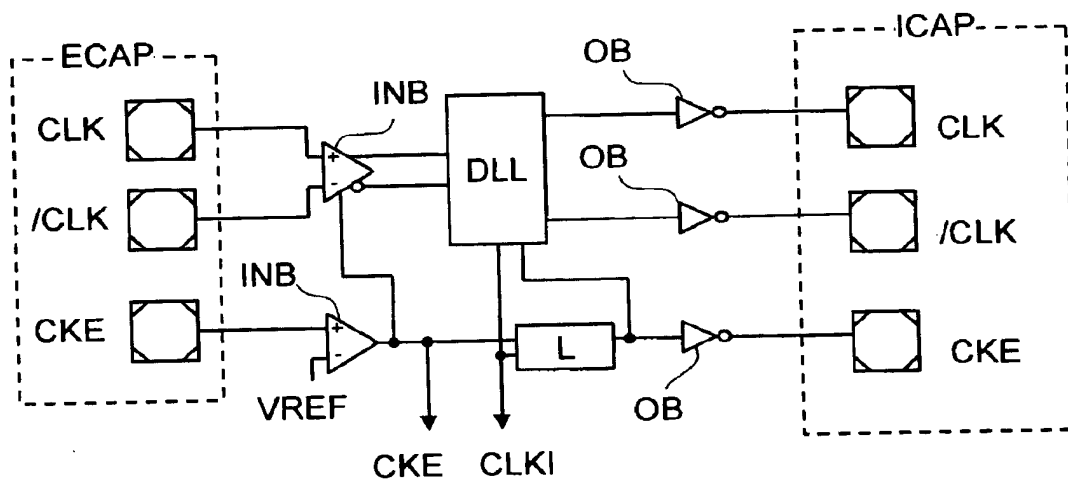


FIG.6

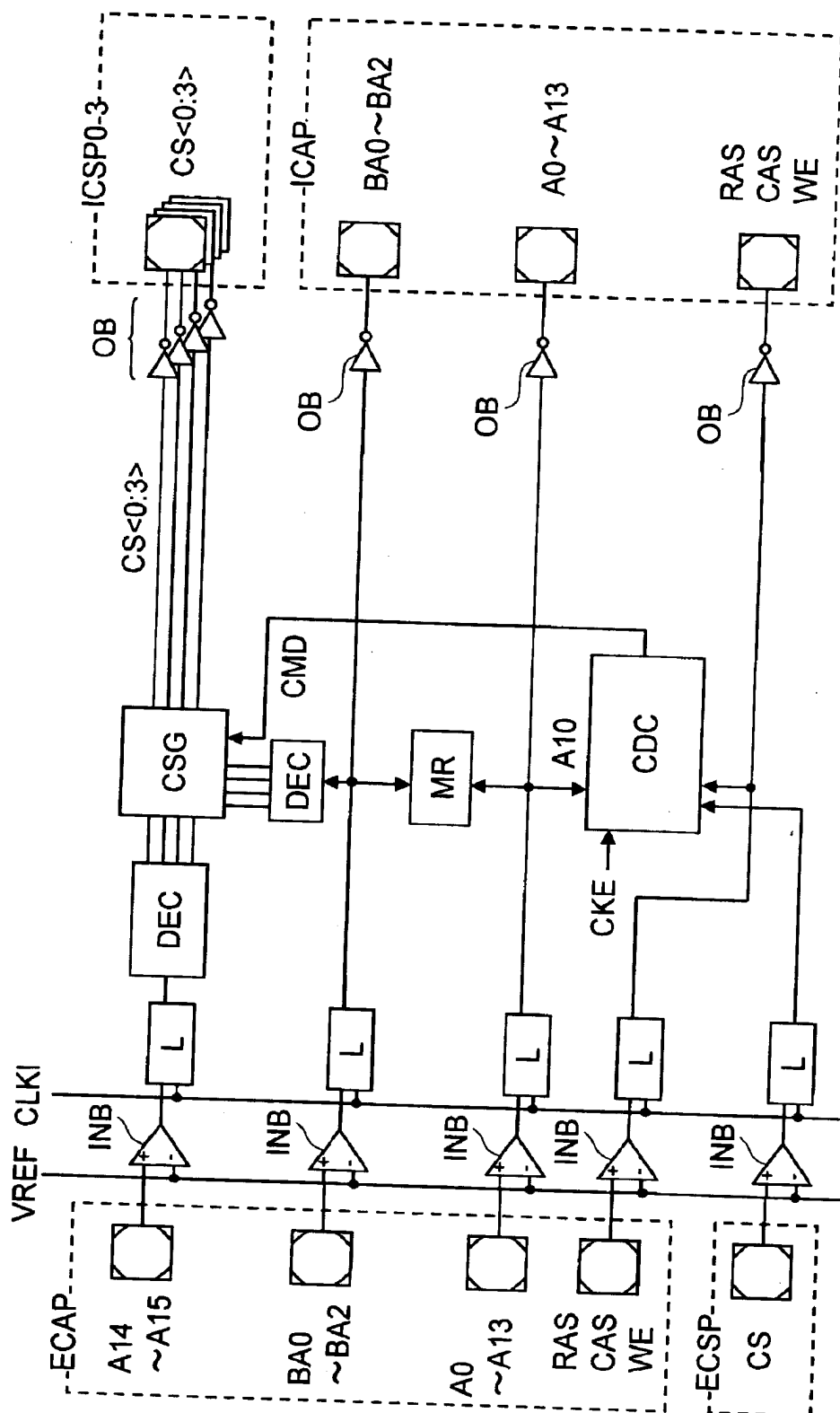


FIG. 7

	COMMAND	SYMBOL	CS PUBLICATION METHOD
INITIALIZATION	Mode register set	MRS	Simultaneously activate all CS
	Extended mode register set	EMRS	
ROW SYSTEM	Auto refresh	REF	Sequentially activate with phase difference
	Self refresh entry	SELF	
	Self refresh exit	SELFEX	Simultaneously activate all CS
	Single bank precharge	PRE	Activate CS stored in the active bank register
	Precharge all banks	PALL	
	Bank activate	ACT	Select CS by decoding A14, 15 and store in active bank register
COLUMN SYSTEM	Write	WRIT	Activate CS stored in the active bank register
	Write with auto precharge	WRITA	
	Read	READ	
	Read with auto precharge	READA	
OTHER	No operation	NOP	Simultaneously activate all CS
	Device deselect	DESL	
	Power down mode entry	PDEN	
	Power down mode exit	PDEX	

FIG.8

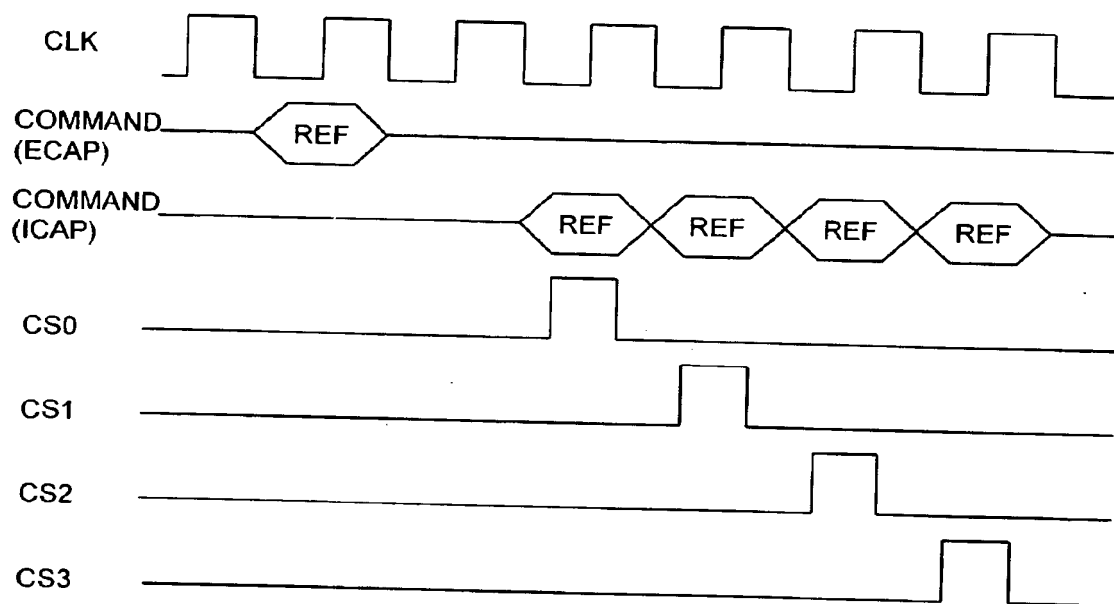


FIG.9

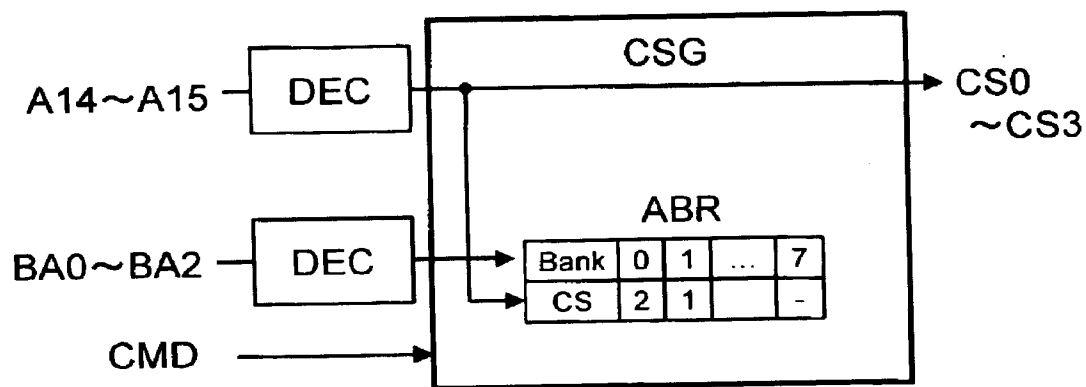


FIG.10A

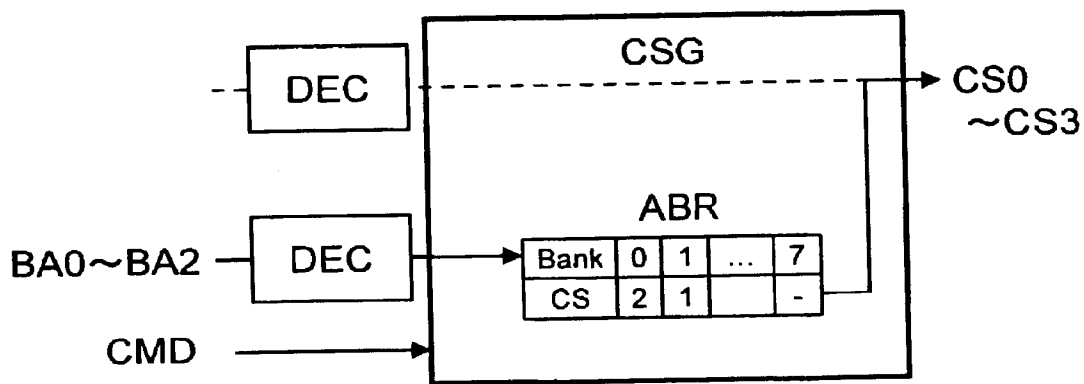


FIG.10B

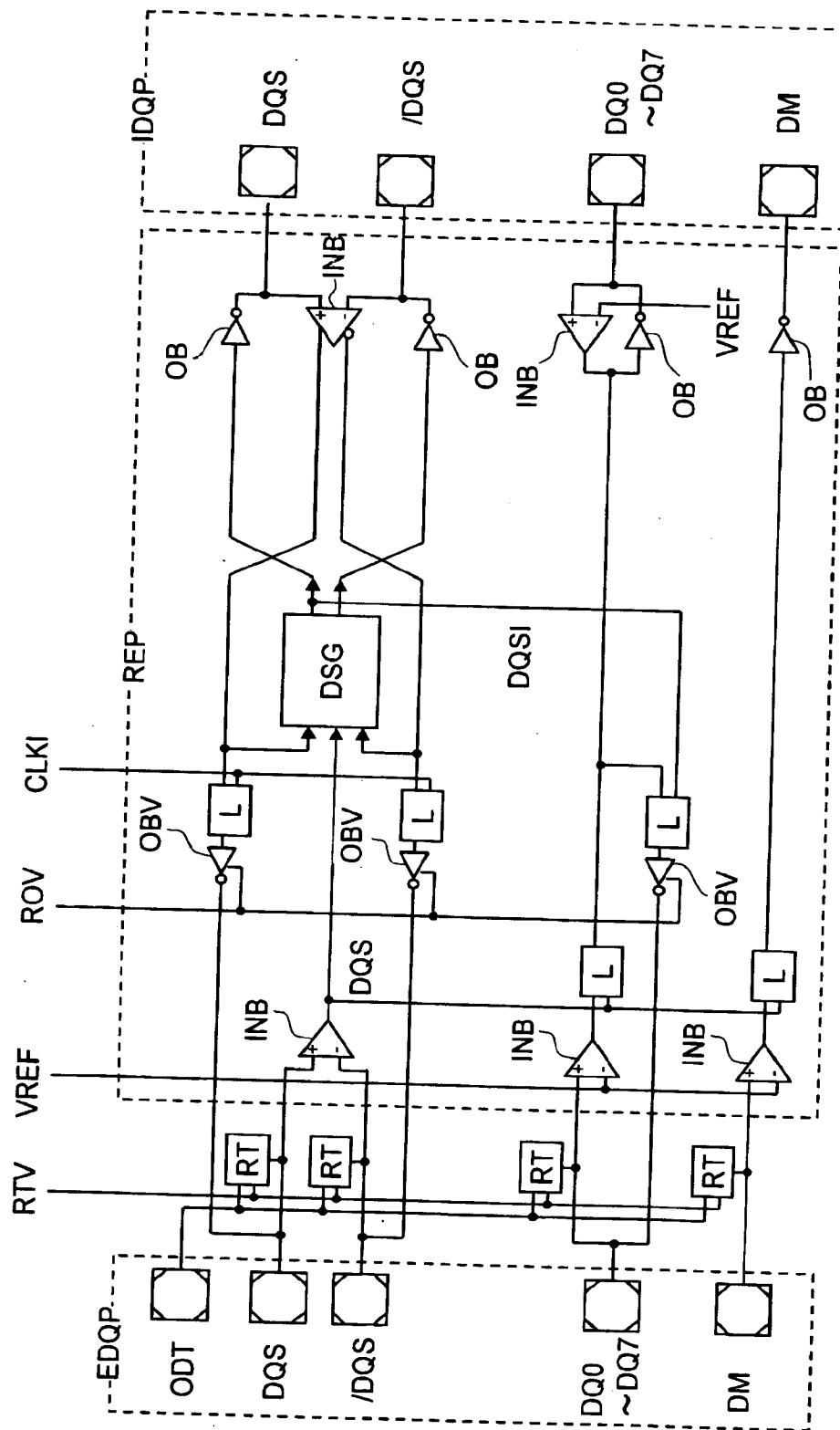


FIG.11

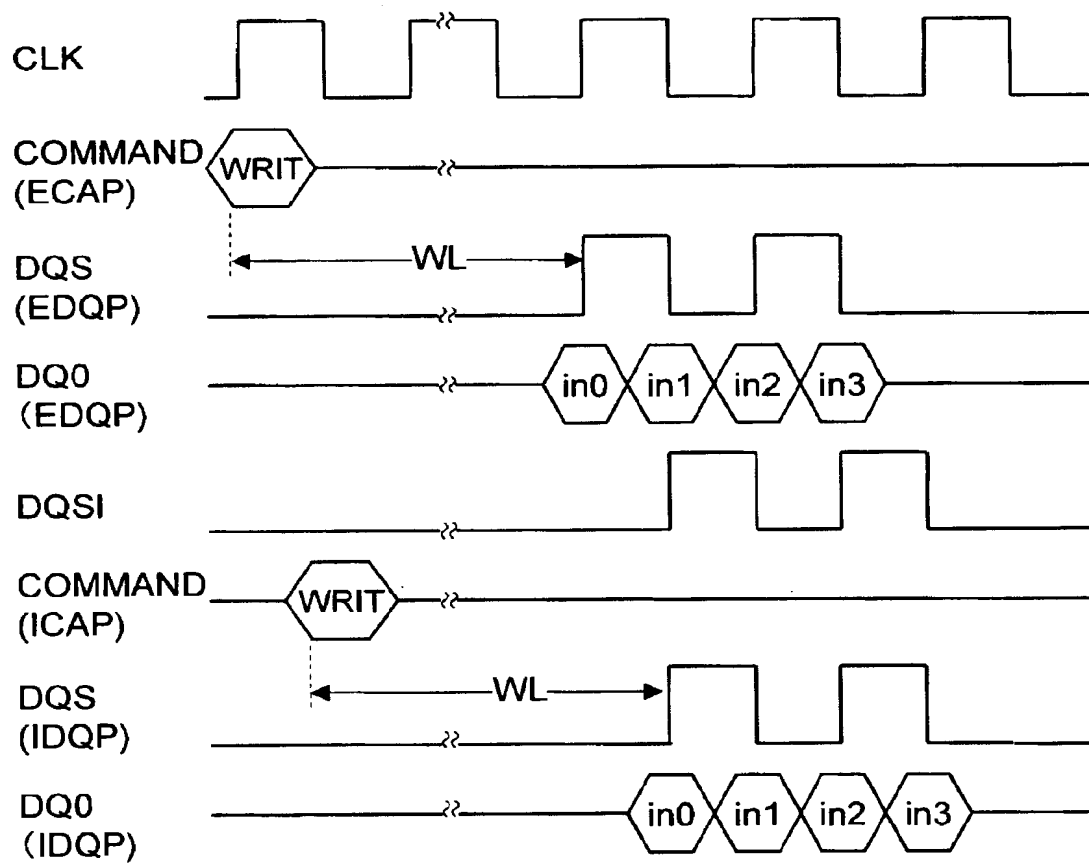


FIG.12

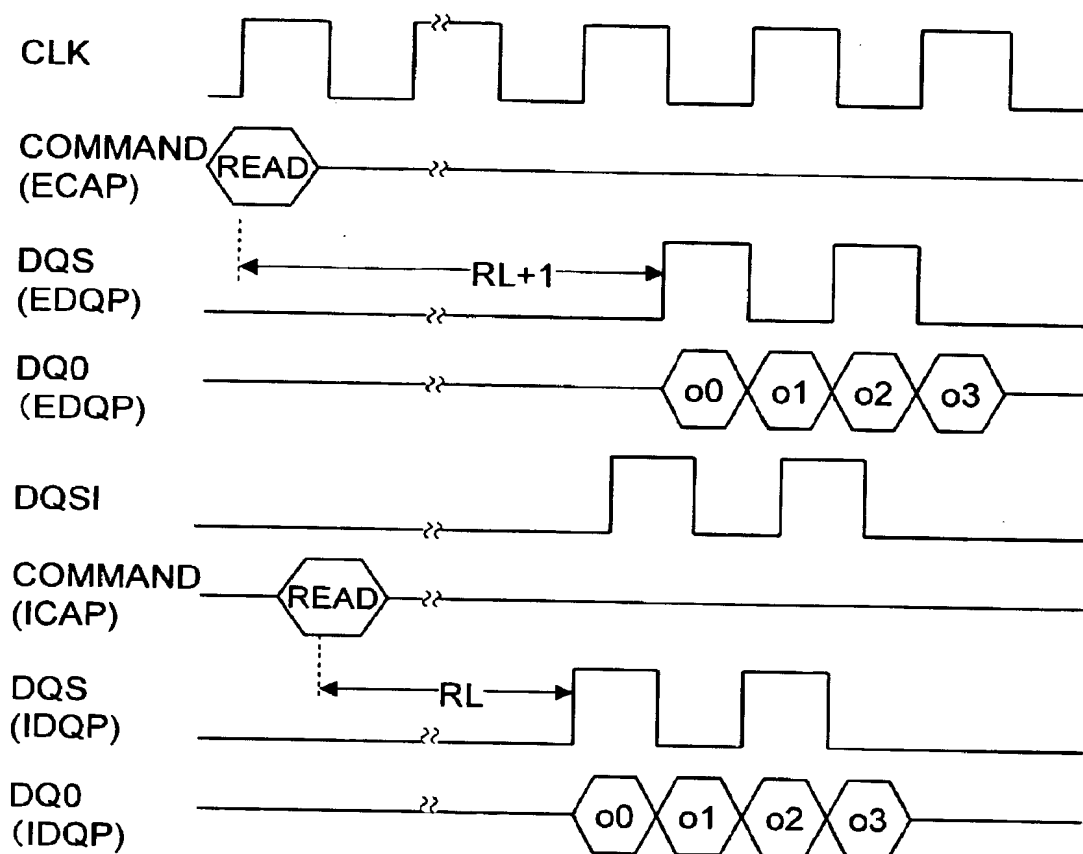


FIG.13

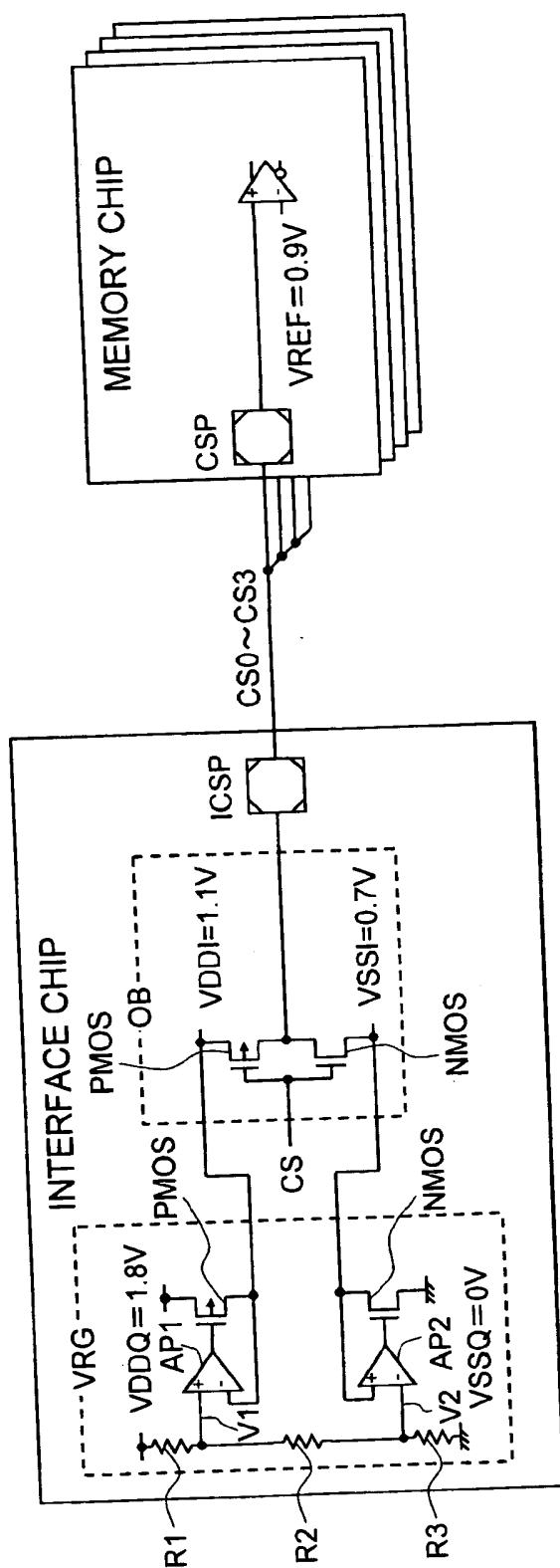


FIG. 14

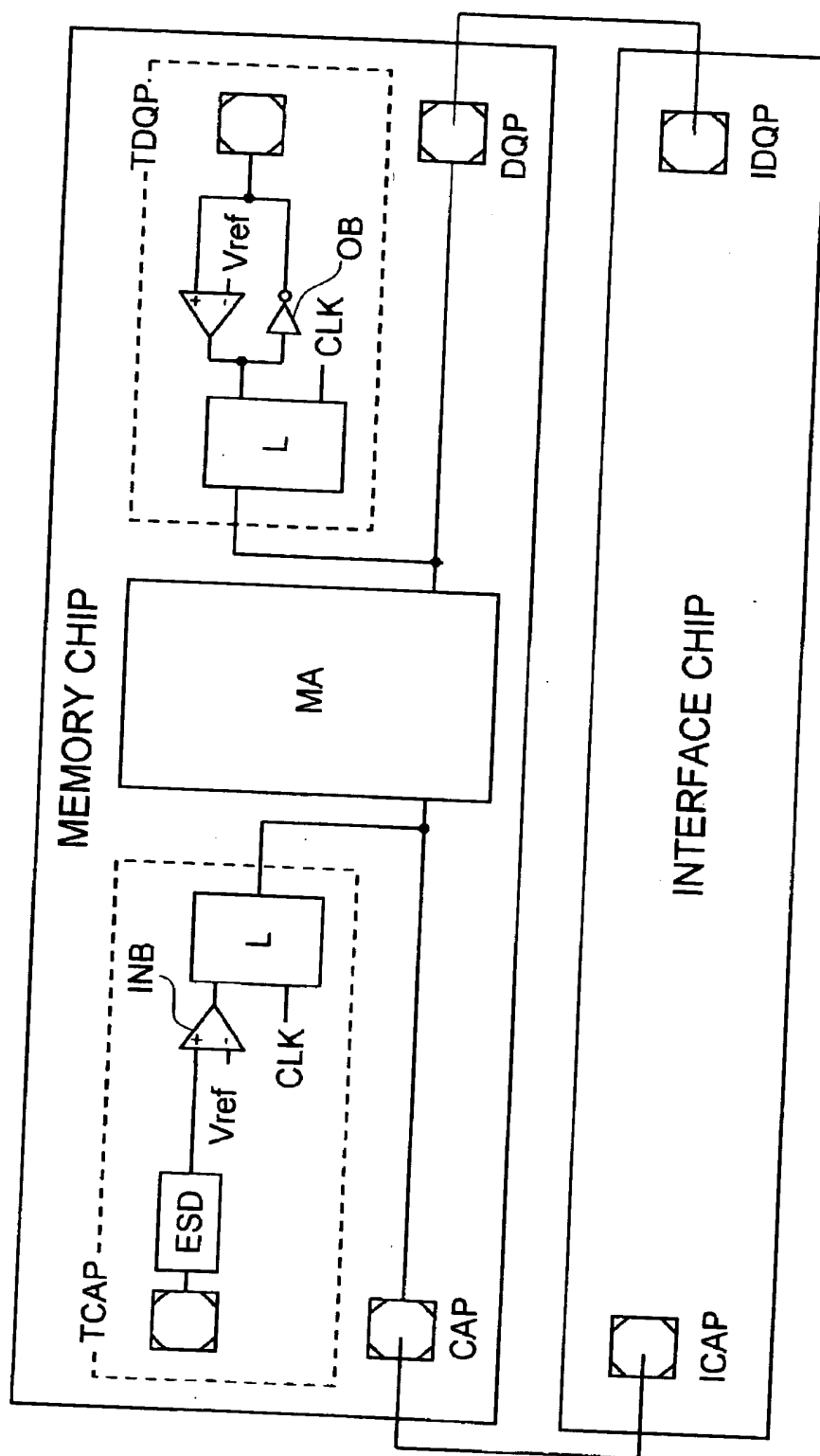


FIG.15

200

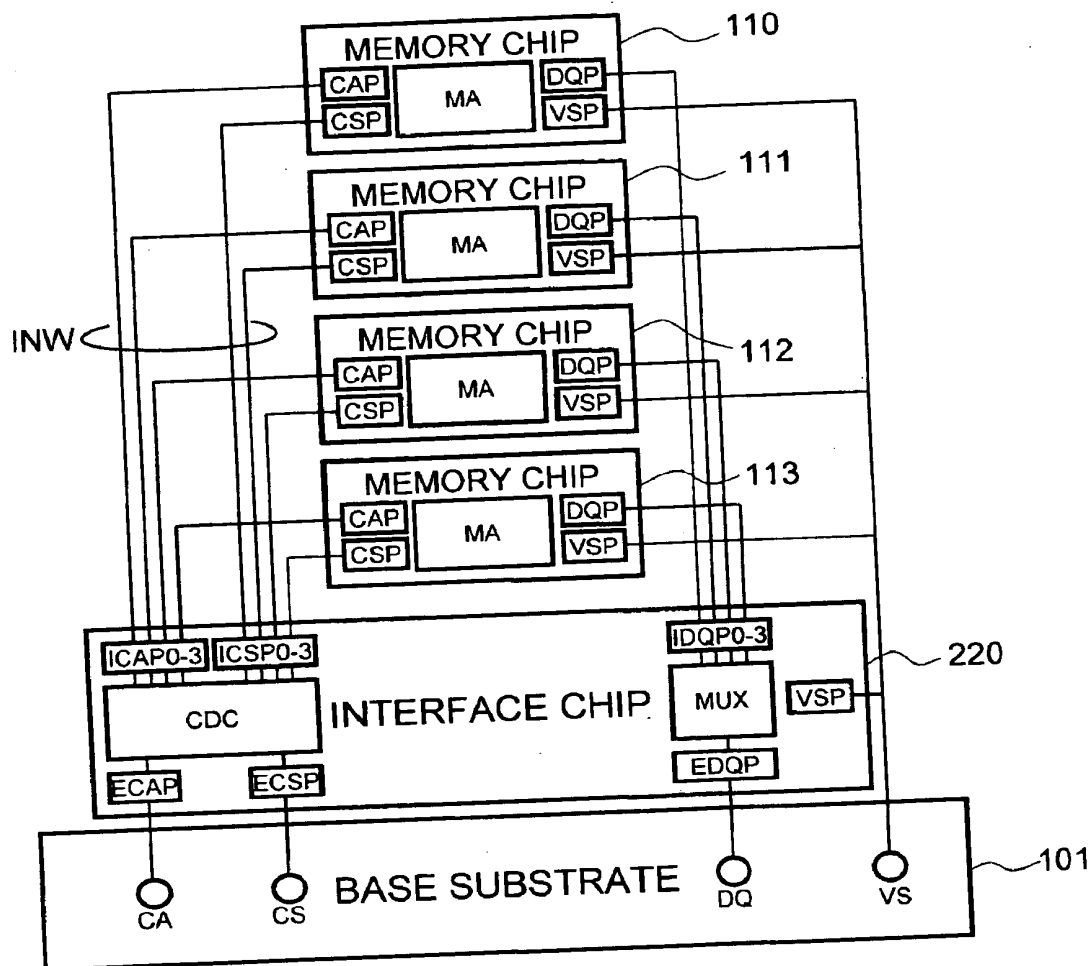


FIG.16

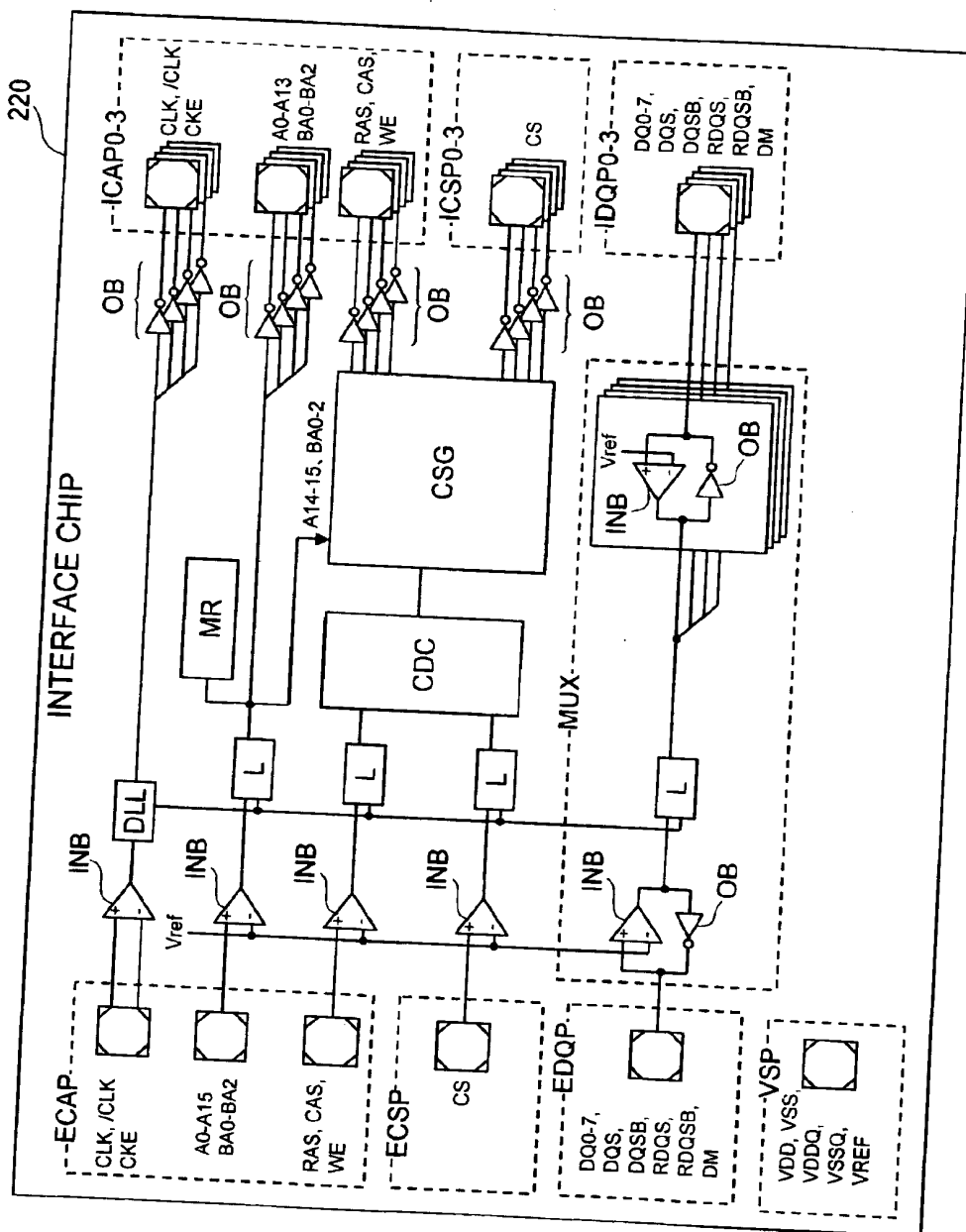


FIG.17

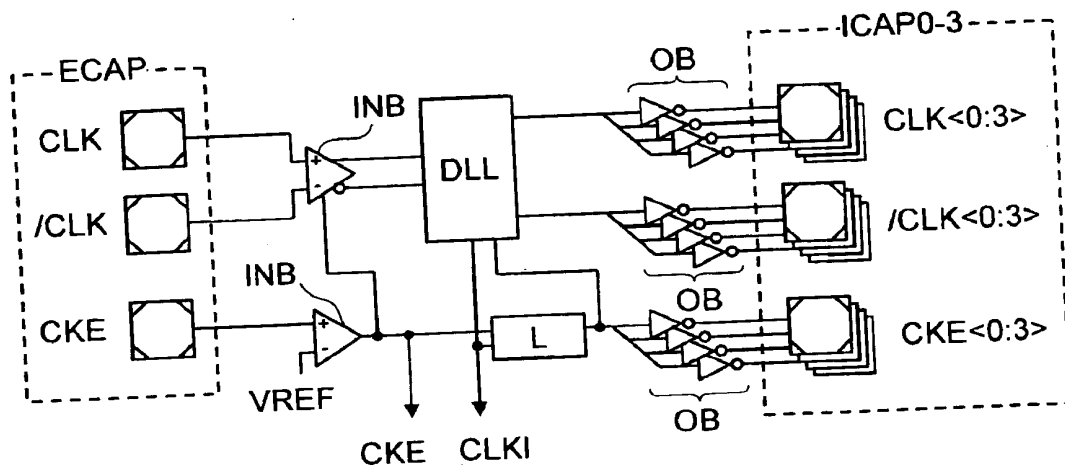


FIG.18

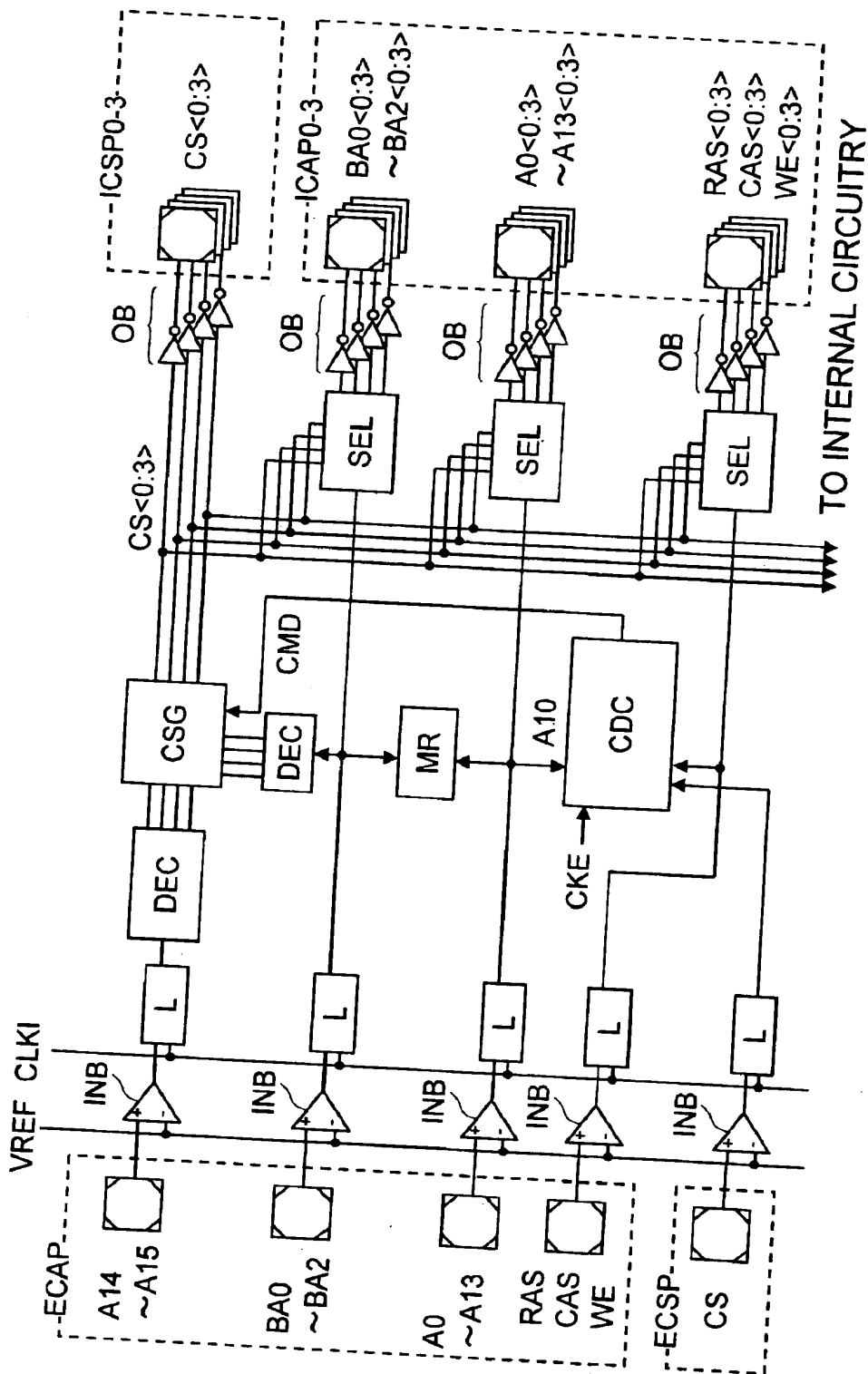


FIG. 19

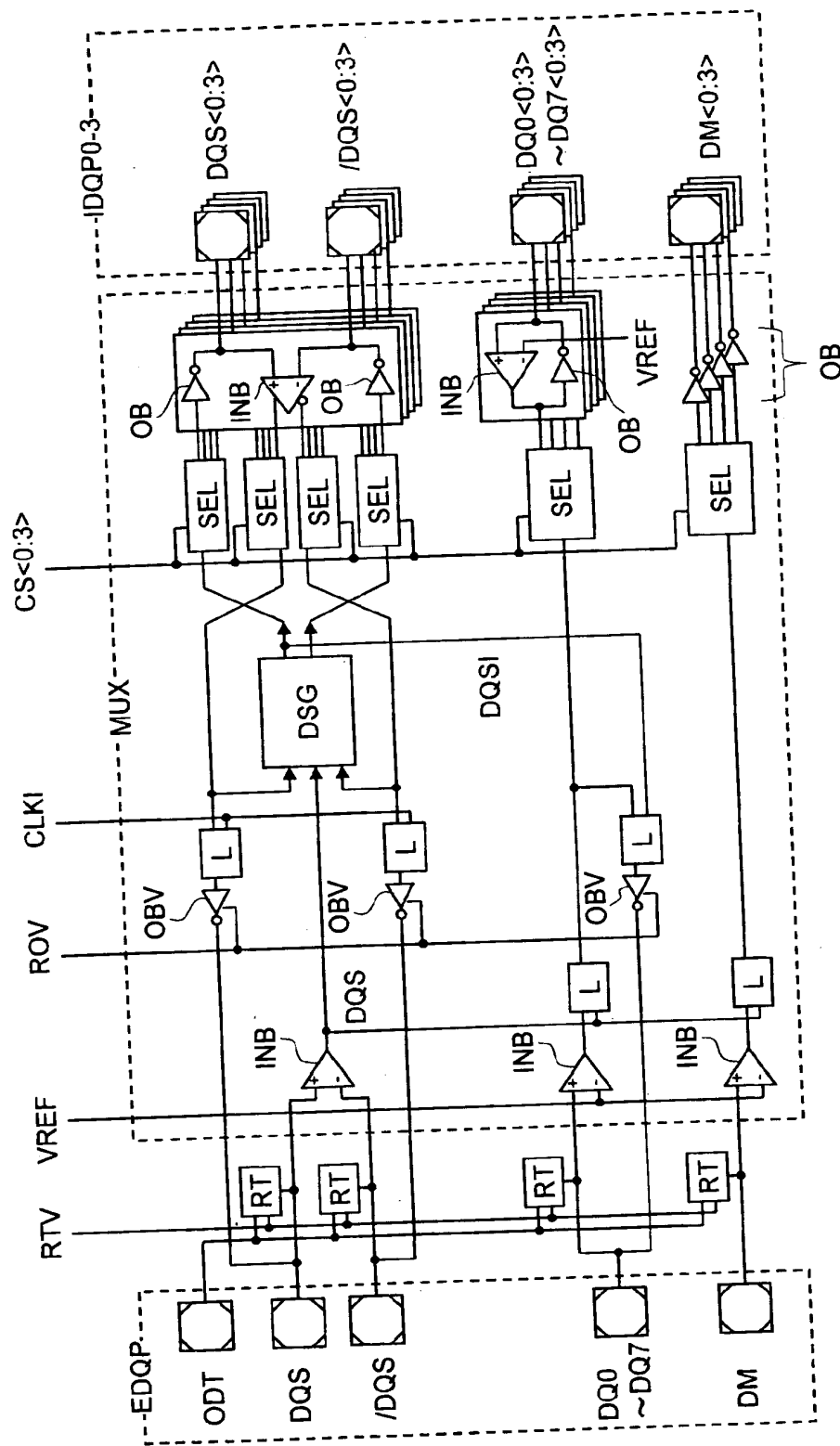


FIG.20

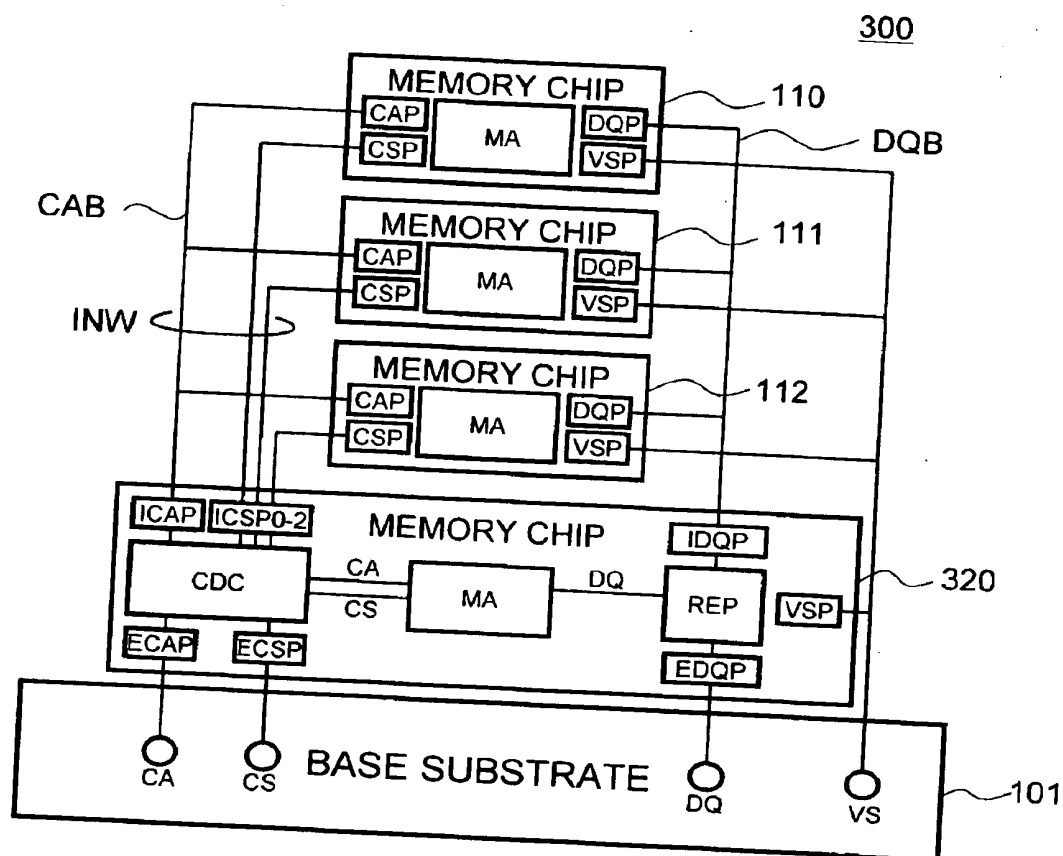


FIG.21

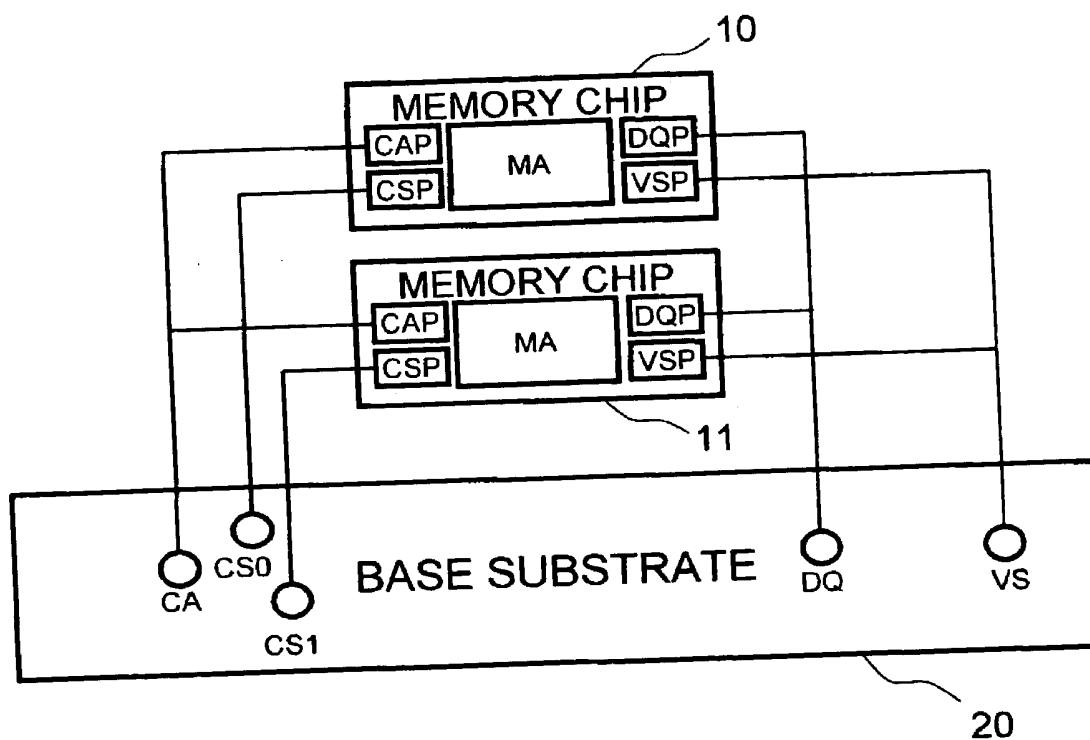


FIG.22

PRIOR ART

SEMICONDUCTOR STORAGE DEVICE HAVING A PLURALITY OF STACKED MEMORY CHIPS

TECHNICAL FIELD

[0001] The present invention relates to a semiconductor storage device, and more specifically relates to a semiconductor storage device that has a plurality of stacked memory chips.

BACKGROUND OF THE INVENTION

[0002] Semiconductor storage devices typified by DRAM (Dynamic Random Access Memory) are widely used as the main memory of servers and other computers, but with the increase in hardware speed and software complexity, the storage capacity required in main memory continues to increase. For this reason, it is difficult to meet the storage capacity needs required in main memory by using only a single memory chip, and a module referred to as DIMM (Dual Inline Memory Module) in which a plurality of memory chips are mounted on a module substrate is ordinarily used.

[0003] However, the number of memory chips that can be mounted is low and the resulting storage capacity is limited when the memory chip is disposed in a flat configuration on a module substrate. Methods have been proposed in which a plurality of memory chips are three-dimensionally stacked and packaged, and the memory chips are mounted on a module substrate. See Japanese Patent Application Laid-Open Nos. H2-290048, 2001-110978, 2001-273755, and 2004-327474.

[0004] FIG. 22 is a schematic diagram of a conventional semiconductor storage device that is three-dimensionally stacked and packaged. The semiconductor storage device shown in FIG. 22 has a structure in which two memory chips 10 and 11 are stacked on the base substrate 20, and the base substrate 20 is mounted on a module substrate (not shown). Each of the memory chips 10 and 11 has a memory array MA, a command/address pad CAP to which commands and addresses are supplied, a chip select signal pad CSP to which a chip select signal is supplied, a data input/output pad DQP for inputting and outputting data, and a power pad VSP for supplying power.

[0005] The command/address pad CAP, data input/output pad DQP, and power pad VSP each have a shared connection with the memory chip 10 and memory chip 11, and are brought together in the command/address external terminal group CA, data input/output external terminal group DQ, and power supply external terminal group VS, respectively, on the base substrate 20. On the contrary, the chip select signal pad CSP does not share a connection, and the chip select signal pad CSP of the memory chip 10 is connected to the chip select signal external terminal CS0, and the chip select signal pad CSP of the memory chip 11 is connected to the chip select signal external terminal CS1. The purpose of this configuration is to allow the memory chips 10 and 11 to be selectively activated.

[0006] Another method has recently been proposed whereby the core unit in which the memory cell is formed and the interface unit in which the peripheral circuit for the memory cell is formed are each placed in a separate chip, and higher storage capacity and higher speeds are made possible by stacking these chips. This means that semicon-

ductor storage devices that are conventionally composed of a single chip are divided into a plurality of chips. This method therefore allows the capacity of the chip (core chip) in which the core unit is formed to be increased, and makes it possible to obtain a very large storage capacity by furthermore stacking the core chips.

[0007] However, since the semiconductor storage device shown in FIG. 22 is configured so that the command/address external terminal group CA and data input/output external terminal group DQ are connected to the memory chips 10 and 11, there is a problem in that the parasitic capacitance of the command/address external terminal group CA and the data input/output external terminal group DQ is considerable in comparison with ordinary semiconductor storage devices in which only a single chip has been packaged.

[0008] Since such parasitic capacitance is the cause of signal waveform disturbance, there is a possibility that an adequate operating margin cannot be assured when the interface speed is very high, that is, when the clock speed exceeds 1 GHz, for example. This problem makes it difficult to make the capacity sufficiently large by using the method shown in FIG. 22 because the problem is made more conspicuous as the number of memory chips to be stacked increases.

[0009] Also, since the semiconductor storage device shown in FIG. 22 is different from an ordinary semiconductor storage device in which only a single chip has been packaged in the sense that two chip select signal external terminals are provided (CS0, CS1) compatibility with ordinary semiconductor storage devices is therefore lost. For this reason, a special-purpose memory controller must be designed in order to use the semiconductor storage device shown in FIG. 22, and the overall cost of the system is liable to increase.

[0010] Using the method described in Japanese Patent Application Laid-Open No. 2004-327474 appears to be able to solve the above-described problems, but this configuration is fundamentally different from the type of semiconductor storage device in which memory chips that are capable of independently carrying out read and write operations are stacked, and is rather a configuration in which core chips that are not capable of independently carrying out read and write operations are stacked, as shown in FIG. 22.

[0011] Therefore, an ordinary memory chip naturally cannot be used and a special-purpose chip must be newly designed. Also, with the configuration described in Japanese Patent Application Laid-Open No. 2004-327474, a through-electrode must be formed in the chips, and other advanced manufacturing techniques that are not sufficiently established must be used. It is therefore believed that manufacturing such chips at low cost under current conditions is difficult.

SUMMARY OF THE INVENTION

[0012] The present invention was contrived in order to solve such problems, and an object of the present invention is to reduce the parasitic capacitance of the command/address external terminal group and the data input/output external terminal group to an amount that is equal to that of a single chip component in a semiconductor storage device having a plurality of stacked memory chips.

[0013] Another object of the present invention is to assure compatibility with ordinary semiconductor storage devices in which a single chip is packaged, in a semiconductor storage device having a plurality of stacked memory chips.

[0014] Yet another object of the present invention is to implement the objects described above with low cost by using memory chips that are each individually capable of read and write operations.

[0015] The above and other objects of the present invention can be accomplished by a semiconductor storage device comprising: a base substrate having a command/address external terminal group to which command signals and address signals are supplied, a data input/output external terminal group for inputting and outputting data signals, and a single chip select external terminal; and a plurality of memory chips that are stacked on the base substrate and are each individually capable of reading and writing operations, wherein a plurality of terminals constituting said command/address external terminal group, a plurality of terminals constituting said data input/output external terminal group, and said single chip select external terminal are connected to a single chip that has an interface function; and the single chip that has the interface function further has at least a chip select signal generation circuit that can individually activate said plurality of memory chips based on said address signals supplied via said command/address external terminal group and based on said chip select signal supplied via said chip select external terminal.

[0016] In accordance with the present invention, since the command/address external terminal group and other external terminals are connected to a single chip, the parasitic capacitance of the terminals can be reduced to an amount that is about equal to that of a single chip component. Also, since the semiconductor storage device of the present invention is provided with only a single chip select external terminal, it is possible to assure compatibility with ordinary semiconductor storage devices in which a single chip is packaged, even though the semiconductor storage device is provided with a plurality of stacked memory chips.

[0017] In addition, the stacked memory chips are memory chips that are individually capable of read and write operations. It is therefore possible to use so-called general-purpose memory, and hence to manufacture these chips at relatively low cost.

[0018] In the present invention, the single chip that has an interface function may be an interface chip that is different than the plurality of memory chips, or may be any one of the plurality of memory chips. With the former, it is possible to use the exact same chip as the memory chip that is used, and with the latter, it is possible to reduce the number of stacked chips by one.

[0019] The command signals, address signals, and data signals may be transmitted among the plurality of memory chips and the single chip that has an interface function by way of wiring with a shared connection to the plurality of memory chips, or may be carried out by way of wiring that is individually connected to the plurality of memory chips. The former allows the number of wires to be reduced, and the latter allows the wiring capacitance to be reduced.

[0020] Thus, in accordance with the present invention, it is possible to reduce the parasitic capacitance of the interface

unit of a semiconductor storage device having a plurality of stacked memory chips to an amount equal to that of a single chip component. The input/output frequency of the command signal, address signal, and data signal can thereby be increased to that equivalent to a single chip component. Also, since the memory chips are three-dimensionally stacked in the present invention, the mounting surface area can be made equal to that of a single chip component.

[0021] Furthermore, since only a single chip select external terminal is provided, it is possible to assure compatibility with ordinary semiconductor storage devices in which a single chip is packaged. For this reason, there is no requirement that a special-purpose memory controller or the like be designed, and cost increases for the entire system can be held in check.

[0022] Since the stacked memory chips are furthermore ordinary memory chips (general-purpose DRAM, for example) that are each individually capable of read and write operations, formation of a through-electrode or other special steps are not required, and an increase in manufacturing costs can be inhibited.

[0023] Since it is also possible to individually activate the plurality of memory chips by using a chip select signal generation circuit, the power consumption of the entire package can also be reduced.

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] The above and other objects, features and advantages of this invention will become more apparent by reference to the following detailed description of the invention taken in conjunction with the accompanying drawings, wherein:

[0025] FIG. 1 is a simplified perspective view that schematically shows a structure of the semiconductor storage device according to the first embodiment of the present invention;

[0026] FIG. 2 is a simplified planar view of the memory chips shown in FIG. 1;

[0027] FIG. 3 is a layout drawing of ball electrodes disposed on the reverse side of the base substrate shown in FIG. 1;

[0028] FIG. 4 is a schematic diagram that describes the connection configuration between the memory chips and the interface chip in the first embodiment;

[0029] FIG. 5 is a simplified circuit diagram showing the circuit configuration of the interface chip used in the first embodiment;

[0030] FIG. 6 is a diagram showing in greater detail the circuit configuration of the clock system circuitry contained in the interface chip shown in FIG. 5;

[0031] FIG. 7 is a diagram showing in greater detail the circuit configuration of the command/address system circuitry contained in the interface chip shown in FIG. 5;

[0032] FIG. 8 is a list showing the commands and symbols that are output by the command decoder shown in FIG. 7, and the methods for issuing a chip select signal CS when the commands are executed;

[0033] FIG. 9 is a timing chart showing the issue timing of the refresh command to each memory chip;

[0034] FIG. 10A is a diagram for describing the function of the active bank register in the case where an activation command is issued;

[0035] FIG. 10B is a diagram for describing the function of the active bank register in the case where a column system command is issued;

[0036] FIG. 11 is a diagram showing in greater detail the circuit configuration of the principal components of the data system circuitry contained in the interface chip shown in FIG. 5;

[0037] FIG. 12 is a timing chart showing the write operation of the semiconductor storage device according to the first embodiment;

[0038] FIG. 13 is a timing chart showing the read operation of the semiconductor storage device according to the first embodiment;

[0039] FIG. 14 is a schematic diagram that describes the method for reducing the amplitude of the inter-chip interface between the memory chips and the interface chip;

[0040] FIG. 15 is a diagram for describing an improved method for connecting the pads and the memory array in the memory chips;

[0041] FIG. 16 is a schematic diagram of the connection configuration between the memory chips and the interface chip of the second embodiment;

[0042] FIG. 17 is a simplified circuit diagram showing the circuit configuration of the interface chip shown in FIG. 16 used in the second embodiment;

[0043] FIG. 18 is a diagram showing in greater detail the circuit configuration of the clock system circuit contained in the interface chip shown in FIG. 17;

[0044] FIG. 19 is a diagram showing in greater detail the circuit configuration of the command/address system circuit contained in the interface chip shown in FIG. 17;

[0045] FIG. 20 is a diagram showing in greater detail the circuit configuration of the data system circuit contained in the interface chip shown in FIG. 17;

[0046] FIG. 21 is a schematic diagram that describes the connection configuration between the memory chips and the memory chip in the semiconductor storage device of the third embodiment of the present invention; and

[0047] FIG. 22 is a schematic diagram of a conventional semiconductor storage device that is three-dimensionally stacked and packaged.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0048] Preferred embodiments of the present invention will now be explained in detail with reference to the drawings.

[0049] FIG. 1 is a simplified perspective view that schematically shows the structure of the semiconductor storage device 100 according to the first embodiment of the present invention.

[0050] The semiconductor storage device 100 of the present embodiment is provided with a base substrate 101, four memory chips 110 to 113 stacked on the base substrate 101, and an interface chip 120 disposed between the base substrate 101 and memory chips 110 to 113, as shown in FIG. 1. In the present embodiment, DRAM is selected as the memory chips 110 to 113, although this selection imposes no particular limitation. The memory chips 110 to 113 are chips having mutually the same configuration, and are DRAM chips that are individually capable of read and write operations, that is to say, the chips are ordinary general-purpose DRAM chips.

[0051] FIG. 2 is a simplified planar view of the memory chips 110 to 113.

[0052] The surface of the memory chips 110 to 113 can be largely classified into a memory array area MA, a peripheral circuit area PERI, and a bonding pad area B, as shown in FIG. 2. The bonding pad area B is disposed in a single direction in the center of the memory chip so as to be disposed between the two peripheral circuit areas PERI, and this area includes power supply pads VSP, a data input/output pad DQP, a command/address pad CAP, and a chip select signal pad CSP.

[0053] The power supply pads VSP have a peripheral circuit operating power supply VDD, a peripheral circuit ground power supply VSS, an output circuit operating power supply VDDQ, an output circuit ground power supply VSSQ, an interface reference power supply VREF, and the like. The data input/output pad DQP has data input/output pads DQ0 to DQ7 (the same applies to the case of x8 components) a data strobe signal pad DQS, a data mask signal pad DM, and the like. The command/address pad CAP has a clock pad CLK, a clock enable pad CKE, a row address strobe pad RAS, a column address strobe pad CAS, a write enable pad WE, a bank address pads BA0 to BA2, an address pads A0 to A13, and the like.

[0054] The pads disposed on the memory chips 110 to 113 are connected to the pads 101a disposed on the base substrate 101 by way of tapes 130 to 133 on which wiring INW is formed, as shown in FIG. 1. More specifically, pads 110a disposed on the memory chip 110 and pads 101a disposed on the base substrate 101 are connected by tape 130 that is disposed so as to cover the memory chip 110, and pads disposed on the memory chips 111 to 113 and the pads 101a disposed on the base substrate 101 are similarly connected by tapes 131 to 133 that are disposed so as to cover the memory chips 111 to 113, respectively.

[0055] Wiring (not shown) is formed on the base substrate 101, and the pads 101a and the pads (not shown) on the memory chip side of interface chip 120 are connected by the wiring. Pads (not shown) on the external circuit side of the interface chip 120 are connected to ball electrodes EXB disposed on the reverse side (mounting side) of the base substrate 101 by way of other wiring (not shown) disposed on the base substrate 101. The ball electrodes EXB are external terminals of the semiconductor storage device 100 of the present embodiment.

[0056] FIG. 3 is a layout drawing of ball electrodes EXB disposed on the reverse side of the base substrate 101.

[0057] Other than the addition of address external terminals A14 and A15, the types of ball electrodes EXB (which

is an external terminal group of the semiconductor storage device 100) match the types of pads disposed on the memory chips 110 to 113, as shown in FIG. 3. Therefore, the chip select signal external terminal CS is always a single terminal. The ball electrodes EXB can be classified into a power supply external terminal group (VS), a data input/output external terminal group (DQ), a command/address external terminal group (CA), and a chip select signal external terminal (CS).

[0058] The peripheral circuit operating power supply VDD, peripheral circuit ground power supply VSS, output circuit operating power supply VDDQ, output circuit ground power supply VSSQ, interface reference power supply VREF, and the like constitute the power external terminal group (VS) in the same manner as the pads disposed on the memory chips 110 to 113. The data input/output terminals DQ0 to DQ7, data strobe signal terminal DQS, data mask signal terminal DM, and the like constitute the data input/output external terminal group (DQ). The clock terminal CLK, clock enable terminal CKE, row address strobe terminal RAS, column address strobe terminal CAS, write enable terminal WE, bank address terminals BA0 to BA2, address terminals A0 to A13, and the like constitute the command/address external terminal group (CA).

[0059] FIG. 4 is a schematic diagram that describes the connection configuration between the memory chips 110 to 113 and the interface chip 120 in the present embodiment.

[0060] In the present embodiment, only the chip select signals are connected in a one-to-one relationship between the memory chips 110 to 113 and the interface chip 120, as shown in FIG. 4. The other signals, that is to say, the command/address signals and data signals, are bus-connected by the command/address bus CAB and the data bus DQB. The power supply is also bus-connected.

[0061] As used herein, the term "one-to-one connection" refers to a connection in which the pads of the memory chips 110 to 113 and the pads of the interface chip 120 are individually connected for each memory chip, and the term "bus-connected" refers to a connection in which the pads of the memory chips 110 to 113 and the pads of the interface chip 120 share connections. Therefore, in the present embodiment, in the set of signals transmitted between the memory chips 110 to 113 and the interface chip 120, only the chip select signal is a signal that is individually provided to the memory chips 110 to 113.

[0062] Described more specifically, the command/address external terminal group CA and the chip select signal external terminal CS are connected to the command/address pad ECAP and the chip select signal pad ECSP, respectively, on the interface chip 120 by way of the ball electrodes EXB on the base substrate 101. These signals are supplied to the command decoder CDC, and the output signals of the decoder are supplied to the command/address pad ICAP and chip select signal pad ICSP, respectively.

[0063] The signals (commands and addresses) that are output to the command/address pad ICAP are supplied in a shared configuration to the command/address pads CAP of the four memory chips 110 to 113 by way of the command/address bus CAB. On the other hand, four chip select signals that are output to the chip select signal pad ICSP are connected in a one-to-one relationship between the interface chip 120 and each of the memory chips 110 to 113.

[0064] The data input/output external terminal group DQ is connected once to the data input/output pad EDQP on the interface chip 120 by way of the ball electrodes EXB on the base substrate 101. The data input/output signals are input to a bidirectional repeater REP, and are connected to the data input/output pads DQP of the memory chips 110 to 113 by way of the data input/output pad IDQP. The data input/output pad IDQP and the data input/output pads DQP of the four memory chips 110 to 113 are connected by a data bus DQB in the form of a 1:4 bus.

[0065] The power supply external terminal group VS is connected to the power supply pads VSP of the memory chips 110 to 113 and to the interface chip 120 directly by way of the ball electrodes EXB on the base substrate 101.

[0066] Thus, in the semiconductor storage device 100 of the present embodiment, the connections between the ball electrodes EXB on the base substrate 101 and the memory chips 110 to 113 (excluding the power supply) are all carried out by way of the interface chip 120. The chip connected to the data input/output external terminal group DQ and the command/address external terminal group CA is only the interface chip 120. Thus, the parasitic capacitance of the external terminals can be reduced to an amount equal to that of a single chip component, and the frequency of the command/address and data input/output can be increased to a level equivalent to that of a single chip component.

[0067] In the present embodiment, since the command/address signals and data input/output signals are connected in the form of a bus inside a package, the number of wires inside the package can be kept low and manufacturing costs can thereby be held in check.

[0068] FIG. 5 is a simplified circuit diagram showing the circuit configuration of the interface chip 120 used in the present embodiment.

[0069] The command/address pad ECAP on the interface chip 120 has pads for three systems: the clock system signals CLK, /CLK, and CKE; address system signals A0 to A15 and BA0 to BA2; and command system signals RAS, CAS, and WE, as shown in FIG. 5. The clock system signals CLK, /CLK, and CKE are supplied from outside of the semiconductor storage device 100 and are supplied to a clock regeneration circuit DLL by way of an input buffer INB. The output of the clock regeneration circuit DLL is supplied to terminals that output, respectively, the clock signals CLK, /CLK, and CKE of the command/address pad ICAP by way of the output buffer OB. The output of the clock regeneration circuit DLL is distributed within the interface chip 120 in order to latch other input signals in the latch circuits L.

[0070] The address system signals A0 to A15 and BA0 to BA2 are latched in the latch circuits L by way of the input buffers INB. The timing signals of the latch circuits L are supplied by the clock regeneration circuit DLL described above. The output of the latch circuits L is supplied to the terminals that output the address signals A0 to A13 and BA0 to BA2 of the command/address pad ICAP by way of the output buffer OB.

[0071] Here, the address signals A0 to A13 and BA0 to BA2 of the address signals A0 to A15 and BA0 to BA2 that are supplied from the exterior are directly fed to the command/address pad ICAP. The address signals A14 and A15 are not supplied to the command/address pad ICAP, but are

rather fed to the chip select signal generation circuit CSG together with the output of the mode register MR and the bank addresses BA0 to BA2. The chip select signal generation circuit CSG is a circuit that activates any or all of the four chip select signal terminals CS supplied to the memory chips 110 to 113.

[0072] The command system signals RAS, CAS, and WE are supplied to the command decoder CDC by way of the input buffers INB and latch circuits L. The output of the command decoder CDC is supplied to the chip select signal generation circuit CSG. The exact same signals as those fed to a general-purpose DRAM can be used as the command system signals RAS, CAS, and WE.

[0073] The chip select signal pad ECSP on the interface chip 120 is a single pad as described above. Therefore, the same chip select signal CS as used in a general-purpose DRAM may be supplied from outside of the semiconductor storage device 100 without consideration for the number of (four) stacked layers of memory chips 110 to 113. The chip select signal CS supplied to the chip select signal pad ECSP is input to the command decoder CDC by way of the input buffer INB and latch circuits L. As described above, the chip select signal generation circuit CSG activates any or all of the four chip select signals CS on the basis of the address signals A14 and A15, the bank addresses BA0 to BA2, and the output of the command decoder CDC.

[0074] The data input/output pad EDQP on the interface chip 120 is a terminal for inputting and outputting the data signals DQ0 to DQ7, data strobe signal DQS, data mask signal DM, and other signals, and is connected to the data input/output pad IDQP by way of the bidirectional repeater REP, as described above.

[0075] In this case, when the memory chips 110 to 113 to be stacked are the double data rate (DDR-800) type, a signal with speed of 400 Mbps is transferred over the command/address external terminal group CA and the command/address bus CAB inside the package. On the other hand, a signal with speed of 800 Mbps is transferred over the data input/output external terminal group DQ and the data bus DQB inside the package.

[0076] Described next in greater detail is the circuit configuration of the interface chip 120 used in the present embodiment.

[0077] FIG. 6 is a diagram showing in greater detail the circuit configuration of the clock system circuitry contained in the interface chip 120.

[0078] The complimentary clock signals CLK and /CLK supplied from the command/address pad ECAP are input to a differential amp input buffer INB, as shown in FIG. 6. The input buffer INB is activated by the output of the differential amp input buffer INB that receives the clock enable signal CKE and the reference voltage VREF. The output of the input buffer INB that receives the complimentary clock signals CLK and /CLK is supplied to the clock regeneration circuit DLL, and the output of the regeneration circuit is supplied to the complimentary clock signals CLK and /CLK of the command/address pad ICAP by way of the output buffers OB. The clock regeneration circuit DLL generates an internal clock signal CLKI, and the internal clock signal CLKI is supplied to the internal circuitry of the interface chip 120.

[0079] The output of the input buffer INB that receives the clock enable signal CKE is latched by the latch circuit L (which uses internal clock signal CLKI as a timing signal), and is supplied to the clock enable terminal CKE of the command/address pad ICAP by way of the output buffer OB. The clock enable signal CKE buffered by the input buffer INB is also supplied to the internal circuitry of interface chip 120. The clockenable signal CKE is fed to the clock regeneration circuit DLL, and when the clock enable signal CKE is activated, the clock feed to the memory chips 110 to 113 is stopped.

[0080] Thus, in this present embodiment, since a clock regeneration circuit DLL is provided inside the interface chip 120, the phase difference between the input and output clocks is reduced and the timing margin of the latch circuits L is increased. The operating margin can therefore be adequately assured even when the operating frequency is higher. When the clock enable signal CKE is deactivated, the operation of the clock regeneration circuit DLL is stopped, and since the clock feed to the memory chips 110 to 113 and the internal circuitry of the interface chip 120 is stopped, the power consumption during standby can be reduced.

[0081] FIG. 7 is a diagram showing in greater detail the circuit configuration of the command/address system circuitry contained in the interface chip 120.

[0082] The command/address system signals A0 to A15, BA0 to BA2, RAS, CAS, and WE are input from outside of the semiconductor storage device 100 to input buffers INB by way of the command/address pad ECAP, as shown in FIG. 7.

[0083] The chip select signal CS is supplied from outside of the semiconductor storage device 100 to an input buffer INB by way of the chip select signal pad ECSP. The input buffers INB are differential amp input buffers in which the reference voltage VREF is used, and the outputs of the buffers are supplied to corresponding latch circuits L. The latch circuits L synchronize with the internal clock signal CLKI generated by the clock regeneration circuit DLL (refer to FIG. 6), and accept the output of the corresponding input buffer INB.

[0084] The address signals A14 and A15 of the address signals A0 to A15 are decoded in the decoder DEC, and decoder output is supplied to the chip select signal generation circuit CSG. The bank addresses BA0 to BA2 are supplied to the command/address pad ICAP and are also supplied to the chip select signal generation circuit CSG and the mode register MR. The address signals A0 to A13 are supplied to the command/address pad ICAP and are also supplied to the mode register MR. The address signal A10 of the address signals A0 to A13 is also supplied to the command decoder CDC together with the command system signals RAS, CAS, and WE, as well as the chip select signal CS and clock enable signal CKE. The command decoder CDC decodes the signals in the same manner as a general-purpose DRAM, and generates a command signal CMD on the basis of the decoding result.

[0085] The command system signals RAS, CAS, and WE are directly supplied to the command/address pad ICAP, and an activate command (Bank activate), read command (Read), write command (Write), precharge command (Precharge), refresh command (Refresh), and other commands are presented to the memory chips 110 to 113.

[0086] The chip select signal generation circuit CSG receives a command signal CMD fed by the command decoder CDC and determines the operating mode. Any or all of the four chip select signals CS0 to CS3 are activated based on the operating mode thus determined. The chip select signals CS0 to CS3 are supplied to the corresponding chip select signal pads ICSP, respectively, by way of output buffers OB.

[0087] Thus, in the present embodiment, since the higher bits A14 and A15 of the address signals are decoded and the chip select signals CS0 to CS3 are activated based on the decoded signals, the memory chip appears from outside of the semiconductor storage device 100 as a single, large memory chip, and only one of the four memory chips 110 to 113 is activated in ordinary operation inside the semiconductor storage device 100. For this reason, an increase in power consumption can be held in check while assuring compatibility with general-purpose DRAM.

[0088] FIG. 8 is a list showing the commands and symbols that are output by the command decoder CDC, and the methods for issuing a chip select signal CS when the commands are executed.

[0089] Commands are input to the command decoder CDC by using the clock system signal CKE, chip select signal CS, address signal A10, and command system signals RAS, CAS, and WE fed from outside the semiconductor storage device 100, and the command decoder CDC generates a command signal CMD by using a combination of the signals. The chip select signal generation circuit CSG that receives the command signal CMD activates the chip select signals CS0 to CS3 as described below in accordance with the type of command.

[0090] First, when a mode set command "Mode register set (MRS)" or "Extended mode register set (EMRS)" classified as an "Initialization" command has been input, all the chip select signals CS0 to CS3 are simultaneously activated.

[0091] When a refresh command "Auto refresh (REF)" or "Self refresh entry (SELF)" classified as a "Row system" command has been input, the chip select signals CS0 to CS3 are sequentially activated with a phase difference. In other words, when synchronization is made with the clock signal CLK from outside the semiconductor storage device 100 and a refresh command (REF or SELF) is input, the interface chip 120 continuously issues a refresh command (REF or SELF) to the memory chips 110 to 113 by way of the command/address pad ICAP, and sequentially activates the chip select signals CS0 to CS3, as shown in FIG. 9.

[0092] The same refresh command (REF or SELF) is thereby sequentially input with a prescribed phase difference to the four memory chips 110 to 113. The phase difference is a single clock cycle as shown in the example of FIG. 9, but no limitation is imposed thereby, and the phase difference may be a plurality of clock cycles.

[0093] Thus, the reason that refresh commands (REF or SELF) are sequentially issued with a phase difference to the memory chips 110 to 113 is that a greater amount of electric current temporarily flows during a refresh operation than during ordinary read and write operations. In other words, when refresh commands (REF and SELF) are simultaneously issued to all the memory chips 110 to 113, the four memory chips 110 to 113 are simultaneously refreshed. As

a result, a considerable amount of electric current flows, and the power supply potential and ground potential are liable to fluctuate.

[0094] In the case that return from self refresh is instructed, that is to say, in the case that a "Self refresh exit (SELFEX)" command has been issued, all of the chip select signals CS0 to CS3 are simultaneously activated. Inputting a precharge command "Single bank precharge (PRE)" or "Precharge all banks (PALL)" activates the chip select signals CS0 to CS3, which correspond to the chip numbers (#0 to #3) stored in the active bank register ABR.

[0095] When an activation command "Back activate (ACT)" has been input, any one of the chip select signals CS0 to CS3 is activated based on the address signals A14 and A15, and the relationship between the activated chip select signal (CS0 to CS3) and the selected bank number (#0 to #7) is stored in the active bank register ABR shown in FIG. 10A. The active bank register ABR is a circuit disposed inside the chip select signal generation circuit CSG, and this register is made up of a group of registers that hold the relationship between the chip numbers (#0 to #3) specified by the higher two bits A14 and A15 of the row address, and the bank numbers (#0 to #7) specified by the bank addresses BA0 to BA2.

[0096] The reason for providing such an active bank register ABR is that since chip selection is carried out by the higher bits A14 and A15 of the row address A0 to A15 and only the bank address BA0 to BA2 and the column address A0 to A9 are fed when a column system command is input, it cannot be determined which chip select signals CS0 to CS3 should be activated when a column system command has been input if the above-described relationship is not stored using the active bank register ABR. However, provided that the bank address is expanded to five bits (BA0 to BA4) in lieu of the row address A14 and A15, that the chip is selected by using the higher two bits (BA0 and BA1), and that the bank inside the chip selected by the lower three bits (BA2 to BA4) is selected, such a problem is solved because the relationship between the bank address and the chip to be selected is fixed.

[0097] Nevertheless, in order to maintain compatibility with common general-purpose DRAM having an 8-bank configuration, the bank address is preferably given a three-bit configuration (BA0 to BA2) and a portion (A14 and A15) of the row address is used rather than the bank address to select a chip, as in the present embodiment. It is for this reason that an active bank register ABR is provided in the present embodiment.

[0098] When a write command "Write (WRIT)" or "Write with auto precharge (WRITA)" classified as a "column system" command, or a read command "Read (READ)" or "Read with auto precharge (READA)" classified as a "column system" command has been input, the bank address BA0 to BA2 is decoded in the decoder DEC, as shown in FIG. 10B, and the chip select signals CS0 to CS3, which correspond to the chip numbers (#0 to #3) stored in the active bank register ABR, are activated.

[0099] When a "No operation (NOP)" command, "Device deselect (DESL)" command, "Power down mode entry (PDEN)" command, or "Power down mode exit (PDEX)" command classified as "another" command has been input, all of the chip select signals CS0 to CS3 are simultaneously activated.

[0100] Thus, in the present embodiment, since the chip number to be activated is stored in the active bank register ABR when an activation command "Bank activate (ACT)" has been input, data can be accessed from outside the semiconductor storage device 100 by inputting to the interface chip 120 only the bank address BA0 to BA2 and the column address A0 to A9 when a read command or write command is input.

[0101] The same address input method as that used for a general-purpose DRAM can therefore be used and compatibility can be assured. Since commands are issued with a phase difference to the memory chips 110 to 113 when a refresh operation is carried out, the peak electric current that flows at the instant the memory chips 110 to 113 carry out a refresh operation can be staggered and power supply noise can be reduced.

[0102] FIG. 11 is a diagram showing in greater detail the circuit configuration of the principal components of the data system circuitry contained in the interface chip 120. In FIG. 11, the RDQS and /RDQS included in the signals of the data system are omitted.

[0103] The data system signals ODT, DQS, /DQS, DQ0 to DQ7, and DM are supplied from outside the semiconductor storage device 100 by way of the data input/output pad EDQP, as shown in FIG. 11. The on-chip resistance activation signal ODT is supplied to terminating resistors RT connected to the terminals, and the resistance value data RTV issued by the mode register is supplied to the terminating resistors RT in a shared configuration.

[0104] The data system signals have a higher frequency than the command/address system signals, and the signal waveform is therefore more easily distorted, but signal reflection can be reduced by carrying out termination inside the chip without any terminating resistance on the substrate. The distortion of the signal waveform can thereby be inhibited, resulting in an increased operating margin. In an ordinary DRAM, a terminating resistance is used inside the memory chip, but in the semiconductor storage device 100 of the present embodiment, a terminating resistance is used inside the interface chip 120.

[0105] The data system signals DQS, /DQS, DQ0 to DQ7, and DM are supplied to the data input/output pad IDQP by way of a bidirectional repeater REP. More specifically, the data signals DQ0 to DQ7 and DM that are input from outside the semiconductor storage device 100 during a write operation are amplified by differential input buffers INB that make a comparison with the reference voltage VREF and are latched in the latch circuits L. The data strobe signals DQS and /DQS input from outside the semiconductor storage device 100 during a write operation are amplified by differential input buffers INB. When signals are output to the memory chips 110 to 113, an internal strobe signal DQSI that is offset in phase by 90° with respect to the data signals DQ0 to DQ7 is applied. The internal strobe signal DQSI is generated by a strobe signal generation circuit DSG.

[0106] Therefore, the data strobe signal DQS on the data input/output pad EDQP is offset in phase by 90° with respect to the data in0, in1, in2, and in3 that are continuously input during a write operation, as shown in FIG. 12. Accordingly, the latch circuits L synchronize with the rise and fall of the data strobe signal DQS, and the data in0, in1, in2, and in3

can be sequentially received. Also, the data in0, in1, in2, and in3 sequentially received in this manner synchronize with the data strobe signal DQS on the data input/output pad IDQP, and are output to the memory chips 110 to 113 selected the chip select signals CS0 to CS3.

[0107] More specifically, synchronization is made with the rising edge of the clock signal CLK during a write operation, as shown in FIG. 12: a write command (WRIT) is input to the command/address pad ECAP; and after a prescribed write latency WL has elapsed, the data signals DQ0 to DQ7 are input to the data input/output pad EDQP. Only a single data DQ0 of the data signals DQ0 to DQ7 is shown in FIG. 12. The data strobe signal DQS (which is offset in phase by 90°) is also input to the data input/output pad EDQP simultaneously with the data signals DQ0 to DQ7 to be written.

[0108] A write command (WRIT) is supplied to the interface chip 120, after which synchronization is made with the subsequent falling clock signal CLK, and the same command (WRIT) is output to the memory chips 110 to 113 by way of the command/address pad ICAP. The data signals DQ0 to DQ7 and data strobe signal DQS are also output from the data input/output pad IDQP after a prescribed write latency WL has elapsed. The write latency WL is a value that is set in advance in the mode register on the memory chips 110 to 113 and on the interface chip 120. In the case of a write operation, the write latency WL is set to the same value in the interface chip 120 and the memory chips 110 to 113.

[0109] The data signals DQ0 to DQ7 output from the memory chips 110 to 113 during a read operation synchronize with the internal strobe signal DQSI that is generated by the strobe signal generation circuit DSG inside the interface chip 120, are received by the latch circuits L, and are output to the outside of the semiconductor storage device 100. Since the data strobe signal DQS fed from the memory chips 110 to 113 during a read operation match the phase of the data signals DQ0 to DQ7, as shown in FIG. 13, the data signals DQ0 to DQ7 cannot be directly received in synchronization with the data strobe signal DQS.

[0110] In view of the above, an internal strobe signal DQSI that is offset in phase by 90° from the data strobe signal DQS is generated by the strobe signal generation circuit DSG, and the read data is received in synchronization with the internal strobe signal. The data signals DQ0 to DQ7 thus received are synchronized with the internal clock signal CLKI (not shown) and output to an impedance variable output buffer OBV. The impedance of the impedance variable output buffer OBV is controlled by the impedance setting value ROV issued from the mode register MR.

[0111] More specifically, the read command (READ) is input to the command/address pad ECAP of the interface chip 120 at the rising edge of the clock signal CLK during a read operation, as shown in FIG. 13. The read command (READ) is thereafter output to the memory chips 110 to 113 at the subsequent falling edge of the clock signal CLK by way of the command/address pad ICAP. After a prescribed read latency RL has elapsed, data signals DQ0 to DQ7 and DQS are output from the memory chips 110 to 113.

[0112] The interface chip 120 receives the data signals by way of the data input/output pad IDQP, synchronizes the signals with the subsequent clock signal CLK, and outputs

the signals to the outside of the semiconductor storage device 100 by way of the data input/output pad EDQP. Since the data strobe signal DQS matches the phase of the data signals DQ0 to DQ7 during a read operation, the signal is directly matched with the clock and is output from the data input/output pad EDQP.

[0113] As viewed from outside of the semiconductor storage device 100, the overall read latency is a value in which a single clock cycle has been added (which is the latency of the interface chip 120 itself) to the read latency RL of the memory chips 110 to 113, that is to say, a value that is equal to $RL+1$, as shown in FIG. 13. Therefore, the value obtained by adding the latency of the interface chip 120 as such to the value of the mode register of the memory chips 110 to 113 is set as the read latency RL in the mode register MR on the interface chip 120.

[0114] Thus, the present embodiment involves receiving data that uses an internal strobe signal DQSI generated by the strobe signal generation circuit DSG, allowing an adequate operating margin to be assured even when the operating frequency is high. Since the value of the terminating resistance and the impedance of the output buffer can be controlled, distortion of the data transfer waveform is controlled, and, as a result, the operating margin can be expanded.

[0115] Next, the signal amplitude between the memory chips 110 to 113 and the interface chip 120 is described.

[0116] FIG. 14 is a schematic diagram that describes the method for reducing the amplitude of the inter-chip interface between the memory chips 110 to 113 and the interface chip 120. As an example, the case is shown in which the amplitude of the chip select signals CS0 to CS3 is reduced, but the same method may be adopted for the other signals as well.

[0117] In the present example, in the output buffer OB contained in the interface chip 120, an inverter with a CMOS structure is used as the output buffer for providing output to the memory chips 110 to 113, as shown in FIG. 14. An internal power supply voltage VDDI on the high level side of the inverter is kept at a lower level than the externally supplied power supply VDDQ, and the internal power supply voltage VSSI on the low level side is kept at a higher level than the externally supplied power supply VSSQ. If, for example, the VDDQ is 1.8 V, the VDDI may be set to about 1.1 V. Also, if the VSSQ is 0 V, the VSSI may be set to about 0.7 V.

[0118] When a signal (the chip select signal CS in FIG. 14) to be output is input to the input terminal of an inverter in which such internal power supply voltages VDDI and VSSI are used, the signal output has reduced amplitude. In this case, the threshold value of the NMOS and PMOS that constitute the inverter is preferably set to be lower than that of an ordinary device in order to increase speed. When set in this fashion, the amplitude of the data signals can be reduced without providing a terminating resistance in the wiring.

[0119] The internal power supply voltages VDDI and VSSI can be generated by a step-down circuit VRG. The step-down circuit VRG is composed of resistors R1 to R3 that produce reference voltages V1 and V2; differential amps AP1 and AP2 in which the reference voltages V1 and V2 are

supplied to a non-inverted input terminal (+) and an inverted input terminal (−) respectively; and a PMOS and NMOS in which the outputs of the differential amps AP1 and AP2 are fed to the gates, respectively. The drains of the PMOS and NMOS are connected to the inverted input terminal (−) of the differential amp AP1 and the non-inverted input terminal (+) of the differential amp AP2, respectively; and the electric potentials are the internal power supply voltages VDDI and VSSI, respectively.

[0120] If such a low-amplitude output buffer is used, the charge and discharge power can be considerably reduced by charging and discharging the parasitic capacitance of the wiring INW and the bonding pad when a command/address or data is transferred.

[0121] Another feature of in the present embodiment is that since four memory chips 110 to 113 are mounted in a single package, there may be a problem in that the temperature inside the package may increase due to the power consumption of the memory chips 110 to 113, but if the power consumption due to signal transfer between the chips is reduced by using a low-amplitude output buffer, the temperature increase inside the package can be inhibited, and, as a result, the data retention time of the memory chips 110 to 113 can be extended.

[0122] The low-amplitude output buffer may be used not only on the interface chip 120 side, but also on the memory chips 110 to 113 side.

[0123] As described above, the semiconductor storage device 100 of the present embodiment has a configuration in which a plurality of memory chips 110 to 113 that are capable of independently carrying out read and write operations are stacked, and the plurality of memory chips 110 to 113 are externally connected by way of an interface chip 120. For this reason, the chip connected to the command/address external terminal group CA and the data input/output external terminal group DQ is a single chip (interface chip 120), and parasitic capacitance of the external terminals can be reduced to an amount that is about equal to that of a single chip component.

[0124] The frequency of the command/address and data input/output can be increased thereby to an amount that is equal to about that of a single chip component. Additionally, since the memory chips 110 to 113 are three-dimensionally stacked, the mounting surface area can be reduced to an amount equal to a single chip component.

[0125] As viewed from external circuitry, the interface chip 120 can be handled as an ordinary general-purpose DRAM, making it possible to assure compatibility with ordinary semiconductor storage devices that are packaged in only a single chip. For this reason, there is no need to design a special-purpose memory controller or the like, and an increase in the cost of the system overall can be held in check.

[0126] Since the stacked memory chips 110 to 113 are ordinary memory chips (general-purpose DRAM) that are each independently capable of read and write operations, a special step for forming a through-electrode or carrying out other processes is not required. For this reason, it is also possible to hold down increases in manufacturing costs.

[0127] In the present embodiment, the command/address signals and data input/output signals are connected in the

form of a bus inside the package, making it possible to keep the number of wires inside the package low and thereby to hold manufacturing costs in check. Since any one of the chip select signals CS0 to CS3 is activated by decoding the higher bits A14 and A15 of the row address during read and write operations, the interface chip 120 can selectively activate the memory chips 110 to 113, and the overall power consumption of the stacked package can be reduced.

[0128] In the embodiment described above, common, general-use DRAM is used for the memory chips 110 to 113, but the data transfer speed between the memory chips 110 to 113 and the interface chip 120 can be further increased by separating the pads for normal use and the pads for testing as the pads of the memory chips 110 to 113. This method is described in FIG. 15.

[0129] FIG. 15 is a diagram for describing an improved method for connecting the pads and the memory array MA in the memory chips 110 to 113. For the sake of simplicity, only the command/address pad CAP and the data input/output pad DQP are depicted, but the chip select signal pad CSP may also be handled in the same manner as the command/address pad CAP.

[0130] In the present example, in addition to being provided with the command/address pad CAP and the data input/output pad DQP for normal use, the memory chips 110 to 113 are also provided with a test command/address pad TCAP and a test data input/output pad TDQP, as shown in FIG. 15. Although not depicted, a test pad that corresponds to the chip select signal pad CSP is also provided.

[0131] The test command/address pad TCAP has an electrostatic protection element ESD, an input buffer INB, and a latch circuit L. The test data input/output pad TDQP has an output buffer OB, an input buffer INB, and a latch circuit L. In contrast to this configuration, the command/address pad CAP and data input/output pad DQP for normal use are not connected to such circuitry and are essentially directly connected to the memory array MA, as shown in FIG. 15.

[0132] The test command/address pad TCAP and the test data input/output pad TDQP are pads that are used when the memory chips 110 to 113 are tested. More specifically, since the parasitic capacitance of the wiring is considerable, a tester used for testing the memory chips 110 to 113 requires an output buffer with a relatively high driving force, an input buffer with high sensitivity, and an electrostatic protection element.

[0133] In contrast, the chip connections are made with very short distance wires in terms of practical use, and since the parasitic capacitance is very low in comparison with the tester, the driving force of the output buffer OB may be fairly low, and there is no requirement that an input buffer INB with such a high sensitivity be used. Also, the necessity to provide an electrostatic protection element ESD is also low. Considering this point, in the present example, the electrostatic protection element ESD, the input buffer INB, and the like are bypassed, and the pads and memory array MA are essentially directly connected.

[0134] Testing can thereby be correctly carried out during the testing procedure, the data transfer speed between the memory chips 110 to 113 and the interface chip 120 can be increased during actual use, and power consumption can be reduced.

[0135] A second preferred embodiment of the present invention is described next.

[0136] The semiconductor storage device 200 of the second embodiment of the present invention is exactly the same as the semiconductor storage device 100 of the first embodiment described above in terms of the external appearance and the memory chips 110 to 113 that are used, but are different in terms of the interface chip that is used and the manner in which the memory chips 110 to 113 and the interface chip are connected. For this reason, a duplicative description of the portions that are the same as the first embodiment is omitted.

[0137] FIG. 16 is a schematic diagram of the connection configuration between the memory chips 110 to 113 and the interface chip 220 of the present embodiment.

[0138] In the present embodiment, all of the signals are connected in a one-to-one relationship between the memory chips 110 to 113 and the interface chip 220, as shown in FIG. 16. In other words, not only are the chip select signals connected in a one-to-one relationship, but the command/address signals and data signals are also connected in such a relationship. The power supply is bus connected. Therefore, in the present embodiment, all of the signals transferred between the memory chips 110 to 113 and the interface chip 220 are individual signals for each memory chips 110 to 113.

[0139] Described more specifically, the command/address external terminal group CA and chip select signal external terminal CS are connected, respectively, to the command/address pad ECAP and the chip select signal pad ECSP on the interface chip 220 by way of the ball electrodes EXB on the base substrate 101. These signals are input to the command decoder CDC, and the four outputs of the decoder are output, respectively, to the command/address pads ICAP0 to ICAP3 and chip select signal pads ICSP0 to ICSP3, which are individually assigned to the memory chips 110 to 113.

[0140] The signals (command and address) output to the command/address pads ICAP0 to ICAP3 are individually fed to the command/address pad CAP of the four memory chips 110 to 113. The four chip select signals that are output to the chip select signal pads ICSP0 to ICSP3 are also connected in a one-to-one relationship between the interface chip 220 and memory chips 110 to 113, respectively.

[0141] The data input/output external terminal group DQ is connected to the data input/output pad EDQP on the interface chip 220 by way of the ball electrodes EXB on the base substrate 101. The data input/output signals are input to a bidirectional multiplexer MUX, and are connected to the data input/output pad DQP of the memory chips 110 to 113, respectively, by way of the data input/output pads IDQP0 to IDQP3 that are individually assigned to the memory chips 110 to 113.

[0142] To be written to the memory chips 110 to 113, the data is input to the interface chip 220 from outside the semiconductor storage device 100 by way of the data input/output external terminal group DQ, and write data is sent to only one of the memory chips 110 to 113 selected by way of the bidirectional multiplexer MUX. When data is read from the memory chips 110 to 113, the read data from the selected chip of the memory chips 110 to 113 is input to

the interface chip 220 and is output to the data input/output external terminal group DQ by way of the bidirectional multiplexer.

[0143] The power supply external terminal group VS is connected directly to the power supply pads VSP of the memory chips 110 to 113 and the interface chip 220 by way of the ball electrodes EXB on the base substrate 101.

[0144] Thus, in the present embodiment, since the command/address signals and data input/output signals are connected in a one-to-one relationship inside the package, only the input/output circuits of two transmitting and receiving channels are connected to the wiring INW inside the package. For this reason, the parasitic capacitance of the wiring INW can be reduced, and the charging and discharging electric current of the parasitic capacitance (which is a problem when high speed signals are transmitted) can be reduced.

[0145] FIG. 17 is a simplified circuit diagram showing the circuit configuration of the interface chip 220 used in the present embodiment.

[0146] The interface chip 220 used in the present embodiment, as shown FIG. 17, is different from the interface chip 120 shown in FIG. 5 in that command/address pads ICAP are individually provided to the memory chips 110 to 113 (ICAP0 to ICAP3), output buffers OB are individually provided in a corresponding relationship, data input/output pads IDQP are individually provided to the memory chips 110 to 113 (IDQP0 to IDQP3), and bidirectional repeaters are substituted for the bidirectional multiplexer MUX in a corresponding relationship. Other than the above, the configuration is the same as the interface chip 120 shown in FIG. 5.

[0147] FIG. 18 is a diagram showing in greater detail the circuit configuration of the clock system circuit contained in the interface chip 220.

[0148] The clock system circuit contained in the interface chip 220, as shown in FIG. 18, is different than the clock system circuit of the interface chip 120 shown in FIG. 6 in that the pads for feeding the complimentary clock signals CLK and /CLK and the clock enable signal CKE are individually provided to the memory chips 110 to 113 (ICAP0 to ICAP3), and output buffers OB are independently provided in a corresponding relationship. Other than the above, the configuration is the same as the clock system circuit shown in FIG. 6.

[0149] FIG. 19 is a diagram showing in greater detail the circuit configuration of the command/address system circuit contained in the interface chip 220.

[0150] The command/address system circuit contained in the interface chip 220, as shown in FIG. 19, is different than the command/address system circuit of the interface chip 120 shown in FIG. 7 in that command/address pads ICAP are individually provided to the memory chips 110 to 113 (ICAP0 to ICAP3), output buffers OB are individually provided in a corresponding relationship, and selectors SEL for selectively activating the output buffers are also provided. Chip select signals CS0 to CS3 are fed to the selectors SEL, and the selectors SEL feed a signal only to the output buffers OB that correspond to the activated chip select

signals CS0 to CS3. Therefore, commands and addresses are not fed to the unselected memory chips 110 to 113.

[0151] FIG. 20 is a diagram showing in greater detail the circuit configuration of the data system circuit contained in the interface chip 220.

[0152] The data system circuit contained in the interface chip 220, as shown in FIG. 20, is different than the data system circuit of the interface chip 120 shown in FIG. 11 in that data input/output pads IDQP are individually provided to the memory chips 110 to 113 (IDQP0 to IDQP3), input buffers INB and output buffers OB are individually provided in a corresponding relationship, and selectors SEL that selectively activate the output buffers are also provided.

[0153] In the same manner as the command/address system circuit, chip select signals CS0 to CS3 are fed to the selectors SEL, the selectors SEL feed a signal only to the output buffers OB that correspond to the activated chip select signals CS0 to CS3, and only the output of the input buffers INB corresponding to the activated chip select signals CS0 to CS3 are allowed to pass. Therefore, data is not fed to the unselected memory chips 110 to 113.

[0154] The semiconductor storage device 200 of the present embodiment can achieve substantially the same effects as the semiconductor storage device 100 of the first embodiment described above, and since the command/address signals and data input/output signals are connected in a one-to-one relationship inside the package, the parasitic capacitance of the wiring INW inside the package can be reduced. The charging and discharging electric current of the parasitic capacitance (which is a problem when high speed signals are transmitted) can thereby be reduced.

[0155] In the present embodiment, both the command/address signals and data input/output signals are connected in a one-to-one relationship, but since the speed of the command/address signals is half the speed of the data input/output signals in a DDR-type DRAM, power consumption does not always become a large problem for command/address signals. Considering this point, the command/address signals are bus-connected in the same manner as in the first embodiment, and the data input/output signals may be connected in a one-to-one relationship. According to this configuration, it is possible to reduce power consumption while keeping the number of wires inside the chip low.

[0156] A third preferred embodiment of the present invention is described next.

[0157] FIG. 21 is a schematic diagram that describes the connection configuration between the memory chips 110 to 112 and the memory chip 320 in the semiconductor storage device 300 of the third embodiment of the present invention.

[0158] In the present embodiment, the special-purpose interface chip is dispensed with. Instead, an interface chip function is provided to the memory chip 320 that is most proximate to the base substrate 101, as shown in FIG. 21. Other than the above, the configuration is the same as the semiconductor storage device 100 of the first embodiment described above. Signals other than the chip select signals CS0 to CS2 are bus-connected using the command/address bus CAB and the data bus DQB between the memory chips 110 to 112 and the memory chip 320. According to the present invention, the number of chips can be reduced by

one, and the thickness of the entire package can therefore be reduced. Also, since the number of wires and the load of the internal wiring INW can be reduced, power consumption can be further lowered.

[0159] In the present embodiment, both the command/address signals and the data input/output signals are bus-connected in the same manner as the semiconductor storage device 100 of the first embodiment, although it is also possible to connect both the command/address signals and the data input/output signals in a one-to-one relationship as in the semiconductor storage device 200 of the second embodiment. Another option is to bus-connect the command/address signals as in the first embodiment, and to connect the data input/output signals in a one-to-one relationship.

[0160] The present invention is in no way limited to the aforementioned embodiments, but rather various modifications are possible within the scope of the invention as recited in the claims, and naturally these modifications are included within the scope of the invention.

[0161] In the embodiments described above, four memory chips were used as an example, but the number of memory chips that may be used is not limited to four, and any number of memory chips may be used as long as this number is two or more.

[0162] In the embodiments described above, DRAM was used as the memory chip, but the present invention is not limited to this type of chip alone, and other types of memory chips may also be used as long as the memory chips are each independently capable of read operations and write operations.

What is claimed is:

1. A semiconductor storage device comprising:

a base substrate having a command/address external terminal group to which command signals and address signals are supplied, a data input/output external terminal group for inputting and outputting data signals, and a single chip select external terminal; and

a plurality of memory chips that are stacked on the base substrate and are each individually capable of reading and writing operations, wherein

a plurality of terminals constituting said command/address external terminal group, a plurality of terminals constituting said data input/output external terminal group, and said single chip select external terminal are connected to a single chip that has an interface function; and

the single chip that has the interface function further has at least a chip select signal generation circuit that can individually activate said plurality of memory chips based on said address signals supplied via said command/address external terminal group and based on said chip select signal supplied via said chip select external terminal.

2. The semiconductor storage device as claimed in claim 1, wherein said single chip that has the interface function is an interface chip that is different from said plurality of memory chips.

3. The semiconductor storage device as claimed in claim 1, wherein said single chip that has the interface function is any one chip from among said plurality of memory chips.

4. The semiconductor storage device as claimed in claim 1, wherein at least one of the signals selected from said command signals, said address signals, and said data signals are transmitted between said plurality of memory chips and said single chip that has the interface function by way of wiring with a shared connection to said plurality of memory chips.

5. The semiconductor storage device as claimed in claim 1, wherein at least one of the signals selected from said command signals, said address signals, and said data signals are transmitted between said plurality of memory chips and said single chip that has the interface function by way of wiring that is individually connected to the plurality of memory chips.

6. The semiconductor storage device as claimed in claim 1, wherein said chip select signal generation circuit activates one of said plurality of memory chips based on a portion of said address signals supplied via said command/address external terminal group which is different from a bank address when an activation command is issued.

7. The semiconductor storage device as claimed in claim 6, wherein said chip select signal generation circuit has an active bank register for storing the relationship between the selected bank and the memory chip that is to be activated, and activates one of said plurality of memory chips by referencing said active bank register when a read command or a write command is issued together with said bank address via said command/address external terminal group.

8. The semiconductor storage device as claimed in claim 1, wherein said chip select signal generation circuit sequentially activates said plurality of memory chips when a refresh command is issued.

9. The semiconductor storage device as claimed in claim 1, wherein said single chip that has the interface function further has a clock regeneration circuit, and a clock regenerated by said clock regeneration circuit is supplied to said plurality of memory chips.

10. The semiconductor storage device as claimed in claim 1, wherein said single chip that has the interface function further has a strobe signal generation circuit, and said chip that has the interface function outputs a data signal that is read from said memory chip via said data input/output external terminal group in synchronism with an output of said strobe signal generation circuit.

11. The semiconductor storage device as claimed in claim 1, wherein said single chip that has the interface function has a terminating resistance that is connected to said data input/output external terminal group.

12. The semiconductor storage device as claimed in claim 1, wherein each of said plurality of memory chips has a bonding pad area, and said plurality of memory chips are connected to said base substrate by way of tape on which wiring is formed.

13. The semiconductor storage device as claimed in claim 1, wherein a first internal power supply voltage that is lower than an externally supplied high-level-side power supply voltage is applied as a high-level-side power supply voltage, and a second internal power supply voltage that is higher than the externally supplied low-level-side power supply voltage is applied as a low-level-side power supply voltage to an output buffer that is designed to provide output to said

plurality of memory chips and is selected from the output buffers included in said single chip that has the interface function.

14. The semiconductor storage device as claimed in claim 1, wherein said plurality of memory chips have a test pad in addition to an ordinary pad designed for actual use, said test pad is connected to a memory array at least via a buffer circuit, and said ordinary pad bypasses said buffer circuit and connects to said memory array.

15. A semiconductor storage device comprising:

- a base substrate having a command/address external terminal group to which command signals and address signals are supplied and a single chip select external terminal to which a chip select signal is supplied;

- an interface chip that is mounted on the base substrate and is connected to the command/address external terminal group and the chip select external terminal; and

- a plurality of memory chips that are stacked on the base substrate and are each individually capable of reading and writing operations,

wherein said interface chip can individually activate said plurality of memory chips based on said address signals and said chip select signal.

16. The semiconductor storage device as claimed in claim 15, wherein at least one of the signals selected from said command signals and said address signals are transmitted between said plurality of memory chips and said interface chip via wiring with a shared connection to said plurality of memory chips.

17. The semiconductor storage device as claimed in claim 15, wherein at least one of the signals selected from said

command signals and said address signals are transmitted between said plurality of memory chips and said interface chip via wiring that is individually connected to the plurality of memory chips.

18. A semiconductor storage device comprising:

- a base substrate having a command/address external terminal group to which command signals and address signals are supplied and a single chip select external terminal to which a chip select signal is supplied; and

- a plurality of memory chips that are stacked on the base substrate and are each individually capable of reading and writing operations,

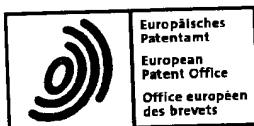
wherein one of said plurality of memory chips is connected to the command/address external terminal group and the chip select external terminal and can individually activate said plurality of memory chips based on said address signals and said chip select signal.

19. The semiconductor storage device as claimed in claim 18, wherein at least one of the signals selected from said command signals and said address signals are transmitted between said one of said plurality of memory chips and a remaining memory chips via wiring with a shared connection to said remaining memory chips.

20. The semiconductor storage device as claimed in claim 18, wherein at least one of the signals selected from said command signals and said address signals are transmitted between said one of said plurality of memory chips and a remaining memory chips via wiring that is individually connected to the remaining memory chips.

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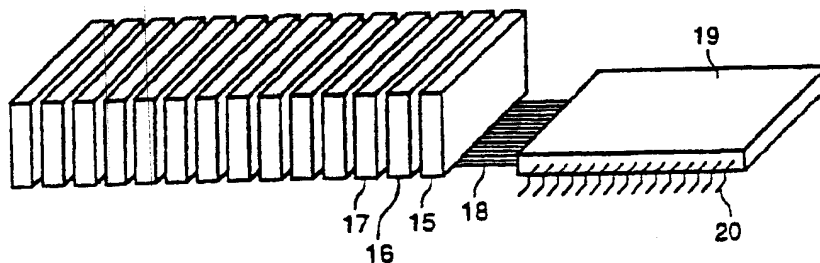
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(54) **Integrated circuit I/O using a high performance bus interface**

(57) The present invention includes a semiconductor device, the semiconductor device comprising connection means adapted to connect the semiconductor device to an external bus which is a part of a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to the external bus, wherein the external bus includes a plurality of bus lines for carrying substantially all address, data, control and device select information needed by the semiconductor device for communication with substantially every other semiconductor device connected to the external bus, and wherein

the external bus has substantially fewer bus lines than the number of bits in a single address, wherein each said bus line is a terminated transmission line, and wherein the external bus carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to the semiconductor device, a bus line driver capable of producing a low-voltage-swing signal on one of said terminated transmission lines, wherein the semiconductor device is a semiconductor transceiver device to interface multiple units to the bus, wherein the multiple units are DRAMs.

**FIG. 3****EP 1 816 570 A2**

Description

FIELD OF THE INVENTION

5 [0001] An integrated circuit bus interface for computer and video systems is described which allows high speed transfer of blocks of data, particularly to and from memory devices, with reduced power consumption and increased system reliability. A new method of physically implementing the bus architecture is also described.

BACKGROUND OF THE INVENTION

10 [0002] Semiconductor computer memories have traditionally been designed and structured to use one memory device for each bit, or small group of bits, of any individual computer word, where the word size is governed by the choice of computer. Typical word sizes range from 4 to 64 bits. Each memory device typically is connected in parallel to a series of address lines and connected to one of a series of data lines. When the computer seeks to read from or write to a specific memory location, an address is put on the address lines and some or all of the memory devices are activated using a separate device select line for each needed device. One or more devices may be connected to each data line but typically only a small number of data lines are connected to a single memory device. Thus data line 0 is connected to device(s) 0, data line 1 is connected to device(s) 1, and so on. Data is thus accessed or provided in parallel for each memory read or write operation. For the system to operate properly, every single memory bit in every memory device must operate dependably and correctly.

20 [0003] To understand the concept of the present invention, it is helpful to review the architecture of conventional memory devices. Internal to nearly all types of memory devices (including the most widely used Dynamic Random Access Memory (DRAM), Static RAM (SRAM) and Read Only Memory (ROM) devices), a large number of bits are accessed in parallel each time the system carries out a memory access cycle. However, only a small percentage of accessed bits which are available internally each time the memory device is cycled ever make it across the device boundary to the external world.

25 [0004] Referring to Fig. 1, all modern DRAM, SRAM and ROM designs have internal architectures with row (word) lines and column (bit) lines to allow the memory cells to tile a two dimensional area. One bit of data is stored at the intersection of each word and bit line. When a particular word line is enabled, all of the corresponding data bits are transferred onto the bit lines. Some prior art DRAMs take advantage of this organization to reduce the number of pins needed to transmit the address. The address of a given memory cell is split into two addresses, row and column, each of which can be multiplexed over a bus only half as wide as the memory cell address of the prior art would have required.

COMPARISON WITH PRIOR ART

35 [0005] Prior art memory systems have attempted to solve the problem of high speed access to memory with limited success. U.S. Patent No. 3,821,715 (Hoff et. al.), was issued to Intel Corporation for the earliest 4-bit micro-processor. That patent describes a bus connecting a single central processing unit (CPU) with multiple RAMs and ROMs. That bus multiplexes addresses and data over a 4-bit wide bus and uses point-to-point control signals to select particular RAMs or ROMs. The access time is fixed and only a single processing element is permitted. There is no block-mode type of operation, and most important, not all of the interface signals between the devices are bused (the ROM and RAM control lines and the RAM select lines are point-to-point).

40 [0006] In U.S. Patent No. 4,315,308 (Jackson), a bus connecting a single CPU to a bus interface unit is described. The invention uses multiplexed address, data, and control information over a single 16-bit wide bus. Block-mode operations are defined, with the length of the block sent as part of the control sequence. In addition, variable access-time operations using a "stretch" cycle signal are provided. There are no multiple processing elements and no capability for multiple outstanding requests, and again, not all of the interface signals are bused.

45 [0007] In U.S. Patent No. 4,449,207 (Kung, et. al.), a DRAM is described which multiplexes address and data on an internal bus. The external interface to this DRAM is conventional, with separate control, address and data connections.

50 [0008] In U.S. Patent Nos. 4,764,846 and 4,706,166 (Go), a 3-D package arrangement of stacked die with connections along a single edge is described. Such packages are difficult to use because of the point-to-point wiring required to interconnect conventional memory devices with processing elements. Both patents describe complex schemes for solving these problems. No attempt is made to solve the problem by changing the interface.

55 [0009] In U.S. Patent No. 3,969,706 (Proebsting, et. al.), the current state-of-the-art DRAM interface is described. The address is two-way multiplexed, and there are separate pins for data and control (RAS, CAS, WE, CS). The number of pins grows with the size of the DRAM, and many of the connections must be made point-to-point in a memory system using such DRAMs.

[0010] There are many backplane buses described in the prior art, but not in the combination described or having the

features of this invention. Many backplane buses multiplex addresses and data on a single bus (e.g., the NU bus). ELXSI and others have implemented split-transaction buses (U.S. Patent No. 4,595,923 and 4,481,625 (Roberts)). ELXSI has also implemented a relatively low-voltage-swing current-mode ECL driver (approximately 1 V swing). Address-space registers are implemented on most backplane buses, as is some form of block mode operation.

5 **[0011]** Nearly all modern backplane buses implement some type of arbitration scheme, but the arbitration scheme used in this invention differs from each of these. U.S. Patent Nos. 4,837,682 (Culler), 4,818,985 (Ikeda), 4,779,089 (Theus) and 4,745,548 (Blahut) describe prior art schemes. All involve either log N extra signals, (Theus, Blahut), where N is the number of potential bus requestors, or additional delay to get control of the bus (Ikeda, Culler). None of the buses described in patents or other literature use only bused connections. All contain some point-to-point connections

10 on the backplane. None of the other aspects of this invention such as power reduction by fetching each data block from a single device or compact and low-cost 3-D packaging even apply to backplane buses.

[0012] The clocking scheme used in this invention has not been used before and in fact would be difficult to implement in backplane buses due to the signal degradation caused by connector stubs. U.S. Patent No. 4,247,817 (Heller) describes a clocking scheme using two clock lines, but relies on ramp-shaped clock signals in contrast to the normal rise-time

15 signals used in the present invention.

[0013] In U.S. Patent No. 4,646,279 (Voss), a video RAM is described which implements a parallel-load, serial-out shift register on the output of a DRAM. This generally allows greatly improved bandwidth (and has been extended to 2, 4 and greater width shift-out paths.) The rest of the interfaces to the DRAM (RAS, CAS, multiplexed address, etc.) remain the same as for conventional DRAMS.

20 **[0014]** One object of the present invention is to use a new bus interface built into semiconductor devices to support high-speed access to large blocks of data from a single memory device by an external user of the data, such as a microprocessor, in an efficient and cost-effective manner.

[0015] Another object of this invention is to provide a clocking scheme to permit high speed clock signals to be sent along the bus with minimal clock skew between devices.

25 **[0016]** Another object of this invention is to allow mapping out defective memory devices or portions of memory devices.

[0017] Another object of this invention is to provide a method for distinguishing otherwise identical devices by assigning a unique identifier to each device.

[0018] Yet another object of this invention is to provide a method for transferring address, data and control information over a relatively narrow bus and to provide a method of bus arbitration when multiple devices seek to use the bus

30 simultaneously.

[0019] Another object of this invention is to provide a method of distributing a high-speed memory cache within the DRAM chips of a memory system which is much more effective than previous cache methods.

[0020] Another object of this invention is to provide devices, especially DRAMS, suitable for use with the bus architecture of the invention.

35 SUMMARY OF INVENTION

[0021] The present invention includes a memory subsystem comprising at least two semiconductor devices, including at least one memory device, connected in parallel to a bus, where the bus includes a plurality of bus lines for carrying

40 substantially all address, data and control information needed by said memory devices, where the control information includes device-select information and the bus has substantially fewer bus lines than the number of bits in a single address, and the bus carries device-select information without the need for separate device-select lines connected directly to individual devices.

[0022] Referring to Fig. 2, a standard DRAM 13, 14, ROM (or SRAM) 12, microprocessor CPU 11, I/O device, disk controller or other special purpose device such as a high speed switch is modified to use a wholly bus-based interface rather than the prior art combination of point-to-point and bus-based wiring used with conventional versions of these devices. The new bus includes clock signals, power and multiplexed address, data and control signals. In a preferred implementation, 8 bus data lines and an AddressValid bus line carry address, data and control information for memory

45 data lines up to 40 bits wide. Persons skilled in the art will recognize that 16 bus data lines or other numbers of bus data lines can be used to implement the teaching of this invention. The new bus is used to connect elements such as memory, peripheral, switch and processing units.

[0023] In the system of this invention, DRAMS and other devices receive address and control information over the bus and transmit or receive requested data over the same bus. Each memory device contains only a single bus interface with no other signal pins. Other devices that may be included in the system can connect to the bus and other non-bus

50 lines, such as input/output lines. The bus supports large data block transfers and split transactions to allow a user to achieve high bus utilization. This ability to rapidly read or write a large block of data to one single device at a time is an important advantage of this invention.

[0024] The DRAMS that connect to this bus differ from conventional DRAMS in a number of ways. Registers are

provided which may store control information, device identification, device-type and other information appropriate for the chip such as the address range for each independent portion of the device. New bus interface circuits must be added and the internals of prior art DRAM devices need to be modified so they can provide and accept data to and from the bus at the peak data rate of the bus. This requires changes to the column access circuitry in the DRAM, with only a minimal increase in die size. A circuit is provided to generate a low skew internal device clock for devices on the bus, and other circuits provide for demultiplexing input and multiplexing output signals.

[0025] High bus bandwidth is achieved by running the bus at a very high clock rate (hundreds of MHz). This high clock rate is made possible by the constrained environment of the bus. The bus lines are controlled-impedance, doubly-terminated lines. For a data rate of 500 MHz, the maximum bus propagation time is less than 1 ns (the physical bus length is about 10 cm). In addition, because of the packaging used, the pitch of the pins can be very close to the pitch of the pads. The loading on the bus resulting from the individual devices is very small. In a preferred implementation, this generally allows stub capacitances of 1-2 pF and inductances of 0.5 - 2 nH. Each device 15, 16, 17, shown in Figure 3, only has pins on one side and these pins connect directly to the bus 18. A transceiver device 19 can be included to interface multiple units to a higher order bus through pins 20.

[0026] A primary result of the architecture of this invention is to increase the bandwidth of DRAM access. The invention also reduces manufacturing and production costs, power consumption, and increases packing density and system reliability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0027]

Figure 1 is a diagram which illustrates the basic 2-D organization of memory devices.

Figure 2 is a schematic block diagram which illustrates the parallel connection of all bus lines and the serial Reset line to each device in the system.

Figure 3 is a perspective view of a system of the invention which illustrates the 3-D packaging of semiconductor devices on the primary bus.

Figure 4 shows the format of a request packet.

Figure 5 shows the format of a retry response from a slave.

Figure 6 shows the bus cycles after a request packet collision occurs on the bus and how arbitration is handled.

Figure 7 shows the timing whereby signals from two devices can overlap temporarily and drive the bus at the same time.

Figure 8 shows the connection and timing between bus clocks and devices on the bus.

Figure 9 is a perspective view showing how transceivers can be used to connect a number of bus units to a transceiver bus. Figure 10 is a block and schematic diagram of input/output circuitry used to connect devices to the bus.

Figure 11 is a schematic diagram of a clocked sense-amplifier used as a bus input receiver.

Figure 12 is a block diagram showing how the internal device clock is generated from two bus clock signals using a set of adjustable delay lines.

Figure 13 is a timing diagram showing the relationship of signals in the block diagram of Figure 12.

Figure 14 is timing diagram of a preferred means of implementing the reset procedure of this invention.

Figure 15 is a diagram illustrating the general organization of a 4 Mbit DRAM divided into 8 subarrays.

DETAILED DESCRIPTION

[0028] The present invention is designed to provide a high speed, multiplexed bus for communication between processing devices and memory devices and to provide devices adapted for use in the bus system. The invention can also be used to connect processing devices and other devices, such as I/O interfaces or disk controllers, with or without memory devices on the bus. The bus consists of a relatively small number of lines connected in parallel to each device on the bus. The bus carries substantially all address, data and control information needed by devices for communication with other devices on the bus. In many systems using the present invention, the bus carries almost every signal between every device in the entire system. There is no need for separate device-select lines since device-select information for each device on the bus is carried over the bus. There is no need for separate address and data lines because address and data information can be sent over the same lines. Using the organization described herein, very large addresses (40 bits in the preferred implementation) and large data blocks (1024 bytes) can be sent over a small number of bus lines (8 plus one control line in the preferred implementation).

[0029] Virtually all of the signals needed by a computer system can be sent over the bus. Persons skilled in the art recognize that certain devices, such as CPUs, may be connected to other signal lines and possibly to independent buses, for example a bus to an independent cache memory, in addition to the bus of this invention. Certain devices, for

example cross-point switches, could be connected to multiple, independent buses of this invention. In the preferred implementation, memory devices are provided that have no connections other than the bus connections described herein and CPUs are provided that use the bus of this invention as the principal, if not exclusive, connection to memory and to other devices on the bus.

[0030] All modern DRAM, SRAM and ROM designs have internal architectures with row (word) and column (bit) lines to efficiently tile a 2-D area. Referring to Fig. 1, one bit of data is stored at the intersection of each word line 5 and bit line 6. When a particular word line is enabled, all of the corresponding data bits are transferred onto the bit lines. This data, about 4000 bits at a time in a 4 MBit DRAM, is then loaded into column sense amplifiers 3 and held for use by the I/O circuits.

[0031] In the invention presented here, the data from the sense amplifiers is enabled 32 bits at a time onto an internal device bus running at approximately 125 MHz. This internal device bus moves the data to the periphery of the devices where the data is multiplexed into an 8-bit wide external bus interface, running at approximately 500 MHz.

[0032] The bus architecture of this invention connects master or bus controller devices, such as CPUs, Direct Memory Access devices (DMAs) or Floating Point Units (FPUs), and slave devices, such as DRAM, SRAM or ROM memory devices. A slave device responds to control signals; a master sends control signals. Persons skilled in the art realize that some devices may behave as both master and slave at various times, depending on the mode of operation and the state of the system. For example, a memory device will typically have only slave functions, while a DMA controller, disk controller or CPU may include both slave and master functions. Many other semiconductor devices, including I/O devices, disk controllers, or other special purpose devices such as high speed switches can be modified for use with the bus of this invention.

[0033] Each semiconductor device contains a set of internal registers, preferably including a device identification (device ID) register, a device-type descriptor register, control registers and other registers containing other information relevant to that type of device. In a preferred implementation, semiconductor devices connected to the bus contain registers which specify the memory addresses contained within that device and access-time registers which store a set of one or more delay times at which the device can or should be available to send or receive data.

[0034] Most of these registers can be modified and preferably are set as part of an initialization sequence that occurs when the system is powered up or reset. During the initialization sequence each device on the bus is assigned a unique device ID number, which is stored in the device ID register. A bus master can then use these device ID numbers to access and set appropriate registers in other devices, including access-time registers, control registers, and memory registers, to configure the system. Each slave may have one or several access-time registers (four in a preferred embodiment). In a preferred embodiment, one access-time register in each slave is permanently or semi-permanently programmed with a fixed value to facilitate certain control functions. A preferred implementation of an initialization sequence is described below in more detail.

[0035] All information sent between master devices and slave devices is sent over the external bus, which, for example, may be 8 bits wide. This is accomplished by defining a protocol whereby a master device, such as a microprocessor, seizes exclusive control of the external bus (i.e., becomes the bus master) and initiates a bus transaction by sending a request packet (a sequence of bytes comprising address and control information) to one or more slave devices on the bus. An address can consist of 16 to 40 or more bits according to the teachings of this invention. Each slave on the bus must decode the request packet to see if that slave needs to respond to the packet. The slave that the packet is directed to must then begin any internal processes needed to carry out the requested bus transaction at the requested time. The requesting master may also need to transact certain internal processes before the bus transaction begins. After a specified access time the slave(s) respond by returning one or more bytes (8 bits) of data or by storing information made available from the bus. More than one access time can be provided to allow different types of responses to occur at different times.

[0036] A request packet and the corresponding bus access are separated by a selected number of bus cycles, allowing the bus to be used in the intervening bus cycles by the same or other masters for additional requests or brief bus accesses. Thus multiple, independent accesses are permitted, allowing maximum utilization of the bus for transfer of short blocks of data. Transfers of long blocks of data use the bus efficiently even without overlap because the overhead due to bus address, control and access times is small compared to the total time to request and transfer the block.

Device Address Mapping

[0037] Another unique aspect of this invention is that each memory device is a complete, independent memory subsystem with all the functionality of a prior art memory board in a conventional backplane-bus computer system. Individual memory devices may contain a single memory section or may be subdivided into more than one discrete memory section. Memory devices preferably include memory address registers for each discrete memory section. A failed memory device (or even a subsection of a device) can be "mapped out" with only the loss of a small fraction of the memory, maintaining essentially full system capability. Mapping out bad devices can be accomplished in two ways, both compatible with this

invention.

[0038] The preferred method uses address registers in each memory device (or independent discrete portion thereof) to store information which defines the range of bus addresses to which this memory device will respond. This is similar to prior art schemes used in memory boards in conventional backplane bus systems. The address registers can include a single pointer, usually pointing to a block of known size, a pointer and a fixed or variable block size value or two pointers, one pointing to the beginning and one to the end (or to the "top" and "bottom") of each memory block. By appropriate settings of the address registers, a series of functional memory devices or discrete memory sections can be made to respond to a contiguous range of addresses, giving the system access to a contiguous block of good memory, limited primarily by the number of good devices connected to the bus. A block of memory in a first memory device or memory section can be assigned a certain range of addresses, then a block of memory in a next memory device or structure) than the last address of the previous block.

[0039] Preferred devices for use in this invention include device-type register information specifying the type of chip, including how much memory is available in what configuration on that device. A master can perform an appropriate memory test, such as reading and writing each memory cell in one or more selected orders, to test proper functioning of each accessible discrete portion of memory (based in part on information like device ID number and device-type) and write address values (up to 40 bits in the preferred embodiment, 10^{12} bytes), preferably contiguous, into device address-space registers. Non-functional or impaired memory sections can be assigned a special address value which the system can interpret to avoid using that memory.

[0040] The second approach puts the burden of avoiding the bad devices on the system master or masters. CPUs and DMA controllers typically have some sort of translation look-aside buffers (TLBs) which map virtual to physical (bus) addresses. With relatively simple software, the TLBs can be programmed to use only working memory (data structures describing functional memories are easily generated). For masters which don't contain TLBs (for example, a video display generator), a small, simple RAM can be used to map a contiguous range of addresses onto the addresses of the functional memory devices.

[0041] Either scheme works and permits a system to have a significant percentage of non-functional devices and still continue to operate with the memory which remains. This means that systems built with this invention will have much improved reliability over existing systems, including the ability to build systems with almost no field failures.

Bus

[0042] The preferred bus architecture of this invention comprises 11 signals: BusData[0:7]; AddrValid; Clk1 and Clk2; plus an input reference level and power and ground lines connected in parallel to each device. Signals are driven onto the bus during conventional bus cycles. The notation "Signal[i:j]" refers to a specific range of signals or lines, for example, BusData[0:7] means BusData0, BusData1, ..., BusData7. The bus lines for BusData[0:7] signals form a byte-wide, multiplexed data/address/control bus. AddrValid is used to indicate when the bus is holding a valid address request, and instructs a slave to decode the bus data as an address and, if the address is included on that slave, to handle the pending request. The two clocks together provide a synchronized, high speed clock for all the devices on the bus. In addition to the bused signals, there is one other line (ResetIn, ResetOut) connecting each device in series for use during initialization to assign every device in the system a unique device ID number (described below in detail).

[0043] To facilitate the extremely high data rate of this external bus relative to the gate delays of the internal logic, the bus cycles are grouped into pairs of even/odd cycles. Note that all devices connected to a bus should preferably use the same even/odd labeling of bus cycles and preferably should begin operations on even cycles. This is enforced by the clocking scheme.

Protocol and Bus Operation

[0044] The bus uses a relatively simple, synchronous, split-transaction, block-oriented protocol for bus transactions. One of the goals of the system is to keep the intelligence concentrated in the masters, thus keeping the slaves as simple as possible (since there are typically many more slaves than masters). To reduce the complexity of the slaves, a slave should preferably respond to a request in a specified time, sufficient to allow the slave to begin or possibly complete a device-internal phase including any internal actions that must precede the subsequent bus access phase. The time for this bus access phase is known to all devices on the bus - each master being responsible for making sure that the bus will be free when the bus access begins. Thus the slaves never worry about arbitrating for the bus. This approach eliminates arbitration in single master systems, and also makes the slave-bus interface simpler.

[0045] In a preferred implementation of the invention, to initiate a bus transfer over the bus, a master sends out a request packet, a contiguous series of bytes containing address and control information. It is preferable to use a request packet containing an even number of bytes and also preferable to start each packet on an even bus cycle.

[0046] The device-select function is handled using the bus data lines. AddrValid is driven, which instructs all slaves to decode the request packet address, determine whether they contain the requested address, and if they do, provide the data back to the master (in the case of a read request) or accept data from the master (in the case of a write request) in a data block transfer. A master can also select a specific device by transmitting a device ID number in a request packet. In a preferred implementation, a special device ID number is chosen to indicate that the packet should be interpreted by all devices on the bus. This allows a master to broadcast a message, for example to set a selected control register of all devices with the same value.

[0047] The data block transfer occurs later at a time specified in the request packet control information, preferably beginning on an even cycle. A device begins a data block transfer almost immediately with a device-internal phase as the device initiates certain functions, such as setting up memory addressing, before the bus access phase begins. The time after which a data block is driven onto the bus lines is selected from values stored in slave access-time registers. The timing of data for reads and writes is preferably the same; the only difference is which device drives the bus. For reads, the slave drives the bus and the master latches the values from the bus. For writes the master drives the bus and the selected slave latches the values from the bus.

[0048] In a preferred implementation of this invention shown in Figure 4, a request packet 22 contains 6 bytes of data -- 4.5 address bytes and 1.5 control bytes. Each request packet uses all nine bits of the multiplexed data/address lines (AddrValid 23 + BusData[0:7] 24) for all six bytes of the request packet. Setting 23 AddrValid = 1 in an otherwise unused even cycle indicates the start of an request packet (control information). In a valid request packet, AddrValid 27 must be 0 in the last byte. Asserting this signal in the last byte invalidates the request packet. This is used for the collision detection and arbitration logic (described below). Bytes 25-26 contain the first 35 address bits, Address[0:35]. The last byte contains AddrValid 27 (the invalidation switch) and 28, the remaining address bits, Address[36:39], and BlockSize [0:3] (control information).

[0049] The first byte contains two 4 bit fields containing control information, AccessType[0:3], an op code (operation code) which, for example, specifies the type of access, and Master[0:3], a position reserved for the master sending the packet to include its master ID number. Only master numbers 1 through 15 are allowed - master number 0 is reserved for special system commands. Any packet with Master[0:3] = 0 is an invalid or special packet and is treated accordingly.

[0050] The AccessType field specifies whether the requested operation is a read or write and the type of access, for example, whether it is to the control registers or other parts of the device, such as memory. In a preferred implementation, AccessType[0] is a Read/Write switch: if it is a 1, then the operation calls for a read from the slave (the slave to read the requested memory block and drive the memory contents onto the bus); if it is a 0, the operation calls for a write into the slave (the slave to read data from the bus and write it to memory). AccessType[1:3] provides up to 8 different access types for a slave. AccessType[1:2] preferably indicates the timing of the response, which is stored in an access-time register, AccessRegN. The choice of access-time register can be selected directly by having a certain op code select that register, or indirectly by having a slave respond to selected op codes with pre-selected access times (see table below). The remaining bit, AccessType[3] may be used to send additional information about the request to the slaves.

[0051] One special type of access is control register access, which involves addressing a selected register in a selected slave. In the preferred implementation of this invention, AccessType[1:3] equal to zero indicates a control register request and the address field of the packet indicates the desired control register. For example, the most significant two bytes can be the device ID number (specifying which slave is being addressed) and the least significant three bytes can specify a register address and may also represent or include data to be loaded into that control register. Control register accesses are used to initialize the access-time registers, so it is preferable to use a fixed response time which can be preprogrammed or even hard wired, for example the value in AccessReg0, preferably 8 cycles. Control register access can also be used to initialize or modify other registers, including address registers.

[0052] The method of this invention provides for access mode control specifically for the DRAMs. One such access mode determines whether the access is page mode or normal RAS access. In normal mode (in conventional DRAMs and in this invention), the DRAM column sense amps or latches have been precharged to a value intermediate between logical 0 and 1. This precharging allows access to a row in the RAM to begin as soon as the access request for either inputs (writes) or outputs (reads) is received and allows the column sense amps to sense data quickly. In page mode (both conventional and in this invention), the DRAM holds the data in the column sense amps or latches from the previous read or write operation. If a subsequent request to access data is directed to the same row, the DRAM does not need to wait for the data to be sensed (it has been sensed already) and access time for this data is much shorter than the normal access time. Page mode generally allows much faster access to data but to a smaller block of data (equal to the number of sense amps). However, if the requested data is not in the selected row, the access time is longer than the normal access time, since the request must wait for the RAM to precharge before the normal mode access can start. Two access-time registers in each DRAM preferably contain the access times to be used for normal and for page-mode accesses, respectively.

[0053] The access mode also determines whether the DRAM should precharge the sense amplifiers or should save the contents of the sense amps for a subsequent page mode access. Typical settings are "precharge after normal

access" and "save after page mode access" but "precharge after page mode access" or "save after normal access" are allowed, selectable modes of operation. The DRAM can also be set to precharge the sense amps if they are not accessed for a selected period of time.

[0054] In page mode, the data stored in the DRAM sense amplifiers may be accessed within much less time than it takes to read out data in normal mode (~10-20 nS vs. 40-100 nS). This data may be kept available for long periods. However, if these sense amps (and hence bit lines) are not precharged after an access, a subsequent access to a different memory word (row) will suffer a precharge time penalty of about 40-100 nS because the sense amps must precharge before latching in a new value.

[0055] The contents of the sense amps thus may be held and used as a cache, allowing faster, repetitive access to small blocks of data. DRAM-based page-mode caches have been attempted in the prior art using conventional DRAM organizations but they are not very effective because several chips are required per computer word. Such a conventional page-mode cache contains many bits (for example, 32 chips x 4Kbits) but has very few independent storage entries. In other words, at any given point in time the sense amps hold only a few different blocks or memory "locales" (a single block of 4K words, in the example above). Simulations have shown that upwards of 100 blocks are required to achieve high hit rates (>90% of requests find the requested data already in cache memory) regardless of the size of each block. See, for example, Anant Agarwal, et. al., "An Analytic Cache Model," ACM Transactions on Computer Systems, Vol. 7 (2), pp. 184-215 (May 1989).

[0056] The organization of memory in the present invention allows each DRAM to hold one or more (4 for 4MBit DRAMS) separately-addressed and independent blocks of data. A personal computer or workstation with 100 such DRAMS (i.e. 400 blocks or locales) can achieve extremely high, very repeatable hit rates (98-99% on average) as compared to the lower (50-80%), widely varying hit rates using DRAMS organized in the conventional fashion. Further, because of the time penalty associated with the deferred precharge on a "miss" of the page-mode cache, the conventional DRAM-based page-mode cache generally has been found to work less well than no cache at all.

[0057] For DRAM slave access, the access types are preferably used in the following way:

AccessType[1:3]	Use	AccessTime
0	Control Register Access	Fixed, 8[AccessReg0]
1	Unused	Fixed, 8[AccessReg0]
2-3	Unused	AccessReg1
4-5	Page Mode DRAM access	AccessReg2
6-7	Normal DRAM access	AccessReg3

Persons skilled in the art will recognize that a series of available bits could be designated as switches for controlling these access modes. For example:

AccessType[2] = page mode/normal switch

AccessType[3] = precharge/save-data switch

BlockSize[0:3] specifies the size of the data block transfer. If BlockSize[0] is 0, the remaining bits are the binary representation of the block size (0-7). If BlockSize[0] is 1, then the remaining bits give the block size as a binary power of 2, from 8 to 1024. A zero-length block can be interpreted as a special command, for example, to refresh a DRAM without returning any data, or to change the DRAM from page mode to normal access mode or vice-versa.

BlockSize[0:2]	Number of Bytes in Block
0-7	0-7 respectively
8	8
9	16
10	32
11	64
12	128
13	256
14	512
15	1024

Persons skilled in the art will recognize that other block size encoding schemes or values can be used.

[0058] In most cases, a slave will respond at the selected access time by reading or writing data from or to the bus over bus lines BusData[0:7] and AddrValid will be at logical 0. In a preferred embodiment, substantially each memory

access will involve only a single memory device, that is, a single block will be read from or written to a single memory device.

Retry Format

[0059] In some cases, a slave may not be able to respond correctly to a request, e.g., for a read or write. In such a situation, the slave should return an error message, sometimes called a N(o)ACK(nowledge) or retry message. The retry message can include information about the condition requiring a retry, but this increases system requirements for circuitry in both slave and masters. A simple message indicating only that an error has occurred allows for a less complex slave, and the master can take whatever action is needed to understand and correct the cause of the error.

[0060] For example, under certain conditions a slave might not be able to supply the requested data. During a page-mode access, the DRAM selected must be in page mode and the requested address must match the address of the data held in the sense amps or latches. Each DRAM can check for this match during a page-mode access. If no match is found, the DRAM begins precharging and returns a retry message to the master during the first cycle of the data block (the rest of the returned block is ignored). The master then must wait for the precharge time (which is set to accommodate the type of slave in question, stored in a special register, PreChargeReg), and then resend the request as a normal DRAM access (AccessType = 6 or 7).

[0061] In the preferred form of the present invention, a slave signals a retry by driving AddrValid true at the time the slave was supposed to begin reading or writing data. A master which expected to write to that slave must monitor AddrValid during the write and take corrective action if it detects a retry message. Figure 5 illustrates the format of a retry message 28 which is useful for read requests, consisting of 23 AddrValid=1 with Master[0:3] = 0 in the first (even) cycle. Note that AddrValid is normally 0 for data block transfers and that there is no master 0 (only 1 through 15 are allowed). All DRAMs and masters can easily recognize such a packet as an invalid request packet, and therefore a retry message. In this type of bus transaction all of the fields except for Mastery[0:3] and AddrValid 23 may be used as information fields, although in the implementation described, the contents are undefined. Persons skilled in the art recognize that another method of signifying a retry message is to add a DataInvalid line and signal to the bus. This signal could be asserted in the case of a NACK.

Bus Arbitration

[0062] In the case of a single master, there are by definition no arbitration problems. The master sends request packets and keeps track of periods when the bus will be busy in response to that packet. The master can schedule multiple requests so that the corresponding data block transfers do not overlap.

[0063] The bus architecture of this invention is also useful in configurations with multiple masters. When two or more masters are on the same bus, each master must keep track of all the pending transactions, so each master knows when it can send a request packet and access the corresponding data block transfer. Situations will arise, however, where two or more masters send a request packet at about the same time and the multiple requests must be detected, then sorted out by some sort of bus arbitration.

[0064] There are many ways for each master to keep track of when the bus is and will be busy. A simple method is for each master to maintain a bus-busy data structure, for example by maintaining two pointers, one to indicate the earliest point in the future when the bus will be busy and the other to indicate the earliest point in the future when the bus will be free, that is, the end of the latest pending data block transfer. Using this information, each master can determine whether and when there is enough time to send a request packet (as described above under Protocol) before the bus becomes busy with another data block transfer and whether the corresponding data block transfer will interfere with pending bus transactions. Thus each master must read every request packet and update its bus-busy data structure to maintain information about when the bus is and will be free.

[0065] With two or more masters on the bus, masters will occasionally transmit independent request packets during the same bus cycle. Those multiple requests will collide as each such master drives the bus simultaneously with different information, resulting in scrambled request information and neither desired data block transfer. In a preferred form of the invention, each device on the bus seeking to write a logical 1 on a BusData or AddrValid line drives that line with a current sufficient to sustain a voltage greater than or equal to the high-logic value for the system. Devices do not drive lines that should have a logical 0; those lines are simply held at a voltage corresponding to a low-logic value. Each master tests the voltage on at least some, preferably all, bus data and the AddrValid lines so the master can detect a logical '1' where the expected level is '0' on a line that it does not drive during a given bus cycle but another master does drive.

[0066] Another way to detect collisions is to select one or more bus lines for collision signalling. Each master sending a request drives that line or lines and monitors the selected lines for more than the normal drive current (or a logical value of ">1"), indicating requests by more than one master. Persons skilled in the art will recognize that this can be implemented with a protocol involving BusData and AddrValid lines or could be implemented using an additional bus line.

[0067] In the preferred form of this invention, each master detects collisions by monitoring lines which it does not drive to see if another master is driving those lines. Referring to Fig. 4, the first byte of the request packet includes the number of each master attempting to use the bus (Master[0:3]). If two masters send packet requests starting at the same point in time, the master numbers will be logical "or"ed together by at least those masters, and thus one or both of the masters, by monitoring the data on the bus and comparing what it sent, can detect a collision. For instance if requests by masters number 2 (0010) and 5 (0101) collide, the bus will be driven with the value Master[0:3]=7 (0010 + 0101 = 0111). Master number 5 will detect that the signal Master[2] = 1 and master 2 will detect that Master[1] and Master[3] = 1, telling both masters that a collision has occurred. Another example is masters 2 and 11, for which the bus will be driven with the value Master[0:3]=11 (0010 + 1011 = 1011), and although master 11 can't readily detect this collision, master 2 can. When any collision is detected, each master detecting a collision drives the value of AddrValid 27 in byte 5 of the request packet 22 to 1, which is detected by all masters, including master 11 in the second example above, and forces a bus arbitration cycle, described below.

[0068] Another collision condition may arise where master A sends a request packet in cycle 0 and master B tries to send a request packet starting in cycle 2 of the first request packet, thereby overlapping the first request packet. This will occur from time to time because the bus operates at high speeds, thus the logic in a second-initiating master may not be fast enough to detect a request initiated by a first master in cycle 0 and to react fast enough by delaying its own request. Master B eventually notices that it wasn't supposed to try to send a request packet (and consequently almost surely destroyed the address that master A was trying to send), and, as in the example above of a simultaneous collision, drives a 1 on AddrValid during byte 5 of the first request packet 27 forcing an arbitration. The logic in the preferred implementation is fast enough that a master should detect a request packet by another master by cycle 3 of the first request packet, so no master is likely to attempt to send a potentially colliding request packet later than cycle 2.

[0069] Slave devices not need to detect a collision directly, but they must wait to do anything irrecoverable until the last byte (byte 5) is read to ensure that the packet is valid. A request packet with Master[0:3] equal to 0 (a retry signal) is ignored and does not cause a collision. The subsequent bytes of such a packet are ignored.

[0070] To begin arbitration after a collision, the masters wait a preselected number of cycles after the aborted request packet (4 cycles in a preferred implementation), then use the next free cycle to arbitrate for the bus (the next available even cycle in the preferred implementation). Each colliding master signals to all other colliding masters that it seeks to send a request packet, a priority is assigned to each of the colliding masters, then each master is allowed to make its request in the order of that priority.

[0071] Figure 6 illustrates one preferred way of implementing this arbitration. Each colliding master signals its intent to send a request packet by driving a single BusData line during a single bus cycle corresponding to its assigned master number (1-15 in the present example). During two-byte arbitration cycle 29, byte 0 is allocated to requests 1-7 from masters 1-7, respectively, (bit 0 is not used) and byte 1 is allocated to requests 8-15 from masters 8-15, respectively. At least one device and preferably each colliding master reads the values on the bus during the arbitration cycles to determine and store which masters desire to use the bus. Persons skilled in the art will recognize that a single byte can be allocated for arbitration requests if the system includes more bus lines than masters. More than 15 masters can be accommodated by using additional bus cycles.

[0072] A fixed priority scheme (preferably using the master numbers, selecting lowest numbers first) is then used to prioritize, then sequence the requests in a bus arbitration queue which is maintained by at least one device. These requests are queued by each master in the bus-busy data structure and no further requests are allowed until the bus arbitration queue is cleared. Persons skilled in the art will recognize that other priority schemes can be used, including assigning priority according to the physical location of each master.

System Configuration/Reset

[0073] In the bus-based system of this invention, a mechanism is provided to give each device on the bus a unique device identifier (device ID) after power-up or under other conditions as desired or needed by the system. A master can then use this device ID to access a specific device, particularly to set or modify registers of the specified device, including the control and address registers. In the preferred embodiment, one master is assigned to carry out the entire system configuration process. The master provides a series of unique device ID numbers for each unique device connected to the bus system. In the preferred embodiment, each device connected to the bus contains a special device-type register which specifies the type of device, for instance CPU, 4 MBit memory, 64 MBit memory or disk controller. The configuration master should check each device, determine the device type and set appropriate control registers, including access-time registers. The configuration master should check each memory device and set all appropriate memory address registers.

[0074] One means to set up unique device ID numbers is to have each device to select a device ID in sequence and store the value in an internal device ID register. For example, a master can pass sequential device ID numbers through shift registers in each of a series of devices, or pass a token from device to device whereby the device with the token

reads in device ID information from another line or lines. In a preferred embodiment, device ID numbers are assigned to devices according to their physical relationship, for instance, their order along the bus.

[0075] In a preferred embodiment of this invention, the device ID setting is accomplished using a pair of pins on each device, ResetIn and ResetOut. These pins handle normal logic signals and are used only during device ID configuration. On each rising edge of the clock, each device copies ResetIn (an input) into a four-stage reset shift register. The output of the reset shift register is connected to ResetOut, which in turn connects to ResetIn for the next sequentially connected device. Substantially all devices on the bus are thereby daisy-chained together. A first reset signal, for example, while ResetIn at a device is a logical 1, or when a selected bit of the reset shift register goes from zero to non-zero, causes the device to hard reset, for example by clearing all internal registers and resetting all state machines. A second reset signal, for example, the falling edge of ResetIn combined with changeable values on the external bus, causes that device to latch the contents of the external bus into the internal device ID register (Device[0:7]).

[0076] To reset all devices on a bus, a master sets the ResetIn line of the first device to a "1" for long enough to ensure that all devices on the bus have been reset (4 cycles times the number of devices -- note that the maximum number of devices on the preferred bus configuration is 256 (8 bits), so that 1024 cycles is always enough time to reset all devices.) Then ResetIn is dropped to "0" and the BusData lines are driven with the first followed by successive device ID numbers, changing after every 4 clock pulses. Successive devices set those device ID numbers into the corresponding device ID register as the falling edge of ResetIn propagates through the shift registers of the daisy-chained devices. Figure 14 shows ResetIn at a first device going low while a master drives a first device ID onto the bus data lines BusData[0:3]. The first device then latches in that first device ID. After four clock cycles, the master changes BusData[0:3] to the next device ID number and ResetOut at the first device goes low, which pulls ResetIn for the next daisy-chained device low, allowing the next device to latch in the next device ID number from BusData[0:3]. In the preferred embodiment, one master is assigned device ID 0 and it is the responsibility of that master to control the ResetIn line and to drive successive device ID numbers onto the bus at the appropriate times. In the preferred embodiment, each device waits two clock cycles after ResetIn goes low before latching in a device ID number from BusData[0:3].

[0077] Persons skilled in the art recognize that longer device ID numbers could be distributed to devices by having each device read in multiple bytes from the bus and latch the values into the device ID register. Persons skilled in the art also recognize that there are alternative ways of getting device ID numbers to unique devices. For instance, a series of sequential numbers could be clocked along the ResetIn line and at a certain time each device could be instructed to latch the current reset shift register value into the device ID register.

[0078] The configuration master should choose and set an access time in each access-time register in each slave to a period sufficiently long to allow the slave to perform an actual, desired memory access. For example, for a normal DRAM access, this time must be longer than the row address strobe (RAS) access time. If this condition is not met, the slave may not deliver the correct data. The value stored in a slave access-time register is preferably one-half the number of bus cycles for which the slave device should wait before using the bus in response to a request. Thus an access time value of '1' would indicate that the slave should not access the bus until at least two cycles after the last byte of the request packet has been received. The value of AccessReg0 is preferably fixed at 8 (cycles) to facilitate access to control registers.

[0079] The bus architecture of this invention can include more than one master device. The reset or initialization sequence should also include a determination of whether there are multiple masters on the bus, and if so to assign unique master ID numbers to each. Persons skilled in the art will recognize that there are many ways of doing this. For instance, the master could poll each device to determine what type of device it is, for example, by reading a special register then, for each master device, write the next available master ID number into a special register.

ECC

[0080] Error detection and correction ("ECC") methods well known in the art can be implemented in this system. ECC information typically is calculated for a block of data at the time that block of data is first written into memory. The data block usually has an integral binary size, e.g. 256 bits, and the ECC information uses significantly fewer bits. A potential problem arises in that each binary data block in prior art schemes typically is stored with the ECC bits appended, resulting in a block size that is not an integral binary power.

[0081] In a preferred embodiment of this invention, ECC information is stored separately from the corresponding data, which can then be stored in blocks having integral binary size. ECC information and corresponding data can be stored, for example, in separate DRAM devices. Data can be read without ECC using a single request packet, but to write or read error-corrected data requires two request packets, one for the data and a second for the corresponding ECC information. ECC information may not always be stored permanently and in some situations the ECC information may be available without sending a request packet or without a bus data block transfer.

[0082] In a preferred embodiment, a standard data block size can be selected for use with ECC, and the ECC method will determine the required number of bits of information in a corresponding ECC block. RAMs containing ECC information

can be programmed to store an access time that is equal to: (1) the access time of the normal RAM (containing data) plus the time to access a standard data block (for corrected data) minus the time to send a request packet (6 bytes); or (2) the access time of a normal RAM minus the time to access a standard ECC block minus the time to send a request packet. To read a data block and the corresponding ECC block, the master simply issues a request for the data immediately followed by a request for the ECC block. The ECC RAM will wait for the selected access time then drive its data onto the bus right after (in case (1) above)) the data RAM has finished driving out the data block. Persons skilled in the art will recognize that the access time described in case (2) above can be used to drive ECC data before the data is driven onto the bus lines and will recognize that writing data can be done by analogy with the method described for a read. Persons skilled in the art will also recognize the adjustments that must be made in the bus-busy structure and the request packet arbitration methods of this invention in order to accommodate these paired ECC requests.

[0083] Since this system is quite flexible, the system designer can choose the size of the data blocks and the number of ECC bits using the memory devices of this invention. Note that the data stream on the bus can be interpreted in various ways. For instance the sequence can be 2^n data bytes followed by 2^m ECC bytes (or vice versa), or the sequence can be 2^k iterations of 8 data bytes plus 1 ECC byte. Other information, such as information used by a directory-based cache coherence scheme, can also be managed this way. See, for example, Anant Agarwal, et al., "Scaleable Directory Schemes for Cache Consistency," 15th International Symposium on Computer Architecture, June 1988, pp. 280-289. Those skilled in the art will recognize alternative methods of implementing ECC schemes that are within the teachings of this invention.

Low Power 3-D Packaging

[0084] Another major advantage of this invention is that it drastically reduces the memory system power consumption. Nearly all the power consumed by a prior art DRAM is dissipated in performing row access. By using a single row access in a single RAM to supply all the bits for a block request (compared to a row-access in each of multiple RAMs in conventional memory systems) the power per bit can be made very small. Since the power dissipated by memory devices using this invention is significantly reduced, the devices potentially can be placed much closer together than with conventional designs.

[0085] The bus architecture of this invention makes possible an innovative 3-D packaging technology. By using a narrow, multiplexed (time-shared) bus, the pin count for an arbitrarily large memory device can be kept quite small - on the order of 20 pins. Moreover, this pin count can be kept constant from one generation of DRAM density to the next. The low power dissipation allows each package to be smaller, with narrower pin pitches (spacing between the IC pins). With current surface mount technology supporting pin pitches as low as 20 mils, all off-device connections can be implemented on a single edge of the memory device. Semiconductor die useful in this invention preferably have connections or pads along one edge of the die which can then be wired or otherwise connected to the package pins with wires having similar lengths. This geometry also allows for very short leads, preferably with an effective lead length of less than 4 mm. Furthermore, this invention uses only bused interconnections, i.e., each pad on each device is connected by the bus to the corresponding pad of each other device.

[0086] The use of a low pin count and an edge-connected bus permits a simple 3-D package, whereby the devices are stacked and the bus is connected along a single edge of the stack. The fact that all of the signals are bused is important for the implementation of a simple 3-D structure. Without this, the complexity of the "backplane" would be too difficult to make cost effectively with current technology. The individual devices in a stack of the present invention can be packed quite tightly because of the low power dissipated by the entire memory system, permitting the devices to be stacked bumper-to-bumper or top to bottom. Conventional plastic-injection molded small outline (SO) packages can be used with a pitch of about 2.5 mm (100 mils), but the ultimate limit would be the device die thickness, which is about an order of magnitude smaller, 0.2-0.5 mm using current wafer technology.

Bus Electrical Description

[0087] By using devices with very low power dissipation and close physical packing, the bus can be made quite short, which in turn allows for short propagation times and high data rates. The bus of a preferred embodiment of the present invention consists of a set of resistor-terminated controlled impedance transmission lines which can operate up to a data rate of 500 MHz (2 ns cycles). The characteristics of the transmission lines are strongly affected by the loading caused by the DRAMs (or other slaves) mounted on the bus. These devices add lumped capacitance to the lines which both lowers the impedance of the lines and decreases the transmission speed. In the loaded environment, the bus impedance is likely to be on the order of 25 ohms and the propagation velocity about $c/4$ (c = the speed of light) or 7.5 cm/ns. To operate at a 2 ns data rate, the transit time on the bus should preferably be kept under 1 ns, to leave 1 ns for the setup and hold time of the input receivers (described below) plus clock skew. Thus the bus lines must be kept quite short, under about 8 cm for maximum performance. Lower performance systems may have much longer lines, e.g. a 4 ns bus

may have 24 cm lines (3 ns transit time, 1 ns setup and hold time).

[0088] In the preferred embodiment, the bus uses current source drivers. Each output must be able to sink 50 mA, which provides an output swing of about 500 mV or more. In the preferred embodiment of this invention, the bus is active low. The unasserted state (the high value) is preferably considered a logical zero, and the asserted value (low state) is therefore a logical 1. Those skilled in the art understand that the method of this invention can also be implemented using the opposite logical relation to voltage. The value of the unasserted state is set by the voltage on the termination resistors, and should be high enough to allow the outputs to act as current sources, while being as low as possible to reduce power dissipation. These constraints may yield a termination voltage about 2V above ground in the preferred implementation. Current source drivers cause the output voltage to be proportional to the sum of the sources driving the bus.

[0089] Referring to Fig.7, although there is no stable condition where two devices drive the bus at the same time, conditions can arise because of propagation delay on the wires where one device, A 41, can start driving its part of the bus 44 while the bus is still being driven by another device, B 42 (already asserting a logical 1 on the bus). In a system using current drivers, when B 42 is driving the bus (before time 46), the value at points 44 and 45 is logical 1. If B 42 switches off at time 46 just when A 41 switches on, the additional drive by device A 41 causes the voltage at the output 44 of A 41 to drop briefly below the normal value. The voltage returns to its normal value at time 47 when the effect of device B 42 turning off is felt. The voltage at point 45 goes to logical 0 when device B 42 turns off, then drops at time 47 when the effect of device A 41 turning on is felt. Since the logical 1 driven by current from device A 41 is propagated irrespective of the previous value on the bus, the value on the bus is guaranteed to settle after one time of flight (t_f) delay, that is, the time it takes a signal to propagate from one end of the bus to the other. If a voltage drive was used (as in ECL wired-ORing), a logical 1 on the bus (from device B 42 being previously driven) would prevent the transition put out by device A 41 being felt at the most remote part of the system, e.g., device 43, until the turnoff waveform from device B 42 reached device A 41 plus one time of flight delay, giving a worst case settling time of twice the time of flight delay.

25 Clocking

[0090] Clocking a high speed bus accurately without introducing error due to propagation delays can be implemented by having each device monitor two bus clock signals and then derive internally a device clock, the true system clock. The bus clock information can be sent on one or two lines to provide a mechanism for each bused device to generate an internal device clock with zero skew relative to all the other device clocks. Referring to Figure 8, in the preferred implementation, a bus clock generator 50 at one end of the bus propagates an early bus clock signal in one direction along the bus, for example on line 53 from left to right, to the far end of the bus. The same clock signal then is passed through the direct connection shown to a second line 54, and returns as a late bus clock signal along the bus from the far end to the origin, propagating from right to left. A single bus clock line can be used if it is left unterminated at the far end of the bus, allowing the early bus clock signal to reflect back along the same line as a late bus clock signal.

[0091] Figure 8b illustrates how each device 51, 52 receives each of the two bus clock signals at a different time (because of propagation delay along the wires), with constant midpoint in time between the two bus clocks along the bus. At each device 51, 52, the rising edge 55 of Clock1 53 is followed by the rising edge 56 of Clock2 54. Similarly, the falling edge 57 of Clock1 53 is followed by the falling edge 58 of Clock2 54. This waveform relationship is observed at all other devices along the bus. Devices which are closer to the clock generator have a greater separation between Clock1 and Clock2 relative to devices farther from the generator because of the longer time required for each clock pulse to traverse the bus and return along line 54, but the midpoint in time 59, 60 between corresponding rising or falling edges is fixed because, for any given device, the length of each clock line between the far end of the bus and that device is equal. Each device must sample the two bus clocks and generate its own internal device clock at the midpoint of the two.

[0092] Clock distribution problems can be further reduced by using a bus clock and device clock rate equal to the bus cycle data rate divided by two, that is, the bus clock period is twice the bus cycle period. Thus a 500 MHz bus preferably uses a 250 MHz clock rate. This reduction in frequency provides two benefits. First it makes all signals on the bus have the same worst case data rates -- data on a 500 MHz bus can only change every 2 ns. Second, clocking at half the bus cycle data rate makes the labeling of the odd and even bus cycles trivial, for example, by defining even cycles to be those when the internal device clock is 0 and odd cycles when the internal device clock is 1.

Multiple Buses

[0093] The limitation on bus length described above restricts the total number of devices that can be placed on a single bus. Using 2.5 mm spacing between devices, a single 8 cm bus will hold about 32 devices. Persons skilled in the art will recognize certain applications of the present invention wherein the overall data rate on the bus is adequate but memory or processing requirements necessitate a much larger number of devices (many more than 32). Larger systems can easily be built using the teachings of this invention by using one or more memory subsystems, designated primary

bus units, each of which consists of two or more devices, typically 32 or close to the maximum allowed by bus design requirements, connected to a transceiver device.

Referring to Figure 9, each primary bus unit can be mounted on a single circuit board 66, sometimes called a memory stick. Each transceiver device 19 in turn connects to a transceiver bus 65, similar or identical in electrical and other respects to the primary bus 18 described at length above. In a preferred implementation, all masters are situated on the transceiver bus so there are no transceiver delays between masters and all memory devices are on primary bus units so that all memory accesses experience an equivalent transceiver delay, but persons skilled in the art will recognize how to implement systems which have masters on more than one bus unit and memory devices on the transceiver bus as well as on primary bus units. In general, each teaching of this invention which refers to a memory device can be practiced using a transceiver device and one or more memory devices on an attached primary bus unit. Other devices, generically referred to as peripheral devices, including disk controllers, video controllers or I/O devices can also be attached to either the transceiver bus or a primary bus unit, as desired. Persons skilled in the art will recognize how to use a single primary bus unit or multiple primary bus units as needed with a transceiver bus in certain system designs.

[0094] The transceivers are quite simple in function. They detect request packets on the transceiver bus and transmit them to their primary bus unit. If the request packet calls for a write to a device on a transceiver's primary bus unit, that transceiver keeps track of the access time and block size and forwards all data from the transceiver bus to the primary bus unit during that time. The transceivers also watch their primary bus unit, forwarding any data that occurs there to the transceiver bus. The high speed of the buses means that the transceivers will need to be pipelined, and will require an additional one or two cycle delay for data to pass through the transceiver in either direction. Access times stored in masters on the transceiver bus must be increased to account for transceiver delay but access times stored in slaves on a primary bus unit should not be modified.

[0095] Persons skilled in the art will recognize that a more sophisticated transceiver can control transmissions to and from primary bus units. An additional control line, *TrncvrRW* can be bused to all devices on the transceiver bus, using that line in conjunction with the *AddrValid* line to indicate to all devices on the transceiver bus that the information on the data lines is: 1) a request packet, 2) valid data to a slave, 3) valid data from a slave, or 4) invalid data (or idle bus). Using this extra control line obviates the need for the transceivers to keep track of when data needs to be forwarded from its primary bus to the transceiver bus - all transceivers send all data from their primary bus to the transceiver bus whenever the control signal indicates condition 2) above. In a preferred implementation of this invention, if *AddrValid* and *TrncvrRW* are both low, there is no bus activity and the transceivers should remain in an idle state. A controller sending a request packet will drive *AddrValid* high, indicating to all devices on the transceiver bus that a request packet is being sent which each transceiver should forward to its primary bus unit. Each controller seeking to write to a slave should drive both *AddrValid* and *TrncvrRW* high, indicating valid data for a slave is present on the data lines. Each transceiver device will then transmit all data from the transceiver bus lines to each primary bus unit. Any controller expecting to receive information from a slave should also drive the *TrncvrRW* line high, but not drive *AddrValid*, thereby indicating to each transceiver to transmit any data coming from any slave on its primary local bus to the transceiver bus. A still more sophisticated transceiver would recognize signals addressed to or coming from its primary bus unit and transmit signals only at requested times.

[0096] An example of the physical mounting of the transceivers is shown in Figure 9. One important feature of this physical arrangement is to integrate the bus of each transceiver 19 with the original bus of DRAMs or other devices 15, 16, 17 on the primary bus unit 66. The transceivers 19 have pins on two sides, and are preferably mounted flat on the primary bus unit with a first set of pins connected to primary bus 18. A second set of transceiver pins 20, preferably orthogonal to the first set of pins, are oriented to allow the transceiver 19 to be attached to the transceiver bus 65 in much the same way as the DRAMs were attached to the primary bus unit. The transceiver bus can be generally planar and in a different plane, preferably orthogonal to the plane of each primary bus unit. The transceiver bus can also be generally circular with primary bus units mounted perpendicular and tangential to the transceiver bus.

[0097] Using this two level scheme allows one to easily build a system that contains over 500 slaves (16 buses of 32 DRAMs each). Persons skilled in the art can modify the device ID scheme described above to accommodate more than 256 devices, for example by using a longer device ID or by using additional registers to hold some of the device ID. This scheme can be extended in yet a third dimension to make a second-order transceiver bus, connecting multiple transceiver buses by aligning transceiver bus units parallel to and on top of each other and busing corresponding signal lines through a suitable transceiver. Using such a second-order transceiver bus, one could connect many thousands of slave devices into what is effectively a single bus.

Device Interface

[0098] The device interface to the high-speed bus can be divided into three main parts. The first part is the electrical interface. This part includes the input receivers, bus drivers and clock generation circuitry. The second part contains the address comparison circuitry and timing registers. This part takes the input request packet and determines if the request

is for this device, and if it is, starts the internal access and delivers the data to the pins at the correct time. The final part, specifically for memory devices such as DRAMs, is the DRAM column access path. This part needs to provide bandwidth into and out of the DRAM sense amps greater than the bandwidth provided by conventional DRAMs. The implementation of the electrical interface and DRAM column access path are described in more detail in the following sections. Persons skilled in the art recognize how to modify prior-art address comparison circuitry and prior-art register circuitry in order to practice the present invention.

Electrical Interface - Input/Output Circuitry

[0099] A block diagram of the preferred input/output circuit for address/data/control lines is shown in Figure 10. This circuitry is particularly well-suited for use in DRAM devices but it can be used or modified by one skilled in the art for use in other devices connected to the bus of this invention. It consists of a set of input receivers 71, 72 and output driver 76 connected to input/output line 69 and pad 75 and circuitry to use the internal clock 73 and internal clock complement 74 to drive the input interface. The clocked input receivers take advantage of the synchronous nature of the bus. To further reduce the performance requirements for device input receivers, each device pin, and thus each bus line, is connected to two clocked receivers, one to sample the even cycle inputs, the other to sample the odd cycle inputs. By thus de-multiplexing the input 70 at the pin, each clocked amplifier is given a full 2 ns cycle to amplify the bus low-voltage-swing signal into a full value CMOS logic signal. Persons skilled in the art will recognize that additional clocked input receivers can be used within the teachings of this invention. For example, four input receivers could be connected to each device pin and clocked by a modified internal device clock to transfer sequential bits from the bus to internal device circuits, allowing still higher external bus speeds or still longer settling times to amplify the bus low-voltage-swing signal into a full value CMOS logic signal.

[0100] The output drivers are quite simple, and consist of a single NMOS pulldown transistor 76. This transistor is sized so that under worst case conditions it can still sink the 50 mA required by the bus. For 0.8 micron CMOS technology, the transistor will need to be about 200 microns long. Overall bus performance can be improved by using feedback techniques to control output transistor current so that the current through the device is roughly 50 mA under all operating conditions, although this is not absolutely necessary for proper bus operation. An example of one of many methods known to persons skilled in the art for using feedback techniques to control current is described in Hans Schumacher, et al., "CMOS Subnanosecond True-ECL Output Buffer," J. Solid State Circuits, Vol. 25 (1), pp. 150-154 (Feb. 1990). Controlling this current improves performance and reduces power dissipation. This output driver which can be operated at 500 MHz, can in turn be controlled by a suitable multiplexer with two or more (preferably four) inputs connected to other internal chip circuitry, all of which can be designed according to well known prior art.

[0101] The input receivers of every slave must be able to operate during every cycle to determine whether the signal on the bus is a valid request packet. This requirement leads to a number of constraints on the input circuitry. In addition to requiring small acquisition and resolution delays, the circuits must take little or no DC power, little AC power and inject very little current back into the input or reference lines. The standard clocked DRAM sense amp shown in Figure 11 satisfies all these requirements except the need for low input currents. When this sense amp goes from sense to sample, the capacitance of the internal nodes 83 and 84 in Figure 11 is discharged through the reference line 68 and input 69, respectively. This particular current is small, but the sum of such currents from all the inputs into the reference lines summed over all devices can be reasonably large.

[0102] The fact that the sign of the current depends upon on the previous received data makes matters worse. One way to solve this problem is to divide the sample period into two phases. During the first phase, the inputs are shorted to a buffered version of the reference level (which may have an offset). During the second phase, the inputs are connected to the true inputs. This scheme does not remove the input current completely, since the input must still charge nodes 83 and 84 from the reference value to the current input value, but it does reduce the total charge required by about a factor of 10 (requiring only a 0.25V change rather than a 2.5V change). Persons skilled in the art will recognize that many other methods can be used to provide a clocked amplifier that will operate on very low input currents.

[0103] One important part of the input/output circuitry generates an internal device clock based on early and late bus clocks. Controlling clock skew (the difference in clock timing between devices) is important in a system running with 2 ns cycles, thus the internal device clock is generated so the input sampler and the output driver operate as close in time as possible to midway between the two bus clocks.

[0104] A block diagram of the internal device clock generating circuit is shown in Figure 12 and the corresponding timing diagram in Figure 13. The basic idea behind this circuit is relatively simple. A DC amplifier 102 is used to convert the small-swing bus clock into a full-swing CMOS signal. This signal is then fed into a variable delay line 103. The output of delay line 103 feeds three additional delay lines: 104 having a fixed delay; 105 having the same fixed delay plus a second variable delay; and 106 having the same fixed delay plus one half of the second variable delay. The outputs 107, 108 of the delay lines 104 and 105 drive clocked input receivers 101 and 111 connected to early and late bus clock inputs 100 and 110, respectively. These input receivers 101 and 111 have the same design as the receivers described

above and shown in Fig. 11. Variable delay lines 103 and 105 are adjusted via feedback lines 116, 115 so that input receivers 101 and 111 sample the bus clocks just as they transition. Delay lines 103 and 105 are adjusted so that the falling edge 120 of output 107 precedes the falling edge 121 of the early bus clock, Clock1 53, by an amount of time 128 equal to the delay in input sampler 101. Delay line 108 is adjusted in the same way so that falling edge 122 precedes the falling edge 123 of late bus clock, Clock2 54, by the delay 128 in input sampler 111.

[0105] Since the outputs 107 and 108 are synchronized with the two bus clocks and the output 73 of the last delay line 106 is midway between outputs 107 and 108, that is, output 73 follows output 107 by the same amount of time 129 that output 73 precedes output 108, output 73 provides an internal device clock midway between the bus clocks. The falling edge 124 of internal device clock 73 precedes the time of actual input sampling 125 by one sampler delay. Note that this circuit organization automatically balances the delay in substantially all device input receivers 71 and 72 (Fig. 10), since outputs 107 and 108 are adjusted so the bus clocks are sampled by input receivers 101 and 111 just as the bus clocks transition.

[0106] In the preferred embodiment, two sets of these delay lines are used, one to generate the true value of the internal device clock 73, and the other to generate the complement 74 without adding any inverter delay. The dual circuit allows generation of truly complementary clocks, with extremely small skew. The complement internal device clock is used to clock the 'even' input receivers to sample at time 127, while the true internal device clock is used to clock the 'odd' input receivers to sample at time 125. The true and complement internal device clocks are also used to select which data is driven to the output drivers. The gate delay between the internal device clock and output circuits driving the bus is slightly greater than the corresponding delay for the input circuits, which means that the new data always will be driven on the bus slightly after the old data has been sampled.

DRAM Column Access Modification

[0107] A block diagram of a conventional 4 MBit DRAM 130 is shown in Figure 15. The DRAM memory array is divided into a number of subarrays 150-157, for example, 8. Each subarray is divided into arrays 148, 149 of memory cells. Row address selection is performed by decoders 146. A column decoder 147A, 147B, including column sense amps on either side of the decoder, runs through the core of each subarray. These column sense amps can be set to precharge or latch the most-recently stored value, as described in detail above. Internal I/O lines connect each set of sense-amps, as gated by corresponding column decoders, to input and output circuitry connected ultimately to the device pins. These internal I/O lines are used to drive the data from the selected bit lines to the data pins (some of pins 131-145), or to take the data from the pins and write the selected bit lines. Such a column access path organized by prior art constraints does not have sufficient bandwidth to interface with a high speed bus. The method of this invention does not require changing the overall method used for column access, but does change implementation details. Many of these details have been implemented selectively in certain fast memory devices, but never in conjunction with the bus architecture of this invention.

[0108] Running the internal I/O lines in the conventional way at high bus cycle rates is not possible. In the preferred method, several (preferably 4) bytes are read or written during each cycle and the column access path is modified to run at a lower rate (the inverse of the number of bytes accessed per cycle, preferably 1/4 of the bus cycle rate). Three different techniques are used to provide the additional internal I/O lines required and to supply data to memory cells at this rate. First, the number of I/O bit lines in each subarray running through the column decoder 147 is increased, for example, to 16, eight for each of the two columns of column sense amps and the column decoder selects one set of columns from the "top" half 148 of subarray 150 and one set of columns from the "bottom" half 149 during each cycle, where the column decoder selects one column sense amp per I/O bit line. Second, each column I/O line is divided into two halves, carrying data independently over separate internal I/O lines from the left half 147A and right half 147B of each subarray (dividing each subarray into quadrants) and the column decoder selects sense amps from each right and left half of the subarray, doubling the number of bits available at each cycle. Thus each column decode selection turns on n column sense amps, where n equals four (top left and right, bottom left and right quadrants) times the number of I/O lines in the bus to each subarray quadrant (8 lines each $\times 4 = 32$ lines in the preferred implementation). Finally, during each RAS cycle, two different subarrays, e.g. 157 and 153, are accessed. This doubles again the available number of I/O lines containing data. Taken together, these changes increase the internal I/O bandwidth by at least a factor of 8. Four internal buses are used to route these internal I/O lines. Increasing the number of I/O lines and then splitting them in the middle greatly reduces the capacitance of each internal I/O line which in turn reduces the column access time, increasing the column access bandwidth even further.

[0109] The multiple, gated input receivers described above allow high speed input from the device pins onto the internal I/O lines and ultimately into memory. The multiplexed output driver described above is used to keep up with the data flow available using these techniques. Control means are provided to select whether information at the device pins should be treated as an address, and therefore to be decoded, or input or output data to be driven onto or read from the internal I/O lines.

[0110] Each subarray can access 32 bits per cycle, 16 bits from the left subarray and 16 from the right subarray. With

8 I/O lines per sense-amplifier column and accessing two subarrays at a time, the DRAM can provide 64 bits per cycle. This extra I/O bandwidth is not needed for reads (and is probably not used), but may be needed for writes. Availability of write bandwidth is a more difficult problem than read bandwidth because over-writing a value in a sense-amplifier may be a slow operation, depending on how the sense amplifier is connected to the bit line. The extra set of internal I/O lines provides some bandwidth margin for write operations.

5 [0111] Persons skilled in the art will recognize that many variations of the teachings of this invention can be practiced that still fall within the claims of this invention which follow.

[0112] It follows a list of further embodiments of the invention:

Embodiment 1. A memory subsystem comprising

10 two memory devices connected in parallel to a bus, said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said memory devices,

said control information including device-select information,

said bus containing substantially fewer bus lines than the number of bits in a single address, and

15 said bus carrying device-select information without the need for separate device-select lines connected directly to individual memory devices.

Embodiment 2. The memory subsystem with the features of embodiment 1 wherein said bus contains at least 8 bus lines adapted to carry at least 16 address bits and at least 8 data bits.

Embodiment 3. The memory subsystem with the features of embodiment 1 wherein said bus also includes parallel lines for clock and power.

Embodiment 4. A system comprising

a memory subsystem with the features of embodiment 1 wherein each bus of said memory subsystem is connected to its own transceiver device,

a transceiver bus connecting said transceiver devices, and

25 a means for transferring information between each of said buses of said memory subsystems and said transceiver bus, whereby memory subsystems may be integrated into a larger system having more memory than an individual memory subsystem.

Embodiment 5. The system with the features of embodiment 4 having a plurality of memory subsystems.

Embodiment 6. The system with the features of embodiment 4 further comprising a master device connected to said transceiver bus.

Embodiment 7. The system with the features of embodiment 6 wherein said master device is selected from the group consisting of a central processing unit, a floating point unit and a direct memory access unit.

Embodiment 8. The system with the features of embodiment 4 further comprising a peripheral device connected to the transceiver bus, said peripheral device adapted for connection to other devices not on the bus.

Embodiment 9. The system with the features of embodiment 8 wherein said peripheral device is selected from the group consisting of an I/O interface port, a video controller and a disk controller.

Embodiment 10. The system with the features of embodiment 5 wherein said transceiver bus is in a different plane than the plane of the bus of each of said memory subsystems.

Embodiment 11. The system with the features of embodiment 5 wherein the bus of each memory subsystem lies substantially in a subsystem bus plane and said transceiver bus lies substantially in a plane orthogonal to said subsystem bus plane.

Embodiment 12. The system with the features of embodiment 4 having at least two transceiver buses, each transceiver bus having a plurality of memory subsystem buses connected through a first transceiver to said transceiver bus, each of said transceiver buses being further connected to a second transceiver adapted to interface to a second-order transceiver bus, whereby each transceiver bus is connected through said second transceiver to form a second-order transceiver bus unit.

Embodiment 13. A semiconductor subsystem bus for interconnecting semiconductor devices comprising a plurality of semiconductor devices connected in parallel to a bus, at least one of said semiconductor devices being a memory device or a transceiver device which in turn is connected to a memory subsystem,

50 said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said semiconductor devices,

said control information including semiconductor device-select information,

said bus containing substantially fewer bus lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual semiconductor devices, and

55 at least one modifiable register in each of the semiconductor devices on said bus, said modifiable registers being accessible from said bus, whereby the subsystem can be configured using signals transmitted on said bus.

Embodiment 14. The semiconductor subsystem bus with the features of embodiment 13 wherein one type of modifiable

register is an access-time register designed to store a time delay after which a device may take some specified action on said bus.

Embodiment 15. The semiconductor subsystem bus with the features of embodiment 13 further comprising a semiconductor device having at least two access-time registers and
 5 one of said access-time registers is permanently programmed to contain a fixed value and at least one of said access-time registers can be modified by information carried on said bus.

Embodiment 16. The semiconductor subsystem bus with the features of embodiment 13 further comprising a memory device having at least one discrete memory section and also having a modifiable address register adapted to store
 10 memory address information which corresponds to each said discrete memory section.

Embodiment 17. The semiconductor subsystem bus with the features of embodiment 16 wherein said memory address information comprises a pointer to said discrete memory section.

Embodiment 18. The semiconductor subsystem bus with the features of embodiment 16 wherein said discrete memory section has a top and a bottom and said memory address information comprises pointers to said top and said bottom.

Embodiment 19. The semiconductor subsystem bus with the features of embodiment 16 wherein said memory address
 15 information comprises

a pointer to said discrete memory section and
 a range value indicating the size of said discrete memory section.

Embodiment 20. The semiconductor subsystem bus with the features of embodiment 16 wherein said address registers
 20 of each of said discrete memory sections of each of said memory devices connected to said bus are set to contain memory address information that is different for each discrete memory section and such that the highest memory address in each discrete memory section is one less than the lowest memory address in another discrete memory section, whereby memory may be organized into one or a small number of contiguous memory blocks.

Embodiment 21. The semiconductor subsystem bus with the features of embodiment 16 further comprising a means for
 25 testing each of said discrete memory sections of each of said memory devices for proper function, and for each non-functional discrete memory section, a means for setting at least one address register which corresponds to said discrete memory section to indicate that said discrete memory section is non-functional,

for each functional discrete memory section, a means for setting at least one address register which corresponds to said discrete memory section to contain such corresponding address information.

Embodiment 22. The semiconductor subsystem bus with the features of embodiment 21 wherein said address registers
 30 corresponding to said discrete memory sections are set to provide one contiguous memory block within the subsystem.

Embodiment 23. The semiconductor subsystem bus with the features of embodiment 13 wherein one of said modifiable registers is a device identification register which can be modified to contain a value unique to that semiconductor device.

Embodiment 24. The semiconductor subsystem bus with the features of embodiment 23 wherein said device identification register is set to contain a unique value which is a function of the physical position of that semiconductor device either
 35 along said bus or in relationship to other semiconductor devices or said bus.

Embodiment 25. A bus subsystem comprising
 40 two semiconductor devices connected in parallel to a bus, wherein one of said semiconductor devices is a master device, said master device including a means for initiating bus transactions, said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said devices,

said control information including device-select information,
 said bus containing substantially fewer lines than the number of bits in a single address, and
 said bus carrying device-select information without the need for separate device-select lines connected directly to individual devices on said bus, whereby said master device initiates bus transactions which transfer information between
 45 said semiconductor devices on said bus.

Embodiment 26. The bus subsystem with the features of embodiment 25 wherein one of said semiconductor devices is a memory device connected to said bus, said memory device having at least one discrete memory section and also having a modifiable address register adapted to store memory address information which corresponds to each said
 50 discrete memory section.

Embodiment 27. The bus subsystem with the features of embodiment 26 wherein one of said semiconductor devices comprises a transceiver device connected in parallel to said bus and connected in parallel to a memory device on a bus
 other than said bus.

28. The bus subsystem with the features of embodiment 26 further including a means Embodiment for said master
 55 device to request said memory device to prepare for a bus transaction by sending a request packet along said bus, said memory device and said master device each having a device-internal means to prepare to begin said bus transaction during a device-internal phase and further having a bus access means to effect said bus transaction during a bus access phase, said request packet including
 a sequence of bytes containing address and control information,

said control information including information about the requested bus transaction and about the access time, which corresponds to a number of bus cycles, which needs to intervene before beginning said bus-access phase, and said address information pointing to at least one memory location within one of said discrete memory sections of said memory device.

Embodiment 29. The bus subsystem with the features of embodiment 28 wherein said memory device includes a means to read said control information and initiate said device-internal means at a time so as to complete said device-internal phase within said access time and begin said bus access phase after said number of bus cycles.

Embodiment 30. The bus subsystem with the features of embodiment 28 wherein said control information comprises an op code.

Embodiment 31. The bus subsystem with the features of embodiment 30 wherein said memory device includes sense amplifiers adapted to hold a bit of information or to precharge after a selected time and a means to transfer a data block during a data block transfer either reading data from said memory device or writing data into said memory device, and wherein said op code instructs said memory device to activate a response means, said response means including a means to

initiate a data block transfer,

select the size of said data block,

select the time to initiate said data block transfer,

access a control register, including reading from or writing to said control register,

precharge said sense amplifiers after each of said data block transfers is complete,

hold a bit of information in each of said sense amplifiers after each of said data block transfers is complete, or

select normal or page-mode access.

Embodiment 32. The bus subsystem with the features of embodiment 31 wherein said data block transfer comprises a read from or a write to memory within a single memory device.

Embodiment 33. The bus subsystem with the features of embodiment 28 further comprising a means for said master device to send control information to a specific one of said semiconductor devices on said bus by including in said request packet a device identification number unique to said semiconductor device.

Embodiment 34. The bus subsystem with the features of embodiment 28 further comprising a means for said master device to send control information to a selected one of said discrete memory portions by including in said request packet a specific memory address.

Embodiment 35. The bus subsystem with the features of embodiment 28 further comprising a means for said master device to send control information to substantially all semiconductor devices on said bus by including in said request packet a special device identification number which is recognized by said semiconductor devices.

Embodiment 36. The bus subsystem with the features of embodiment 28 wherein said control information specifies directly or indirectly the number of bus cycles for said master device and said memory device to wait before beginning said bus access phase.

Embodiment 37. The bus subsystem with the features of embodiment 36 wherein, for a data block transfer, said master device and said memory device use the same access time and same data block size regardless of whether said data block transfer is a read or write operation.

Embodiment 38. The bus subsystem with the features of embodiment 28 wherein said control information further includes a block-size value that encodes and specifies the size of the block of data to be transferred.

Embodiment 39. The bus subsystem with the features of embodiment 38 wherein said block-size value is encoded as a linear value for relatively small block sizes values and is encoded as a logarithmic value for relatively larger block sizes.

Embodiment 40. The bus subsystem with the features of embodiment 38 wherein said block-size value is encoded using four bits, and where the encoded value is

Encoded Value	Block Size (Bytes)
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	16

(continued)

	Encoded Value	Block Size (Bytes)
5	10	32
	11	64
	12	128
	13	256
	14	512
10	15	1024

Embodiment 41. The bus subsystem with the features of embodiment 26 wherein said memory device is a DRAM device containing

a plurality of sense amplifiers,

15 a means to hold said sense amplifiers in an unmodified state after a read or write operation, leaving the device in page mode,

a means to precharge said sense amplifiers and

a means for selecting whether to precharge said sense amplifiers or to hold said sense amplifiers in an unmodified state.

Embodiment 42. The bus subsystem with the features of embodiment 28 wherein said request packet comprises an

even number of bytes.

Embodiment 43. The bus subsystem with the features of embodiment 28 further including a means for generating and controlling a plurality of bus cycles, during which said bus carries said address, data and control information, and wherein alternate said bus cycles are designated odd cycles and even cycles, respectively, and wherein said request packet begins only on an even cycle.

Embodiment 44. The bus subsystem with the features of embodiment 28 further including a means for generating ECC information corresponding to a block of data and a means for using said ECC information to correct errors in storing or reading said block of data, wherein said ECC information may be stored separately from said block of data.

Embodiment 45. The bus subsystem with the features of embodiment 44 further comprising at least two of said memory devices wherein said ECC information and said corresponding block of data are stored in a first and a second said memory device, respectively, and said master device includes a means to write or read said block of data with error correction by sending separate ones of said request packets for said ECC information and for said corresponding block of data.

Embodiment 46. A bus subsystem comprising

a memory device and a master device connected in parallel on a bus,

35 a means for said master device to send a request packet and initiate a bus transaction and

a means for said master device to keep track of current and pending bus transactions,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by said memory devices,

said bus containing substantially fewer lines than the number of bits in a single address, and

40 said bus carrying device-select information without the need for separate device-select lines connected directly to individual devices on said bus, whereby said master device initiates bus transactions which transfer information between devices on said bus and collisions on said bus are avoided because said master device avoids initiating bus transactions which would conflict with current or pending bus transactions.

Embodiment 47. The bus subsystem with the features of embodiment 46 having at least two of said master devices and including

45 a collision detecting means whereby a first said master device sending a first said request packet can detect a second said master device sending one of said colliding request packets, where one of said said colliding request packet may be sent simultaneous with the initial sending of or overlapping the sending of said first request packet, and

an arbitration means whereby said first and said second master devices select a priority order in which each of said

50 master devices will be allowed to access said bus sequentially.

Embodiment 48. The bus subsystem with the features of embodiment 47 wherein each of said master devices has a master ID number and each of said request packets includes a master ID position which is a predetermined number of bits in a predetermined position in said request packet, and wherein said collision detection means comprises

a means included in each master device for sending a request packet including said master ID number of said master device in said master ID position of said request packet and

55 a means to detect a collision and invoke said arbitration means if any master device detects any other master ID number in said master ID position.

Embodiment 49. The bus subsystem with the features of embodiment 47 wherein each of said master devices includes

a means for sending a request packet,
 a means for driving a selected bus line or lines during at least one selected bus cycle while said request packet is being sent,
 a means for monitoring said selected bus line or lines to see if a said master device is sending a colliding request packet and
 a means for informing all other master devices that a collision has occurred and for invoking said arbitration means.

5 Embodiment 50. The bus subsystem with the features of embodiment 47 wherein each of said master devices includes
 a means, when sending a request packet, to drive a selected bus line or lines with a certain current during at least one
 selected bus cycle,

a means for monitoring said selected bus line or lines for a greater than normal current to see if another master device
 is driving that line or lines,

10 a means for detecting said greater than normal current, and

a means for informing all said master devices that a collision has occurred and for invoking said arbitration means.

Embodiment 51. The bus subsystem with the features of embodiment 47 wherein said arbitration means comprises

a means for initiating an arbitration cycle,

a means for allocating a single bus line to each master device during at least one selected bus cycle relative to the start
 15 of said arbitration cycle,

a means for allocating each master device to a single bus line during one of said selected bus cycles if there are more
 master devices than available bus lines,

a means for each of said master devices which sent a colliding request packet to drive said bus line allocated to said
 master device during said selected bus cycle, and

20 a means in at least one of said master devices for storing information about which master devices sent a colliding request
 packet,

whereby said master devices can monitor selected bus lines during said arbitration cycle and identify each said master
 device which sent a colliding request packet.

Embodiment 52. The bus subsystem with the features of embodiment 47 wherein said arbitration means comprises

25 a means included in a first one of said master devices which sent colliding request packets for identifying each of said
 master devices which sent colliding request packets,

a means for assigning a priority to each said master device which sent a colliding request packet, and

a means for allowing each said master device which sent a colliding request packet to access the bus sequentially
 according to that priority.

30 Embodiment 53. The bus subsystem with the features of embodiment 52 wherein said priority is based on the physical
 location of each of said master devices.

Embodiment 54. The bus subsystem with the features of embodiment 52 wherein said priority is based on said master
 ID number of said master devices.

Embodiment 55. The bus subsystem with the features of embodiment 52 wherein each of said master devices includes

35 a means, when sending a colliding request packet, for deciding which master device can send the next request packet
 in what order or at what time, whereby no master device may send a new request packet until responses to each pending
 request packet have been completed or scheduled.

Embodiment 56. A bus subsystem comprising

40 a plurality of semiconductor devices connected in parallel to a bus,

said bus including a plurality of bus lines for carrying substantially all address, data and control information needed by
 said semiconductor devices,

said control information including device-select information,

said bus containing substantially fewer lines than the number of bits in a single address,

said bus carrying said device-select information without the need for separate device-select lines connected directly to
 45 individual semiconductor devices,

said semiconductor devices including a reset means having an input and an output, the output of the reset means of
 one semiconductor device being connected to the input of the reset means of the next semiconductor device in series.

Embodiment 57. The bus subsystem with the features of embodiment 56 further including system reset means comprising

a means for generating a first and a second reset signal,

50 a means for passing said first reset signal to a first of said semiconductor devices and then to subsequent ones of said
 semiconductor devices in series and

a means for passing a second reset signal to said first semiconductor device and then to said subsequent semiconductor
 devices in series,

said bus subsystem including one of said semiconductor devices containing

55 a device identification register adapted to contain a number unique to said semiconductor device within said bus sub-
 system,

a device identification register setting means, and

a device reset means for resetting said semiconductor device to some desired, known reset state in response to said

first reset signal and for setting said device identification register in response to said second reset signal, whereby said bus subsystem can be reset to a known reset state with a unique device identification value in said device identification register of each of said semiconductor devices.

Embodiment 58. The bus subsystem with the features of embodiment 57 wherein said desired, known reset state is where all registers in the semiconductor device are cleared and the state machines are reset.

Embodiment 59. The bus subsystem with the features of embodiment 57 wherein said device identification register setting means comprises

a means for detecting said second reset signal,

a means for reading a device identification number from said bus lines at a specific time relative to said second reset signal and

a means for storing said device identification number in said device identification register of said semiconductor device.

Embodiment 60. The bus subsystem with the features of embodiment 57 wherein said second reset signal comprises multiple pulse sequences and wherein said device identification setting means includes

a means for interpreting said pulse sequences as a device identification number and

a means for storing said device identification number in said device identification register of said semiconductor device.

Embodiment 61. The bus subsystem with the features of embodiment 57 wherein said device reset means comprises an n -stage shift register capable of storing n -bit values, wherein said device reset means interprets a specific value in said shift register as said first reset signal and interprets a specific value in said shift register as said second reset signal.

Embodiment 62. The bus subsystem with the features of embodiment 57 wherein one of said semiconductor devices is a master device, said master device including a means for generating said first and said second reset signals.

Embodiment 63. The bus subsystem with the features of embodiment 57 wherein one of said semiconductor devices is a master device, said master device including

a master ID register,

a means for assigning a master ID number to said master device and

a means for storing said master ID number in said master ID register.

Embodiment 64. The bus subsystem with the features of embodiment 63 further comprising a second one of said master devices, and a means for a first one of said master devices to assign a master ID number to substantially all other said master devices, whereby said first master device assigns one of said master ID numbers to each of said master devices on said bus subsystem and each said master device stores said assigned master ID number in said master ID register.

Embodiment 65. The bus subsystem with the features of embodiment 57 wherein one of said semiconductor devices includes a device-type register adapted to contain an identifier characteristic of that type of semiconductor device, and one or more modifiable registers, at least one of which is an access-time register adapted for storing access times.

Embodiment 66. The bus subsystem with the features of embodiment 65 wherein one of said semiconductor devices is a master device having

a means for selecting a semiconductor device,

a means for reading said device-type register of said selected semiconductor device,

a means for determining the device type of said selected semiconductor device,

a means for determining access-time values appropriate for said selected semiconductor device and for storing said access-time values in said access-time registers of said selected semiconductor device, and

a means for selecting and storing other values appropriate for said selected semiconductor device in corresponding registers of said selected semiconductor device,

whereby said master device can select a semiconductor device, determine what type it is, and set said access-time and other registers to contain appropriate values.

Embodiment 67. The bus subsystem with the features of embodiment 66 further comprising a memory device having at least one discrete memory section and at least one modifiable address register adapted to store memory address information which corresponds to each of said discrete memory sections, and

said master device further comprising a means for selecting and testing each of said discrete memory sections and a means for storing address information in said address registers corresponding to each of said discrete memory sections, whereby said master device can test all said discrete memory sections and assign unique address values thereto.

Embodiment 68. A bus subsystem comprising

two semiconductor devices connected in parallel to a bus, one of said semiconductor devices being a master device, said bus including a plurality of bus data lines for carrying substantially all address, data and control information needed by said semiconductor devices,

said control information including device-select information,

said bus containing substantially fewer of said bus data lines than the number of bits in a single address, and

said bus carrying device-select information without the need for separate device-select lines connected directly to individual semiconductor devices,

wherein all of said bus data lines are terminated transmission lines and all of said address, data and control information

is carried on said bus data lines as a sequential series of bits in the form of low-voltage-swing signals.

Embodiment 69. The bus subsystem with the features of embodiment 68 further comprising a semiconductor device including a current-mode driver connected to drive one of said bus data lines.

Embodiment 70. The bus subsystem with the features of embodiment 69 further comprising a semiconductor device having a means to measure the voltage of said low-voltage-swing signals on a selected one of said bus data lines, whereby said semiconductor device can determine whether zero, one, or more than one of said current-mode drivers are driving said selected bus data line.

Embodiment 71. The bus subsystem with the features of embodiment 70 further comprising a semiconductor device having

a plurality of input receivers connected to one of said bus data lines, and a selection means for selecting said input receivers one by one to sense and store, one at a time, the bits of said sequential series of bits.

Embodiment 72. The bus subsystem with the features of embodiment 70 further comprising a semiconductor device having two input receivers connected to one of said bus data lines.

Embodiment 73. A bus subsystem comprising two semiconductor devices connected in parallel to a bus having a first and a second end, said bus including a bus clock line, said bus clock line having first and second ends corresponding to said first and second ends of said bus, respectively, a clock generator connected to said first end of said bus clock line to generate early bus clock signals with a normal rise time, and

signal return means at said second end of said bus clock line to return said early bus clock signals to said first end of said bus as corresponding late bus clock signals, whereby each of said early bus clock signals will propagate from said clock generator along said clock line starting from said first end to said second end of said bus and then return at a later time to said first end of said bus as a corresponding late bus clock signal, whereby each semiconductor device on said bus can detect said early bus clock signals and said corresponding late bus clock signals.

Embodiment 74. The bus subsystem with the features of embodiment 73 further comprising a first and a second said bus clock line having first and second ends at said first and said second ends of said bus, respectively, wherein said signal return means directly connects said second ends of said first and said second bus clock lines whereby each of said early bus clock signals will propagate from said clock generator at said first end of said bus along said first bus clock line to said second end of said bus and then return on said second bus clock line to said first end of said bus as one of said corresponding late bus clock signals.

Embodiment 75. The bus subsystem with the features of embodiment 73 wherein said signal return means comprises said first bus clock line without a line terminator at said second end thereof whereby each of said early bus clock signals reaching said second end of said first bus clock line will be reflected back along said first bus clock line as said corresponding late bus clock signals.

Embodiment 76. The bus subsystem with the features of embodiment 73 further comprising a means for operating said bus in bus cycles timed to have a certain bus cycle frequency and a corresponding bus cycle period and

a means for operating said clock generator with a period of twice the bus cycle period.

Embodiment 77. The bus subsystem with the features of embodiment 76 wherein said bus cycle frequency is greater than approximately 50 MHz and less than or equal to approximately 500 MHz.

Embodiment 78. The bus subsystem with the features of embodiment 73 further including a semiconductor device having an internal device clock generating means to derive the midpoint time between said early and corresponding late bus clock signals and to generate an internal device clock synchronized to said midpoint time.

Embodiment 79. The bus subsystem with the features of embodiment 73 further including a semiconductor device having a low-skew clock generator circuit comprising a first delay line having an input, an output and a basic delay and means for synchronizing the output of said first delay line with said early bus clock signal,

a second delay line having said basic delay plus a variable delay, said second delay line having an output and a means for synchronizing the output of said second delay line with said late bus clock signal, and a third delay line having a third delay and a means to set said third delay midway between the delays of said first and second delay lines, said third delay line having an output which provides an internal device clock signal synchronized to a time halfway between said early and said late bus clock signals.

Embodiment 80. The bus subsystem with the features of embodiment 73 wherein said early and said late bus clock signals are low-voltage-swing signals that transition cyclically between low and high logical values, and further including a semiconductor device having a low-skew clock generator circuit comprising a DC amplifier to convert said early and said late bus clock signals into full-swing logic signals, a first variable delay line having a first variable delay and an input and an output, the input of said first variable delay

line being connected to said DC amplifier

a first, a second and a third additional delay line, each having an input and an output, the input of each of said additional delay lines being connected to the output of said first delay line,

said first additional delay line having a fixed delay,

said second additional delay line having said fixed delay plus a second variable delay, and

said third additional delay line having said fixed delay plus one half of said second variable delay,

a first clocked input receiver connected to sample said early bus clock signal and gated by said output of said first additional delay line,

a means for adjusting said first variable delay so said first clocked input receiver samples said early bus clock signal just as said early bus clock signal transitions,

a second clocked input receiver connected to sample said late bus clock signal and gated by said output of said second additional delay line,

a means for adjusting said second variable delay so said second clocked input receiver samples said late bus clock signal just as said late bus clock signal transitions,

whereby said output of said third additional delay line is synchronized to a time halfway between said outputs of said first and said second additional delay lines, and said output of said third additional delay line provides an internal device clock signal.

Embodiment 81. The bus subsystem with the features of embodiment 80 further comprising a semiconductor device having

a first one of said low-skew clock generator circuits which generates a "true" internal device clock signal and

a second one of said low-skew clock generator circuits connected to generate a "complement" internal device clock signal synchronized with but opposite in logical value to said "true" internal device clock signal.

Embodiment 82. A DRAM device designed to be connected to an external bus having a plurality of bus lines for carrying substantially all address, data and control information needed by said DRAM device as a sequential series of bits, said

control information including device-select information, said external bus containing substantially fewer said bus lines than the number of bits in a single address, and said bus carrying device-select information without the need for separate

device-select lines connected directly to said DRAM device, said DRAM device comprising

an array of memory cells connected in rows and columns, each of said memory cells adapted to store one of said bits, a row address selection means for selecting one of said rows,

a column sense amp connected to each of said columns, each of said column sense amps adapted to latch one of said bits as a binary logical value or to precharge to a selected state,

a column decoding means connected to each of said column sense amps for selecting a plurality of said column sense amps for inputting one of said bits to or outputting one of said bits from said memory cells,

an internal I/O bus having a plurality of internal I/O lines wherein each of said internal I/O lines is connected to a plurality of said column sense amps, and

a plurality of bus connection means designed to connect said internal I/O lines to said external bus,

whereby a selected bit of said sequential series of bits can be transferred from said external bus to a selected one of said memory cells or said bit contained in a selected one of said memory cells can be transferred to said external bus.

Embodiment 83. The DRAM device with the features of embodiment 82 further comprising

an output driver connected to one said bus connection means,

an output multiplexer having an output connected to said output driver and a plurality of inputs, each of said inputs being connected to one of said internal I/O lines, and

a control means to select whether said output driver can drive said external bus,

whereby a plurality of memory cells are selected using said row address selection means and said column decoding means and a plurality of bits contained in said plurality of memory cells are output through said column sense amps to

said internal I/O bus to said output multiplexer to said output driver to said external bus.

Embodiment 84. The DRAM device with the features of embodiment 82 further comprising

a plurality of input receivers connected to one of said bus data lines and to said internal I/O bus,

a selection means for selecting said input receivers one by one to sense and store, one at a time, the bits of said sequential series of bits, and

a control means to select whether an input receiver can drive said internal I/O bus, whereby a bit of said sequential series of bits is input from said external bus through one of said input receivers to one of said internal I/O lines to one

of said column sense amps to one of said memory cells.

Embodiment 85. The DRAM device with the features of embodiment 82 further comprising

a first and a second half-array of said memory cells wherein each said row of said array of said memory cells is subdivided into two parts,

a first and a second one of said internal I/O buses connected to said column sense amps in said first and said second half-arrays, respectively, and

a column decoder means to gate selected ones of said column sense amps connected to said memory cells in a selected row of said first and said second half-arrays simultaneously.

Embodiment 86. The DRAM device with the features of embodiment 85 wherein said column decoder means selects sixteen column sense amps at a time.

Embodiment 87. The DRAM device with the features of embodiment 82 wherein said external bus operates at a certain speed and wherein said DRAM device includes four of said internal I/O buses, each of which operates at one-fourth the speed of said external bus.

Embodiment 88. The DRAM device with the features of embodiment 82 further comprising a means for precharging one of said column sense amps to a precharged state from which a binary logical value can quickly be loaded into said column sense amp, if said column sense amp contains a binary logical value, a means for latching the logical value currently contained in said column sense amp and a means for instructing said DRAM device to precharge said column sense amp or latch said binary logical value in said column sense amp.

Embodiment 89. The DRAM device with the features of embodiment 88 further comprising a means for instructing said DRAM device to precharge said column sense amp without further instruction whenever said row address selection means selects a different one of said rows.

Embodiment 90. The DRAM device with the features of embodiment 88 further comprising a means for instructing said DRAM device to precharge said column sense amp without further instruction at a first or a second preselected time after latching the latest said binary logical value, said first preselected time being long enough for said DRAM to latch said binary logical value into said column sense amp and transfer said binary logical value into memory or onto one of said internal I/O lines, and said second preselected time being a variable which can be stored in said DRAM device whereby said DRAM can latch a binary logical value into said column sense amp for transferring said binary logical value into or out of a selected said memory cell, then precharge to allow a faster subsequent read or write.

Embodiment 91. A package containing a semiconductor die having a side, circuitry and a plurality of connecting areas positioned along or near said side, spaced at a selected pitch and connected to said circuitry, said package comprising a plurality of bus connecting means for connecting to a plurality of external bus lines, each of said external bus lines corresponding to one of said connecting areas, each of said bus connecting means being positioned on a first side of said package, connected to one said external bus line and to said corresponding connecting area on said semiconductor die, and spaced at a pitch substantially identical to said selected pitch of said connecting areas, whereby each of said external bus lines can be connected to said corresponding connecting area on said semiconductor die by bus connection means positioned along a single side of said package.

Embodiment 92. The package with the features of embodiment 91 further comprising a plurality of said bus connecting means wherein each of said bus connecting means includes a pin adapted for connection to one of said external bus lines and a wire connecting said pin to one of said connecting areas on said semiconductor die, said wire having an effective lead length less than about 4 millimeters and wherein the effective lead length of said wire of each of said bus connection means for said package is approximately equal.

Embodiment 93. A plurality of packages with the features of embodiment 91 wherein at least two of said semiconductor die are memory devices, each of said packages being generally flat, having a top and a bottom, and wherein said packages are physically secured adjacent and parallel to each other in a stack, where a first one of said packages is adjacent to a second one of said packages in said stack, said top of said first package is substantially aligned with said bottom of said second package, and said bus connecting means of each of said packages are substantially aligned and are lying substantially in a plane.

Embodiment 94. The plurality of packages with the features of embodiment 93 further comprising a plurality of stacks wherein each of said bus connecting means can be electrically connected to corresponding said bus connecting means in each of said stacks.

Embodiment 95. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and at least one modifiable identification register accessible to said bus through said connection means, whereby data may

be transmitted to said register via said bus and enable said device thereafter to be uniquely identified.

Embodiment 96. The semiconductor device with the features of embodiment 95 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

5 Embodiment 97. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

10 connection means adapted to connect said semiconductor device to said bus, and at least one modifiable register to hold device address information, said modifiable register accessible to said bus through said connection means, whereby data may be transmitted to said register via said bus which enables said device thereafter to respond to a predetermined range of addresses.

15 Embodiment 98. The semiconductor device with the features of embodiment 97 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

20 Embodiment 99. The semiconductor device with the features of embodiment 98 wherein said memory device has at least one discrete memory section and also has at least one modifiable address register adapted to store memory address information which corresponds to each said discrete memory section.

Embodiment 100. The semiconductor device with the features of embodiment 99 wherein said memory address information comprises a pointer to said discrete memory section.

25 Embodiment 101. The semiconductor device with the features of embodiment 100 wherein said discrete memory section has a top and a bottom and said memory address information comprises pointers to said top and said bottom.

Embodiment 102. The semiconductor device with the features of embodiment 100 wherein said memory address information comprises

a pointer to said discrete memory section and
a range value indicating the size of said discrete memory section.

30 Embodiment 103. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data and control information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, said semiconductor device comprising

35 connection means adapted to connect said semiconductor device to said bus, and
at least one modifiable access-time register accessible to said bus through said connection means, whereby data may be transmitted to said register via said bus which establishes a predetermined amount of time that said semiconductor device thereafter must wait before using said bus in response to a request.

40 Embodiment 104. The semiconductor device with the features of embodiment 103 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 105. The semiconductor device with the features of embodiment 103 further comprising at least two access-time registers and one of said access-time registers is permanently programmed to contain a fixed value and at least one of said access-time registers can be modified by information carried on said bus.

45 Embodiment 106. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, and wherein each said bus line is a terminated transmission line, said semiconductor device comprising

50 connection means adapted to connect said semiconductor device to said bus, and
a bus line driver capable of producing a low-voltage-swing signal on one of said terminated transmission lines.

55 Embodiment 107. The semiconductor device with the features of embodiment 106 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 108. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying

substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said bus further including at least one bus clock line for carrying early and late bus clock signals, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, and an internal device clock generating means which generates an internal device clock synchronized to a time halfway between said early and said late bus clock signals.

Embodiment 109. The semiconductor device with the features of embodiment 108 wherein said bus further includes a first and a second one of said bus clock lines, said first bus clock line carries said early bus clock signal and said second bus clock line carries said late bus clock signal, said semiconductor device further comprising a means to detect said early bus clock signal on said first bus clock line and a means to detect said late bus clock signal on said second bus clock line.

Embodiment 110. The semiconductor device with the features of embodiment 109 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 111. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying a sequential series of bits substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus, a plurality of input receivers connected to one of said bus data lines and a selection means for selecting said input receivers one by one to sense and store, one at a time, the bits of said sequential series of bits.

Embodiment 112. The semiconductor device with the features of embodiment 111 wherein said semiconductor device is a memory device which connects substantially only to said bus and sends and receives substantially all address, data and control information over said bus.

Embodiment 113. The semiconductor device with the features of embodiment 112 wherein two input receivers are connected to one of said bus lines.

Embodiment 114. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said bus system includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said system bus, an internal input/output bus within said semiconductor device having more lines than said system bus, and a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus.

Embodiment 115. The semiconductor device with the features of embodiment 114 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 116. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said system bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said system bus, an internal input/output bus within said semiconductor device having more lines than said system bus, a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus, and

at least one modifiable identification register accessible to said system bus through said connection means, whereby data may be transmitted to said register via said system bus and which enables said device thereafter to be uniquely identified.

Embodiment 117. The semiconductor device with the features of embodiment 116 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 118. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said system bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, said semiconductor device comprising connection means adapted to connect said semiconductor device to said system bus, an internal input/output bus within said semiconductor device having more lines than said system bus, a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus, and at least one modifiable register to hold device address information, said modifiable register accessible to said system bus through said connection means, whereby data may be transmitted to said register via said system bus which enables said device thereafter to respond to a predetermined range of addresses.

Embodiment 119. The semiconductor device with the features of embodiment 118 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 120. The semiconductor device with the features of embodiment 119 wherein said memory device has at least one discrete memory section and also has at least one modifiable address register adapted to store memory address information which corresponds to each said discrete memory section.

Embodiment 121. A semiconductor device capable of use in an architecture for a semiconductor system bus including a plurality of semiconductor devices connected in parallel to a bus wherein said system bus includes a plurality of bus lines for carrying substantially all address, data and control information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said system bus, and has substantially fewer bus lines than the number of bits in a single address, said semiconductor device comprising connection means adapted to connect said semiconductor device to said system bus, an internal input/output bus within said semiconductor device having more lines than said system bus, a means for multiplexing the lines of said internal bus to the lines of said system bus, whereby said system bus can run at a higher speed than said internal bus, and at least one modifiable access-time register accessible to said system bus through said connection means, whereby data may be transmitted to said register via said system bus which establishes a predetermined amount of time that said semiconductor device thereafter must wait before using said system bus in response to a request.

Embodiment 122. The semiconductor device with the features of embodiment 121 wherein said semiconductor device is a memory device which connects substantially only to said system bus and sends and receives substantially all address, data and control information over said system bus.

Embodiment 123. The semiconductor device with the features of embodiment 121 further comprising at least two access-time registers and one of said access-time registers is permanently programmed to contain a fixed value and at least one of said access-time registers can be modified by information carried on said system bus.

Embodiment 124. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying substantially all address, data, control and device-select information needed by said semiconductor device for communication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus lines than the number of bits in a single address, and carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to said individual semiconductor device, wherein said address, data, control and device-select information is carried over said bus in the form of request packets and bus transactions, said semiconductor device comprising connection means adapted to connect said semiconductor device to said bus, a means to receive said request packets over said bus, a means to decode information in said request packets, and a means to respond to said information in said request packets.

Embodiment 125. The semiconductor device with the features of embodiment 124 wherein said means to decode information in said request packet further comprises

a means to identify and decode said control information in said request packet,
 a means to identify and decode said device-select information in said request packet,
 a means to identify and decode said address information in said request packet and
 a means to determine whether said control information or said address information instructs said semiconductor device
 5 to begin a response.

Embodiment 126. The semiconductor device with the features of embodiment 124 wherein each of said bus transactions
 is carried out in response to said address and said control information in one of said request packets, and wherein said
 means to identify and decode information in said request packets includes a means to identify a sequence of bytes on
 said bus as one of said request packets containing said address and said control information, said control information
 10 including information about the type of said bus transaction being requested and the access time which needs to intervene
 before beginning said bus transaction over said bus and said address and said control information includes device-
 select information instructing one or more said semiconductor devices to respond to said address and said control
 information.

Embodiment 127. The semiconductor device with the features of embodiment 124 further comprising
 15 a plurality of sense amplifiers adapted to precharge to a selected state or to latch a bit of information,
 a means to hold said sense amplifiers in an unmodified state after latching one of said bits of information,
 a means to precharge said sense amplifiers and
 a means for selecting whether said semiconductor device should precharge said sense amplifiers or should hold said
 sense amplifiers in an unmodified state.

Embodiment 128. The semiconductor device 124 wherein said means to respond to
 20 said information, where said information is control information, further comprises a means to
 transfer a data block during a data block transfer, further including a means to
 read data from said semiconductor device and
 write data into said semiconductor device, and

25 initiate a data block transfer,
 transfer a data block of a selected size,
 transfer a data block at a selected time,
 access a control register, including a means to read from or write to said control register, or
 select normal or page-mode access.

Embodiment 129. The semiconductor device with the features of embodiment 124 further comprising a means to respond
 30 to said information in said request packet if said information includes a device identification number unique to said
 semiconductor device.

Embodiment 130. The semiconductor device with the features of embodiment 124 further comprising a means to respond
 to said information in said request packet if said information includes a special device identification number which calls
 35 for said semiconductor device to respond.

Embodiment 131. The semiconductor device with the features of embodiment 124 further comprising a means to respond
 to said information in said request packet if said information includes an address unique to said semiconductor device.

Embodiment 132. The semiconductor device with the features of embodiment 124 further comprising a means to interpret
 said control information and decode the time to wait before beginning said bus transaction over said bus.

Embodiment 133. The semiconductor device with the features of embodiment 124 further comprising a means to interpret
 40 said control information and decode the size of a data block to transfer during one of said bus transactions.

Embodiment 134. The semiconductor device with the features of embodiment 124, 125, 126, 127, 128, 129, 130, 131,
 132 or 133 wherein said semiconductor device is a memory device which connects substantially only to said bus and
 sends and receives substantially all address, data and control information over said bus.

Embodiment 135. A semiconductor device capable of use in a semiconductor bus architecture including a plurality of
 45 semiconductor devices connected in parallel to a bus wherein said bus includes a plurality of bus lines for carrying
 substantially all address, data, control and device-select information needed by said semiconductor device for commu-
 nication with substantially every other semiconductor device connected to said bus, and has substantially fewer bus
 lines than the number of bits in a single address, and carries device-select information for said semiconductor device
 50 without the need for a separate device-select line connected directly to said individual semiconductor device, wherein
 said address, data, control and device-select information is carried over said bus in the form of request packets and bus
 transactions, said semiconductor device comprising

connection means adapted to connect said semiconductor device to said bus,
 a means to encode address and control information in said request packets and
 55 a means to send said request packets over said bus.

Embodiment 136. The semiconductor device 135 further comprising a means to request a bus transaction wherein each
 of said bus transactions is carried out in response to said address and said control information in one of said request
 packets, and wherein said means to encode information in said request packets includes a means to mark a sequence

of bytes on said bus as one of said request packets, said control information including information about the type of said bus transaction being requested and the access time which needs to intervene before beginning said bus transaction over said bus and said address and said control information includes device-select information instructing one or more said semiconductor devices to respond to said address and said control information.

5 Embodiment 137. The semiconductor device with the features of embodiment 135 wherein one or more of said plurality of semiconductor devices has a unique device identification number, said semiconductor device further comprising a means to send control information to a specific one of said plurality of semiconductor devices by including in said request packet a selected said device identification number.

10 Embodiment 138. The semiconductor device with the features of embodiment 135 wherein each of said plurality of semiconductor devices is adapted to respond to a special device identification number, said semiconductor device further comprising a means to send control information to each of said plurality of semiconductor devices by including in said request packet said special device identification number.

15 Embodiment 139. The semiconductor device with the features of embodiment 135 wherein one or more of said plurality of semiconductor devices is a memory device having a plurality of addresses, said semiconductor device further comprising a means to send control information to a specific address or range of addresses in one of said plurality of semiconductor devices by including said specific address or range of addresses in said request packet.

Embodiment 140. The semiconductor device with the features of embodiment 135 wherein at least one of said request packets is a request packet requesting a bus transaction which is followed by a corresponding one of said bus transactions, said semiconductor device further comprising a means to encode said control information to specify directly or indirectly the time between the end of said request packet requesting a bus transaction and said corresponding bus transaction over said bus.

20 Embodiment 141. The semiconductor device with the features of embodiment 140 wherein one type of said bus transactions is a transfer of a data block, said semiconductor device further comprising a means to encode said control information to specify the size of said data block to transfer.

25 Embodiment 142. The semiconductor device with the features of embodiment 140 further comprising a means to keep track of current and pending bus transactions, whereby collisions on said bus are avoided because said semiconductor device avoids initiating bus transactions which would conflict with current or pending bus transactions.

Embodiment 143. The semiconductor device with the features of embodiment 135 wherein said semiconductor device is a first master device and one of said plurality of semiconductor devices is a second master device, further comprising a collision detecting means whereby said first master device when sending a first one of said request packets can detect said second master device sending a colliding one of said request packets, where said colliding request packet may be sent simultaneously with the initial sending of or overlapping the sending of said first request packet, and an arbitration means whereby said first and said second master devices select a priority order in which each of said master devices will be allowed to access said bus sequentially.

30 Embodiment 144. The semiconductor device with the features of embodiment 143 wherein said semiconductor device is a master device and at least one of said plurality of semiconductor devices is a master device, each of said master devices has a master ID number and each of said request packets includes a master ID position which is a predetermined number of bits in a predetermined position in said request packet, and wherein said collision detection means comprises a means for said semiconductor device to send its master ID number in said request packet and

40 a means to detect a collision and invoke said arbitration means if said semiconductor device detects any other master ID number in said master ID position.

Embodiment 145. The semiconductor device with the features of embodiment 144 wherein said system bus architecture includes a means for carrying information on said bus during bus cycles, said semiconductor device further comprising a means for driving a selected bus line or lines during at least one selected bus cycle while sending each said request packet,

45 a means for monitoring said selected bus line or lines to see if another said master device is sending one of said colliding request packets and

a means for informing all said master devices that a collision has occurred and for invoking said arbitration means.

50 Embodiment 146. The semiconductor device with the features of embodiment 145 further comprising a means, when sending a request packet, for driving a selected bus line or lines with a certain current during at least one selected bus cycle,

a means for monitoring said selected bus line or lines for a greater than normal current to see if another said master device is driving that line or lines,

a means for detecting said greater than normal current, and

55 a means for informing all said master devices that a collision has occurred and for invoking said arbitration means.

Embodiment 147. The semiconductor device with the features of embodiment 143 wherein said arbitration means comprises

a means for initiating an arbitration cycle,

a means for allocating a single bus line to each said master device during at least one selected bus cycle relative to the start of said arbitration cycle,
a means for allocating each said master device to a single bus line during one of said selected bus cycles if there are more master devices than available bus lines,
5 a means for each of said master devices which sent one of said colliding request packets to drive said bus line allocated to said master device during said selected bus cycle, and
a means in at least one of said master devices for storing information about which master devices sent one of said colliding request packets,
whereby said master devices can monitor selected bus lines during said arbitration cycle and identify each said master
10 device which sent one of said colliding request packets.
Embodiment 148. The semiconductor device with the features of embodiment 143 wherein said arbitration means comprises
a means for identifying each of said master devices which sent one of said colliding request packets,
a means for assigning a priority to each said master device which sent one of said colliding request packets, and
15 a means for allowing each said master device which sent one of said colliding request packets to access the bus sequentially according to that priority.
Embodiment 149. The semiconductor device with the features of embodiment 143 wherein said priority is based on the physical location of each of said master devices.
Embodiment 150. The semiconductor device with the features of embodiment 143 wherein said priority is based on said
20 master ID number of said master devices.

Claims

25 1. A semiconductor device, the semiconductor device comprising:

connection means adapted to connect the semiconductor device to an external bus which is a part of a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to the external bus, wherein the external bus includes a plurality of bus lines for carrying substantially all address, data, control
30 and device select information needed by the semiconductor device for communication with substantially every other semiconductor device connected to the external bus, and wherein the external bus has substantially fewer bus lines than the number of bits in a single address, wherein each bus line of the plurality of bus lines is a terminated transmission line, and wherein the external bus carries device-select information for the semiconductor device without the need for a separate device-select line connected directly to the semiconductor device,
35 a bus line driver capable of producing a low-voltage-swing signal on one of the terminated transmission lines,

wherein the semiconductor device is a semiconductor transceiver device to interface multiple units to the bus,
wherein the multiple units are DRAMs.

40 2. The semiconductor device of claim 1,
wherein the bus line driver is capable of producing a low-voltage-swing signal having a voltage swing of not more than 1 Volt on one of said terminated transmission lines.

45 3. The semiconductor device of claim 1 or 2,
wherein the bus line driver is capable of producing a low-voltage-swing signal having a voltage swing of at least 500 mV on one of said terminated transmission lines.

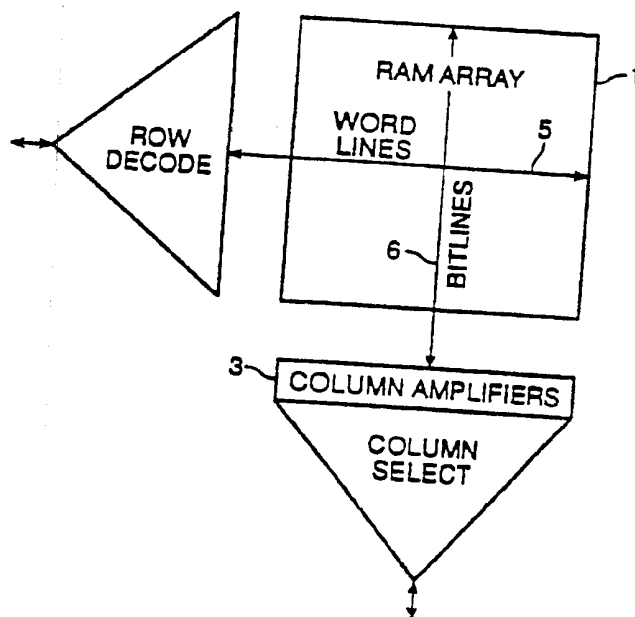


FIG. 1

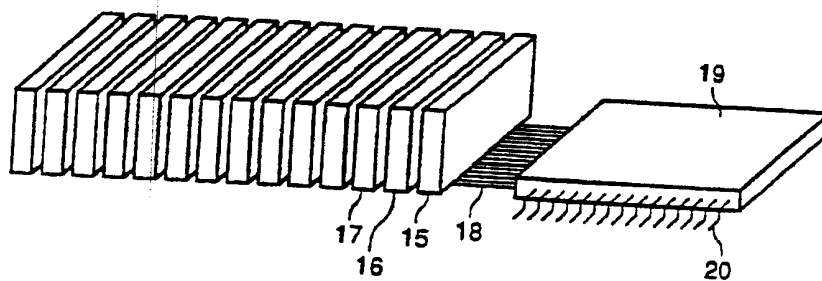


FIG. 3

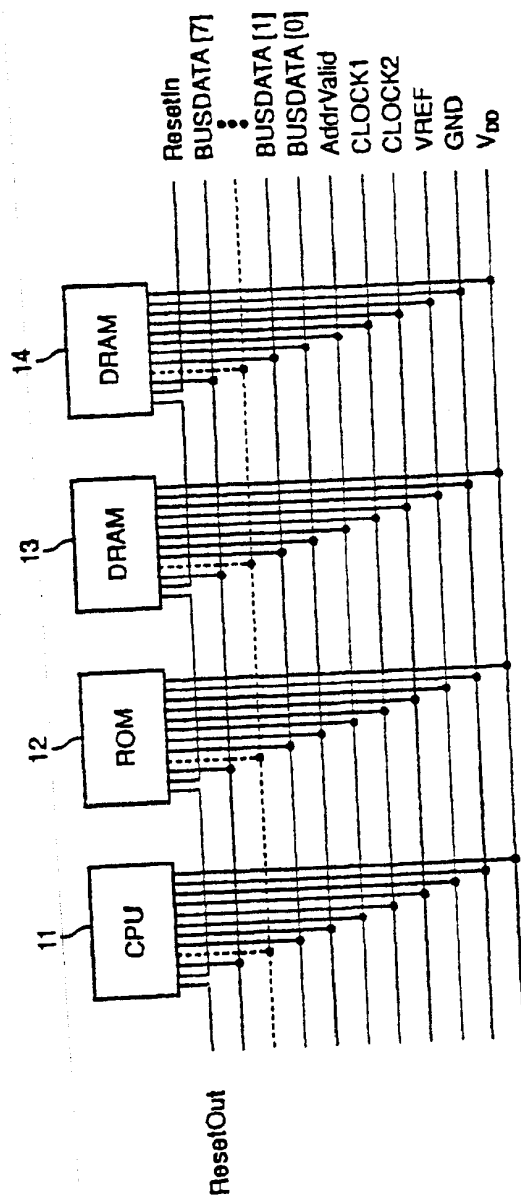


FIG. 2

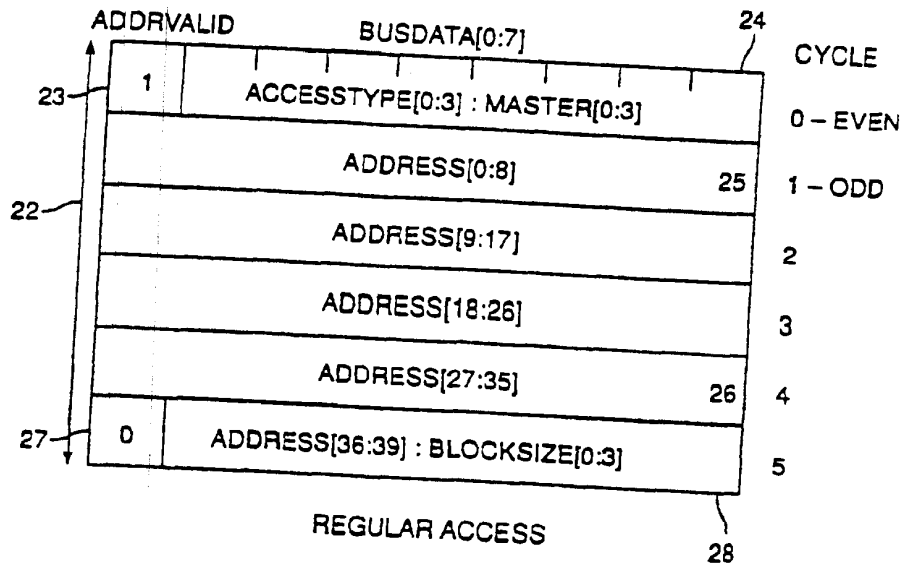


FIG. 4

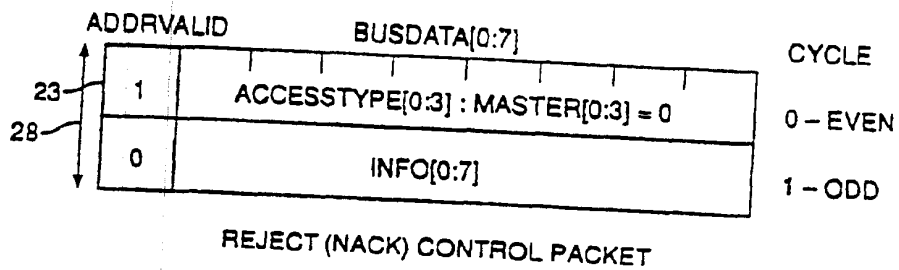


FIG. 5

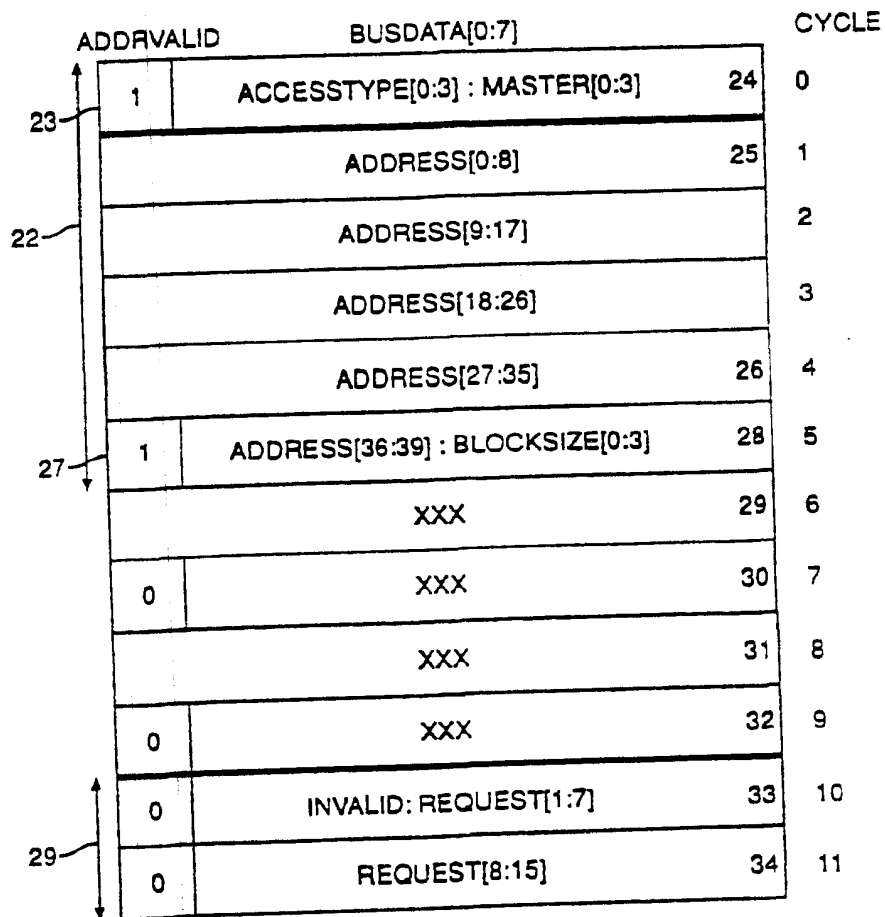


FIG. 6



FIG. 7a

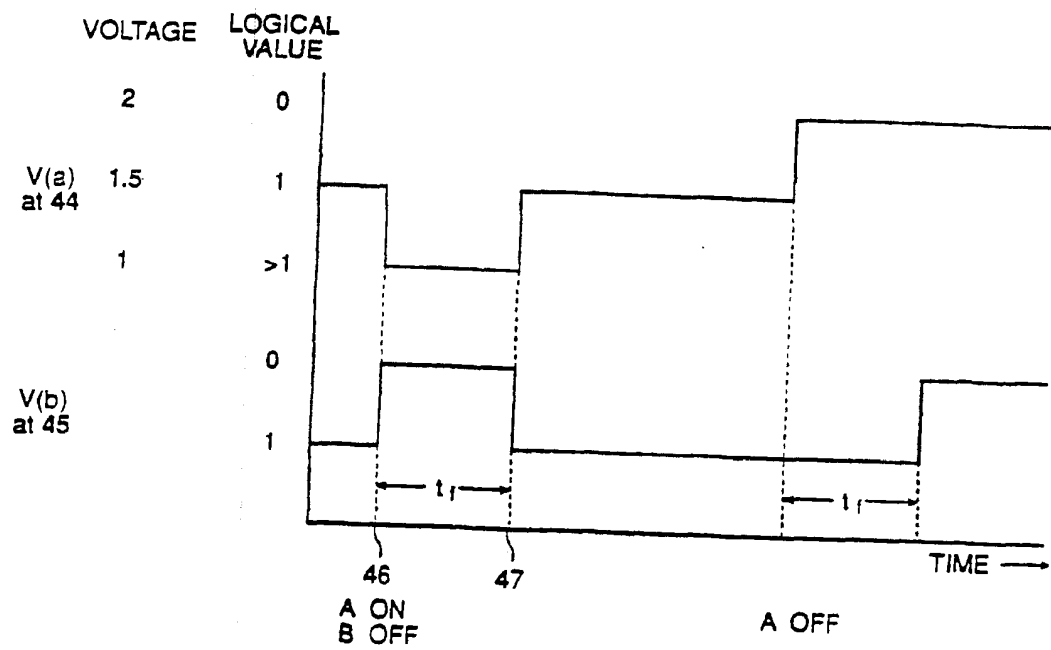


FIG. 7b

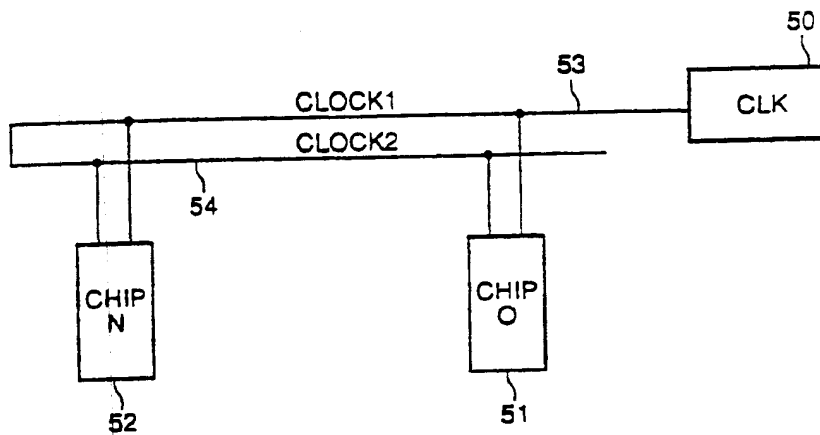


FIG. 8a

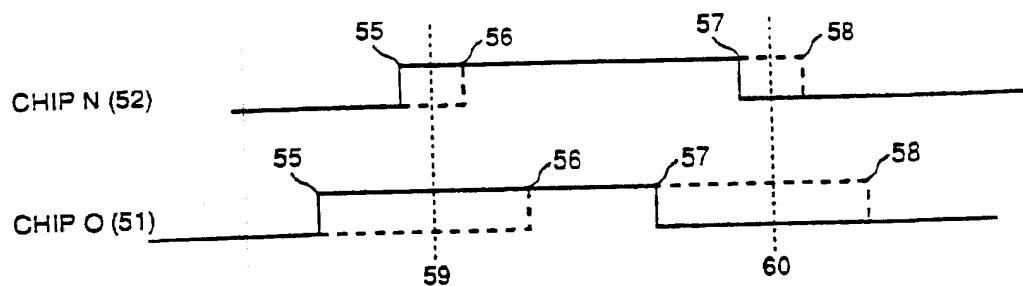


FIG. 8b

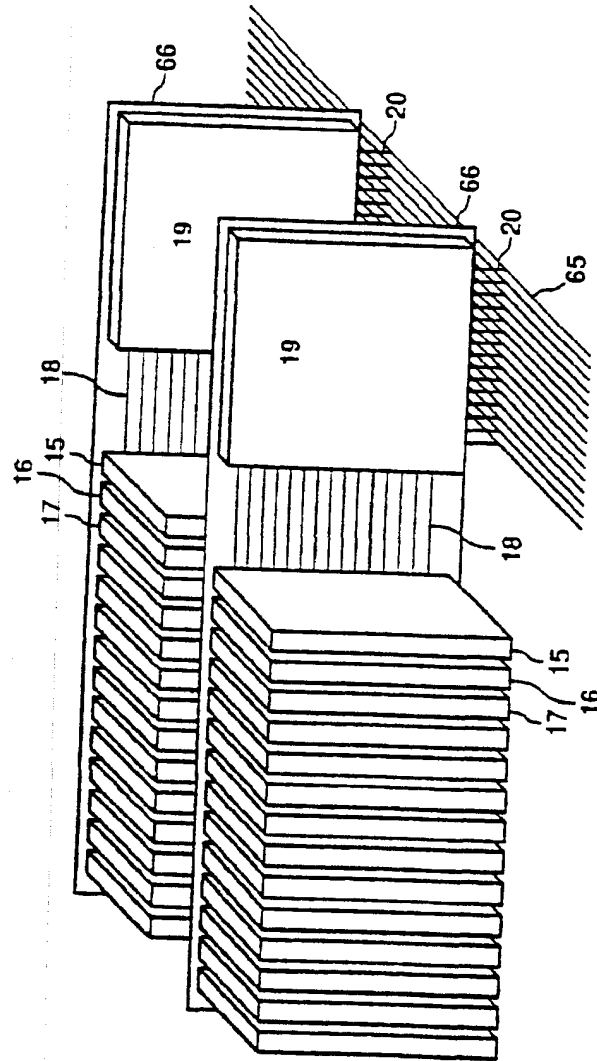


FIG. 9

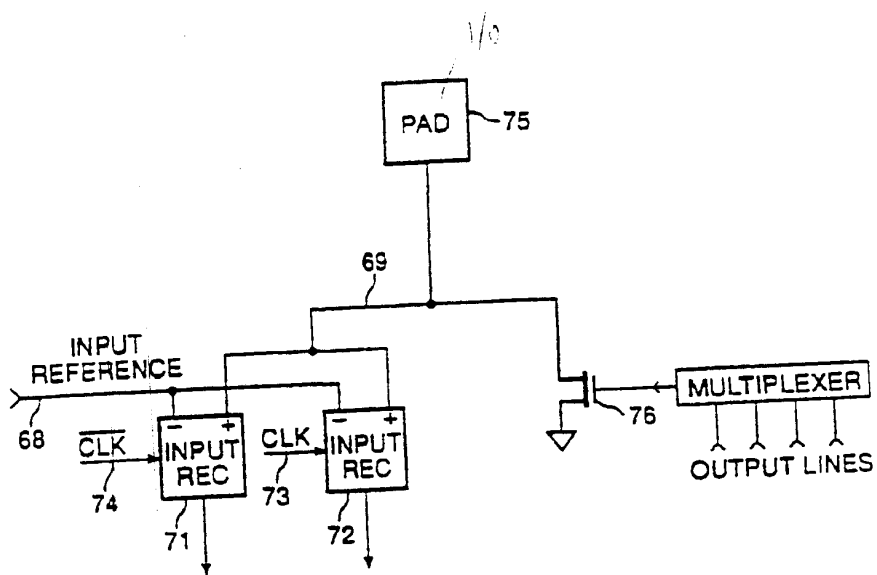


FIG. 10

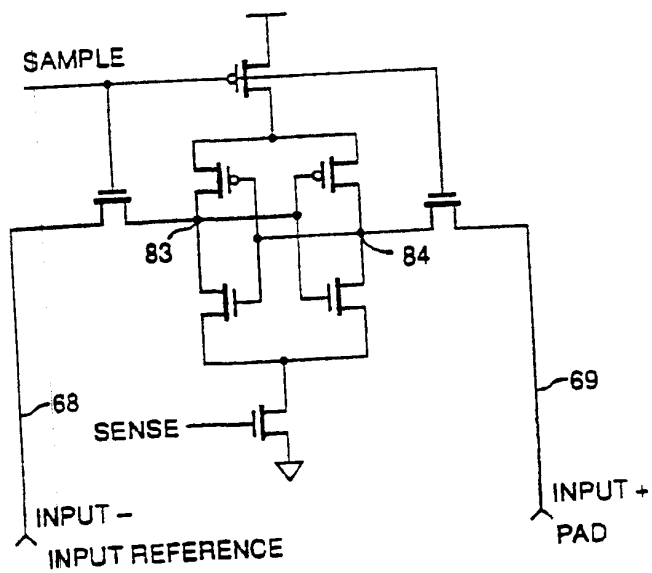


FIG. 11

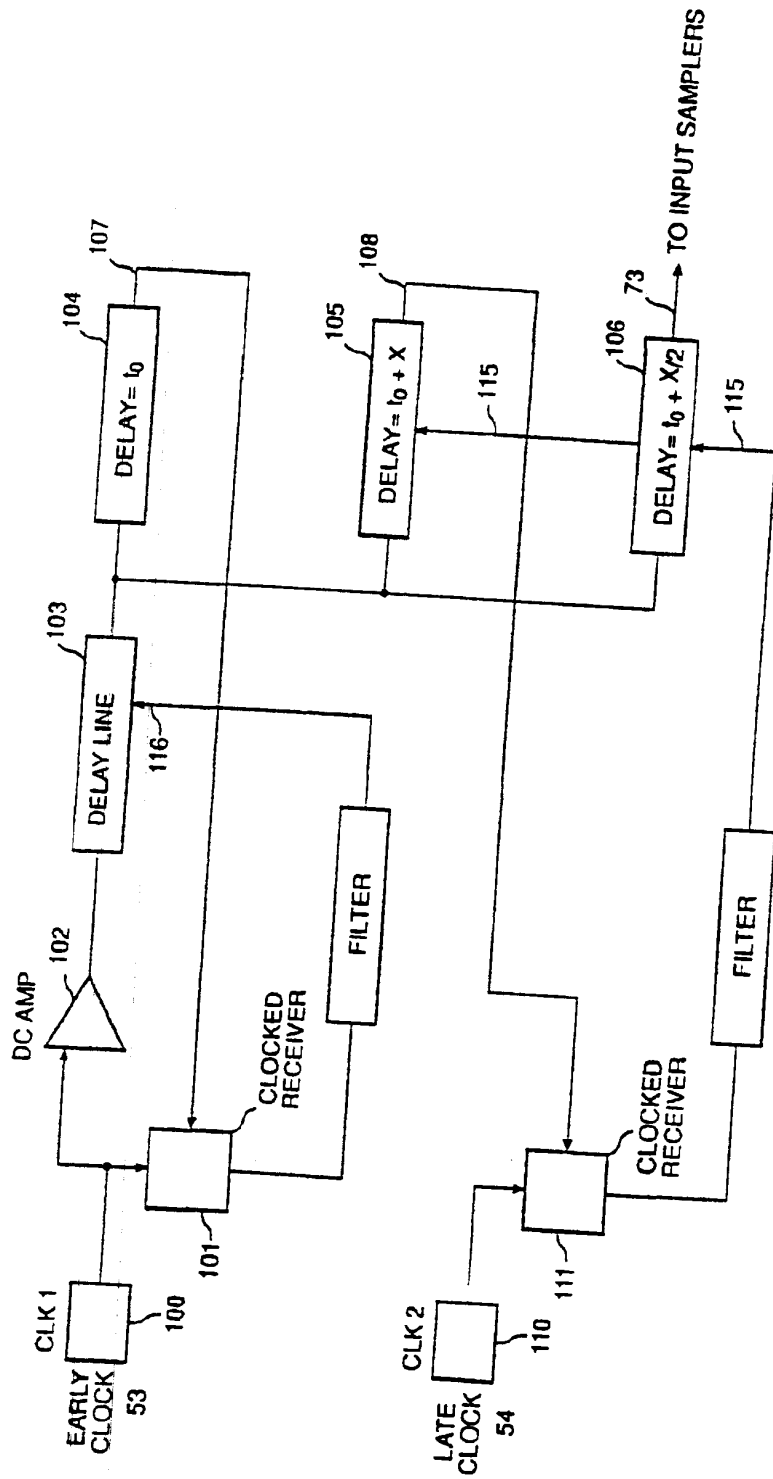


FIG. 12

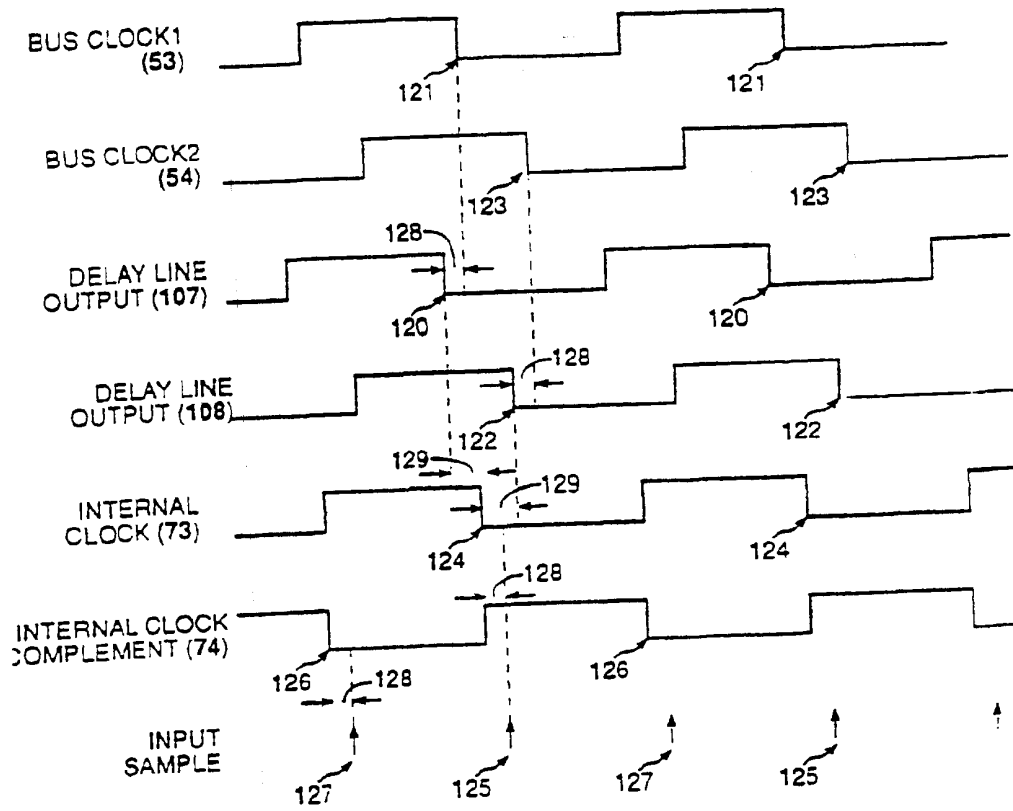


FIG. 13

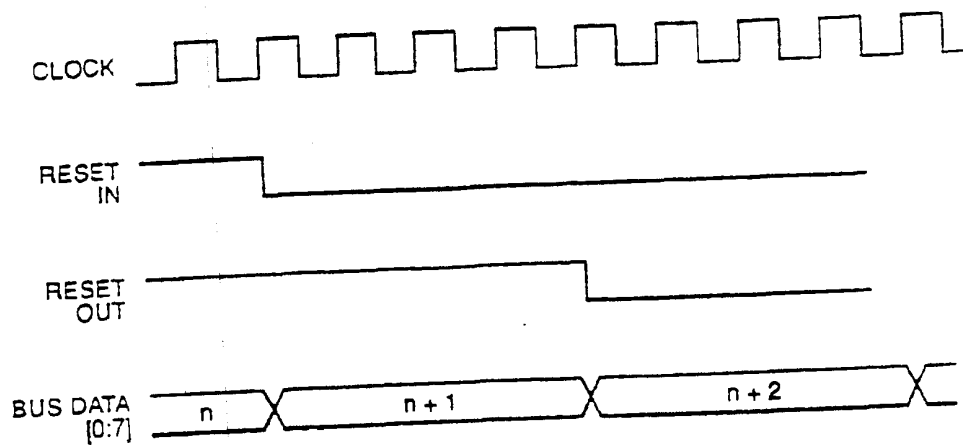


FIG. 14

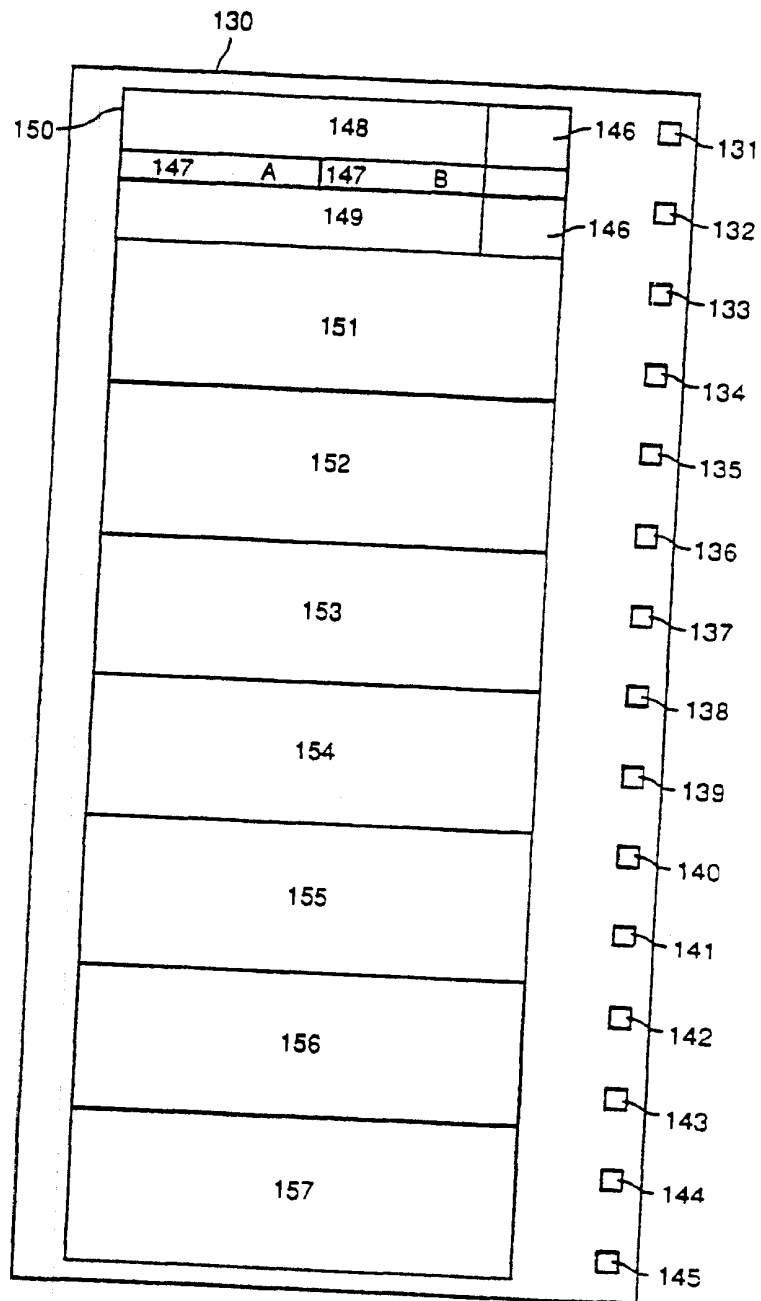


FIG. 15

REFERENCES CITED IN THE DESCRIPTION

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(19)



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G11C 7/00 (2006.01)

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00100018.1 / 1 022 641
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(54) **Integrated circuit I/O using a high performance bus interface**

(57) The present invention includes a semiconductor device, the semiconductor device comprising connection means adapted to connect the semiconductor device to an external bus which is a part of a semiconductor bus architecture including a plurality of semiconductor devices connected in parallel to the external bus, wherein the external bus includes a plurality of bus lines for carrying substantially all address, data, control and device select information needed by the semiconductor device for communication with substantially every other semiconductor device connected to the external bus, and wherein

the external bus has substantially fewer bus lines than the number of bits in a single address, wherein each said bus line is a terminated transmission line, and wherein the external bus carries device-select information for said semiconductor device without the need for a separate device-select line connected directly to the semiconductor device, a bus line driver capable of producing a low-voltage-swing signal on one of said terminated transmission lines, wherein the semiconductor device is a semiconductor transceiver device to interface multiple units to the bus, wherein the multiple units are DRAMs.

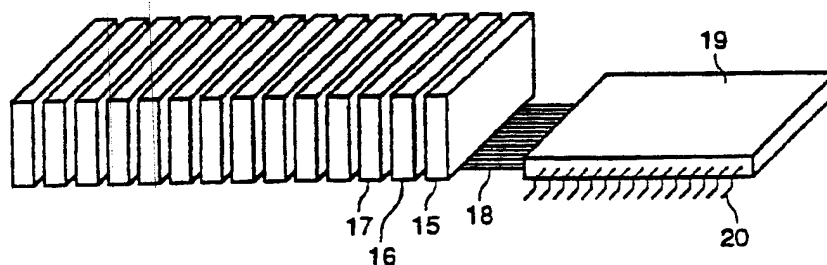


FIG. 3

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European Patent
Office

EUROPEAN SEARCH REPORT

Application Number
EP 06 12 5954

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			TECHNICAL FIELDS SEARCHED (IPC)
			G06F G11C
4 The present search report has been drawn up for all claims			
Place of search The Hague		Date of completion of the search 10 July 2007	Examiner NGUYEN XUAN HIEP, C
CATEGORY OF CITED DOCUMENTS X: particularly relevant if taken alone Y: particularly relevant if combined with another document of the same category A: technological background O: non-written disclosure P: intermediate document T: theory or principle underlying the invention E: earlier patent document, but published on, or after the filing date D: document cited in the application L: document cited for other reasons & member of the same patent family, corresponding document			

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Electronic Acknowledgement Receipt

EFS ID:	21226449
Application Number:	14324990
International Application Number:	
Confirmation Number:	1022
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C2
Receipt Date:	16-JAN-2015
Filing Date:	07-JUL-2014
Time Stamp:	02:35:38
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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Information:

57	Non Patent Literature	Exhibit_G1_G6.pdf	814795	no	79
			f3816f131e664f7c7386b336a28043f2ba245fa3		

Warnings:

Information:

58	Non Patent Literature	2003_ALTERA_acex_iK.pdf	3876406	no	86
			3f5c5ab2b41558a9baef0259106af34eb9cbaa8f		

Warnings:

Information:

59	Non Patent Literature	1989_Horowitz_Art_of_Electronics.pdf	804824	no	15
			4fe58ae9db8168f345983f24ffe9b752877cc2c4		

Warnings:

Information:

60	Non Patent Literature	Exhibit_H_Diablo.pdf	45048	no	3
			81f1e4da57ee373a3d87d4c1e42a28642560bda9		

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Total Files Size (in bytes):			232407025		
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	1	of	12	Attorney Docket Number	NETL.045C2

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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	212.	JP09237492	09-09-1997	Toshiba Corp.	Machine Translation	
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				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C2
Sheet	8	of	12		

	227.	WO2006/055497	05-26-2006	Intel Corporation	
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Examiner Initials *	Cite No. ¹	Application Number	Filing Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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	230.	13/971,231	8/20/2013	Netlist, Inc.	Entire Prosecution History
	231.	14/229,844	3/29/2014	Netlist, Inc.	Entire Prosecution History
	232.	14/337,168	7/21/2014	Netlist, Inc.	Entire Prosecution History

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
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	234.	Written Opinion of the International Preliminary Examining Authority, PCT/US2010/040826, October 24, 2011	
	235.	International Preliminary Report on Patentability, PCT/US2010/040826, November 28, 2011	
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	239.	Response to Non-Final Office Action mailed September 13, 2012, for U.S. Patent Application No. 12/761,179, filed March 13, 2013, 16 pages.	
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	241.	Inter Partes Review Case No. IPR2014-01029, Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on June 24, 2014.	
	242.	Inter Partes Review Case No. IPR2014-01029, Exhibit 1008 to Petition for Inter Partes Review, "Declaration of Charles J. Neuhauser, Ph.D. under 37 C.F.R. § 1.68," filed on June 24, 2014.	
	243.	Inter Partes Review Case No. IPR2014-01029, Supplemental Petition for Inter Partes Review of U.S. Patent No. 8,516,185, filed on July 23, 2014.	
	244.	Inter Partes Review Case No. IPR2014-01029, Patent Owner's Preliminary Response pursuant to 37 C.F.R. § 42.107, filed on October 17, 2014.	
	245.	Inter Partes Review Case No. IPR2014-01029, Decision Denying Institution of Inter Partes Review 37 C.F.R. § 42.108, issued December 16, 2014.	

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				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	9	of	12	Attorney Docket Number	NETL.045C2

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
	246.	Inter Partes Review Case No. IPR2014-01369, Corrected Petition for Inter Partes Review of Claims 1-19 of U.S. Patent No. 8,516,185, filed on September 22, 2014.		
	247.	Inter Partes Review Case No. IPR2014-01369, Exhibit 1008 to Corrected Petition for Inter Partes Review, “Declaration of Dr. Nader Bagherzadeh under 37 C.F.R. § 1.68,” filed on September 22, 2014.		
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	251.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-00882, Exhibit 1007 to Petition for Inter Partes Review, “Declaration of Dr. Srinivasan Jagannathan,” filed on June 22, 2014.		
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	254.	Inter Partes Review of U.S. Patent No. 8,081,536, IPR Case No. IPR2014-00883, Corrected Petition for Inter Partes Review of U.S. Patent No. 8,081,536, filed on July 8, 2014.		
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	256.	Inter Partes Review of U.S. Patent No. 8,081,536, IPR Case No. IPR2014-00883, Patent Owner’s Preliminary Response Pursuant to 37 C.F.R. § 42.107, filed October 7, 2014		
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				Application Number	14/324,990
				Filing Date	July 7, 2014
				First Named Inventor	Hyun Lee
				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	10	of	12	Attorney Docket Number	NETL.045C2

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.		T ²
	262.	Inter Partes Review of U.S. Patent No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 3001 to Decision – Institution of Inter Partes Review, Excerpts from IEEE Dictionary, issued December 16, 2014		
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	267.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.’s Invalidity Contentions, dated June 6, 2014		
	268.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to “Smart Storage Systems, Inc.’s Invalidity Contentions,” dated June 6, 2014.		
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				Art Unit	2184
				Examiner Name	BANSAL, GURTEJ
Sheet	11	of	12	Attorney Docket Number	NETL.045C2

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Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.		T ²
	275.	U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G-1 to G-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated June 6, 2014.		
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	297.	Miles J. Murdocca et al., "Principles of Computer Architecture", Prentice Hall, 2000, pp. 249-251		
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Electronic Acknowledgement Receipt

EFS ID:	21226480
Application Number:	14324990
International Application Number:	
Confirmation Number:	1022
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C2
Receipt Date:	16-JAN-2015
Filing Date:	07-JUL-2014
Time Stamp:	03:14:05
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Non Patent Literature	2001_Huang_BIST_Method_VI SI_Design.pdf	3496109 6bc20bd6ff74761ea00ffdc2e5032ebb5bdf1321	no	17

Warnings:

Information:

2	Non Patent Literature	2002-12-10_Jacob_SDRAM_Architectures.pdf	5974132	no	22
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Information:					
3	Non Patent Literature	2001_JEDEC_21-C_4-5-7_168_Pin_RSDRAM.pdf	801227	no	8
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4	Non Patent Literature	1997_JEDEC_21C_4-6-1-278_Pin.pdf	4025672	no	24
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Warnings:					
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Warnings:					
Information:					
11	Non Patent Literature	2013-12-31_NFOA_NETL075A_13411344.pdf	971619	no	28
			b278719d3b2dddffc1637e378af88207fd28f63		
Warnings:					
Information:					
12	Non Patent Literature	2011-11-17_Office_Action_NETL018CPCCC2_13473413.pdf	2305035	no	46
			cbd6a7c2bba9d79504ef99af9de7c3613344c86c		
Warnings:					
Information:					
13	Non Patent Literature	2014-01-13_Amendment_NFOA_NETL071A_13288850.pdf	168355	no	15
			b33ca64fdc3320438b5b3ac2786db1fd6b4608e0		
Warnings:					
Information:					
14	Non Patent Literature	2014-03-31_Amendment_NFOA_NETL075A_13411344.pdf	140357	no	12
			61b4323ec431a93c0b8ea73482dfab3ec42d7d25		
Warnings:					
Information:					
15	Non Patent Literature	2013-02-12_Notice_of_Allowance_048A.pdf	2287416	no	52
			0c5794115a0cd49f569c122cc0850ba4169798a5		
Warnings:					
Information:					
16	Non Patent Literature	2014-01-02_NFOA_NETL018P1C6.pdf	2348292	no	42
			127ee6aa81f352113c242f877f736f1ff7ac948b		
Warnings:					
Information:					
17	Non Patent Literature	2014-04-02_Rsps_NFOA_NETL018P1C6.pdf	146023	no	12
			fc425b898e830757f902cd1d25ae548fddbda50		
Warnings:					
Information:					
18	Non Patent Literature	2014-04-02_Office_Action_NT001JP.pdf	215922	no	7
			7c6abc5db1e611da113d9158b8152e1564d0fc6		
Warnings:					

Information:					
19	Non Patent Literature	2000_Murdocca_Principles_of_Comp_Arch.pdf	216398 c5695cd6c6e94460d33a58f7e03bb10fe1b7588d	no	5
Warnings:					
Information:					
20	Non Patent Literature	2006-10_Intel_6400-6402_Adv_Mem_Buff_Datasheet.pdf	21327672 37950adab1b9c9d431c3711414b06c87c5845b0c	no	250
Warnings:					
Information:					
21	Non Patent Literature	2003-03_Jedec_DDR_SDRAM_Spec_JESD79C.pdf	6291832 404f7d8951a88990fd4f45fdac578edc532eef4	no	82
Warnings:					
Information:					
22	Non Patent Literature	2002-02_Jedec_Design_Spec_PC133_SDRAM_Reg_DIMM_Rev1-4.pdf	2385368 39d57c14167aaf682fa574824caffa89d0a43852	no	75
Warnings:					
Information:					
23	Non Patent Literature	2002-JEDEC_PC2100_PC1600_DDR_SDRAM_Reg_Design.pdf	3571816 abf1a0c4e0c639a658bcd212408e8d334984b844	no	82
Warnings:					
Information:					
24	Non Patent Literature	2003-05_Jedec_Def_of_the_SS_TV16859_2V_JESD82-4B.pdf	813559 7cf36d2c62b7f85787ffa07f96b0f96a12c0704d	no	17
Warnings:					
Information:					
25	Non Patent Literature	2002_Journal_of_Sys_Arch_A_banked-promotion_trans_lookaside_buff_sys.pdf	1112497 42166274ded228f33d03d54269982e9ef7ec7587	no	13
Warnings:					
Information:					
26	Non Patent Literature	2000-09-01_onchip_cache_compress_tech_overhead_and_design_complex.pdf	1940557 a1fd5188b2047efb97df437a34f00fbc63770e14	no	18
Warnings:					
Information:					
27	Non Patent Literature	2004-01_Jedec_DDR_SDRAM_Spec_JESD79D.pdf	5081472 70637eeb494a681c2bb5ca236e386a311bb77662	no	83
Warnings:					

Information:					
28	Transmittal Letter	2015-01-15_IDS_Cover_NETL045C2.pdf	58070 bf736e982c18374a7915542973a98bb27e0efffd	no	2
Warnings:					
Information:					
29	Information Disclosure Statement (IDS) Form (SB08)	2015-01-15_IDS_NETL045C2.pdf	391475 0a3bfa4b50c6f0c15bfcaf972962db8170389af5	no	12
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
Total Files Size (in bytes):			94139870		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: NETLIST, INC.
App. No.	: 14/324,990
Filed	: July 7, 2014
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: BANSAL, GURTEJ
Art Unit	: 2189
Conf No.	: 1022

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 305 references. Copies of foreign documents and/or their translation and non-patent literature are submitted herewith.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits

Application No.: 14/324,990

Filing Date: July 7, 2014

was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 50-5963.

Respectfully submitted,

Dated: January 15, 2015

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 273-6008

51167

(Reg. No.)

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 14/324,990	Filing Date 07/07/2014	☐ To be Mailed
---	---	----------------------------------	---------------------------------------

ENTITY: ☐ LARGE ☒ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	01/16/2015	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$40 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	X \$210 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
TOTAL ADD'L FEE						0	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

LIE
/CHRISTINE V. MOORE/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/324,990	07/07/2014	Hyun Lee	NETL.045C2

CONFIRMATION NO. 1022

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

NOTICE



Date Mailed: 10/01/2014

INFORMATIONAL NOTICE TO APPLICANT

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

The item(s) indicated below are also required and should be submitted with any reply to this notice to avoid further processing delays.

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

Hyun Lee
Jayesh R. Bhakta
Jeffrey C. Solomon
Mario Jesus Martinez
Chi-She Chen



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/324,990	07/07/2014	2189	800	NETL.045C2	12	3

CONFIRMATION NO. 1022

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

UPDATED FILING RECEIPT



Date Mailed: 10/01/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 07/18/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/324,990**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Circuit Providing Load Isolation and Noise Reduction

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/324,990

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)
FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(j))	12 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

* If the difference in column 1 is less than zero, enter "0" in column 2.

SMALL ENTITY

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

OR

OTHER THAN SMALL ENTITY

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)	(Column 2)	(Column 3)
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus **	=
Independent (37 CFR 1.16(h))	*	Minus ***	=
Application Size Fee (37 CFR 1.16(s))			
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			

SMALL ENTITY

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

OTHER THAN SMALL ENTITY

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

	(Column 1)	(Column 2)	(Column 3)
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus **	=
Independent (37 CFR 1.16(h))	*	Minus ***	=
Application Size Fee (37 CFR 1.16(s))			
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))			

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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UNITED STATES DEPARTMENT OF COMMERCE
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/324,990	07/07/2014	2827	800	NETL.045C2	12	3

CONFIRMATION NO. 1022

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

UPDATED FILING RECEIPT



Date Mailed: 09/29/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 07/18/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/324,990**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Circuit Providing Load Isolation and Noise Reduction

Preliminary Class

365

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/324,990

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	70		N/A	
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	300		N/A	
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	360		N/A	
TOTAL CLAIMS (37 CFR 1.16(j))	12	minus 20 = *	x 40 =	0.00	OR		
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3	minus 3 = *	x 210 =	0.00			
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			0.00			
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				0.00			
			TOTAL	730		TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

		(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/324,990	07/07/2014	Hyun Lee	NETL.045C2

CONFIRMATION NO. 1022

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

NOTICE



Date Mailed: 09/29/2014

INFORMATIONAL NOTICE TO APPLICANT

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

The item(s) indicated below are also required and should be submitted with any reply to this notice to avoid further processing delays.

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

Hyun Lee
Jayesh R. Bhakta
Jeffrey C. Solomon
Mario Jesus Martinez
Chi-She Chen

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/324,990
Filed : July 7, 2014
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Examiner : Unknown
Art Unit : 2827
Conf. No. : 1022
Docket No. : NETL.045C2

CERTIFICATE OF EFS WEB
TRANSMISSION

I hereby certify that this correspondence and
any other attachment noted on the Automated
Acknowledgement Receipt, is being transmitted
from within the Pacific Time zone to the
Commissioner for Patents via the EFS Web
Server on:

September 22, 2014

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

RESPONSE TO NOTICE TO FILE MISSING PARTS DATED JULY 22, 2014
AND
CLAIM OF SMALL ENTITY STATUS

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

In response to the Notice to File Missing Parts of Nonprovisional Application dated July 22, 2014, Applicant hereby submits the following items in the above identified patent application:

- The statutory basic filing fee for small entity,
- The application search fee for small entity,
- The application examination fee for small entity, and
- Surcharge as set forth in 37 CFR 1.16(f) for small entity.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

The Assignee of all rights and title in the above-identified patent application, Netlist, Inc., is a small entity concern as defined in 37 C.F.R. 1.27(a). Assertion of entitlement to small entity status under 37 C.F.R. 1.27(c) is hereby made in this application.

Applicants acknowledge the duty to file notification of a loss of entitlement to small entity status in the patent prior to paying, or at the time of paying, any maintenance fee due after the date on which status as a small entity is no longer appropriate.

Respectfully submitted,

Date: September 22, 2014

By: /Jamie J. Zheng/
Jamie J. Zheng
Registration No. 51,167
Customer No. 79,141
(650) 273-6008

Electronic Patent Application Fee Transmittal

Application Number:	14324990			
Filing Date:	07-Jul-2014			
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.045C2			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360
Pages:				
Claims:				
Miscellaneous-Filing:				
Late Filing Fee for Oath or Declaration	2051	1	70	70
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				800

Electronic Acknowledgement Receipt

EFS ID:	20212503
Application Number:	14324990
International Application Number:	
Confirmation Number:	1022
Title of Invention:	Circuit Providing Load Isolation and Noise Reduction
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C2
Receipt Date:	23-SEP-2014
Filing Date:	07-JUL-2014
Time Stamp:	00:39:33
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$800
RAM confirmation Number	8879
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Transmittal Letter	2014-09-22_Missing_Parts_NET L045C2.pdf	68944 a5f24d9e6e5702da830f3028ab563e52273 a2f90	no	2
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	36467 51b7623e88e9413b3120c9d5b74a6787f38 76a5c	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			105411		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Netlist, Inc.
App. No. : 14/324,990
Filed : July 7, 2014
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Examiner : Unknown
Art Unit : 2827
Conf. No. : 1022
Docket No. : NETL.045C2

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Commissioner for Patents via the EFS Web
Server on:

September 22, 2014

(Date)

/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

RESPONSE TO NOTICE TO FILE MISSING PARTS DATED JULY 22, 2014
AND
CLAIM OF SMALL ENTITY STATUS

Commissioner for Patents
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Alexandria, VA 22313-1450

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- The application examination fee for small entity, and
- Surcharge as set forth in 37 CFR 1.16(f) for small entity.

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Respectfully submitted,

Date: September 22, 2014

By: /Jamie J. Zheng/
Jamie J. Zheng
Registration No. 51,167
Customer No. 79,141
(650) 273-6008

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
14/324,990

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	12 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	70
N/A	300
N/A	360
x 40 =	0.00
x 210 =	0.00
	0.00
	0.00
TOTAL	730

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
14/324,990	07/07/2014	2827	0.00	NETL.045C2	12	3

CONFIRMATION NO. 1022

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

FILING RECEIPT



Date Mailed: 07/22/2014

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Applicant(s)

Netlist, Inc., Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This application is a CON of 13/412,243 03/05/2012 PAT 8782350
which is a CON of 12/422,853 04/13/2009 PAT 8154901
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

If Required, Foreign Filing License Granted: 07/18/2014

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 14/324,990**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

**** SMALL ENTITY ****

Title

Circuit Providing Load Isolation and Noise Reduction

Preliminary Class

365

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
14/324,990	07/07/2014	Hyun Lee	NETL.045C2

CONFIRMATION NO. 1022

FORMALITIES LETTER



79141
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Date Mailed: 07/22/2014

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. Small entity discount is in effect. If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **70** basic filing fee.
- \$ **70** surcharge.
- \$ **300** search fee.
- \$ **360** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **800** TOTAL FEE BALANCE DUE.

Items Required To Avoid Processing Delays:

Applicant is notified that the above-identified application contains the deficiencies noted below. No period for reply is set forth in this notice for correction of these deficiencies. However, if a deficiency relates to the inventor's

oath or declaration, the applicant must file an oath or declaration in compliance with 37 CFR 1.63, or a substitute statement in compliance with 37 CFR 1.64, executed by or with respect to each actual inventor no later than the expiration of the time period set in the "Notice of Allowability" to avoid abandonment. See 37 CFR 1.53(f).

- A properly executed inventor's oath or declaration has not been received for the following inventor(s):

Hyun Lee
Jayesh R. Bhakta
Jeffrey C. Solomon
Mario Jesus Martinez
Chi-She Chen

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/aabranys/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence ⁱ	US
Mailing Address of Inventor:					
Address 1		21 Thalia			
Address 2					
City	Ladera Ranch	State/Province	CA		
Postal Code	92694	Country ⁱ	US		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jayesh	R.	Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
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Mailing Address of Inventor:					
Address 1		12220 Rose Street			
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country ⁱ	US		
Inventor 3					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

City	Irvine	State/Province	CA	Country of Residence i	US
------	--------	----------------	----	------------------------	----

Mailing Address of Inventor:

Address 1	16 Silver Fir				
Address 2					
City	Irvine	State/Province	CA		
Postal Code	92604	Country i	US		

Inventor 4

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Mario	Jesus	Martinez	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Laguna Niguel	State/Province	CA	Country of Residence i	US
------	---------------	----------------	----	------------------------	----

Mailing Address of Inventor:

Address 1	19 Lindall Street				
Address 2					
City	Laguna Niguel	State/Province	CA		
Postal Code	92677	Country i	US		

Inventor 5

Remove

Legal Name

Prefix	Given Name	Middle Name	Family Name	Suffix
	Chi-She		Chen	

Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service

City	Walnut	State/Province	CA	Country of Residence i	US
------	--------	----------------	----	------------------------	----

Mailing Address of Inventor:

Address 1	944 Crystal Water Lane				
Address 2					
City	Walnut	State/Province	CA		
Postal Code	91789	Country i	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the **Add** button.

Add

Correspondence Information:Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).☐ An Address is being provided for the correspondence Information of this application.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2	
		Application Number		
Title of Invention	Circuit Providing Load Isolation and Noise Reduction			
Customer Number	79141			
Email Address	jz@jzpatent.com		Add Email	Remove Email

Application Information:

Title of the Invention	Circuit Providing Load Isolation and Noise Reduction			
Attorney Docket Number	NETL.045C2	Small Entity Status Claimed		☒
Application Type	Nonprovisional			
Subject Matter	Utility			
Suggested Class (if any)		Sub Class (if any)		
Suggested Technology Center (if any)				
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)		

Publication Information:

☐	Request Early Publication (Fee required at time of Request 37 CFR 1.219)
☒	Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application has not and will not be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.			
Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	79141		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.			
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	13412243	2012-03-05
Prior Application Status	Patented	Remove	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
13412243	Continuation of	12422853	2009-04-13	8154901	2012-04-10
Prior Application Status	Expired		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12422853	non provisional of	61044839	2008-04-14		
Prior Application Status	Expired		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12422853	non provisional of	61044825	2008-04-14		
Prior Application Status	Expired		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)		
12422853	non provisional of	61044801	2008-04-14		
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).			
			Remove
Application Number	Country i	Filing Date (YYYY-MM-DD)	Priority Claimed
			☒ Yes ☐ No
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			Add

Authorization to Permit Access:

☐ Authorization to Permit Access to the Instant Application by the Participating Offices

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the World Intellectual Property Office (WIPO), and any other intellectual property offices in which a foreign application claiming priority to the instant patent application is filed access to the instant patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, WIPO, or other intellectual property office in which a foreign application claiming priority to the instant patent application is filed to have access to the instant patent application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the instant patent application with respect to: 1) the instant patent application-as-filed; 2) any foreign application to which the instant patent application claims priority under 35 U.S.C. 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the instant patent application; and 3) any U.S. application-as-filed from which benefit is sought in the instant patent application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing this Authorization.

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor :			
If the Applicant is an Organization check here. ☒			
Organization Name	Netlist, Inc.		
Mailing Address Information:			
Address 1	175 Technology, Suite 150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		
Email Address	nwhitley@netlist.com		
Additional Applicant Data may be generated within this form by selecting the Add button.			Add

Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Assignee 1				
Complete this section only if non-applicant assignee information is desired to be included on the patent application publication in accordance with 37 CFR 1.215(b). Do not include in this section an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest), as the patent application publication will include the name of the applicant(s).				
				Remove
If the Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information:				
Address 1				
Address 2				
City		State/Province		
Country i		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee Data may be generated within this form by selecting the Add button.				Add

Signature:

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications				
Signature	/Jamie J. Zheng/		Date (YYYY-MM-DD)	2014-07-07
First Name	Jamie	Last Name	Zheng	Registration Number
				51167
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C2
		Application Number	
Title of Invention	Circuit Providing Load Isolation and Noise Reduction		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
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3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is a continuation of U.S. Pat. Appl. No. 13/412,243, filed March 5, 2012, which is a continuation of U.S. Pat. Appl. No. 12/422,853, filed April 13, 2009, now U.S. Patent No. 8,154,901, which claims the benefit of priority from U.S. Provisional Appl. No. 61/1044,839, filed April 14, 2008, from U.S. Provisional Appl. No. 61/044,825, filed April 14, 2008, and from U.S. Provisional Appl. No. 61/044,801, filed April 14, 2008, each of which are incorporated in its entirety by reference herein. This application is also related to U.S. Pat. Appl. No. 12/422,912, filed on April 13, 2009, now abandoned, and U.S. Pat. Appl. No. 12/422,925, filed on April 13, 2009, now U.S. Pat. No. 8,001,434, both of which are incorporated in their entirety by reference herein. This application is further related to U.S. Pat. Appl. No. 13/183,253, filed July 14, 2011, now U.S. Pat. No. 8,359,501, U.S. Pat. Appl. No. 13/745,790, filed January 19, 2013, now U.S. Pat. Appl. No. 8,689,064, and U.S. Pat. Appl. No. 14/229,844, filed March 29, 2014. This application is further related to U.S. Pat. Appl. No. 12/422,912, filed on April 13, 2009, now abandoned, U.S. Pat. Appl. No. 12/422,925, filed on April 13, 2009, now U.S. Pat. No. 8,001,434, U.S. Pat. Appl. No. 13/183,253, filed July 14, 2011, now U.S. Pat. No. 8,359,501, U.S. Pat. Appl. No. 13/745,790, filed January 19, 2013, now U.S. Pat. No. 8,689,064, U.S. Pat. Appl. No. 14/299,844, filed March 29, 2014, U.S. Pat. Appl. No. 12/504,131, filed July 16, 2009, now U.S. Pat. No. 8,417,870, U.S. Pat. Appl. No. 12/761,179, filed April 15, 2010, now U.S. Pat. No. 8,516,185, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, U.S. Pat. Appl. No. 13/287,042, filed November 1, 2011, now, U.S. Pat. No. 8,756,364, U.S. Pat. Appl. No. 13/971,231, filed August 20, 2013, and U.S. Appl. No. 13/288,850, filed November 3, 2011.

BACKGROUND

[0002] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity. Description of the Related Art

[0003] As the demand on the performance of electronic systems increase, the systems include higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0004] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system 10 which incorporates a multiplexer-demultiplexer 12 to isolate unused subsystems of a plurality of subsystems 16, 18. This arrangement can reduce the load on each driver 14 when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver 14, the receiver 20 and the multiple destinations. However, the multiplexer and/or demultiplexer 12 can present transmission line discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with electrical lengths longer than 1/4 of the wavelength of the operational frequency or the signal transition rate). In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

[0005] In memory applications, some systems control "chip select" or device select signals, along with the memory address and control signals, based on the device select signals, in order to disable memory devices when they are not being accessed. Figure 2 schematically illustrates an example of such a system 20 which isolates unused subsystems of a plurality of subsystems 16, 18 using device select signals 22, 24 (e.g., chip select signals). The devices are disabled by not activating (asserting) the address, control, and/or device select signals, thus logically isolating the memory devices which are not accessed. This arrangement generally requires a device select signal for each physical memory rank and the address and device select signals are generally controllable based on whether the corresponding memory device or group of memory devices is being accessed. Generating controllable address and device select signals can introduce timing delays on the device select, address, and control paths. In addition, this arrangement does not address the power dissipation and may degrade

performance degradation by, for example, requiring that secondary control signals be generated for the memory devices.

SUMMARY

[0006] Certain embodiments described herein include a memory module comprising a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module comprises a plurality of memory devices on the printed circuit board and a circuit. The circuit comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The memory module further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector. The memory module of certain embodiments comprises a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0007] A method of using a memory module with a computer system is described in certain embodiments. The method includes providing a memory module comprising a printed circuit board having at least one connector configured to be operatively coupled to a memory controller of a computer system and to a plurality of memory devices. The 'memory module further including a circuit which comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The circuit further comprises a second set of ports comprising one or more bi-directional ports. Each port of the second set of ports is operatively coupled to the at least one connector. The circuit is configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports in some embodiments. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit. The correction

circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports. The method includes activating the circuit to selectively operatively couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports. The method also includes activating the circuit to reduce noise in the one or more signals.

[0008] Certain embodiments are described herein provide a circuit comprising a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports may be configured to be operatively coupled to at least one memory device of a plurality of memory devices on a printed circuit board. The circuit further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one connector on the printed circuit board.

The at least one connector is configured to be operatively coupled to a memory controller of a computer system. The circuit further comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduce noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0009] In certain embodiments, a circuit comprises a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports can be configured to be operatively coupled to at least one subsystem of a plurality of subsystems. The circuit may comprise a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one subsystem of one or more subsystems. In certain embodiments, the circuit comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 schematically illustrates an example system which isolates unused subsystems and incorporates a multiplexer-demultiplexer.

[0011] Figure 2 schematically illustrates an example system which isolates unused subsystems using device select signals.

[0012] Figure 3 schematically illustrates an example memory system incorporating a memory module in accordance with certain embodiments described herein.

[0013] Figure 4 schematically illustrates an example correction circuit in accordance with certain embodiments described herein.

[0014] Figure 5 schematically illustrates an example programmable impedance matching circuit in accordance with certain embodiments described herein.

[0015] Figure 6 schematically illustrates an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0016] Figure 7 schematically illustrates example configuration including an example filter element connected to an example repeater/amplifier in accordance with certain embodiments described herein.

[0017] Figure 8 schematically illustrates an example system having an example circuit connecting the system with one or more other systems in accordance with certain embodiments described herein.

[0018] Figure 9 schematically illustrates an example system including an example circuit in accordance with certain embodiments described herein.

[0019] Figure 10 is a flowchart of an example method 400 of using a memory module with a computer system in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0020] Electronic systems incorporating devices and methods in accordance with

embodiments described herein may include memory subsystems that contain multiple memory boards or modules that are driven by a single memory controller which synchronizes memory access and manages memory coherency. However, having a single memory controller may gradually degrade system performance and increase power consumption as the physical and the logical size of the memory subsystem increases. Increases in memory speed may further increase the power consumption. Additionally, impedance mismatches between the system board, the memory boards, and the memory devices may also substantially degrade the performance and power efficiency of traditional memory systems. Such memory systems may be used with a power supply which helps to stabilize the voltage level, and/or a thermo-mechanical component to moderate memory subsystem temperature, and/or software for throttling the processor to lower the memory access rate.

[0021] Embodiments described herein address these speed and power issues in memory systems. For example, a circuit of certain embodiments can comprise an isolation switch and can logically isolate inactive subsystems and reduce signal reflection and/or other noise at, for example, the interface between the system board and the memory board and at the interface between the memory board and the memory devices.

[0022] Certain embodiments described herein present a circuit for isolating inactive subsystems and maintaining and improving signal integrity in a variety of systems. Certain embodiments described herein are used to increase the performance of electronic systems without increasing the power dissipation or the memory access time. In certain embodiments, a plurality of circuits are configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits and a plurality of corresponding devices (e.g., memory devices) of a system or subsystem (e.g., a memory module of a computer system). For example, the data width of each of the plurality of circuits may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example.

[0023] Certain embodiments described herein describe a circuit for use in high performance electronic systems which may contain transmission lines and relatively heavy fan-outs. A circuit is described herein which, in certain embodiments, includes at least some of the following components: a signal router, a programmable coarse impedance matching circuit, a

self-adjusting damper circuit, a filter and/or a repeater/amplifier.

[0024] Figure 3 schematically illustrates an example system incorporating a memory module 100 in accordance with certain embodiments described herein. The memory module 100 comprises a printed circuit board (PCB) 102. The PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108. The memory module 100 includes a plurality of memory devices 110 on the PCB 102.

[0025] The memory module 100 of certain embodiments further includes a circuit 112 comprising a first set of ports 122 comprising a plurality of bi-directional ports 124, each port 124 of the first set of ports 122 operatively coupled to at least one memory device 110 of the plurality of memory devices 110. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports 128, each port 128 of the second set of ports 126 operatively coupled to the at least one connector 104. The circuit 112 further comprises a switching sub-circuit 140 configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises a correction circuit 150 (see, e.g., Figure 4) which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0026] In certain embodiments, the memory module 100 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, or higher or lower. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bits, 8 bits, 16 bits, 32 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 102 has an industry-standard form factor. For example, the PCB 102 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 102 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 102 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other

embodiments, certain non-DIMM form factors are possible such as, for example, single inline memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0027] In certain embodiments, the plurality of memory devices 110 of the memory module 100 may be arranged as one or more ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 111 of the plurality of memory devices 110 that can be accessed by the memory controller 106 to access the full bit-width of the memory bus of the memory module 100. For example, a memory module 100 in which each rank of the memory module is 64 bits wide is described as having an "x 64" organization. Similarly, a memory module 100 having 72-bit-wide ranks is described as having an "x 72" organization. The number of memory devices 111 of a memory module 100 can be increased by increasing the number of memory devices 111 per rank or by increasing the number of ranks of the memory module 100. For example, a memory module with four ranks with each rank having N 512-MB memory devices 111 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 111 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices.

[0028] During operation, one or more ranks of a memory module 100 may be selected or activated from the plurality of ranks by control signals that are received from a component of the system (e.g., a system memory controller 106 or a local memory controller of the memory module 100). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 100 comprises only one rank of memory devices 111.

[0029] As discussed, the PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108, such as a host computer system 108. For example, the connector 104 is electrically coupled, logically coupled, or both, with the memory controller 106 in some other manner. In certain embodiments, the connector 104 comprises a wireless radio frequency (RF) interface or a portion thereof and is wirelessly connected (e.g., via one or more RF antennas) to the memory controller 106. In some embodiments, some other connection mechanism is used besides the connector 104. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal

computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 106 may comprise a disk controller of the computer system 108, for example. The memory controller 106 of certain embodiments may be mounted on a system board 109 of the host computer 108. The connector 104 can comprise a plurality of edge connections distributed along one or more edges of the PCB 102, which fit into a corresponding slot connector of the host system. The connector 104 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 100 and the host system 108. For example, the connector 104 can comprise a standard 240-pin DDR2 edge connector. Additionally, in certain embodiments, more than one memory module 100 is coupled to the host system 108.

[0030] The plurality of memory devices 110 on the PCB 102 may include one or more volatile memory components. For example, the plurality of memory devices 110 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 111. For example, in the block diagram of Figure 3, the plurality of memory devices 110 comprises eight DRAM devices 111. Types of DRAM devices 111 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). The memory devices 111 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 111 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 111 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro, BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0031] The first set of ports 122 of the circuit 112 comprises a plurality of bidirectional ports 124 where each port of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. In addition, the second set of ports 126 of the connector 104 comprises one or more bi-directional ports 128 where each port of the second set of ports 126 is operatively coupled to the at least one connector 104. The switching sub-circuit 140 is configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The operative coupling of the first set of ports 122 to the memory devices 110, the second set of ports 126 to the connector 104, and the selective operative coupling between the first and

second set of ports 122, 126 may comprise electrical coupling, logical coupling, or both. In other embodiments, optical coupling or some other manner of coupling may be employed.

[0032] In certain embodiments, each bi-directional port 128 of the second set of ports 126 may correspond to a signal of a memory bus 130 which operatively couples the memory controller 106 and the memory module 100 via the connector 104. For example, the memory controller 106 may generally read data from and/or write data to the plurality of memory devices 110 on the memory module 100 through the memory bus 130 and through the bi-directional ports 124, 128. In certain embodiments, the number of bi-directional ports 128 of the second set of ports 126 may correspond to the number of data bits in the memory bus 130 (i.e., the bit-width of the memory bus 130).

[0033] In certain embodiments, each of the bi-directional ports 124 in the first set of ports 122 may correspond to and be operatively coupled to a signal port of one of the memory devices 111 of the plurality of memory devices 110. The bi-directional ports 124 correspond to the data signal ports, for example, of the plurality of memory devices 110. Because each memory device 111 includes associated data signal ports, the number of bidirectional ports 124 in the first set of ports of certain embodiments may be larger than the number of bi-directional ports 128 in the second set of ports 122 (e.g., the number of bits in the memory bus 130). In general, the overall number of ports of the plurality of memory devices 110 and corresponding bi-directional ports 124 in the first set of ports 126 may depend on the organization of the memory module. For example, the number of ports may depend on the number of memory devices and/or the number of ranks in the memory module. In certain embodiments, the memory module 100 includes a plurality of circuits 112. The circuits 112 may be configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits 112 and a plurality of corresponding memory devices 111 of memory module 100. For example, the data width of each of the plurality of circuits 112 may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example. Other data widths are possible in other embodiments. In some embodiments, the data width of each circuit 112 corresponds to the data width of each of the memory devices 111 such that each circuit 112 is operatively coupled to one of the plurality of memory devices 110. In other embodiments, the

data width of each circuit 112 corresponds to a multiple of the data width of each memory device 111 such that each circuit 112 may be operatively coupled to the data bits of more than one memory device 111. In various embodiments, the data width of each circuit 112 corresponds to a segment of the data width of each memory device 111 or to segments of multiple memory devices 111. As such, in some embodiments, each circuit 112 may be operatively coupled to a segment of the data bits of one memory device 111 or to segments of more than one memory device 111. Other configurations are possible which optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In addition, in certain embodiments, the location of each of the plurality of circuits 112 on the PCB 102 is selected to optimize the interconnection, timing, power consumption, and signal integrity.

[0034] In other embodiments, the bi-directional ports 124 correspond to one or more other bits of the memory bus 130 instead of, or in addition to, the data bits. For example, one or more of the ports 124 may correspond to address, chip select, control (e.g., read/write) signals. Moreover, in some embodiments, one or more ports of the first set of ports 122 is not bi-directional. For example, in one embodiment, signals corresponding to address, chip select and/or control signals from the first set of ports 122 to the plurality of memory devices 110 and not vice versa. For example, one or more of the address, chip select, and control signals may be used to control the routing of the bi-directional ports 124 which correspond to data bits.

[0035] As discussed, the switching sub-circuit 140 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the switching sub-circuit 112 may be capable of interconnecting any number of the second set of ports 126 to any number of the first set of ports 122 simultaneously. In certain embodiments, the switching sub-circuit 140 can pass signals (e.g., information such as the data signals described above) from one port 128 of the second set of ports 126 to an operatively coupled port 124 of the first set of ports 122 and vice versa. Additionally, the switching sub-circuit 140 of some embodiments can pass signals from one port 128 of the second set of ports 126 to multiple operatively coupled ports 124 of the first set of ports 122 and vice versa. In certain embodiments, the switching sub-circuit 140 isolates and provides a known static value to all un-driven ports, thereby reducing any dynamic power consumption associated with floating or toggling electrical nodes.

[0036] The switching sub-circuit 140 of certain embodiments can function as an isolation switch which isolates and/or inactivates individual subsystems not actively transferring information. For example, in an example embodiment, the plurality of memory devices 110 comprises multiple independently accessible ranks. When the memory controller accesses one of the ranks of the memory module 100, the switching sub-circuit 140 isolates and/or inactivates at least the one or more other, un-accessed ranks. For example, the switching sub-circuit 140 does not operatively couple ports from the second of set of ports 126 to the first set of ports 122 which are coupled to the memory devices 111 of the unaccessed one or more ranks of memory devices 110. In certain embodiments, the switching sub-circuit 140 may function as or be characterized as a scalable isolation switch router which routes the information from one subsystem to another (e.g., host computer 108 to the memory module 100 via the memory controller 106), establishing a one-to-one connection for transferring data from point-to-point, and/or establishing a one-to-many connection for broadcasting data. The switching sub-circuit 140 comprises or is characterized as a bidirectional multiplexer-demultiplexer in certain embodiments. In other embodiments, the switching sub-circuit 140 comprises or is characterized as a router.

[0037] In certain embodiments, the memory module 100 further includes circuitry (not shown) for controlling the operation of the circuit 112 and the components therein (e.g., the switching sub-circuit 140 and the correction circuit 150, as described more fully below). For example, in one embodiment, the memory module 100 comprises a local memory controller which is operatively coupled to one or more control inputs (not shown) of the circuit 112 to control the selective coupling of the second set of ports 126 to the first set of ports 122. In certain other embodiments, some other component of the memory module 100 controls the operation of the circuit 112 or the circuit 112 comprises internal control circuitry. In various embodiments, the circuitry for controlling the circuit 112 includes discrete logic, one or more microprocessors, a field-programmable gate array (FPGA), or a computer programmable logic device (CPLD). In yet other configurations, an external component controls the operation of the circuit 112.

[0038] The circuit 112 of certain embodiments comprises one physically integral component (e.g., one integrated circuit package mounted on the memory module 100).

Various alternative configurations for the physical organization of the circuit 112 are

possible, however. In certain other embodiments, the circuit 112 comprises multiple, physically separate components. For example, in one embodiment, the circuit 112 comprises one physically separate component per memory device 111 (e.g., integrate circuit packages mounted on the memory module 100). In another embodiment, the circuit 112 comprises one physically separate component per memory rank. In some embodiments, the circuit 112 comprises one physically separate component corresponding to multiple memory devices 111.

[0039] Figure 4 schematically illustrates an example correction circuit 150 in accordance with certain embodiments described herein. As shown, in certain embodiments, the correction circuit 150 comprises one or more correction elements (e.g., at least one coarse correction element 152 to provide coarse correction of the noise and at least one fine correction fine correction element 154 to provide fine correction of the noise). The correction circuit 150 of certain embodiments further comprises at least one filter element 156. In certain embodiments, the correction circuit 150 further comprises at least one repeater/amplifier element 158, which improves the signal quality of the one or more signals.

[0040] Referring to Figures 3 and 4, the correction circuit 150 reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. For example, the correction circuit 150 may reduce noise due to impedance mismatches (e.g., due to discontinuity regions) at nodes between the memory controller 106 and the plurality of memory devices 110. In certain embodiments, the noise may comprise signal reflections. For example, in certain embodiments, the signal paths between the host computer 108 and the plurality of memory devices 110 or portions thereof may be characterized as transmission lines if they are greater than or about equal to a particular fraction of the wavelength of the signal, such as, for example, greater than or equal to about 1/4 of the wavelength of the signal. In other configurations, the signal paths may be characterized as transmission lines when the signal path is greater than or equal to some other fraction of the wavelength of the signal such as, for example, 1/2, 1/8, or 1/10 of the wavelength. In certain embodiments, the system (e.g., memory module 100) incorporating the circuit 112 has signal delays which are related to signal characteristics such as the signal transition rate and/or the rise/fall time of the signals. For example, the system incorporating the circuit 112 may include signal delays which are more than some fraction (e.g., 1/2) of the

signal transition rate and/or the rise/fall time of the signals. In other embodiments, the signal delays are related in some other way to the signal transition rate and/or the rise/fall time of the signals (e.g., are more than 1/4, 1/8 of those values) or are related to some other signal property.

[0041] The reflections may be caused by one or more discontinuity regions along the signal paths. For example, discontinuity regions comprise points or regions at which the characteristic impedance of the corresponding signal path changes. For example, the interface between the system board 109 of the host computer 108 and the memory module 100 (e.g., the connector 104) may comprise a discontinuity region along each of the signal paths. The interfaces between the connector 104 of the memory module 100 and each of the memory devices 111 mounted on the PCB 102 may comprise one or more discontinuity regions along each of the signal paths.

[0042] The noise may comprise cross-coupling between the signal paths in certain embodiments. For example, a cross-coupling impedance may be present between one or more of the signal paths or portions thereof. In certain embodiments, the cross-coupling impedance may include cross-coupling due to cross-coupling capacitance, inductance (e.g., mutual inductance), transconductance, or a combination thereof. Additionally, a variety of other noise are possible such as, for example, electromagnetic interference or other types of noise.

[0043] Referring to Figure 4, in certain embodiments, the correction circuit 150 includes one or more elements to provide for coarse correction of signal noise due to impedance mismatches (e.g., discontinuity regions) in the system. The at least one coarse correction element 152 comprises at least one programmable impedance matching circuit in certain embodiments. Figure 5 schematically illustrates an example programmable impedance matching circuit 160 in accordance with certain embodiments described herein. For example, the programmable impedance matching circuit 160 may comprises at least one programmable resistor 162. The resistance of the programmable resistor 162 may be controllable by a control signal 164. In certain embodiments, the coarse correction element 152 may be integrated with the switching sub-circuit 140, the fine correction element 154, the filter element 156, and/or the repeater/amplifier element 158. Alternatively, in other embodiments, the coarse correction element 152 may be implemented with discrete components.

[0044] In other embodiments, the coarse correction element 152 comprises a network of resistors 162 which can be configured to provide programmable impedance matching. In some embodiments, the coarse correction element 152 may comprise a set of programmable coarse impedance matching circuits 160 that are connected to each of the ports 124, 128 of the first set of ports 122 and the ports 128 of the second set of ports 126. Advantageously, the impedance values of the coarse correction element 152 may be dynamically controllable, configurable, and/or programmable, so as to match the characteristic impedances of the transmission lines (e.g., signal paths) in the system. The values of these impedances may differ (e.g., may be programmed differently) based on which particular signal path is activated (e.g., by the switching sub-circuit 140). Where the coarse correction element 152 comprises at least one impedance matching circuit, the granularities of the configurable impedance values may be configured based on the application of the particular system. In some embodiments, the impedance matching circuit of the coarse correction element 152 may include a real time controllable impedance control input for preprogramming and/or dynamically controlling the impedance matching circuit. In this manner, the impedance matching circuit 152 may, in certain embodiments, provide dynamically controllable variable impedance which can be pre-set for matching the impedance of the subsystem that drives the circuit 112.

[0045] In certain embodiments, the at least one fine correction element 154 of the correction circuit 150 further reduces signal reflections due to impedance mismatches and/or reduces other noise. In certain embodiments, for example, the coarse correction element 152 may be characterized as a first refining stage and the fine correction element 154 may be referred to as a second refining stage. The fine correction element 154 may smooth out the signal after it has undergone coarse correction (e.g., impedance matching) and reduces or removes transmission line characteristics from the received signal. Referring again to Figure 4, the fine correction element 154 can be connected, for example, in parallel with the programmable coarse correction element 152. In some embodiments, the fine correction element 154 operates to reduce the impedance mismatches by introducing a current source that damps to the slew rate of the input signal, reducing signal reflection. The fine correction element 154 actively counteracts the noise in certain embodiments. In addition, the fine correction element 154 of certain embodiments is dynamically responsive to one or more changes in the noise, such as, for example, changes in the frequency of the noise component.

[0046] The fine correction element 154 may include at least one damper circuit. The damper circuit includes at least one self-adjusting damper circuit 170 in certain embodiments, such as the example self-adjusting damper circuit 170 schematically illustrated by Figure 6. In certain embodiments, the self-adjusting damper circuit 170 comprises an input terminal 172 adapted to receive an input signal 174 comprising an input noise component and an input signal component. The damper circuit 170 further comprises an output terminal 176. In certain embodiments, the damper circuit 170 includes a first sub-circuit 184 adapted to generate a first signal 187 responsive to at least a portion of the noise component. For example, the first sub-circuit 184 may generate the first signal 187 in response to the portion of the noise component not reduced by the coarse correction element 152 in certain embodiments. The damper circuit 170 of certain embodiments further comprises a second sub-circuit 188 adapted to generate a second signal 190 corresponding to a delayed version of at least a portion of the input signal. The damper circuit 170 combines the first signal 187 with the second signal 190 to generate an output signal 108 on the output terminal 176 wherein the output signal comprises an output noise component smaller than the input noise component. For example, the first signal 187 and the second signal 190 may be combined at the node 182 of the damper circuit 170. The output signal 176 of certain embodiments is substantially representative of the signal component of the input signal 174. The first subcircuit 184 of certain embodiments comprises one or more current sources 185, 186. For example, the current sources 185, 186 may each comprise a controlled current source such as a voltage-controlled current source 185, 186 in certain embodiments. In some embodiments, the first sub-circuit 184 may comprise or be characterized as a negative feedback current source. In various embodiments, the self-adjusting damper circuit 170 includes an enable control input (not shown) for enabling the self-adjusting damper circuit 170. Examples of self-adjusting damper circuits 170 in accordance with certain embodiments described herein are described in U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and copending U.S. Application No. 12/422,912, filed by the assignee of the present application on April 13, 2009. Both applications are incorporated in their entirety by reference herein.

[0047] In certain embodiments, the at least one filter element 156 and the at least one repeater/amplifier element 158 may also reduce noise from other sources such as, for example, signal coupling (e.g., cross-coupling). The filter element 156 may reduce noise from the one or more signals where the noise includes a predetermined range of frequencies.

For example, the filter element 156 may include at least one band-pass filter and may reduce any low and/or high frequency noise components. In other embodiments, the filter element 156 comprises at least one other type of a filter such as a notch filter, a low-pass filter, or a high-pass filter.

[0048] One or more of the filter element 156 and the repeater/amplifier element 158 of certain embodiments enhance signal slew rate to reduce the transient power dissipation at target devices or components (e.g., one or more of the memory devices 110). Figure 7 schematically illustrates example configuration including an example filter element 156 connected to an example repeater/amplifier element 158 in accordance with certain embodiments described herein. The repeater/amplifier element 158 includes an output enable control input 159 for controlling the bi-directional signal path. In various embodiments one or more of the filter element 156 and repeater/amplifier element 158 are programmable and/or dynamically controllable. The repeater/amplifier element may further include a reference input 157. In certain embodiments, the repeater/amplifier element 158 may include one or more operational amplifiers, for example. Referring again to Figure 4, in certain embodiments, only filter element 156, only the repeater, only the amplifier, or any combination of the three elements may be included in the circuit.

[0049] The filter element 156 and/or the repeater/amplifier element 158 of certain embodiments provide wave reshaping functions and generally reduces signal noise. For example, the filter element 156 and/or the repeater/amplifier element 158 may reduce signal ringing due to reflection (e.g., reflections at one or more interfaces along the signal paths such as interfaces between the circuit 112, the connector 104, the memory devices 110 and/or the system board 109 of Figure 3). In certain embodiments, the repeater/amplifier element 158 reshapes the signal, which may be deteriorated due to transmission over lossy transmission lines having discontinuity regions. The repeater/amplifier element 158 may be configured to repeat the input signal, generating an output signal with earlier transition trigger points and faster slew rates in order to regain speed and to reduce the transient power.

[0050] In certain embodiments, the circuit 112 or portions thereof (e.g., the switching sub-circuit 140, the correction circuit 150 or elements thereof, etc.) is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device,

and a complex programmable-logic device (CPLD). For example, in certain embodiments, the circuit 112 or portions of thereof is implemented in one or more of a PLD, ASIC, FPGA, or CPLD. In other embodiments, the circuit 112 or portions thereof is implemented using some other technology, such as, for example, using discrete components.

[0051] In various embodiments, the elements of the circuit 112 such as the switching sub-circuit 140, the coarse correction element 152, the fine correction element 154, the filter element 156, and the repeater/amplifier element 158, comprise or may be characterized as independent, fully controllable functional blocks. Additionally, in certain embodiments, any subset of one or more of these functional blocks may be implemented in order to enhance performance, reduce power dissipation, or provide other advantages.

[0052] Figure 8 schematically illustrates an example system 200 having an example circuit 112 connecting a system 200 with one or more other systems 220 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 3, for example. The system 200 of Figure 8 includes a driver 202, a receiver 201, and two subsystems 203, 204. For illustration purposes, the system 200 of Figure 8 is described as an open system that communicates with other systems 220 via the driver 202 and the receiver 201. Furthermore, in some embodiments, a connector (not shown) connects the system 200 with other systems by means of transmission lines. For example, the system 200 may be a memory system and the receiver 201 and the driver 202 of Figure 8 may represent memory signals (e.g., data, address and/or control signals) transmitted between the system 220 (e.g., a memory controller and the subsystems 203, 204 (e.g., two sets of independently addressable memory devices such as DRAM devices). In certain embodiments, for example, the circuit 112 may be disposed on a memory board that is populated with sets of memory devices 203, 204.

[0053] In various embodiments, the receiver 201 may or may not contain any active circuits. For example, the receiver 201 may be a connector such as a socket or DIMM interface with a system board (e.g., the connector 104 described above in regards to Figure 3). The output signal from the receiver 201 may exhibit the characteristics of the signal on a transmission line such as, for example, wave dispersion, attenuation, and reflection. In certain embodiments, the output signal from the receiver 201 goes through a number of refining, filtering and/or amplification stages in circuit 112 as described herein to improve the

performance of the overall system. For example, the signal quality is enhanced by introducing at least one circuit 112 including at least one correction circuit 150 comprising one or more coarse correction elements 152 (e.g., impedance matching circuits), fine correction elements 154 (e.g., self-adjusting dampers), filter elements 156 (e.g., bandpass filters) and/or repeater/amplifier elements 158 as described herein on the signal paths (e.g., from receiver 101 to subsystem 103). For example, the circuit 112 may reduce power consumption by selectively isolating the inactive (or un-accessed) subsystems and may enhance the access rate of a targeted subsystem. For example, in one embodiment, the system 200 comprises a memory module 100 such as one of the memory modules described herein, the subsystems 203 and 204 comprise memory devices such as the memory devices 111 described herein, and the other system 220 comprises a host computer 108 including a memory controller 106 as described herein.

[0054] In certain embodiments, the performance is enhanced and power dissipation is reduced by removing the load associated with other (e.g., unused or inactive) subsystems (e.g., the subsystem 204) and reshaping the signal waveform using a filter element 156 and/or the repeater/amplifier element 158. In some embodiments, there are two refining stages for reducing signal reflections due to impedance mismatches. In some embodiments, for example, the first refining stage is the coarse correction element 152 (e.g., an impedance matching circuit) that matches the impedance of the transmission line (e.g., wire) prior to a receiver 201 with the characteristic impedance of the transmission line (e.g., wire) in the system 100. The second stage may be the fine correction element 154 (e.g., a self adjusting damper for further reducing signal reflections due to impedance mismatches and/or reducing other noise.

[0055] Figure 9 schematically illustrates an example system 300 including an example circuit 112 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 3, for example. The circuit 112 comprises a first set of ports 122 comprising a plurality of bi-directional ports, each port of the first set of ports 122 configured to be operatively coupled to at least one subsystem 311 of a plurality of subsystems 310. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports, each port of the second set of ports 126 configured to be operatively coupled to at least

one subsystem 321 of one or more subsystems 320. In certain embodiments, the circuit 112 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126 each comprises a correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. The correction circuit 150 of certain embodiments may comprise one of the correction circuits 150 described herein such as the correction circuits 150 described above with respect to Figures 4-7 and may include/ at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158, or any combination or sub-combination thereof. In other embodiments, the fine correction element 154 may be placed after the switching sub-circuit 140 in order to achieve better results. For example, the fine correction element 154 may be connected after the filter element 156 (e.g., between the filter 156 and the repeater/amplifier element 158) in certain embodiments.

[0056] In certain embodiments, the system 300 includes a first set of interfaces 330 positioned between (e.g., logically and/or physically between) and operatively coupled to the first set of ports 122 and to the plurality of subsystems 310. In addition, the system 300 of certain embodiments may include a second set of interfaces 340 positioned between (e.g., logically and/or physically between) and operatively coupled to the second set of ports 126 and the one or more subsystems 320. For example, the first and/or second sets of interfaces 330, 340 may comprise port connections.

[0057] In an example configuration, the plurality of subsystems 310 include a plurality of memory devices 311 (e.g., DRAM devices) mounted on a memory module and the first set of interfaces 330 comprise interfaces between the circuit 112 and the plurality of memory devices 311. For example, each of the first set of ports 122 may correspond to a memory signal (e.g., a data, address, and/or control signal) and each of the first set of interfaces 330 is operatively coupled to a corresponding memory signal port of a corresponding memory device 311. The one or more subsystems 320 of the example configuration comprise a host computer system and the second set of interfaces 340 comprises memory board (e.g., a PCB of a memory module) interfaces to the host system. For example, the second set of interfaces

340 form part of a connector. Each of the second set of ports 126 and corresponding interfaces of the second set of interfaces 340 corresponds to a memory signal (e.g., a data, address, and/or control signal) of at least one memory bus between the host system and the memory module.

[0058] The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 126 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the system board and the memory board. The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 122 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the circuit 112, the memory board and/or the plurality of memory devices. The filter elements 156 (e.g., a bandpass filter) reduce or filter out cross-coupling between signals, for example. For example, the filter elements 158 reduce or filter out cross-coupling between data, address, and/or control signals (e.g., wires).

[0059] Figure 10 is a flowchart of an example method 400 of using a memory module 100 with a computer system in accordance with certain embodiments described herein. While the method 400 is described herein by reference to the memory module 100 schematically illustrated by Figure 3, other memory modules, electronic systems or subsystems, and/or circuits are also compatible with the embodiments described herein. In certain embodiments, at operational block 402, the method 400 comprises providing a memory module 100 comprising a printed circuit board 102 having at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108 and to a plurality of memory devices 110. The memory module 100 further comprises a circuit 112 including a first set of ports 122 comprising a plurality of bi-directional ports 124. In certain embodiments, each port 124 of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. The circuit of certain embodiments further includes a second set of ports 126 comprising one or more bidirectional ports 128' where each port 128 of the second set of ports 126 is operatively coupled to the at least one connector 104.

[0060] The circuit 112 of certain embodiments is configured to selectively operatively couple

one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises at least one correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0061] The method 400 of certain embodiments further includes activating the circuit 112 at operational block 404 to selectively operatively couple at least one of the one or more ports 128 of the second set of ports 126 to at least one of the one or more ports 124 of the first set of ports 122. For example, in certain embodiments, activating the circuit 112 to perform the selective operative coupling may include sending the appropriate control signal or combination of control signals to a switching sub-circuit 140 of the circuit 112. In certain embodiments, the method 400 further includes activating the circuit 112 to reduce noise in the one or more signals at operational block 406. For example, in one embodiment, activating the circuit 112 to reduce noise comprises activating one or more of the elements of the correction circuit 150 (e.g., at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158) of ports 124, 128 of the first and second sets of ports 122, 126 corresponding to active signal paths between the memory controller 106 and the memory devices 110. Activating one or more of the elements 152, 154, 156, 158 of the correction circuit 150 may include sending control signals (e.g., enable signals) to the appropriate elements 152, 154, 156, 158 and/or receiving the control signals (e.g., enable signals) at the appropriate elements 152, 154, 156, 158.

[0062] The operational blocks 402, 404, 406 of the method 400 may be performed in a different order than provided or one or more of the operational blocks may be performed substantially at the same time or during overlapping time periods. For example, in certain embodiments, at least the operational block 404 and the operational block 406 are performed substantially simultaneously. In addition, in some embodiments, one or more of the operational blocks 402, 404, 406 may not be included, an additional operational block may be included, or both. The devices and methods described herein may be incorporated in any system (e.g. a board level system) or device that contains one or more connectors or wires.

[0063] Although certain preferred embodiments and examples are discussed above, it is

understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory module operable to communicate signals with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;

a plurality of memory devices on the printed circuit board;

a circuit coupled between the at least one connector and the plurality of memory devices, the circuit capable of selectively coupling a one or more first memory devices among the plurality of memory devices to the system memory controller via the at least one connector while isolating one or more second memory devices among the plurality of memory devices from the system memory controller, the circuit comprising first programmable impedance matching circuits corresponding to respective ones of the plurality of edge connections, each of the first programmable impedance matching circuits including at least one variable impedance that can be programmed to match a characteristic impedance of a corresponding signal path in the system; and

control circuitry configured to control the selective coupling in the circuit, and to program the programmable impedance matching circuits in response to one or more signals in the address and control signals.

2. The memory module of claim 1, wherein the circuit comprises a set of ports coupled to the plurality of memory devices, and second programmable impedance matching circuits corresponding to respective ones of the set of ports, the second programmable impedance matching circuits being pre-set for matching impedance associated with the plurality of memory devices.

3. The memory module of claim 2, wherein a number of ports in the set of ports is larger than a number of edge connections in the edge connector.

4. The memory module of claim 2, wherein the circuit further comprises signal waveform reshaping circuits corresponding to respective ones of the set of ports.

5. The memory module of claim 1, wherein the first programmable impedance matching circuits comprises at least one network of programmable resistors that is dynamically controllable.
6. The memory module of claim 1, wherein the circuit further comprises self-adjusting damper circuits corresponding to respective ones of the plurality of edge connections.
7. The memory module of claim 1, wherein the circuit further comprises signal waveform reshaping circuits corresponding to respective ones of the plurality of edge connections.
8. A memory module operable to communicate signals with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:
 - a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus, the at least one connector including a plurality of edge connections distributed along one or more edges of the printed circuit board;
 - a plurality of memory devices on the printed circuit board;
 - a circuit coupled between the at least one connector and the plurality of memory devices, the circuit capable of selectively coupling one or more first memory devices among the plurality of memory devices to the system memory controller via the at least one connector while isolating one or more second memory devices among the plurality of memory devices from the system memory controller, the circuit further capable of reshaping data signals between the system memory controller and the one or more first memory devices; and
 - control circuitry configured to control the selective coupling and the reshaping of the data signals in response to one or more signals in the address and control signals.
9. The memory module of claim 8, wherein the circuit comprises first dynamically programmable repeater/amplifier elements corresponding to respective ones of the plurality of edge connections.

10. The memory module of claim 8, wherein the circuit comprises a set of ports coupled to the plurality of memory devices, and second dynamically programmable repeater/amplifier elements corresponding to respective ones of the set of ports.
11. The memory module of claim 10, wherein a number of ports in the set of ports is larger than a number of edge connections in the edge connector.
12. A memory module operable to communicate signals with a system memory controller via a memory bus in response to address and control signals from the system memory controller, comprising:
- a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus;
 - a plurality of memory devices on the printed circuit board;
 - a circuit coupled to the at least one connector, the circuit including a first set of ports coupled to one or more first memory devices among the plurality of memory devices, a second set of ports coupled to one or more second memory devices among the plurality of memory devices, and a switching circuit capable of selectively coupling the one or more first memory devices to the system memory controller via the at least one connector while isolating the one or more second memory devices from the system memory controller, the circuit being further configured to provide a known static value to the second set of ports; and
 - control circuitry configured to control the selective coupling in the circuit.

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

ABSTRACT

Certain embodiments described herein include a memory module having a printed circuit board including at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module further includes a plurality of memory devices on the printed circuit board and a circuit including a first set of ports operatively coupled to at least one memory device. The circuit further includes a second set of ports operatively coupled to the at least one connector. The circuit includes a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. Each port of the first set and the second set comprises a correction circuit which reduces noise in one or more signals transmitted between the first set of ports and the second set of ports.

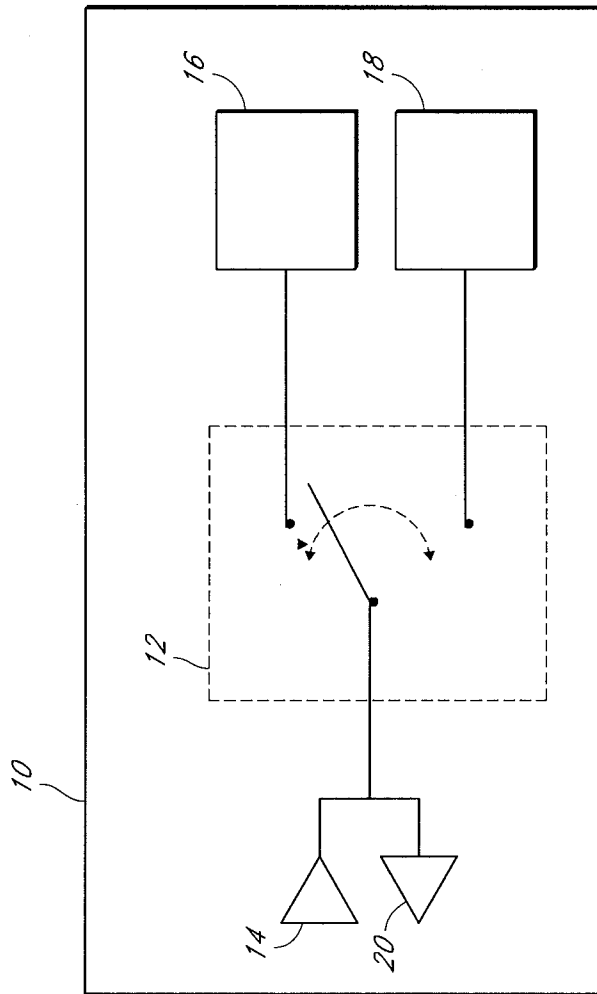


FIG. 1

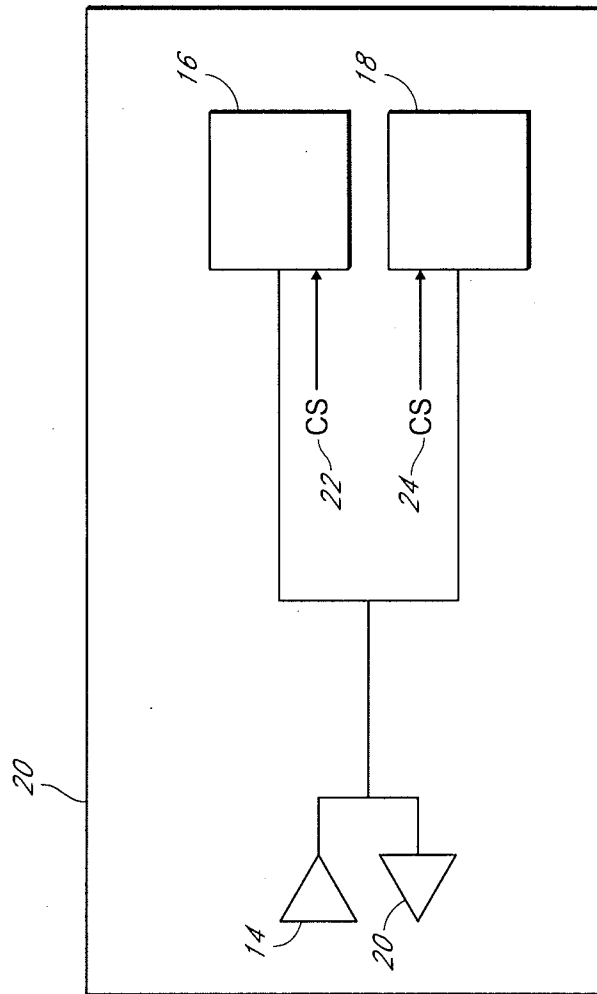


FIG. 2

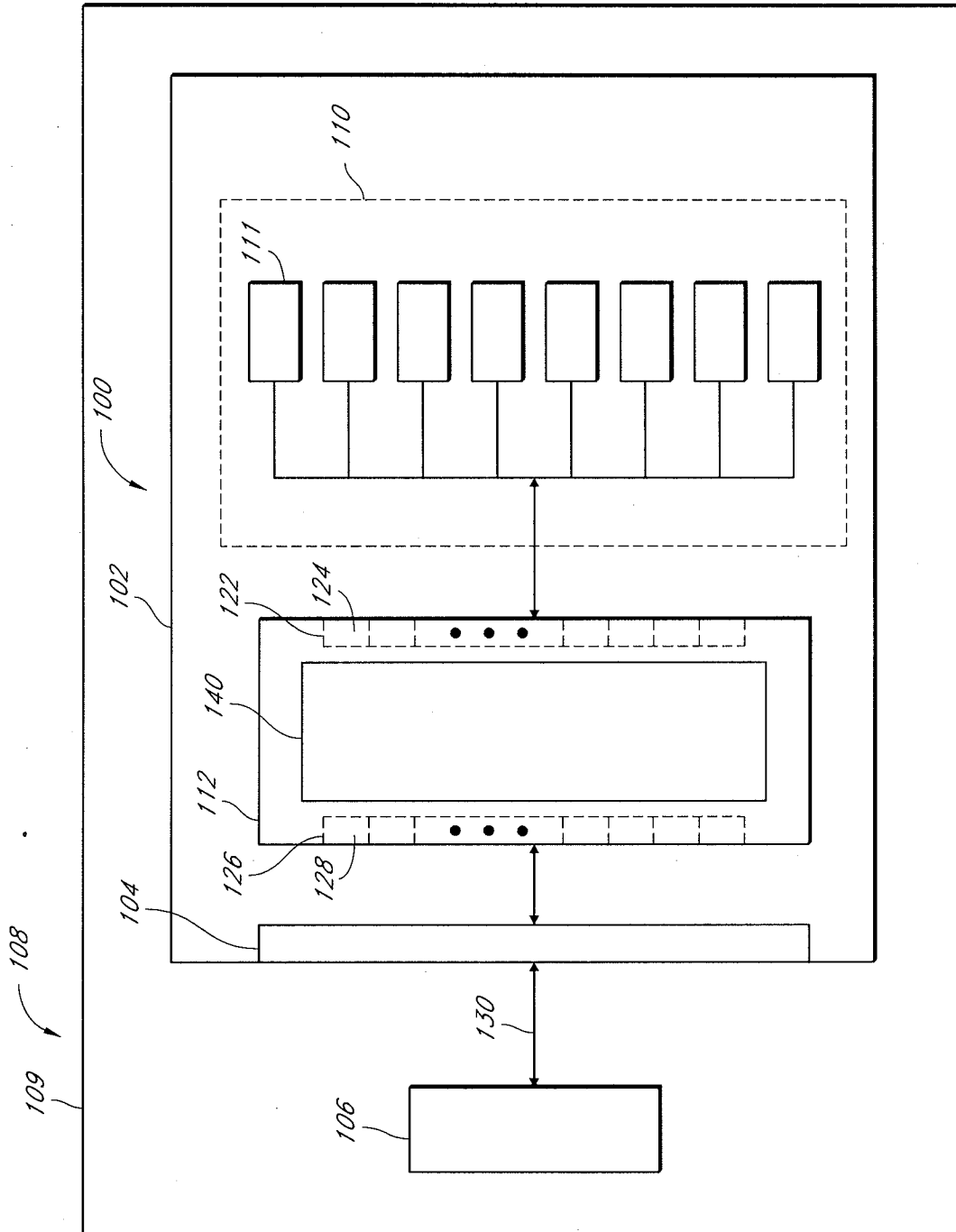


FIG. 3

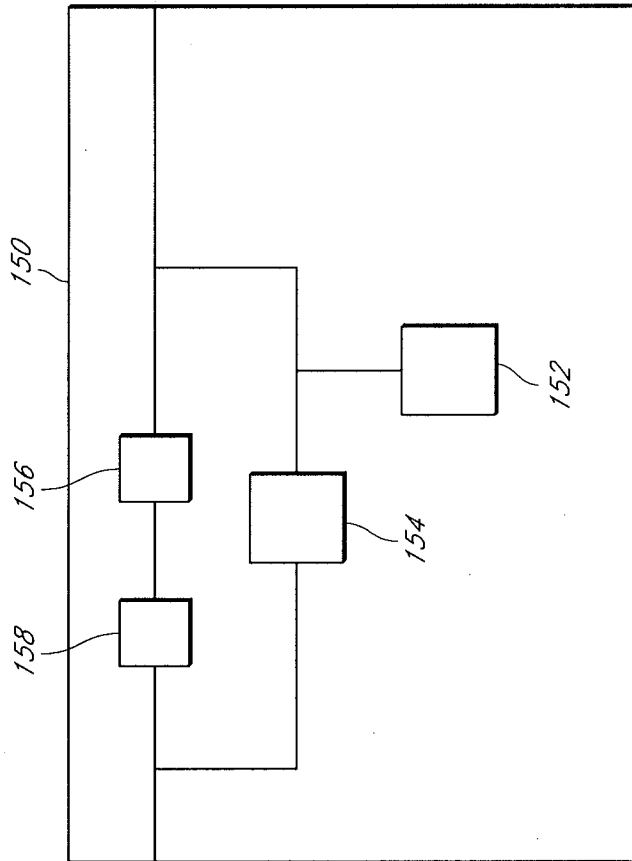


FIG. 4

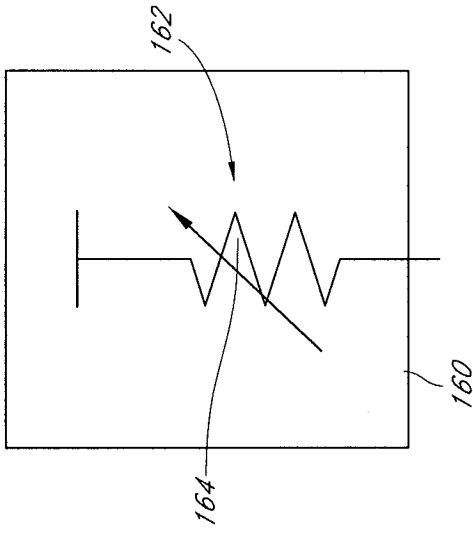


FIG. 5

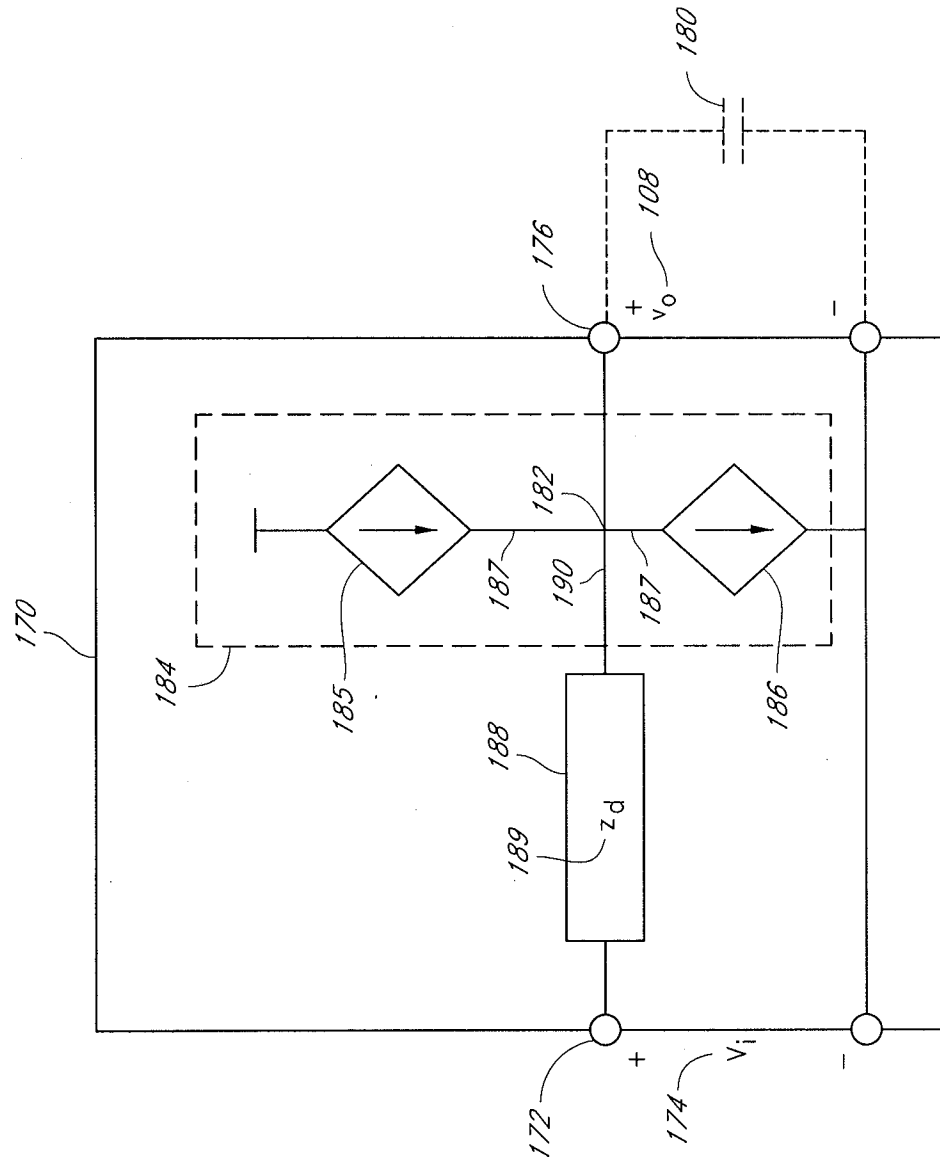


FIG. 6

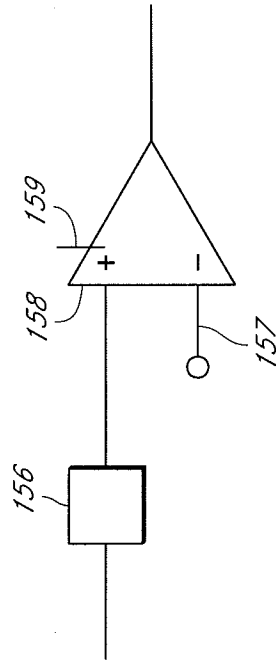


FIG. 7

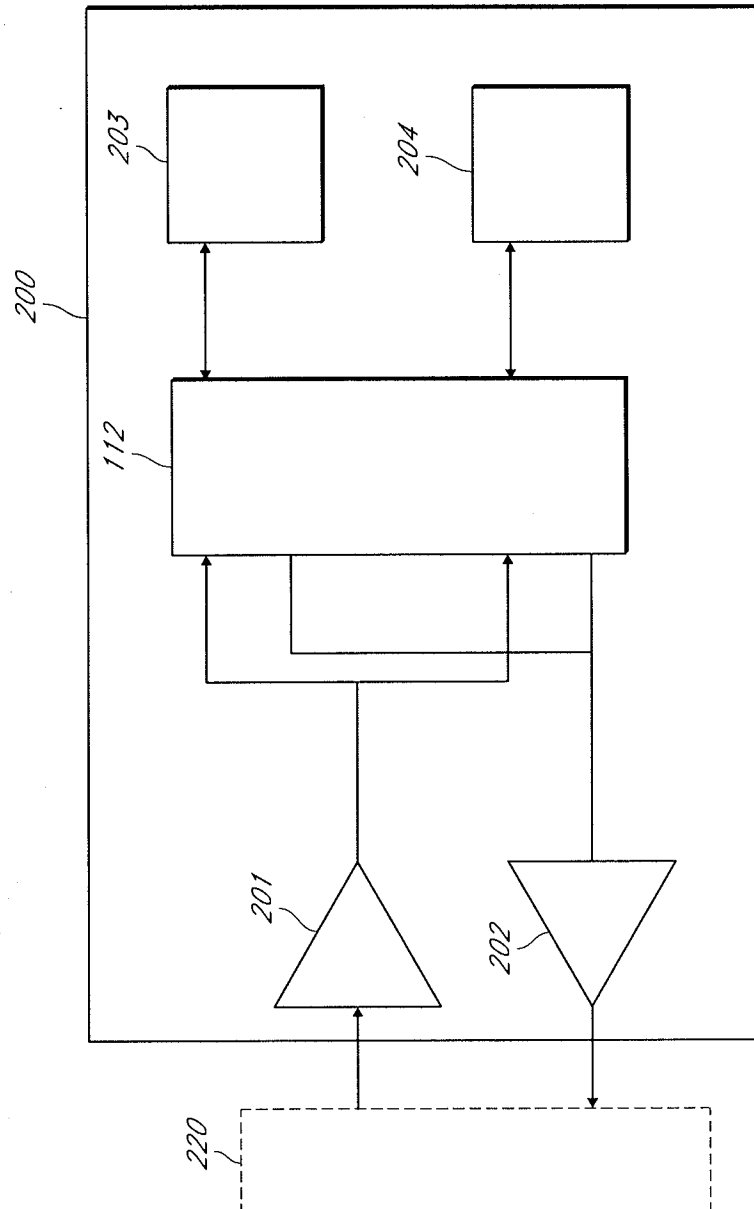


FIG. 8

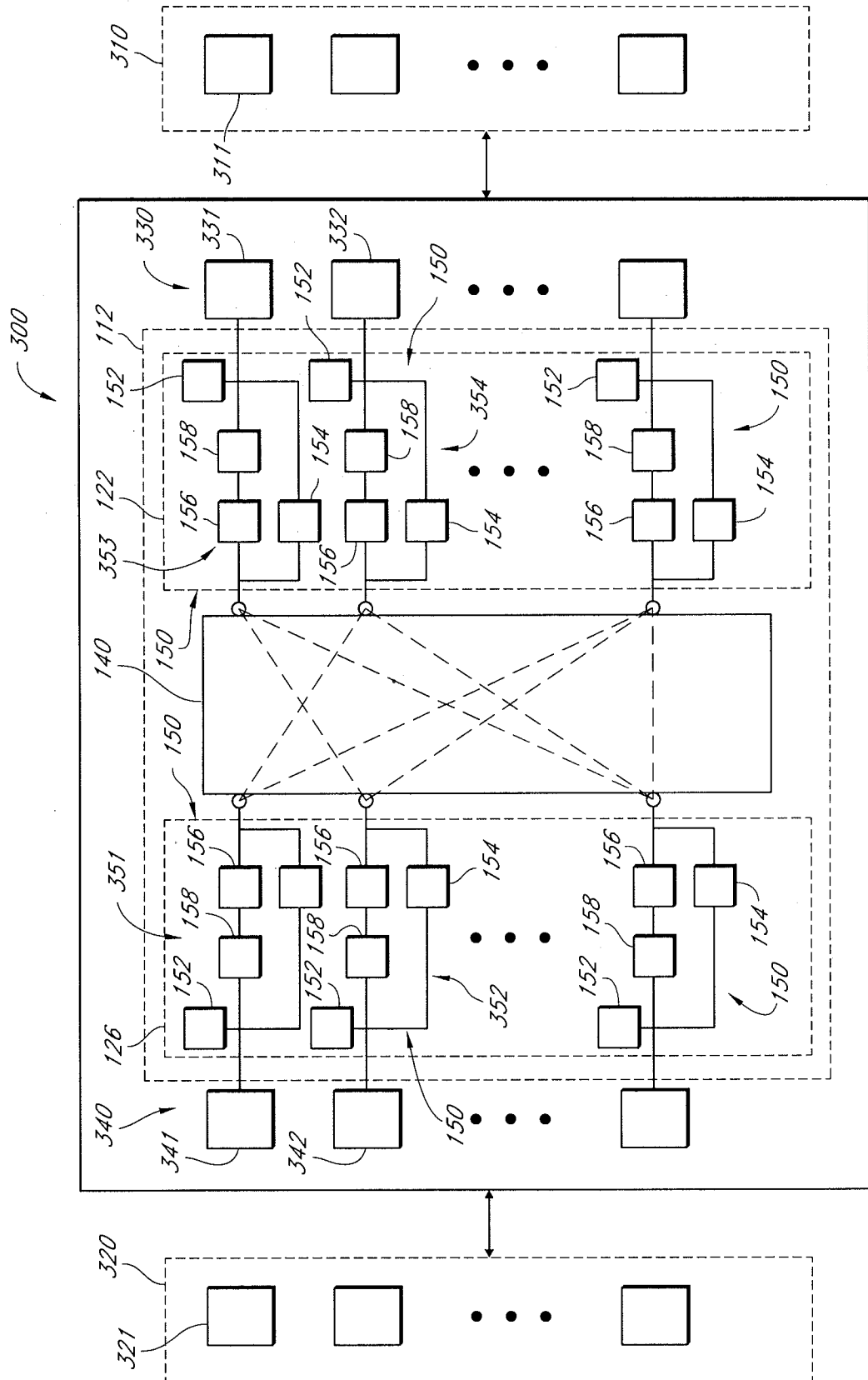


FIG. 9

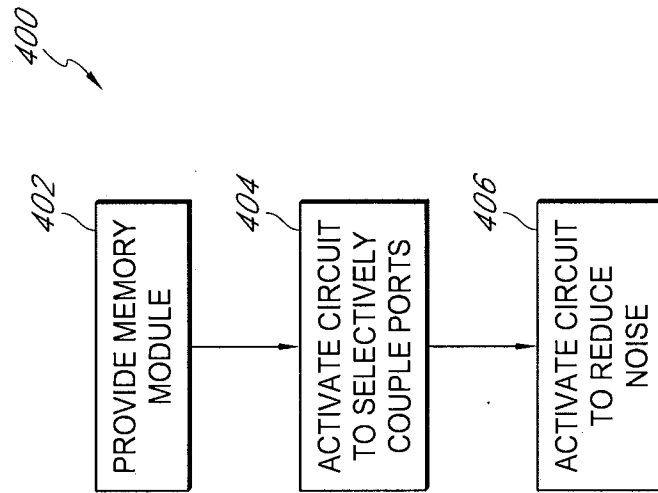


FIG. 10

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Jamie Jie Zheng

Attorney Docket Number:

NETL.045C2

Filed as Small Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility filing Fee (Electronic filing)	4011	1	70	70
Utility Search Fee	2111	1	300	300
Utility Examination Fee	2311	1	360	360

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				730

Electronic Acknowledgement Receipt

EFS ID:	19509611
Application Number:	14324990
International Application Number:	
Confirmation Number:	1022
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C2
Receipt Date:	07-JUL-2014
Filing Date:	
Time Stamp:	17:52:32
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	2014-07-07_ApplicationDataSheet_NETL045C2.pdf	1170641 179649f8d70559cc94da6d8084aafc90abd5584	no	8

Warnings:

Information:

2		2014-07-07_Specification_NETL045C2.pdf	250784 5d83c2319d5f1e3e050f49318a97476f49e59d9f	yes	27
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	Document Description		Start	End	
	Specification		1	23	
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3	Drawings-only black and white line drawings	2014-07-07_Drawings_NETL045C2.pdf	83908 86f673602c169c98c4c8fc880c1e7ffd027fb92	no	9
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4	Fee Worksheet (SB06)	fee-info.pdf	32703 8be202403cc4716d83c93891f99bc3b7924d9036	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			1538036		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

**REVOCATION
AND
GENERAL POWER OF ATTORNEY**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

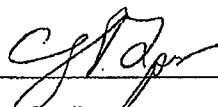
The undersigned is an empowered representative of the Assignee and hereby appoints the registrants of Knobbe, Martens, Olson & Bear, LLP, **Customer No. 20,995**, as attorneys and agents to represent the Assignee before the United States Patent and Trademark Office (USPTO) in connection with any and all patent applications assigned to the Assignee according to the USPTO assignment records or assignment documents supplied with an accompanying Statement Under 37 CFR § 3.73(b). This appointment is to be to the exclusion of the inventor(s) and his attorney(s) in accordance with the provisions of 37 CFR § 3.71.

Submission of this paper in connection with any matter of the below named assignee, together with a statement under 37 CFR 3.73(b), shall serve to revoke any previous powers of attorney in that matter.

A Statement Under 37 CFR § 3.73(b), signed by a registrant of Knobbe, Martens, Olson & Bear, LLP, is attached setting forth a full chain of title for the subject application owned by the Assignee named below.

Please recognize or change the correspondence address for the application identified in the attached Statement to **Customer No. 20,995**.

By:



Date:

6/7/07

Name:

Christopher Lopes

Title: Vice President

Assignee: NETLIST, INC.

475 Goddard, Suite 100

Address: Irvine, CA 92618

Electronic Acknowledgement Receipt

EFS ID:	12225289
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Robin Leu
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045C1
Receipt Date:	05-MAR-2012
Filing Date:	
Time Stamp:	16:40:12
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	ADS.pdf	334685 1e5a9ac290ca89bce74f67253e364399e960afb2	no	6

Warnings:

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C1
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
				☐ Party of Interest under 35 U.S.C. 118
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Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
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Postal Code	92694	Country	US	
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Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
				☐ Party of Interest under 35 U.S.C. 118
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Citizenship under 37 CFR 1.41(b)		US		
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Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
				☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name	Suffix
	Jeffrey	C.	Solomon	
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City	Irvine	State/Province	CA	Country of Residence US

Application Data Sheet 37 CFR 1.76		Attorney Docket Number		NETL.045C1	
		Application Number			
Title of Invention		CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
Citizenship under 37 CFR 1.41(b)		US			
Mailing Address of Applicant:					
Address 1		16 Silver Fir			
Address 2					
City	Irvine	State/Province		CA	
Postal Code		92604	Country	US	
Applicant 4					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C1
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Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
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Application Information:

Title of the Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
Attorney Docket Number	NETL.045C1	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	9	Suggested Figure for Publication (if any)	

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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	Continuation of	12/422853	2009-04-13
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Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/422853	non provisional of	61/044839	2008-04-14
Prior Application Status	Expired	Remove	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C1
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/422853	non provisional of	61/044825	2008-04-14
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
12/422853	non provisional of	61/044801	2008-04-14

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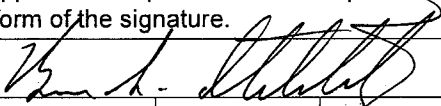
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045C1
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

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CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application is a continuation of U.S. Pat. Appl. No. 12/422,853, filed April 13, 2009, and incorporated in its entirety by reference herein, which claims the benefit of priority from U.S. Provisional Appl. No. 61/044,839, filed April 14, 2008, from U.S. Provisional Appl. No. 61/044,825, filed April 14, 2008, and from U.S. Provisional Appl. No. 61/044,801, filed April 14, 2008, each of which are incorporated in their entirety by reference herein. This application is also related to U.S. Pat. Appl. No. 12/422,912, filed on April 13, 2009, and U.S. Pat. Appl. No. 12/422,925, filed on April 13, 2009, now U.S. Pat. No. 8,001,434, both of which are incorporated in their entirety by reference herein, and U.S. Pat. Appl. No. 13/183,253, filed July 14, 2011.

BACKGROUNDField

[0002] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0003] As the demand on the performance of electronic systems increase, the systems include higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0004] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system 10 which incorporates a multiplexer-demultiplexer 12 to isolate unused subsystems of a plurality of subsystems 16, 18. This arrangement can reduce the load on each driver 14 when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver 14, the receiver 20 and the multiple destinations. However, the multiplexer and/or demultiplexer 12 can present transmission line

discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with electrical lengths longer than 1/4 of the wavelength of the operational frequency or the signal transition rate). In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

[0005] In memory applications, some systems control “chip select” or device select signals, along with the memory address and control signals, based on the device select signals, in order to disable memory devices when they are not being accessed. Figure 2 schematically illustrates an example of such a system 20 which isolates unused subsystems of a plurality of subsystems 16, 18 using device select signals 22, 24 (e.g., chip select signals). The devices are disabled by not activating (asserting) the address, control, and/or device select signals, thus logically isolating the memory devices which are not accessed. This arrangement generally requires a device select signal for each physical memory rank and the address and device select signals are generally controllable based on whether the corresponding memory device or group of memory devices is being accessed. Generating controllable address and device select signals can introduce timing delays on the device select, address, and control paths. In addition, this arrangement does not address the power dissipation and may degrade performance degradation by, for example, requiring that secondary control signals be generated for the memory devices.

SUMMARY

[0006] Certain embodiments described herein include a memory module comprising a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module comprises a plurality of memory devices on the printed circuit board and a circuit. The circuit comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The memory module further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one

connector. The memory module of certain embodiments comprises a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0007] A method of using a memory module with a computer system is described in certain embodiments. The method includes providing a memory module comprising a printed circuit board having at least one connector configured to be operatively coupled to a memory controller of a computer system and to a plurality of memory devices. The memory module further including a circuit which comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The circuit further comprises a second set of ports comprising one or more bi-directional ports. Each port of the second set of ports is operatively coupled to the at least one connector. The circuit is configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports in some embodiments. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit. The correction circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports. The method includes activating the circuit to selectively operatively couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports. The method also includes activating the circuit to reduce noise in the one or more signals.

[0008] Certain embodiments are described herein provide a circuit comprising a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports may be configured to be operatively coupled to at least one memory device of a plurality of memory devices on a printed circuit board. The circuit further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one connector on the printed circuit board.

The at least one connector is configured to be operatively coupled to a memory controller of a computer system. The circuit further comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduce noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0009] In certain embodiments, a circuit comprises a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports can be configured to be operatively coupled to at least one subsystem of a plurality of subsystems. The circuit may comprise a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one subsystem of one or more subsystems. In certain embodiments, the circuit comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 schematically illustrates an example system which isolates unused subsystems and incorporates a multiplexer-demultiplexer.

[0011] Figure 2 schematically illustrates an example system which isolates unused subsystems using device select signals.

[0012] Figure 3 schematically illustrates an example memory system incorporating a memory module in accordance with certain embodiments described herein.

[0013] Figure 4 schematically illustrates an example correction circuit in accordance with certain embodiments described herein.

[0014] Figure 5 schematically illustrates an example programmable impedance matching circuit in accordance with certain embodiments described herein.

[0015] Figure 6 schematically illustrates an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0016] Figure 7 schematically illustrates example configuration including an example filter element connected to an example repeater/amplifier in accordance with certain embodiments described herein.

[0017] Figure 8 schematically illustrates an example system having an example circuit connecting the system with one or more other systems in accordance with certain embodiments described herein.

[0018] Figure 9 schematically illustrates an example system including an example circuit in accordance with certain embodiments described herein.

[0019] Figure 10 is a flowchart of an example method 400 of using a memory module with a computer system in accordance with certain embodiments described herein.

DETAILED DESCRIPTION

[0020] Electronic systems incorporating devices and methods in accordance with embodiments described herein may include memory subsystems that contain multiple memory boards or modules that are driven by a single memory controller which synchronizes memory access and manages memory coherency. However, having a single memory controller may gradually degrade system performance and increase power consumption as the physical and the logical size of the memory subsystem increases. Increases in memory speed may further increase the power consumption. Additionally, impedance mismatches between the system board, the memory boards, and the memory devices may also substantially degrade the performance and power efficiency of traditional memory systems. Such memory systems may be used with a power supply which helps to stabilize the voltage level, and/or a thermo-mechanical component to moderate memory subsystem temperature, and/or software for throttling the processor to lower the memory access rate.

[0021] Embodiments described herein address these speed and power issues in memory systems. For example, a circuit of certain embodiments can comprise an isolation switch and can logically isolate inactive subsystems and reduce signal reflection and/or other

noise at, for example, the interface between the system board and the memory board and at the interface between the memory board and the memory devices.

[0022] Certain embodiments described herein present a circuit for isolating inactive sub-systems and maintaining and improving signal integrity in a variety of systems. Certain embodiments described herein are used to increase the performance of electronic systems without increasing the power dissipation or the memory access time. In certain embodiments, a plurality of circuits are configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits and a plurality of corresponding devices (e.g., memory devices) of a system or subsystem (e.g., a memory module of a computer system). For example, the data width of each of the plurality of circuits may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example.

[0023] Certain embodiments described herein describe a circuit for use in high performance electronic systems which may contain transmission lines and relatively heavy fan-outs. A circuit is described herein which, in certain embodiments, includes at least some of the following components: a signal router, a programmable coarse impedance matching circuit, a self-adjusting damper circuit, a filter and/or a repeater/amplifier.

[0024] Figure 3 schematically illustrates an example system incorporating a memory module 100 in accordance with certain embodiments described herein. The memory module 100 comprises a printed circuit board (PCB) 102. The PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108. The memory module 100 includes a plurality of memory devices 110 on the PCB 102.

[0025] The memory module 100 of certain embodiments further includes a circuit 112 comprising a first set of ports 122 comprising a plurality of bi-directional ports 124, each port 124 of the first set of ports 122 operatively coupled to at least one memory device 110 of the plurality of memory devices 110. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports 128, each port 128 of the second set of ports 126 operatively coupled to the at least one connector 104. The circuit 112 further comprises a

switching sub-circuit 140 configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises a correction circuit 150 (see, e.g., Figure 4) which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0026] In certain embodiments, the memory module 100 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, or higher or lower. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bits, 8 bits, 16 bits, 32 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 102 has an industry-standard form factor. For example, the PCB 102 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 102 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 102 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0027] In certain embodiments, the plurality of memory devices 110 of the memory module 100 may be arranged as one or more ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 111 of the plurality of memory devices 110 that can be accessed by the memory controller 106 to access the full bit-width of the memory bus of the memory module 100. For example, a memory module 100 in which each rank of the memory module is 64 bits wide is described as having an “x 64” organization. Similarly, a memory module 100 having 72-bit-wide ranks is described as having an “x 72” organization. The number of

memory devices 111 of a memory module 100 can be increased by increasing the number of memory devices 111 per rank or by increasing the number of ranks of the memory module 100. For example, a memory module with four ranks with each rank having N 512-MB memory devices 111 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 111 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices.

[0028] During operation, one or more ranks of a memory module 100 may be selected or activated from the plurality of ranks by control signals that are received from a component of the system (e.g., a system memory controller 106 or a local memory controller of the memory module 100). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 100 comprises only one rank of memory devices 111.

[0029] As discussed, the PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108, such as a host computer system 108. For example, the connector 104 is electrically coupled, logically coupled, or both, with the memory controller 106 in some other manner. In certain embodiments, the connector 104 comprises a wireless radio frequency (RF) interface or a portion thereof and is wirelessly connected (e.g., via one or more RF antennas) to the memory controller 106. In some embodiments, some other connection mechanism is used besides the connector 104. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 106 may comprise a disk controller of the computer system 108, for example. The memory controller 106 of certain embodiments may be mounted on a system board 109 of the host computer 108. The connector 104 can comprise a plurality of edge connections distributed along one or more edges of the PCB 102 which fit into a corresponding slot connector of the host system. The connector 104 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 100 and the host system 108. For example, the connector 104 can comprise a standard 240-pin DDR2 edge connector.

Additionally, in certain embodiments, more than one memory module 100 is coupled to the host system 108.

[0030] The plurality of memory devices 110 on the PCB 102 may include one or more volatile memory components. For example, the plurality of memory devices 110 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 111. For example, in the block diagram of Figure 3, the plurality of memory devices 110 comprises eight DRAM devices 111. Types of DRAM devices 111 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). The memory devices 111 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 111 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 111 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0031] The first set of ports 122 of the circuit 112 comprises a plurality of bi-directional ports 124 where each port of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. In addition, the second set of ports 126 of the connector 104 comprises one or more bi-directional ports 128 where each port of the second set of ports 126 is operatively coupled to the at least one connector 104. The switching sub-circuit 140 is configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The operative coupling of the first set of ports 122 to the memory devices 110, the second set of ports 126 to the connector 104, and the selective operative coupling between the first and second set of ports 122, 126 may comprise electrical coupling, logical coupling, or both. In other embodiments, optical coupling or some other manner of coupling may be employed.

[0032] In certain embodiments, each bi-directional port 128 of the second set of ports 126 may correspond to a signal of a memory bus 130 which operatively couples the memory controller 106 and the memory module 100 via the connector 104. For example, the

memory controller 106 may generally read data from and/or write data to the plurality of memory devices 110 on the memory module 100 through the memory bus 130 and through the bi-directional ports 124, 128. In certain embodiments, the number of bi-directional ports 128 of the second set of ports 126 may correspond to the number of data bits in the memory bus 130 (i.e., the bit-width of the memory bus 130).

[0033] In certain embodiments, each of the bi-directional ports 124 in the first set of ports 122 may correspond to and be operatively coupled to a signal port of one of the memory devices 111 of the plurality of memory devices 110. The bi-directional ports 124 correspond to the data signal ports, for example, of the plurality of memory devices 110. Because each memory device 111 includes associated data signal ports, the number of bi-directional ports 124 in the first set of ports of certain embodiments may be larger than the number of bi-directional ports 128 in the second set of ports 122 (e.g., the number of bits in the memory bus 130). In general, the overall number of ports of the plurality of memory devices 110 and corresponding bi-directional ports 124 in the first set of ports 126 may depend on the organization of the memory module. For example, the number of ports may depend on the number of memory devices and/or the number of ranks in the memory module. In certain embodiments, the memory module 100 includes a plurality of circuits 112. The circuits 112 may be configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits 112 and a plurality of corresponding memory devices 111 of memory module 100. For example, the data width of each of the plurality of circuits 112 may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example. Other data widths are possible in other embodiments. In some embodiments, the data width of each circuit 112 corresponds to the data width of each of the memory devices 111 such that each circuit 112 is operatively coupled to one of the plurality of memory devices 110. In other embodiments, the data width of each circuit 112 corresponds to a multiple of the data width of each memory device 111 such that each circuit 112 may be operatively coupled to the data bits of more than one memory device 111. In various embodiments, the data width of each circuit 112 corresponds to a segment of the data width

of each memory device 111 or to segments of multiple memory devices 111. As such, in some embodiments, each circuit 112 may be operatively coupled to a segment of the data bits of one memory device 111 or to segments of more than one memory device 111. Other configurations are possible which optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In addition, in certain embodiments, the location of each of the plurality of circuits 112 on the PCB 102 is selected to optimize the interconnection, timing, power consumption, and signal integrity.

[0034] In other embodiments, the bi-directional ports 124 correspond to one or more other bits of the memory bus 130 instead of, or in addition to, the data bits. For example, one or more of the ports 124 may correspond to address, chip select, control (e.g., read/write) signals. Moreover, in some embodiments, one or more ports of the first set of ports 122 is not bi-directional. For example, in one embodiment, signals corresponding to address, chip select and/or control signals from the first set of ports 122 to the plurality of memory devices 110 and not vice versa. For example, one or more of the address, chip select, and control signals may be used to control the routing of the bi-directional ports 124 which correspond to data bits.

[0035] As discussed, the switching sub-circuit 140 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the switching sub-circuit 112 may be capable of interconnecting any number of the second set of ports 126 to any number of the first set of ports 122 simultaneously. In certain embodiments, the switching sub-circuit 140 can pass signals (e.g., information such as the data signals described above) from one port 128 of the second set of ports 126 to an operatively coupled port 124 of the first set of ports 122 and vice versa. Additionally, the switching sub-circuit 140 of some embodiments can pass signals from one port 128 of the second set of ports 126 to multiple operatively coupled ports 124 of the first set of ports 122 and vice versa. In certain embodiments, the switching sub-circuit 140 isolates and provides a known static value to all un-driven ports, thereby reducing any dynamic power consumption associated with floating or toggling electrical nodes.

[0036] The switching sub-circuit 140 of certain embodiments can function as an isolation switch which isolates and/or inactivates individual subsystems not actively transferring information. For example, in an example embodiment, the plurality of memory devices 110 comprises multiple independently accessible ranks. When the memory controller accesses one of the ranks of the memory module 100, the switching sub-circuit 140 isolates and/or inactivates at least the one or more other, un-accessed ranks. For example, the switching sub-circuit 140 does not operatively couple ports from the second of set of ports 126 to the first set of ports 122 which are coupled to the memory devices 111 of the un-accessed one or more ranks of memory devices 110. In certain embodiments, the switching sub-circuit 140 may function as or be characterized as a scalable isolation switch router which routes the information from one subsystem to another (e.g., host computer 108 to the memory module 100 via the memory controller 106), establishing a one-to-one connection for transferring data from point-to-point, and/or establishing a one-to-many connection for broadcasting data. The switching sub-circuit 140 comprises or is characterized as a bi-directional multiplexer-demultiplexer in certain embodiments. In other embodiments, the switching sub-circuit 140 comprises or is characterized as a router.

[0037] In certain embodiments, the memory module 100 further includes circuitry (not shown) for controlling the operation of the circuit 112 and the components therein (e.g., the switching sub-circuit 140 and the correction circuit 150, as described more fully below). For example, in one embodiment, the memory module 100 comprises a local memory controller which is operatively coupled to one or more control inputs (not shown) of the circuit 112 to control the selective coupling of the second set of ports 126 to the first set of ports 122. In certain other embodiments, some other component of the memory module 100 controls the operation of the circuit 112 or the circuit 112 comprises internal control circuitry. In various embodiments, the circuitry for controlling the circuit 112 includes discrete logic, one or more microprocessors, a field-programmable gate array (FPGA), or a computer-programmable logic device (CPLD). In yet other configurations, an external component controls the operation of the circuit 112.

[0038] The circuit 112 of certain embodiments comprises one physically integral component (e.g., one integrated circuit package mounted on the memory module 100).

Various alternative configurations for the physical organization of the circuit 112 are possible, however. In certain other embodiments, the circuit 112 comprises multiple, physically separate components. For example, in one embodiment, the circuit 112 comprises one physically separate component per memory device 111 (e.g., integrate circuit packages mounted on the memory module 100). In another embodiment, the circuit 112 comprises one physically separate component per memory rank. In some embodiments, the circuit 112 comprises one physically separate component corresponding to multiple memory devices 111.

[0039] Figure 4 schematically illustrates an example correction circuit 150 in accordance with certain embodiments described herein. As shown, in certain embodiments, the correction circuit 150 comprises one or more correction elements (e.g., at least one coarse correction element 152 to provide coarse correction of the noise and at least one fine correction element 154 to provide fine correction of the noise). The correction circuit 150 of certain embodiments further comprises at least one filter element 156. In certain embodiments, the correction circuit 150 further comprises at least one repeater/amplifier element 158 which improves the signal quality of the one or more signals.

[0040] Referring to Figures 3 and 4, the correction circuit 150 reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. For example, the correction circuit 150 may reduce noise due to impedance mismatches (e.g., due to discontinuity regions) at nodes between the memory controller 106 and the plurality of memory devices 110. In certain embodiments, the noise may comprise signal reflections. For example, in certain embodiments, the signal paths between the host computer 108 and the plurality of memory devices 110 or portions thereof may be characterized as transmission lines if they are greater than or about equal to a particular fraction of the wavelength of the signal, such as, for example, greater than or equal to about 1/4 of the wavelength of the signal. In other configurations, the signal paths may be characterized as transmission lines when the signal path is greater than or equal to some other fraction of the wavelength of the signal such as, for example, 1/2, 1/8, or 1/10 of the wavelength. In certain embodiments, the system (e.g., memory module 100) incorporating the circuit 112 has signal delays which are related to

signal characteristics such as the signal transition rate and/or the rise/fall time of the signals. For example, the system incorporating the circuit 112 may include signal delays which are more than some fraction (e.g., 1/2) of the signal transition rate and/or the rise/fall time of the signals. In other embodiments, the signal delays are related in some other way to the signal transition rate and/or the rise/fall time of the signals (e.g., are more than 1/4, 1/8 of those values) or are related to some other signal property.

[0041] The reflections may be caused by one or more discontinuity regions along the signal paths. For example, discontinuity regions comprise points or regions at which the characteristic impedance of the corresponding signal path changes. For example, the interface between the system board 109 of the host computer 108 and the memory module 100 (e.g., the connector 104) may comprise a discontinuity region along each of the signal paths. The interfaces between the connector 104 of the memory module 100 and each of the memory devices 111 mounted on the PCB 102 may comprise one or more discontinuity regions along each of the signal paths.

[0042] The noise may comprise cross-coupling between the signal paths in certain embodiments. For example, a cross-coupling impedance may be present between one or more of the signal paths or portions thereof. In certain embodiments, the cross-coupling impedance may include cross-coupling due to cross-coupling capacitance, inductance (e.g., mutual inductance), transconductance, or a combination thereof. Additionally, a variety of other noise are possible such as, for example, electromagnetic interference or other types of noise.

[0043] Referring to Figure 4, in certain embodiments, the correction circuit 150 includes one or more elements to provide for coarse correction of signal noise due to impedance mismatches (e.g., discontinuity regions) in the system. The at least one coarse correction element 152 comprises at least one programmable impedance matching circuit in certain embodiments. Figure 5 schematically illustrates an example programmable impedance matching circuit 160 in accordance with certain embodiments described herein. For example, the programmable impedance matching circuit 160 may comprises at least one programmable resistor 162. The resistance of the programmable resistor 162 may be controllable by a control signal 164. In certain embodiments, the coarse correction element 152 may be integrated with the switching sub-circuit 140, the fine correction element 154, the filter

element 156, and/or the repeater/amplifier element 158. Alternatively, in other embodiments, the coarse correction element 152 may be implemented with discrete components.

[0044] In other embodiments, the coarse correction element 152 comprises a network of resistors 162 which can be configured to provide programmable impedance matching. In some embodiments, the coarse correction element 152 may comprise a set of programmable coarse impedance matching circuits 160 that are connected to each of the ports 124, 128 of the first set of ports 122 and the ports 128 of the second set of ports 126. Advantageously, the impedance values of the coarse correction element 152 may be dynamically controllable, configurable, and/or programmable, so as to match the characteristic impedances of the transmission lines (e.g., signal paths) in the system. The values of these impedances may differ (e.g., may be programmed differently) based on which particular signal path is activated (e.g., by the switching sub-circuit 140). Where the coarse correction element 152 comprises at least one impedance matching circuit, the granularities of the configurable impedance values may be configured based on the application of the particular system. In some embodiments, the impedance matching circuit of the coarse correction element 152 may include a real time controllable impedance control input for pre-programming and/or dynamically controlling the impedance matching circuit. In this manner, the impedance matching circuit 152 may, in certain embodiments, provide dynamically controllable variable impedance which can be pre-set for matching the impedance of the subsystem that drives the circuit 112.

[0045] In certain embodiments, the at least one fine correction element 154 of the correction circuit 150 further reduces signal reflections due to impedance mismatches and/or reduces other noise. In certain embodiments, for example, the coarse correction element 152 may be characterized as a first refining stage and the fine correction element 154 may be referred to as a second refining stage. The fine correction element 154 may smooth out the signal after it has undergone coarse correction (e.g., impedance matching) and reduces or removes transmission line characteristics from the received signal. Referring again to Figure 4, the fine correction element 154 can be connected, for example, in parallel with the programmable coarse correction element 152. In some embodiments, the fine correction element 154 operates to reduce the impedance mismatches by introducing a current source

that damps to the slew rate of the input signal, reducing signal reflection. The fine correction element 154 actively counteracts the noise in certain embodiments. In addition, the fine correction element 154 of certain embodiments is dynamically responsive to one or more changes in the noise, such as, for example, changes in the frequency of the noise component.

[0046] The fine correction element 154 may include at least one damper circuit. The damper circuit includes at least one self-adjusting damper circuit 170 in certain embodiments, such as the example self-adjusting damper circuit 170 schematically illustrated by Figure 6. In certain embodiments, the self-adjusting damper circuit 170 comprises an input terminal 172 adapted to receive an input signal 174 comprising an input noise component and an input signal component. The damper circuit 170 further comprises an output terminal 176. In certain embodiments, the damper circuit 170 includes a first sub-circuit 184 adapted to generate a first signal 187 responsive to at least a portion of the noise component. For example, the first sub-circuit 184 may generate the first signal 187 in response to the portion of the noise component not reduced by the coarse correction element 152 in certain embodiments. The damper circuit 170 of certain embodiments further comprises a second sub-circuit 188 adapted to generate a second signal 190 corresponding to a delayed version of at least a portion of the input signal. The damper circuit 170 combines the first signal 187 with the second signal 190 to generate an output signal 108 on the output terminal 176 wherein the output signal comprises an output noise component smaller than the input noise component. For example, the first signal 187 and the second signal 190 may be combined at the node 182 of the damper circuit 170. The output signal 176 of certain embodiments is substantially representative of the signal component of the input signal 174. The first sub-circuit 184 of certain embodiments comprises one or more current sources 185, 186. For example, the current sources 185, 186 may each comprise a controlled current source such as a voltage-controlled current source 185, 186 in certain embodiments. In some embodiments, the first sub-circuit 184 may comprise or be characterized as a negative feedback current source. In various embodiments, the self-adjusting damper circuit 170 includes an enable control input (not shown) for enabling the self-adjusting damper circuit 170. Examples of self-adjusting damper circuits 170 in accordance with certain embodiments described herein are described in U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and co-

pending U.S. Application No. 12/422,912, filed by the assignee of the present application on April 13, 2009. Both applications are incorporated in their entirety by reference herein.

[0047] In certain embodiments, the at least one filter element 156 and the at least one repeater/amplifier element 158 may also reduce noise from other sources such as, for example, signal coupling (e.g., cross-coupling). The filter element 156 may reduce noise from the one or more signals where the noise includes a predetermined range of frequencies. For example, the filter element 156 may include at least one band-pass filter and may reduce any low and/or high frequency noise components. In other embodiments, the filter element 156 comprises at least one other type of a filter such as a notch filter, a low-pass filter, or a high-pass filter.

[0048] One or more of the filter element 156 and the repeater/amplifier element 158 of certain embodiments enhance signal slew rate to reduce the transient power dissipation at target devices or components (e.g., one or more of the memory devices 110). Figure 7 schematically illustrates example configuration including an example filter element 156 connected to an example repeater/amplifier element 158 in accordance with certain embodiments described herein. The repeater/amplifier element 158 includes an output enable control input 159 for controlling the bi-directional signal path. In various embodiments one or more of the filter element 156 and repeater/amplifier element 158 are programmable and/or dynamically controllable. The repeater/amplifier element may further include a reference input 157. In certain embodiments, the repeater/amplifier element 158 may include one or more operational amplifiers, for example. Referring again to Figure 4, in certain embodiments, only filter element 156, only the repeater, only the amplifier, or any combination of the three elements may be included in the circuit.

[0049] The filter element 156 and/or the repeater/amplifier element 158 of certain embodiments provide wave reshaping functions and generally reduces signal noise. For example, the filter element 156 and/or the repeater/amplifier element 158 may reduce signal ringing due to reflection (e.g., reflections at one or more interfaces along the signal paths such as interfaces between the circuit 112, the connector 104, the memory devices 110 and/or the system board 109 of Figure 3). In certain embodiments, the repeater/amplifier element 158 reshapes the signal which may be deteriorated due to transmission over lossy

transmission lines having discontinuity regions. The repeater/amplifier element 158 may be configured to repeat the input signal, generating an output signal with earlier transition trigger points and faster slew rates in order to regain speed and to reduce the transient power.

[0050] In certain embodiments, the circuit 112 or portions thereof (e.g., the switching sub-circuit 140, the correction circuit 150 or elements thereof, etc.) is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD). For example, in certain embodiments, the circuit 112 or portions thereof is implemented in one or more of a PLD, ASIC, FPGA, or CPLD. In other embodiments, the circuit 112 or portions thereof is implemented using some other technology, such as, for example, using discrete components.

[0051] In various embodiments, the elements of the circuit 112 such as the switching sub-circuit 140, the coarse correction element 152, the fine correction element 154, the filter element 156, and the repeater/amplifier element 158, comprise or may be characterized as independent, fully controllable functional blocks. Additionally, in certain embodiments, any subset of one or more of these functional blocks may be implemented in order to enhance performance, reduce power dissipation, or provide other advantages.

[0052] Figure 8 schematically illustrates an example system 200 having an example circuit 112 connecting a system 200 with one or more other systems 220 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 8, for example. The system 200 of Figure 8 includes a driver 202, a receiver 201, and two subsystems 203, 204. For illustration purposes, the system 200 of Figure 8 is described as an open system that communicates with other systems 220 via the driver 202 and the receiver 201. Furthermore, in some embodiments, a connector (not shown) connects the system 200 with other systems by means of transmission lines. For example, the system 200 may be a memory system and the receiver 201 and the driver 202 of Figure 8 may represent memory signals (e.g., data, address and/or control signals) transmitted between the system 220 (e.g., a memory controller and the subsystems 203, 204 (e.g., two sets of independently addressable memory devices such as DRAM devices)). In certain embodiments, for example, the circuit

112 may be disposed on a memory board that is populated with sets of memory devices 203, 204.

[0053] In various embodiments, the receiver 201 may or may not contain any active circuits. For example, the receiver 201 may be a connector such as a socket or DIMM interface with a system board (e.g., the connector 104 described above in regards to Figure 3). The output signal from the receiver 201 may exhibit the characteristics of the signal on a transmission line such as, for example, wave dispersion, attenuation, and reflection. In certain embodiments, the output signal from the receiver 201 goes through a number of refining, filtering and/or amplification stages in circuit 112 as described herein to improve the performance of the overall system. For example, the signal quality is enhanced by introducing at least one circuit 112 including at least one correction circuit 150 comprising one or more coarse correction elements 152 (e.g., impedance matching circuits), fine correction elements 154 (e.g., self-adjusting dampers), filter elements 156 (e.g., bandpass filters) and/or repeater/amplifier elements 158 as described herein on the signal paths (e.g., from receiver 101 to subsystem 103). For example, the circuit 112 may reduce power consumption by selectively isolating the inactive (or un-accessed) subsystems and may enhance the access rate of a targeted subsystem. For example, in one embodiment, the system 200 comprises a memory module 100 such as one of the memory modules described herein, the subsystems 203 and 204 comprise memory devices such as the memory devices 111 described herein, and the other system 220 comprises a host computer 108 including a memory controller 106 as described herein.

[0054] In certain embodiments, the performance is enhanced and power dissipation is reduced by removing the load associated with other (e.g., unused or inactive) subsystems (e.g., the subsystem 204) and reshaping the signal waveform using a filter element 156 and/or the repeater/amplifier element 158. In some embodiments, there are two refining stages for reducing signal reflections due to impedance mismatches. In some embodiments, for example, the first refining stage is the coarse correction element 152 (e.g., an impedance matching circuit) that matches the impedance of the transmission line (e.g., wire) prior to a receiver 101 with the characteristic impedance of the transmission line (e.g., wire) in the system 100. The second stage may be the fine correction element 154 (e.g., a self-

adjusting damper for further reducing signal reflections due to impedance mismatches and/or reducing other noise.

[0055] Figure 9 schematically illustrates an example system 300 including an example circuit 112 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 3, for example. The circuit 112 comprises a first set of ports 122 comprising a plurality of bi-directional ports, each port of the first set of ports 122 configured to be operatively coupled to at least one subsystem 311 of a plurality of subsystems 310. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports, each port of the second set of ports 126 configured to be operatively coupled to at least one subsystem 321 of one or more subsystems 320. In certain embodiments, the circuit 112 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126 each comprises a correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. The correction circuit 150 of certain embodiments may comprise one of the correction circuits 150 described herein such as the correction circuits 150 described above with respect to Figures 4-7 and may include at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158, or any combination or sub-combination thereof. In other embodiments, the fine correction element 154 may be placed after the switching sub-circuit 140 in order to achieve better results. For example, the fine correction element 154 may be connected after the filter element 156 (e.g., between the filter 156 and the repeater/amplifier element 158) in certain embodiments.

[0056] In certain embodiments, the system 300 includes a first set of interfaces 330 positioned between (e.g., logically and/or physically between) and operatively coupled to the first set of ports 122 and to the plurality of subsystems 310. In addition, the system 300 of certain embodiments may include a second set of interfaces 340 positioned between (e.g., logically and/or physically between) and operatively coupled to the second set of ports 126

and the one or more subsystems 320. For example, the first and/or second sets of interfaces 330, 340 may comprise port connections.

[0057] In an example configuration, the plurality of subsystems 310 include a plurality of memory devices 311 (e.g., DRAM devices) mounted on a memory module and the first set of interfaces 330 comprise interfaces between the circuit 112 and the plurality of memory devices 311. For example, each of the first set of ports 122 may correspond to a memory signal (e.g., a data, address, and/or control signal) and each of the first set of interfaces 330 is operatively coupled to a corresponding memory signal port of a corresponding memory device 311. The one or more subsystems 320 of the example configuration comprise a host computer system and the second set of interfaces 340 comprises memory board (e.g., a PCB of a memory module) interfaces to the host system. For example, the second set of interfaces 340 form part of a connector. Each of the second set of ports 126 and corresponding interfaces of the second set of interfaces 340 corresponds to a memory signal (e.g., a data, address, and/or control signal) of at least one memory bus between the host system and the memory module.

[0058] The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 126 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the system board and the memory board. The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 122 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the circuit 112, the memory board and/or the plurality of memory devices. The filter elements 156 (e.g., a bandpass filter) reduce or filter out cross-coupling between signals, for example. For example, the filter elements 158 reduce or filter out cross-coupling between data, address, and/or control signals (e.g., wires).

[0059] Figure 10 is a flowchart of an example method 400 of using a memory module 100 with a computer system in accordance with certain embodiments described herein. While the method 400 is described herein by reference to the memory module 100 schematically illustrated by Figure 3, other memory modules, electronic systems or

subsystems, and/or circuits are also compatible with the embodiments described herein. In certain embodiments, at operational block 402, the method 400 comprises providing a memory module 100 comprising a printed circuit board 102 having at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108 and to a plurality of memory devices 110. The memory module 100 further comprises a circuit 112 including a first set of ports 122 comprising a plurality of bi-directional ports 124. In certain embodiments, each port 124 of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. The circuit of certain embodiments further includes a second set of ports 126 comprising one or more bi-directional ports 128 where each port 128 of the second set of ports 126 is operatively coupled to the at least one connector 104.

[0060] The circuit 112 of certain embodiments is configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises at least one correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0061] The method 400 of certain embodiments further includes activating the circuit 112 at operational block 404 to selectively operatively couple at least one of the one or more ports 128 of the second set of ports 126 to at least one of the one or more ports 124 of the first set of ports 122. For example, in certain embodiments, activating the circuit 112 to perform the selective operative coupling may include sending the appropriate control signal or combination of control signals to a switching sub-circuit 140 of the circuit 112. In certain embodiments, the method 400 further includes activating the circuit 112 to reduce noise in the one or more signals at operational block 406. For example, in one embodiment, activating the circuit 112 to reduce noise comprises activating one or more of the elements of the correction circuit 150 (e.g., at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158) of ports 124, 128 of the first and second sets of ports 122, 126 corresponding to

active signal paths between the memory controller 106 and the memory devices 110. Activating one or more of the elements 152, 154, 156, 158 of the correction circuit 150 may include sending control signals (e.g., enable signals) to the appropriate elements 152, 154, 156, 158 and/or receiving the control signals (e.g., enable signals) at the appropriate elements 152, 154, 156, 158.

[0062] The operational blocks 402, 404, 406 of the method 400 may be performed in a different order than provided or one or more of the operational blocks may be performed substantially at the same time or during overlapping time periods. For example, in certain embodiments, at least the operational block 404 and the operational block 406 are performed substantially simultaneously. In addition, in some embodiments, one or more of the operational blocks 402, 404, 406 may not be included, an additional operational block may be included, or both. The devices and methods described herein may be incorporated in any system (e.g. a board level system) or device that contains one or more connectors or wires.

[0063] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory module comprising:
 - a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board;
 - a circuit, comprising:
 - a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and
 - a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and
 - a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

ABSTRACT

Certain embodiments described herein include a memory module having a printed circuit board including at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module further includes a plurality of memory devices on the printed circuit board and a circuit including a first set of ports operatively coupled to at least one memory device. The circuit further includes a second set of ports operatively coupled to the at least one connector. The circuit includes a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. Each port of the first set and the second set comprises a correction circuit which reduces noise in one or more signals transmitted between the first set of ports and the second set of ports.

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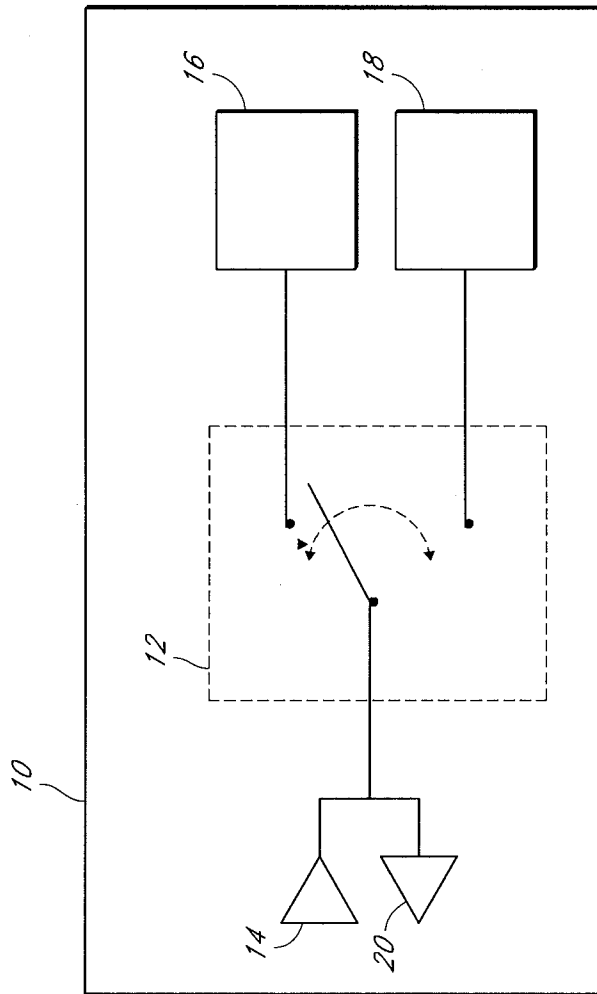


FIG. 1

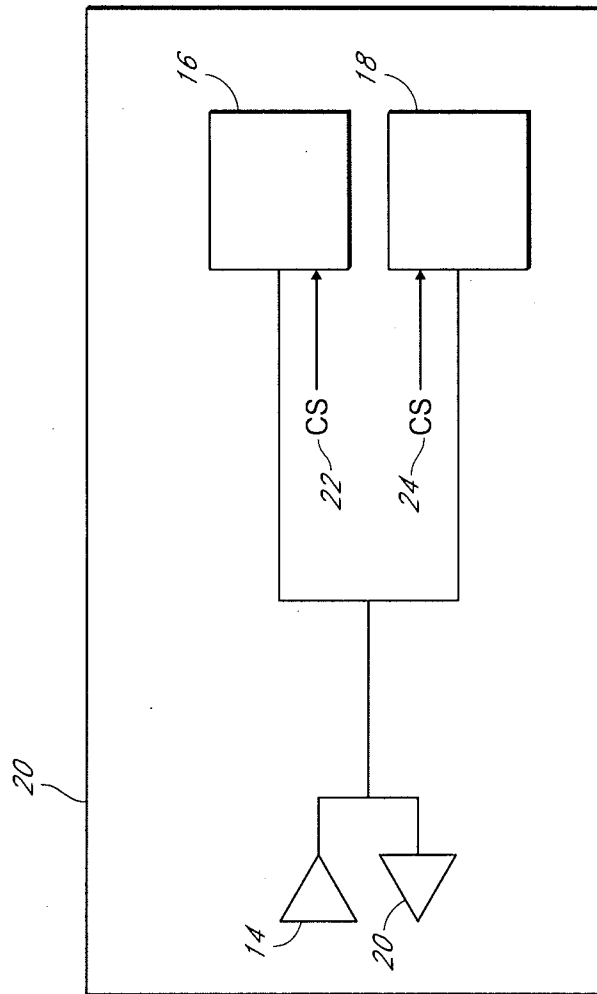


FIG. 2

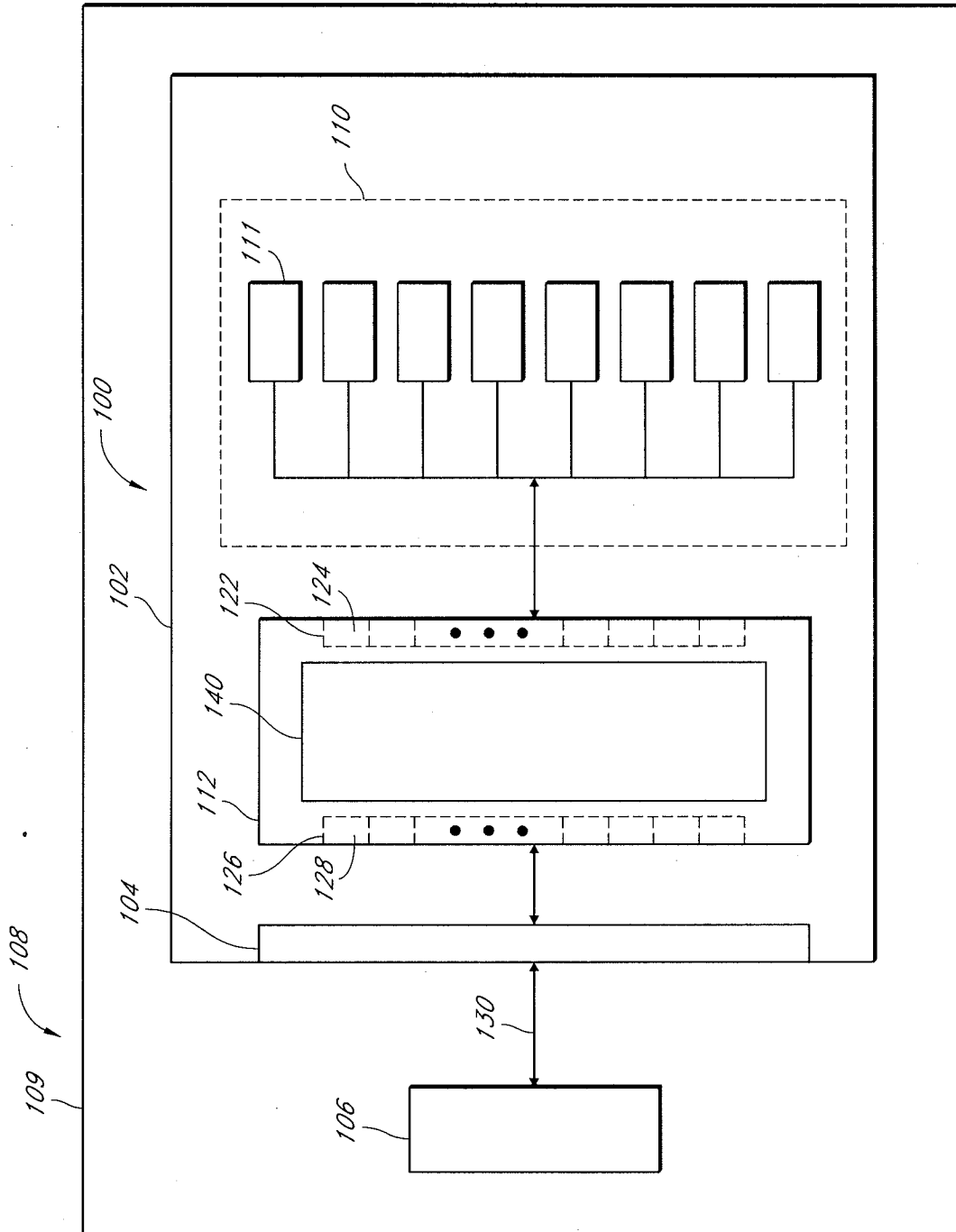


FIG. 3

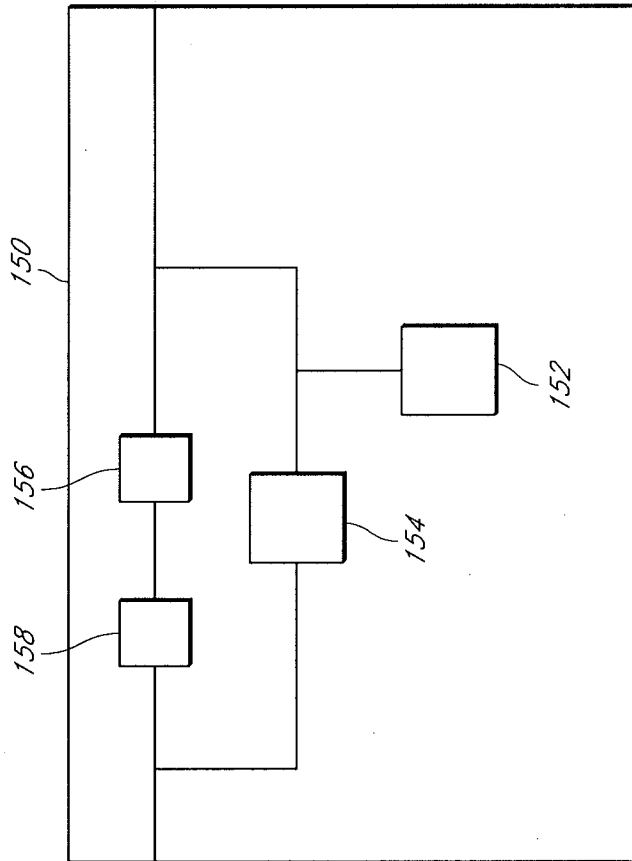


FIG. 4

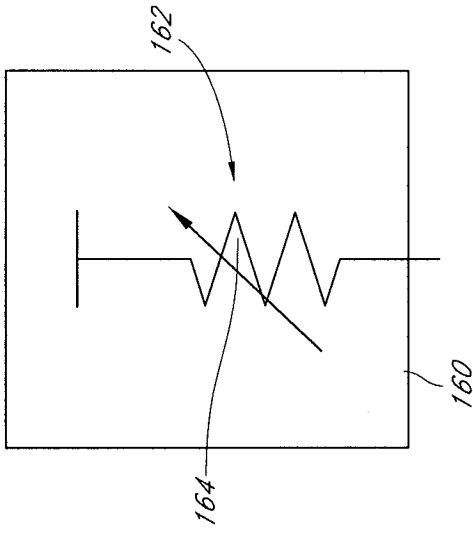


FIG. 5

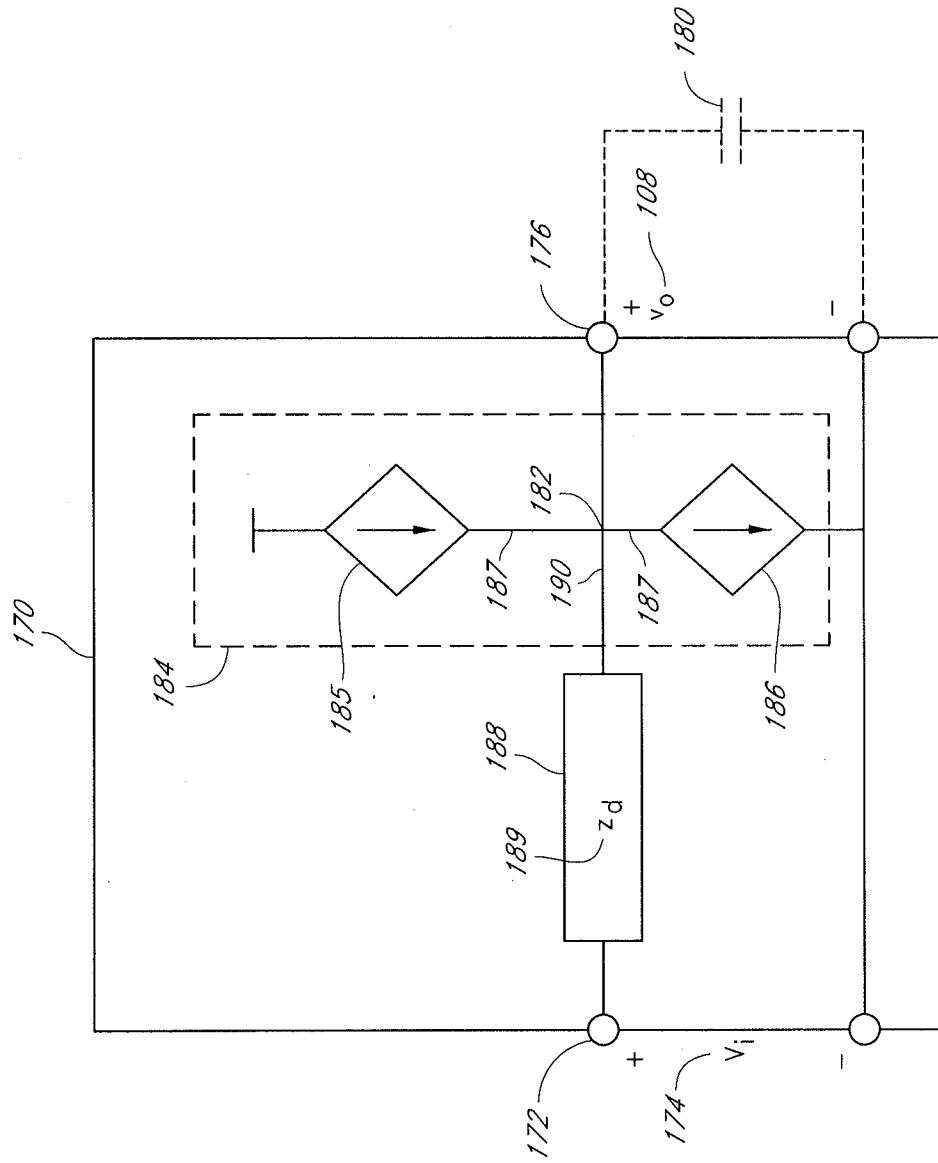


FIG. 6

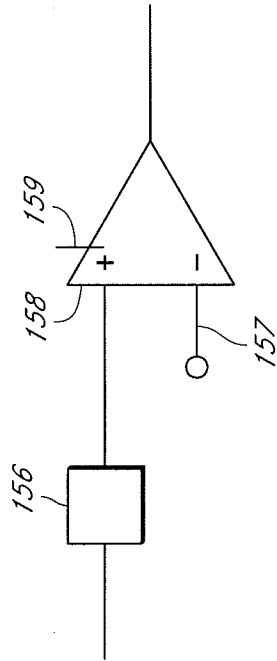


FIG. 7

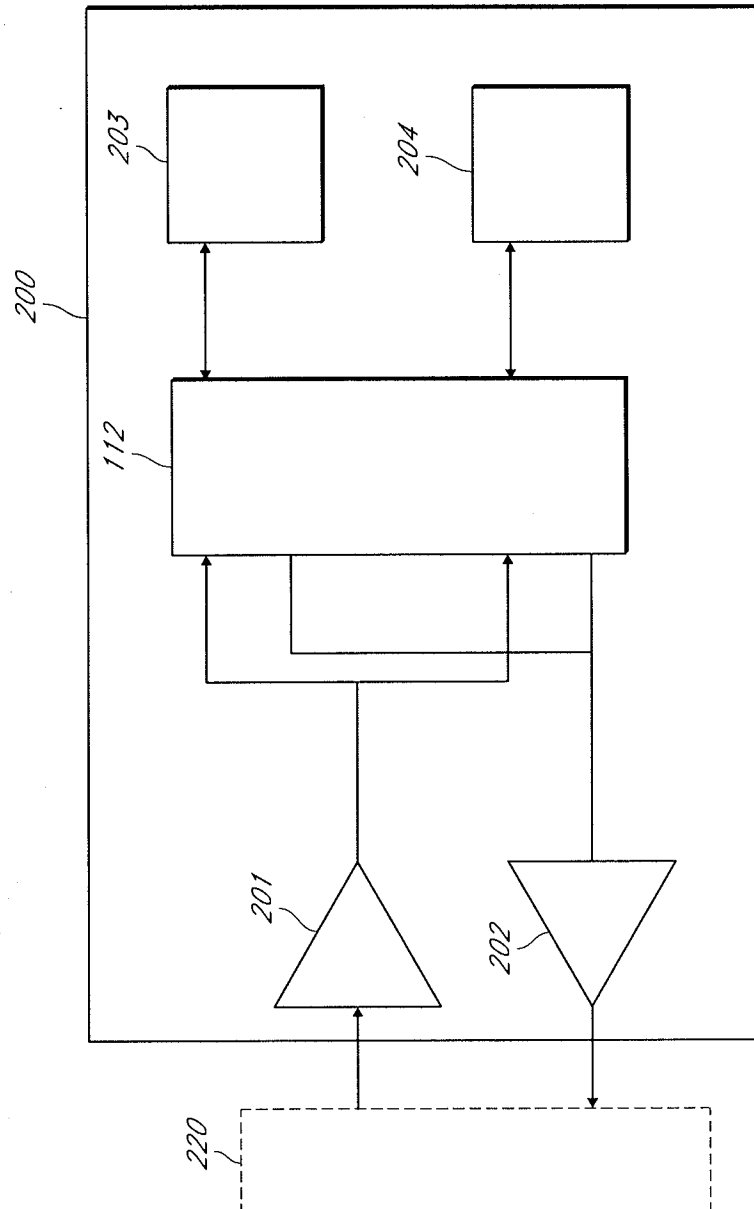


FIG. 8

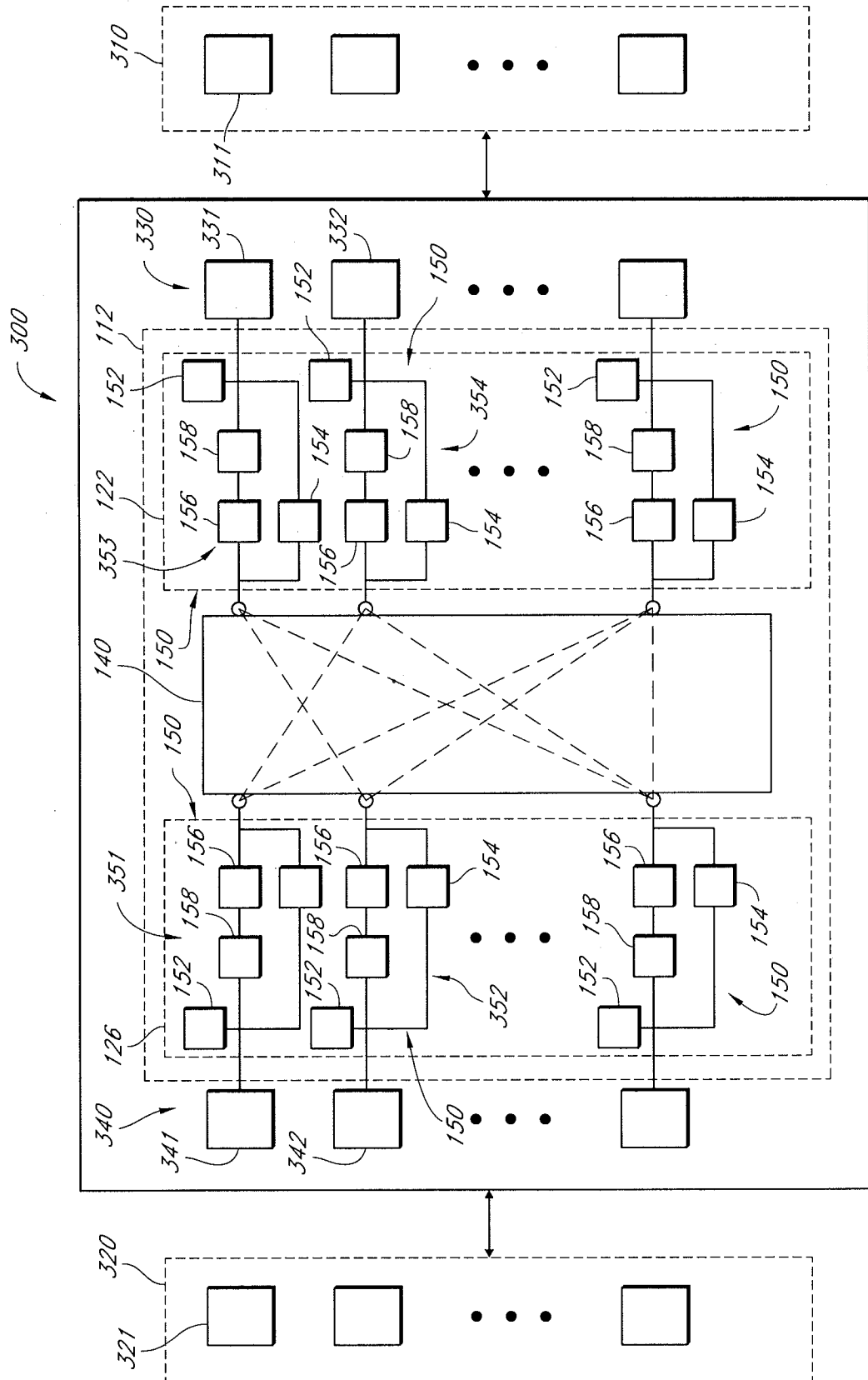


FIG. 9

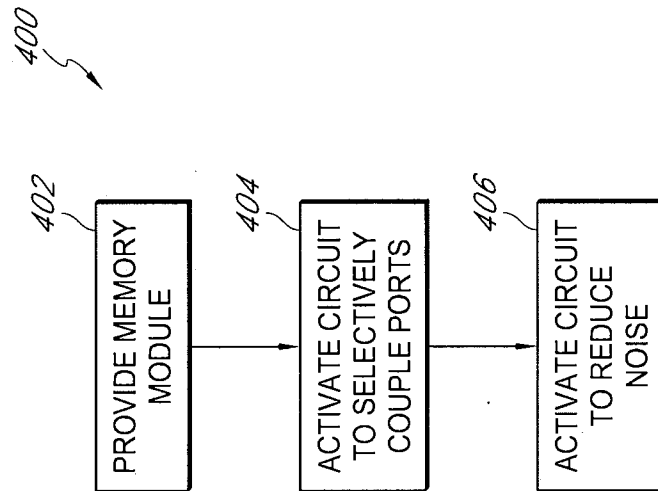


FIG. 10

STATEMENT UNDER 37 CFR § 3.73(b)
ESTABLISHMENT OF ASSIGNEE

Applicant	:	Lee et al.
App. No.	:	Unknown
Filed	:	Herewith
For	:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	:	Unknown
Group Art Unit	:	Unknown

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

This document is being filed with a copy of a Power of Attorney signed by the Assignee. This Statement sets forth the chain of title of the above-identified application.

Netlist, Inc., a Corporation, is the Assignee of the entire right, title, and interest of the above-referenced application by virtue of:

The Assignment from the inventor(s) to the Assignee recorded in the United States Patent and Trademark Office on November 8, 2010, at Reel 025301, and Frame 0961.

The undersigned is an agent of Customer Number 20995 and is authorized to act on behalf of the Assignee. Please recognize or change the correspondence address for the above-identified application to **Customer No. 20995.**

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: _____

3/5/12

By: _____

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Please Direct All Correspondence to Customer Number 20995

RESCISSION OF ANY PRIOR DISCLAIMERS AND REQUEST TO REVISIT ART

Applicant	: Lee et al.
App. No	: 13/412,243
Filed	: March 5, 2012
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: Unknown
Art Unit	: 2827
Conf #	: 2878

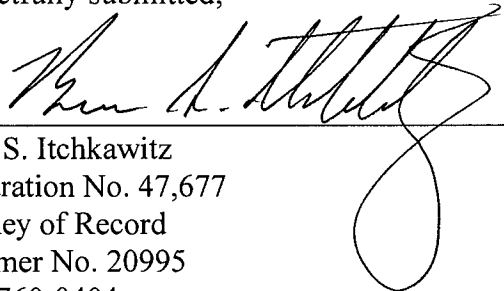
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

The claims of the present application are different and possibly broader in scope than the claims pursued in the parent application(s). To the extent any prior amendments or characterizations of the scope of any claim or referenced art could be construed as a disclaimer of any subject matter supported by the present disclosure, Applicant hereby rescinds and retracts such disclaimer. Accordingly, the references previously considered in the parent application(s) may need to be re-visited.

Please charge any additional fees, including any fees for additional extension of time, or credit overpayment to Deposit Account No. 11-1410.

Respectfully submitted,



Dated: March 20, 2012

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Electronic Acknowledgement Receipt

EFS ID:	12350016
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Erica Van Sciver
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045C1
Receipt Date:	20-MAR-2012
Filing Date:	
Time Stamp:	18:02:09
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Miscellaneous Incoming Letter	rescission.pdf	39083 f8f6ed71f16faa8de2ca4f454c02153a131efdeb	no	1

Warnings:

Information:

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/412,243

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	1 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	380
N/A	620
N/A	250
x 60 =	0.00
x 250 =	0.00
	0.00
	0.00
TOTAL	1250

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus	**	=	
Independent (37 CFR 1.16(h))	*	Minus	***	=	
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
Total (37 CFR 1.16(i))	*	Minus	**	=	
Independent (37 CFR 1.16(h))	*	Minus	***	=	
Application Size Fee (37 CFR 1.16(s))					
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/412,243	03/05/2012	2827	0.00	NETL.045C1	1	1

CONFIRMATION NO. 2878

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000053881426

Date Mailed: 04/25/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 20995

Domestic Priority data as claimed by applicant

This application is a CON of 12/422,853 04/13/2009 PAT 8154901 *
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008
(*)Data provided by applicant is not consistent with PTO records.

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 04/23/2012

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/412,243**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No
Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

365

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where

the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage, facilitate, and accelerate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.



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United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/412,243	03/05/2012	Hyun Lee	NETL.045C1

CONFIRMATION NO. 2878

POA ACCEPTANCE LETTER

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614



Date Mailed: 04/25/2012

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 03/05/2012.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/hngo/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/412,243	03/05/2012	Hyun Lee	NETL.045C1

CONFIRMATION NO. 2878

FORMALITIES LETTER



OC000000053881427

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/25/2012

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
Applicant must submit \$380 to complete the basic filing fee for a non-small entity. If appropriate, applicant may make a written assertion of entitlement to small entity status and pay the small entity filing fee (37 CFR 1.27).
- The oath or declaration is missing.
A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.
Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- A surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of \$130 for a non-small entity, must be submitted.

SUMMARY OF FEES DUE:

Total fee(s) required within **TWO MONTHS** from the date of this Notice is \$1380 for a non-small entity

- \$380 Statutory basic filing fee.
- \$130 Surcharge.
- The application search fee has not been paid. Applicant must submit \$620 to complete the search fee.
- The application examination fee has not been paid. Applicant must submit \$250 to complete the examination fee for a non-small entity.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/tnegash/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/412,243
	Filing Date	March 05, 2012
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 1 OF 4	Attorney Docket No.	NETL.045C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
	11	4,910,597	03-20-1990	Harada et al.	
	12	4,942,556	07-17-1990	Sasaki et al.	
	13	4,987,321	01-22-1991	Toohy	
	14	5,138,619	08-11-1992	Fasang et al.	
	15	5,173,906	12-22-1992	Dreibelbis et al.	
	16	5,222,066	06-22-1993	Gula et al.	
	17	5,241,503	08-31-1993	Cheng	
	18	5,304,856	04-19-1994	Rainal	
	19	5,337,254	08-09-1994	Knee et al.	
	20	5,359,235	10-25-1994	Coyle et al.	
	21	5,394,037	02-28-1995	Josephson et al.	
	22	5,430,335	07-04-1995	Tanoi	
	23	5,525,917	06-11-1996	Wong et al.	
	24	5,841,296	11-24-1998	Churcher et al.	
	25	5,914,543	06-22-1999	Scherpenberg, et al.	
	26	6,000,048	12-07-1999	Krishna et al.	
	27	6,044,481	03-28-2000	Kornachuk et al.	
	28	6,070,217	05-30-2000	Connolly et al.	
	29	6,169,696	01-02-2001	Bissey	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/412,243
	Filing Date	March 05, 2012
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 2 OF 4	Attorney Docket No.	NETL.045C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	30	6,194,959	02-27-2001	Kamoshida et al.	
	31	6,216,240	04-10-2001	Won et al.	
	32	6,467,056	10-15-2002	Satou et al.	
	33	6,560,740	05-06-2003	Zuraski, Jr. et al.	
	34	6,614,702	09-02-2003	Lee	
	35	6,721,150	04-13-2004	Guerrero, Jr.	
	36	6,812,869	11-02-2004	Rahman et al.	
	37	6,829,728	12-07-2004	Cheng et al.	
	38	6,918,072	07-12-2005	Cowels et al.	
	39	6,928,024	08-09-2005	Pfeiffer et al.	
	40	6,928,593	08-09-2005	Halbert et al.	
	41	6,930,509	08-16-2005	Banik	
	42	6,934,900	08-23-2005	Cheng et al.	
	43	7,036,064	04-25-2006	Kebichi et al.	
	44	7,053,470	05-30-2006	Sellers et al.	
	45	7,062,696	06-13-2006	Barry et al.	
	46	7,184,915	02-27-2007	Hansquine et al.	
	47	7,190,210	03-13-2007	Azrai et al.	
	48	7,203,873	04-10-2007	Adams	
	49	7,253,652	08-07-2007	Azimi et al.	
	50	7,284,166	10-16-2007	Zappa et al.	
	51	2001/0048342	12-06-2001	Yoshida et al.	
	52	2002/0000847	01-03-2002	Taguchi	
	53	2002/0131535	09-19-2002	Huber	
	54	2002/0140523	10-03-2002	Park et al.	
	55	2003/0098742	05-29-2003	Nakagawa et al.	
	56	2003/0126346	07-03-2003	Kuo	
	57	2003/0197797	10-23-2003	Segura	
	58	2003/0218491	11-27-2003	Nagasue	

Examiner Signature	Date Considered
--------------------	-----------------

***Examiner:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/412,243
	Filing Date	March 05, 2012
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 3 OF 4	Attorney Docket No.	NETL.045C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	59	2004/0155702	08-12-2004	Danielsson	
	60	2004/0199843	10-07-2004	Hansquine et al.	
	61	2005/0093620	05-05-2005	Ho et al.	
	62	2005/0127989	06-16-2005	Miyagi	
	63	2005/0289423	12-29-2005	Yabuta	
	64	2006/0082383	04-20-2006	Choi	
	65	2006/0107156	05-18-2006	Lee et al.	
	66	2006/0140015	06-26-2006	Kasamsetty	
	67	2006/0144015	07-06-2006	Cash et al.	
	68	2006/0147217	07-06-2006	Hahin et al.	
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	72	2006/0271748	11-30-2006	Jain et al.	
	73	2007/0030814	02-08-2007	Shin et al.	
	74	2007/0058471	03-15-2007	Rajan et al.	
	75	2007/0079199	04-05-2007	Chorn et al.	
	76	2007/0109707	05-17-2007	Honda	
	77	2007/0152743	07-05-2007	Keeth et al.	
	78	2007/0204075	08-30-2007	Rajan et al.	
	79	2007/0223296	09-27-2007	Miller et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹

Examiner Signature	Date Considered
--------------------	-----------------

***Examiner:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	13/412,243
	Filing Date	March 05, 2012
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 4 OF 4	Attorney Docket No.	NETL.045C1

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
	80	DER-CHANG ET AL. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, April 2002, Volume 21, Issue 4, Pages 449-465	
	81	BAYARD "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
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	83	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

13357008/052912

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

Electronic Acknowledgement Receipt

EFS ID:	12907208
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Adriana Perez
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045C1
Receipt Date:	31-MAY-2012
Filing Date:	05-MAR-2012
Time Stamp:	19:37:31
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		IDS.pdf	272449 463feb074630d1225863931ef182cf5c818a6345	yes	5

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Transmittal Letter	1	1
	Information Disclosure Statement (IDS) Form (SB08)	2	5
Warnings:			
Information:			
Total Files Size (in bytes):		272449	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

INFORMATION DISCLOSURE STATEMENT

Applicant : Hyun Lee, et al.
App. No. : 13/412243
Filed : March 5, 2012
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Examiner : Unknown
Art Unit : 2827
Conf. No. : 2878

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

References and Listing

Submitted herewith in the above-identified application is an Information Disclosure Statement listing references for consideration. Listed references are of record in U.S. patent application No. 12/422,853, filed April 13, 2009, which is the parent of this non-provisional application, and is relied upon for an earlier filing date under 35 USC 120. Copies of the references are not submitted pursuant to 37 CFR 1.98(d).

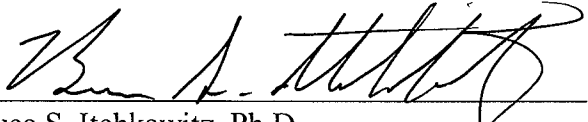
Timing of Disclosure

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 CFR 1.17(p) to Deposit Account No. 11-1410.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: 5/31/12

By: 

Bruce S. Itchkawitz, Ph.D.
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Please Direct All Correspondence to Customer Number 20995

RESPONSE TO FORMALITIES NOTICE

Applicant	:	Hyun Lee, et al.
App. No.	:	13/412,243
Filed	:	March 5, 2012
For	:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Art Unit	:	2827
Conf No.	:	2878

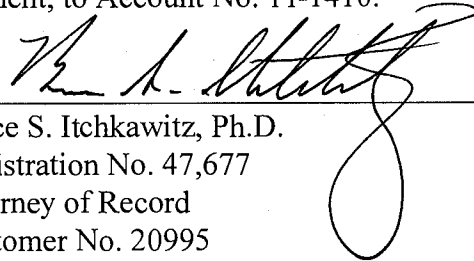
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without a Declaration and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following.

- (X) A Declaration in 6 pages.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.



Bruce S. Itchkawitz, Ph.D.
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Doc Code: OATH

Document Description: Oath or declaration filed

PTO/SB/01 (10-08)

Approved for use through 06/30/2010. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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**DECLARATION FOR UTILITY OR
DESIGN
PATENT APPLICATION
(37 CFR 1.63)**



Declaration
Submitted
With Initial
Filing

OR



Declaration
Submitted after Initial
Filing (surcharge
(37 CFR 1.16 (f))
required)

Attorney Docket
Number

NETL.045A

First Named Inventor

Hyun Lee

COMPLETE IF KNOWN

Application Number

12/422,853

Filing Date

April 13, 2009

Art Unit

Unknown

Examiner Name

Unknown

I hereby declare that: (1) Each inventor's residence, mailing address, and citizenship are as stated below next to their name; and (2) I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is claimed and for which a patent is sought on the invention entitled:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

(Title of the Invention)

the application of which



is attached hereto

OR



was filed on (MM/DD/YYYY)

04/13/2009

as United States Application Number or PCT International

Application Number

12/422,853

and was amended on (MM/DD/YYYY)

(if applicable).

I hereby state that I have reviewed and understand the contents of the above identified application, including the claims, as amended by any amendment specifically referred to above.

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56, including for continuation-in-part applications, material information which became available between the filing date of the prior application and the national or PCT international filing date of the continuation-in-part application.

Authorization To Permit Access To Application by Participating Offices



If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), and any other intellectual property offices in which a foreign application claiming priority to the above-identified application is filed access to the above-identified patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, or other intellectual property office in which a foreign application claiming priority to the above-identified application is filed to have access to the application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the application-as-filed with respect to: 1) the above-identified application, 2) any foreign application to which the above-identified application claims priority under 35 USC 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the above-identified US application, and 3) any U.S. application from which benefit is sought in the above-identified application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing the Authorization to Permit Access to Application by Participating Offices.

[Page 1 of 3]

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

DECLARATION — Utility or Design Patent Application

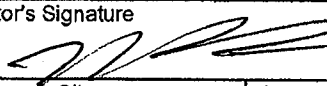
Claim of Foreign Priority Benefits

I hereby claim foreign priority benefits under 35 U.S.C. 119(a)-(d) or (f), or 365(b) of any foreign application(s) for patent, inventor's or plant breeder's rights certificate(s), or 365(a) of any PCT international application which designated at least one country other than the United States of America, listed below and have also identified below, by checking the box, any foreign application for patent, inventor's or plant breeder's rights certificate(s), or any PCT international application having a filing date before that of the application on which priority is claimed.

Prior Foreign Application Number(s)	Country	Foreign Filing Date (MM/DD/YYYY)	Priority Not Claimed	Certified Copy Attached?	
				YES	NO
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐

☐ Additional foreign application numbers are listed on a supplemental priority data sheet PTO/SB/02B attached hereto.

DECLARATION — Utility or Design Patent Application

Direct all correspondence to:	☒ The address associated with Customer Number.	20995	OR	☐ Correspondence address below
Name				
Address				
City State			ZIP	
Country T	elephone	Email		
WARNING: Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available. Petitioner/applicant is advised that documents which form the record of a patent application (such as the PTO/SB/01) are placed into the Privacy Act system of records DEPARTMENT OF COMMERCE, COMMERCE-PAT-7, System name: <i>Patent Application Files</i> . Documents not retained in an application file (such as the PTO-2038) are placed into the Privacy Act system of COMMERCE/PAT-TM-10, System name: <i>Deposit Accounts and Electronic Funds Transfer Profiles</i> .				
I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under 18 U.S.C. 1001 and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.				
NAME OF SOLE OR FIRST INVENTOR:		☐ A petition has been filed for this unsigned inventor		
Given Name (first and middle [if any])		Family Name or Surname		
Hyun		Lee		
Inventor's Signature			Date	
			7/8/09	
Residence: City	State	Country	Citizenship	
Ladera Ranch	CA	US	US	
Mailing Address				
21 Thalia				
City State		Zip	Country	
Ladera Ranch	CA	92694	US	
☒ Additional inventors or a legal representative are being named on the 2 supplemental sheet(s) PTO/SB/02A or 02LR attached hereto.				

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

DECLARATION**ADDITIONAL INVENTOR(S)**
Supplemental SheetPage 1 of 2

Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Jayesh R.		Bhakta	
Inventor's Signature <i>Jayesh Bhakta</i>		Date <i>7-7-09</i>	
Residence: City	Cerritos	State	CA
		Country	US
Citizenship US			
Mailing Address 12220 Rose Street			
City	Cerritos	State	CA
		Zip	90703
		Country	US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Jeffrey C. <i>Jeffrey C. Solomon</i>		Solomon	
Inventor's Signature <i>Jeffrey C. Solomon</i>		Date	
Residence: City	Irvine	State	CA
		Country	US
Citizenship US			
Mailing Address 16 Silver Fir			
City	Irvine	State	CA
		Zip	92604
		Country	US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Mario Jesus		Martinez	
Inventor's Signature <i>Mario Jesus Martinez</i>		Date <i>7/9/09</i>	
Residence: City	Laguna Niguel	State	CA
		Country	US
Citizenship US			
Mailing Address 19 Lindall Street			
City	Laguna Niguel	State	CA
		Zip	92677
		Country	US

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786-9199) and select option 2.

DECLARATION	ADDITIONAL INVENTOR(S) Supplemental Sheet
Page <u>2</u> of <u>2</u>	

Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Chi-She		Chen	
Inventor's Signature		Date <u>07-07-09</u>	
Residence: City <u>Walnut</u>	State <u>CA</u>	Country <u>US</u>	Citizenship <u>TW</u>
Mailing Address <u>944 Crystal Water Lane</u>			
City <u>Walnut</u>	State <u>CA</u>	Zip <u>91789</u>	Country <u>US</u>
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Inventor's Signature		Date	
Residence: City	State	Country	Citizenship
Mailing Address			
City	State	Zip	Country
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Inventor's Signature		Date	
Residence: City	State	Country	Citizenship
Mailing Address			
City	State	Zip	Country

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786-9199) and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13412243			
Filing Date:	05-Mar-2012			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Bruce S. Itchkawitz/Nicole Jevsevar			
Attorney Docket Number:	NETL.045C1			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	380	380
Utility Search Fee	1111	1	620	620
Utility Examination Fee	1311	1	250	250
Pages:				
Claims:				
Miscellaneous-Filing:				
Late filing fee for oath or declaration	1051	1	130	130
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1380

Electronic Acknowledgement Receipt

EFS ID:	13065861
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Nino Lopez
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045C1
Receipt Date:	20-JUN-2012
Filing Date:	05-MAR-2012
Time Stamp:	20:18:02
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1380
RAM confirmation Number	8708
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	RESPONSE.pdf	32187	no	1
			ba3c1ee5da1d515a590c12baebcc9abcaaba99d8		
Warnings:					
Information:					
2	Oath or Declaration filed	DECLARATION.pdf	383549	no	6
			04474894e49ef847b23dec64585ea923cd35450b		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	37107	no	2
			c7ece5a004f88e6b8694327aad82b81c47f0cc43		
Warnings:					
Information:					
Total Files Size (in bytes):			452843		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
13/412,243	03/05/2012	2827	1380	NETL.045C1	1	1

CONFIRMATION NO. 2878

UPDATED FILING RECEIPT



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/28/2012

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Assignment For Published Patent Application

NETLIST, INC., Irvine, CA

Power of Attorney: The patent practitioners associated with Customer Number 20995

Domestic Priority data as claimed by applicant

This application is a CON of 12/422,853 04/13/2009 PAT 8154901 *
which claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008
(*)Data provided by applicant is not consistent with PTO records.

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)

If Required, Foreign Filing License Granted: 04/23/2012

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 13/412,243**

Projected Publication Date: 10/04/2012

Non-Publication Request: No

Early Publication Request: No
Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

365

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

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the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
13/412,243

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(i))	1 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	380
N/A	620
N/A	250
x 60 =	0.00
x 250 =	0.00
	0.00
	0.00
TOTAL	1250

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

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POWER OF ATTORNEY OR REVOCATION OF POWER OF ATTORNEY WITH A NEW POWER OF ATTORNEY AND CHANGE OF CORRESPONDENCE ADDRESS	Application Number	13/412243
	Filing Date	5-Mar-2012
	First Named Inventor	Lee, Hyun
	Title	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE
	Art Unit	2827
	Examiner Name	Unknown
	Attorney Docket Number	NETL.045C1

I hereby revoke all previous powers of attorney given in the above-identified application.

☐ A Power of Attorney is submitted herewith.

OR

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

79141

OR

☐ I hereby appoint Practitioner(s) named below as my/our attorney(s) or agent(s) to prosecute the application identified above, and to transact all business in the United States Patent and Trademark Office connected therewith:

Practitioner(s) Name	Registration Number

Please recognize or change the correspondence address for the above-identified application to:

☒ The address associated with the above-mentioned Customer Number.

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the:

☐ Applicant/Inventor.

OR

☒ Assignee of record of the entire interest. See 37 CFR 3.71.
Statement under 37 CFR 3.73(b) (Form PTO/SB/96) submitted herewith or filed on _____.

SIGNATURE of Applicant or Assignee of Record

Signature

Date

September 20, 2012

Name

Gail Sasaki

Telephone

949-435-0025

Title and Company

VP and CFO

NOTE: Signatures of all the inventors or assignees of record of the entire interest or their representative(s) are required. Submit multiple forms if more than one signature is required, see below*.☐ *Total of _____ forms are submitted.

This collection of information is required by 37 CFR 1.31, 1.32 and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 13/412,243Filed/Issue Date: 5-Mar-2012Titled: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Netlist, Inc. _____, a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)

the patent application/patent identified above, by virtue of either:

- A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 025301, Frame 0961, or for which a copy therefore is attached.

OR

- B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at
Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

- ☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

Signature

Gail Sasaki

Printed or Typed Name

SEPTEMBER 20, 2012

Date

VP and CFO

Title

Electronic Acknowledgement Receipt

EFS ID:	13837096
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C1
Receipt Date:	25-SEP-2012
Filing Date:	05-MAR-2012
Time Stamp:	19:43:58
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	POA045C1.pdf	347187 bc83d5d01a319458842be03bd2ecd36e46575953	no	2

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Information:

Total Files Size (in bytes):

347187

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/412,243	03/05/2012	Hyun Lee	NETL.045C1

CONFIRMATION NO. 2878

POA ACCEPTANCE LETTER

79141

The Law Office of Jamie Zheng, Ph.D Esq.

P.O. Box 60573

Palo Alto, CA 94306



OC000000056847033

Date Mailed: 10/03/2012

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 09/25/2012.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/tnnguyen/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/412,243	03/05/2012	Hyun Lee	NETL.045C1

CONFIRMATION NO. 2878

POWER OF ATTORNEY NOTICE



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 10/03/2012

NOTICE REGARDING CHANGE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 09/25/2012.

- The Power of Attorney to you in this application has been revoked by the assignee who has intervened as provided by 37 CFR 3.71. Future correspondence will be mailed to the new address of record(37 CFR 1.33).

/tnnguyen/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
13/412,243	03/05/2012	Hyun Lee	NETL.045C1

CONFIRMATION NO. 2878

PUBLICATION NOTICE



OC000000056914824

79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Title:CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Publication No.US-2012-0250386-A1

Publication Date:10/04/2012

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Office of Public Records. The Office of Public Records can be reached by telephone at (703) 308-9726 or (800) 972-6382, by facsimile at (703) 305-8759, by mail addressed to the United States Patent and Trademark Office, Office of Public Records, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <http://pair.uspto.gov/>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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Alexandria, Virginia 22313-1450
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/412,243

03/05/2012

Hyun Lee

NETL.045C1

2878

79141

7590

01/02/2014

The Law Office of Jamie Zheng, Ph.D Esq.

P.O. Box 60573

Palo Alto, CA 94306

EXAMINER

BANSAL, GURTEJ

ART UNIT

PAPER NUMBER

2189

NOTIFICATION DATE

DELIVERY MODE

01/02/2014

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jz@jzpatent.com

Office Action Summary	Application No. 13/412,243	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2189	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 03/05/2012.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1 is/are pending in the application.
5a) Of the above claim(s) ____ is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 1 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☒ The drawing(s) filed on 03/05/2012 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- | | |
|--|---|
| 1) ☒ Notice of References Cited (PTO-892) | 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. ____. |
| 2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date <u>05/31/2012</u> . | 4) ☐ Other: ____. |

DETAILED ACTION

Remarks

The present application is being examined under the pre-AIA first to invent provisions.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the claims at issue are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); and *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on a nonstatutory double patenting ground provided the reference application or patent either is shown to be commonly owned with this application, or claims an invention made as a result of

activities undertaken within the scope of a joint research agreement. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO internet Web site contains terminal disclaimer forms which may be used. Please visit <http://www.uspto.gov/forms/>. The filing date of the application will determine what form should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to <http://www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp>.

"A later patent claim is not patentably distinct from an earlier patent claim if the later claim is obvious over, or anticipated by, the earlier claim. In re Longi, 759 F.2d at 896,225 USPQ at 651 (affirming a holding of obviousness-type double patenting because the claims at issue were obvious over claims in four prior art patents); In re Berq, 140 F.3d at 1437, 46 USPQ2d at 1233 (Fed. Cir. 1998) (affirming a holding of obviousness-type double patenting where a patent application claim to a genus is anticipated by a patent claim to a species within that genus). " ELI LILLY AND COMPANY v BARR LABORATORIES, INC., United States Court of Appeals for the Federal Circuit, ON PETITION FOR REHEARING EN BANC (DECIDED: May 30, 2001).

1. Claim 1 is rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1 of U.S. Patent No. 8,154,901 (hereafter '901). Although the

claims at issue are not identical, they are not patentably distinct from each other because claim 1 of '901 teaches all the limitations of claim 1 of the present application.

Claim Rejections - 35 USC § 103

The following is a quotation of pre-AIA 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains. Patentability shall not be negated by the manner in which the invention was made.

2. Claim 1 is rejected under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Haddock et al. (US 7,130,308) (hereafter Haddock), in view of Aekins (US 2007/0293094) (hereafter Aekins).

As per claim 1, Haddock teaches a memory module comprising:

an apparatus comprising at least one connector (fig. 2A, 111a/b, 112 a/b, 113 a/b, 114 a/b)) configured to be operatively coupled to a memory controller of a computer system (fig. 1, CTRL Memory 170);

a plurality of memory devices on the apparatus (fig. 2A, 130a~n);

a circuit (fig. 2a), comprising:

a first set of ports (fig. 2b ,121a~n) comprising a plurality of bi-directional ports (As illustrated in fig. 2b, with the arrow indicating that data can be received and transmitted) each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices (fig. 2b, 130a~n); and

a second set of ports (fig. 2a/b, the termination points that allow lines 111a/b; 112a/b; 113a/b and 114a/b to transmit/receive data) comprising one or more bi-directional ports (As illustrated in fig. 2a/b these ports are bi-directional), each port of the second set of ports operatively coupled to the at least one connector (As illustrated in fig. 2a/b these ports are bi-directional); and

a switching circuit (fig. 2a/b, 120) configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports (fig. 2a/b, 120 couples these ports).

Haddock does not explicitly teach wherein the apparatus is a printed circuit board and wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

However, Aekins teaches wherein the apparatus is a printed circuit board ([0016]) and wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports ([0018] and [0021] describes how the BEMI is used to reduce port to port noise).

It would have been obvious to a person of ordinary skill in the art at the time of the invention to have combined the printed circuit board and correction circuit of Aekins

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with the memory modules of Haddock because they limit the amount of noise that occurs from port to port coupling ([0021])).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to GURTEJ BANSAL whose telephone number is (571)270-5588. The examiner can normally be reached on Monday - Friday, 7:30 a.m. - 5:00 p.m., EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Reginald Bragdon can be reached on (571)272-4204. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

GURTEJ BANSAL
Examiner
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Examiner, Art Unit 2189

Notice of References Cited	Application/Control No. 13/412,243	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2189	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
*	A	US-7,130,308	10-2006	Haddock et al.	370/423
*	B	US-2007/0293094	12-2007	Aekins, Robert A.	439/676
	C	US-			
	D	US-			
	E	US-			
	F	US-			
	G	US-			
	H	US-			
	I	US-			
	J	US-			
	K	US-			
	L	US-			
	M	US-			

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	7	"5559970".pn. "6700823".pn. "6812869".pn. "6981089".pn. "7136958".pn. "7184916".pn. "20070013429".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 15:58
S2	496	bidirectional with port with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S3	89	bidirectional with port with (crossbar switch) same memory	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S4	366	bidirectional with ports with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:05
S5	48	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:15
S6	2546	bi\$1directional near2 ports	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:20
S7	666	bi\$1directional near2 ports same (crossbar switch\$3)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:40
S8	1	"7130308".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 17:14
S9	786	711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:03
S10	78	port with noise with reduc\$4 and PCB	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:16

EAST Search History (Interference)

<This search history is empty>

12/ 28/ 2013 10:28:44 PM

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	13/412,243
	Filing Date	March 05, 2012
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
SHEET 1 OF 4		Examiner Unknown
		Attorney Docket No. NETL.045C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
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	12	4,942,556	07-17-1990	Sasaki et al.	
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	14	5,138,619	08-11-1992	Fasang et al.	
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	16	5,222,066	06-22-1993	Gula et al.	
	17	5,241,503	08-31-1993	Cheng	
	18	5,304,856	04-19-1994	Rainal	
	19	5,337,254	08-09-1994	Knee et al.	
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	26	6,000,048	12-07-1999	Krishna et al.	
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	28	6,070,217	05-30-2000	Connolly et al.	
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Examiner Signature	/Gurtej Bansal/ (12/28/2013)	Date Considered	12/28/2013
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.			

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i> SHEET 2 OF 4	Application No.	13/412,243
	Filing Date	March 05, 2012
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
	Examiner	Unknown
	Attorney Docket No.	NETL.045C1

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
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	46	7,184,915	02-27-2007	Hansquine et al.	
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Examiner Signature	/Gurtej Bansal/ (12/28/2013)	Date Considered	12/28/2013
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	Art Unit	2827
SHEET 3 OF 4		Examiner Unknown
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	65	2006/0107156	05-18-2006	Lee et al.	
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Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

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(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 4 OF 4	Attorney Docket No.	NETL.045C1

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
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	81	BAYARD "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
	82	MUTOH et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, November 2-6, 2004, Volume 1, pages 963-968	
	83	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

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noise reduction port bidirectional



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Articles

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Since 2012

Since 2009

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Sort by date

☒ include patents☒ include citations

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Optical signal noise reduction for fiber optic gyroscopes

SXY Huang, N Sarma, KM Kilian... - US Patent 5,393,436, 1999 - Google Patents

... It would be desirable to **reduce** the random walk coefficient of **noise** in fiber optic gyroscopes without sacrificing ... 4 is a schematic block diagram of an alternative gyroscope with **reduced** relative intensity **noise** having an ... If desired, common mode **noise reduction** can be pro- vided ...

Cited by 13 Related articles All 2 versions Cite Save

Bi-directional balance low noise communication interface

RA Askins - US Patent 6,593,618, 2003 - Google Patents

... compensation method with the appropriate RJ45 modular housing, substantial cross-talk **noise reduction** is achieved up ... result 10 is an inter-matable device that provides lower NEXT **noises** at higher ... For an EIA T568B RJ45 plug, the worse case **noise** is typically between pairs 1 ...

Cited by 62 Related articles All 2 versions Cite Save

Settop terminal controlled return path filter for minimizing noise ingress on bidirectional cable systems

L Vince, G Goffin - US Patent 5,987,330, 1999 - Google Patents

... Therefore, this method of **reducing** ingress **noise** has diminishing benefits. ... A **bi-directional** CATV network including the ingress **noise reduction** device of the present invention is shown in ... These alterna- tive locations **reduce** the cost since there are fewer SRPF's 144 per tap **port** ...

Cited by 28 Related articles All 2 versions Cite Save

Rayleigh backscattering noise-eliminated 115-km long-reach bidirectional centralized WDM-PON with 10-Gb/s DPSK downstream and remodulated 2.5-Gb/s OCS- ...

A Chowdhury, HC Chien, MF Huang... - Photonics ... , 2008 - IEEE Xplore

... fiber for both DS and US trans- mission in a full-duplex fashion is essential to further **reduce** the cost ... before and after the remod- ulation at IM and the US signal after passing through the even **port** of the IL ... [12] CW Chow, G. Talli, and PD Townsend, "Rayleigh **noise reduction** in 10 ...

Cited by 25 Related articles All 3 versions Cite Save

Crosstalk noise reduction connector for telecommunication system

RA Askins - US Patent 5,618,185, 1997 - Google Patents

... The connector will **reduce** crosstalk **noise** regardless of whether the induced crosstalk results from magnetic ... To achieve crosstalk **reduction**, it is important to know where the greatest source of ... signal, ie, E1 to E2, to an adjacent transmission line whose **noise** originated from ...

Cited by 79 Related articles All 2 versions Cite Save

Challenges and recent developments in hearing aids part I. Speech understanding in noise, microphone technologies and noise reduction algorithms

K Chung - Trends in Amplification, 2004 - tia.sagepub.com

... One of the most challenging tasks of engi- neers who design hearing aids is to **reduce** back- ground **noise** ... **Noise Reduction** Strategy No. ... order directional microphones and array microphones with higher directional effects are available to further attenuate **noise** originating from ...

Cited by 106 Related articles All 5 versions Cite Save

(PDF) from psu.edu

Bidirectional power converter modules and power system using paralleled modules

FA Linkowsky, JD Singley - US Patent 5,602,464, 1997 - Google Patents

... of each module tends to attenu- ate the current components, but sufficient integration to remove sufficient **noise** also tends to **reduce** the bandwidth ... This bandwidth **reduction** is disadvantageous, because it tends to bring the bandwidth of the local feedback loop toward the ...

Cited by 37 Related articles All 2 versions Cite Save

Bi-directional balance low noise communication interface

RA Askins - US Patent 6,340,816, 2005 - Google Patents

... stage compensation method with the appropriate RJ45 modular housing, substantial cross-talk **noise reduction** is achieved ... B2 5 is an inter-matable device that provides loWer NEXT **noises** at higher ... For an EIA T568B RJ45 plug, the Worse case **noise** is typically betWeen pairs ...

Cited by 30 Related articles All 4 versions Cite Save

Suppression of coherent rayleigh noise in bidirectional communication systems

L Boivin, RD Feldman, MC Nuss - US Patent 6,197,611, 2000 - Google Patents

... Rayleigh **noise** fall outside the bandwidth of the detector, so as to **reduce** coherent Rayleigh ... The substantial **reduction** in coherent Rayleigh **noise** (CRN) occurs because the broad band optical power source ... The CRN is related to so that the signal-to-**noise** ratio increases as the ...

Cited by 13 Related articles All 2 versions Cite Save

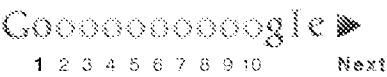
A 900 Mb/s bidirectional signaling scheme

R Mooney, C Dike, S Borkar - Solid-State Circuits, IEEE ... , 1995 - IEEE Xplore

... this relationship is quite linear, a 10% difference in supply voltage results in a 10% **reduction**

in link ... This spreads the middle ternary level, subtracting directly from the **noise** margin of the link. This effect can be **reduced** substantially by matching the driver to an external resistor. ...
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Search Notes 	Application/Control No. 13412243	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2189

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	149	12/28/2013	GB

SEARCH NOTES		
Search Notes	Date	Examiner
Inventor Search in eDAN	12/28/2013	GB
Text Search in EAST (See Attached)	12/28/2013	GB
NPL Search in Google Scholar (See Attached)	12/28/2013	GB

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

/GURTEJ BANSAL/ Examiner.Art Unit 2189	
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<i>Index of Claims</i> 	Application/Control No. 13412243	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2189

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	12/28/2013							
	1	✓							



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BIB DATA SHEET

CONFIRMATION NO. 2878

SERIAL NUMBER 13/412,243	FILING or 371(c) DATE 03/05/2012 RULE	CLASS 711	GROUP ART UNIT 2189	ATTORNEY DOCKET NO. NETL.045C1
APPLICANTS INVENTORS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Jeffrey C. Solomon, Irvine, CA; Mario Jesus Martinez, Laguna Niguel, CA; Chi-She Chen, Walnut, CA;				
** CONTINUING DATA ***** This application is a CON of 12/422,853 04/13/2009 PAT 8154901 * which claims benefit of 61/044,839 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,801 04/14/2008 (*)Data provided by applicant is not consistent with PTO records.				
** FOREIGN APPLICATIONS *****				
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 04/23/2012				
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /GURTEJ BANSAL/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 9	TOTAL CLAIMS 1
INDEPENDENT CLAIMS 1				
ADDRESS The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306 UNITED STATES				
TITLE CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION				
FILING FEE RECEIVED 1380	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit	

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant : Lee, et al.
App. No. : 13/412,243
Filed : March 5, 2012
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Examiner : BANSAL, GURTEJ
Art Unit : 2189
Conf. No. : 2878
Docket No. : NETL.045C1

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May 2, 2014

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/Jamie J. Zheng/

Jamie J. Zheng, Reg. No. 51167

AMENDMENT RESPONSE TO OFFICE ACTION DATED JANUARY 2, 2014**Mail Stop Amendment**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

In response to the Office Action dated January 2, 2014, Applicant hereby submits the following amendment and remarks.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 50-5963.

Amendments to the Claims: Begin on page 2 of this paper.

Remarks: Begin on page 7 of this paper.

AMENDMENTS TO THE CLAIMS

A complete listing of all claims is presented below with insertions underlined (e.g., insertion), and deletions struckthrough or in double brackets (e.g., ~~deletion~~ or [[deletion]]):

1. (Currently Amended) A memory module operable to communicate signals with a system memory controller, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the system ~~a memory controller of a computer system~~;

a plurality of memory devices on the printed circuit board;

a circuit, comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

a switching circuit configured to selectively and operatively couple one or more ports of the second set of ports to one or more selected ports of the first set of ports, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, wherein the switching circuit is further configured to reshape signals between the system memory controller and the one or more first memory devices so as to ameliorate signal deterioration due to transmission over lossy transmission lines having discontinuity regions between the system memory controller and the one or more first memory devices ~~wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.~~

2. (New) The memory module of claim 1, wherein the switching circuit is further configured to and to selectively isolate one or more un-driven ports of the first set of ports from the second set of ports provide a known static value to the un-driven ports, thereby reducing dynamic power consumption associated with floating or toggling electrical nodes.

3. (New) The memory module of claim 1, wherein the switching circuit includes at least one router circuit and at least one signal correction circuit for each of a plurality of signal paths between the system memory controller and the plurality of memory devices, the at least one signal correction circuit including a repeater/amplifier element that can be enabled to generate a respective output signal for a respective input signal, the respective output signal having an earlier transition trigger point and a faster slew rate than the respective input signal.
4. (New) The memory module of claim 3, wherein the at least one router circuit can be enabled to operatively couple a corresponding one of the first set of ports to a corresponding one of the second set of ports or disabled to isolate the corresponding one of the first set of ports from the corresponding one of the second set of ports.
5. (New) The memory module of claim 3, wherein the at least one signal correction circuit further includes a filter, a programmable impedance matching circuit and a self-adjusting damper circuit.
6. (New) The memory module of claim 5, wherein the programmable impedance matching circuit includes a network of programmable resistors that are dynamically controllable to match a characteristic impedance of a corresponding signal path.
7. (New) The memory module of claim 5, wherein the self-adjusting damper circuit includes an input terminal adapted to receive an input signal comprising an input noise component and an input signal component, a first sub-circuit adapted to generate a first signal responsive to at least a portion of the noise component, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of at least a portion of the input signal, the self-adjusting damper circuit combining the first signal with the second signal to generate an output signal.
8. (New) The memory module of claim 1, wherein the switching circuit comprises a plurality of circuits each corresponding to a subset of the plurality of devices and each having a data width that is a fraction of a data width of the memory module, the fraction being selected to optimize interconnection characteristics selected from the group consisting of timing, power consumption and signal integrity between the switching circuit and the plurality of memory devices.

9. (New) The memory module of claim 1, further comprising a local memory controller to control the selective and operative coupling and the selective isolating in the switching circuit based on control signals from the system memory controller.

10. (New) The memory module of claim 1, wherein the switching circuit establishes a one-to-one signal path for transferring data from each port of the second set of ports to a selected corresponding port of the first set of ports.

11. (New) The memory module of claim 1, wherein the switching circuit establishes a one-to-many connection for broadcasting data from each port of the second set of ports to selected corresponding ports of the first set of ports.

12. (New) A method of operating a memory module to communicate signals with a system memory controller via a plurality of signal transmission lines, the memory module including a plurality of memory devices on a module board, the module board including at least one connector configured to be operatively coupled to the plurality of signal transmission lines, the method comprising:

selectively and operatively coupling one or more ports of a second set of ports to one or more selected ports of a first set of ports, the first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, the second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

reshaping signals between the memory controller and the one or more first memory devices so as to ameliorate signal deterioration due to transmission over lossy transmission lines having discontinuity regions between the system memory controller and the one or more first memory devices.

13. (New) The method of claim 12, further comprising:

selectively isolating one or more un-driven ports of the first set of ports from the second set of ports, and the one or more un-driven ports being operatively coupled to one or more second memory devices that are not being accessed by the system memory controller; and

providing a known static value to the un-driven ports, thereby reducing dynamic power

consumption associated with floating or toggling electrical nodes.

14. (New) The method of claim 12, wherein reshaping the input signals to generate reshaped signals comprises generating a respective output signal for a respective input signal, the respective output signal having an earlier transition trigger point and a faster slew rate than the respective input signal.

15. (New) The method of claim 12, further comprising dynamically controlling a network of programmable resistors associated with each active signal path between the system memory controller and the plurality of memory devices so as to match a characteristic impedance of the each active signal path.

16. (New) The method of claim 12, wherein each input signal comprises an input noise component and an input signal component, the method further comprising, for each input signal:
generating a first signal responsive to at least a portion of the noise component and a second signal corresponding to a delayed version of at least a portion of the input signal; and
combining the first signal with the second signal to generate an output signal.

17. (New) The method of claim 12, further comprising controlling the selective and operative coupling and the selective isolating in the switching circuit based on control signals from the system memory controller.

18. (New) A memory module operable to communicate signals with a system memory controller via a plurality of signal transmission lines, comprising:
a printed circuit board comprising at least one connector configured to be operatively coupled to the plurality of signal transmission lines;
a plurality of memory devices on the printed circuit board;
a circuit, comprising:
a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices;
a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and
a switching circuit configured to selectively and operatively couple one or more ports of the second set of ports to one or more selected ports of the first set of ports and to selectively

isolate one or more un-driven ports of the first set of ports from the second set of ports, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, and the one or more un-driven ports being operatively coupled to one or more second memory devices that are not being accessed by the system memory controller, wherein the switching circuit is further configured to provide a known static value to the un-driven ports, thereby reducing dynamic power consumption associated with floating or toggling electrical nodes.

19. (New) The memory module of claim 18, wherein the switching circuit includes at least one router circuit and at least one signal correction circuit for each of a plurality of signal paths between the system memory controller and the plurality of memory devices, the at least one signal correction circuit including a repeater/amplifier element that can be enabled to generate a respective output signal for a respective input signal, the respective output signal having an earlier transition trigger point and a faster slew rate than the respective input signal.

20. (New) The memory module of claim 18, wherein the switching circuit includes at least one programmable impedance matching circuit for each of a plurality of signal paths between the system memory controller and the plurality of memory devices, the programmable impedance matching circuit having a network of programmable resistors that are dynamically controllable to match a characteristic impedance of a corresponding signal path.

REMARKS

This amendment responds to the non-final Office Action dated January 2, 2014, in which the Examiner:

- rejected claims 1 on the ground of nonstatutory double patenting as being unpatentable over claim 1 of U.S. Patent No. 8,154,901 (hereafter “the ‘901 patent”); and
- rejected claim 1 under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Haddock et al. (US 7,130,308) (hereafter Haddock), in view of Aekins (US 2007/0293094) (hereafter Aekins).

I. MCKESSON STATEMENT

This application is related to U.S. Pat. Appl. No. 12/422,912, filed on April 13, 2009, now abandoned, U.S. Pat. Appl. No. 12/422,925, filed on April 13, 2009, now U.S. Pat. No. 8,001,434, U.S. Pat. Appl. No. 13/183,253, filed July 14, 2011, now U.S. Pat. No. 8,359,501, U.S. Pat. Appl. No. 13/745,790, filed January 19, 2013, now U.S. Pat. No. 8,689,064, U.S. Pat. Appl. No. 14/299,844, filed March 29, 2014, U.S. Pat. Appl. No. 12/504,131, filed July 16, 2009, now U.S. Pat. No. 8,417,870, U.S. Pat. Appl. No. 12/761,179, filed April 15, 2010, now U.S. Pat. No. 8,516,185, U.S. Pat. Appl. No. 13/970,606, filed August 20, 2013, U.S. Pat. Appl. No. 13/287,042, filed November 1, 2011, U.S. Pat. Appl. No. 13/971,231, filed August 20, 2013, and U.S. Pat. Appl. No. 13/288,850, filed November 3, 2011. The Examiner is encouraged to review the art made of record, Office Actions, responses to Office Actions, and the Notice of Allowance in the above-mentioned related applications, all of which are available on PAIR.

II. CLAIM AMENDMENT

Claim 1 is amended and new claims 2-20 are presented. No new matter is added.

III. REJECTION OF CLAIMS 1 ON THE GROUND OF NONSTATUTORY DOUBLE PATENTING AS BEING UNPATENTABLE OVER CLAIM 1 OF THE ‘901 PATENT

Claim 1 as amended is now patentably distinct from claim 1 or any other claim of the ‘901 patent. The rejection thereto should therefore be withdrawn.

IV. REJECTION OF CLAIM 1 UNDER PRE-AIA 35 U.S.C. 103(A) AS BEING UNPATENTABLE OVER HADDOCK IN VIEW OF AEKINS

Claim 1 as amended recites:

A memory module operable to communicate signals with a system memory controller, comprising:
a printed circuit board comprising at least one connector configured to be operatively coupled to the system memory controller;
a plurality of memory devices on the printed circuit board;
a circuit, comprising:
a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and
a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector;
a switching circuit configured to selectively and operatively couple one or more ports of the second set of ports to one or more selected ports of the first set of ports, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, wherein the switching circuit is further configured to reshape signals between the system memory controller and the one or more first memory devices so as to ameliorate signal deterioration due to transmission over lossy transmission lines having discontinuity regions between the system memory controller and the one or more first memory devices.

Thus, claim 1 includes the limitations of a switching circuit that is capable of selectively and operatively couple one or more first memory devices that are being accessed by the system memory controller to the memory controller while reshaping signals between the system memory controller and the one or more first memory devices so as to ameliorate signal deterioration due to transmission over lossy transmission lines having discontinuity regions between the system memory controller and the one or more first memory devices. Neither Haddock nor Aekins teaches or discloses such a switching circuit on a memory module and coupled between a system memory controller and a plurality of memory devices on the memory module. Therefore, claim 1 as amended is patentable over Haddock in view of Aekins.

V. NEW CLAIMS 2-20

New claims 2-11 depend from claim 1 as amended and include further limitations in addition to the limitations in claim 1 as amended. Therefore, new claims 2-11 are patentable for at least the same reasons claim 1 as amended is patentable.

The argument regarding claim 1 as amended applies to new claims 12-17. Therefore, new claims 12-17 are also patentable.

New claim 18 recites:

A memory module operable to communicate signals with a system memory controller via a plurality of signal transmission lines, comprising:
a printed circuit board comprising at least one connector configured to be operatively coupled to the plurality of signal transmission lines;
a plurality of memory devices on the printed circuit board;
a circuit, comprising:
a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices;
a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and
a switching circuit configured to selectively and operatively couple one or more ports of the second set of ports to one or more selected ports of the first set of ports and to selectively isolate one or more un-driven ports of the first set of ports from the second set of ports, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, and the one or more un-driven ports being operatively coupled to one or more second memory devices that are not being accessed by the system memory controller, wherein the switching circuit is further configured to provide a known static value to the un-driven ports, thereby reducing dynamic power consumption associated with floating or toggling electrical nodes.

Since neither Haddock nor Aekins teaches or discloses such a switching circuit in new claim 18, new claim 18 is patentable over Haddock in view Aekins.

New claims 19-20 depend from new claim 18 and include further limitations in addition to the limitations in new claim 18. Therefore, new claims 19-20 are patentable for at least the same reasons new claim 18 is patentable.

The Examiner is invited to call the undersigned if such phone call would help resolve any remaining issues.

Respectfully submitted,

Date: May 2, 2014

By: /Jamie J. Zheng/
Jamie J. Zheng
Registration No. 51,167
Customer No. 79,141
(650) 273-6008

ASSERTION OF ENTITLEMENT TO SMALL ENTITY STATUS UNDER 37 CFR 1.27

Applicant	: Lee, et al.
App. No.	: 13/412,243
Filed	: March 5, 2012
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: BANSAL, GURTEJ
Art Unit	: 2189
Conf. No.	: 2878
Docket No.	: NETL.045C1

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

The Assignee of all rights and title in the above-identified patent, Netlist, Inc., is a small entity concern as defined in 37 C.F.R. 1.27(a). Assertion of entitlement to small entity status under 37 C.F.R. 1.27(c) is hereby made in this application.

Applicants acknowledge the duty to file notification of a loss of entitlement to small entity status in the patent prior to paying, or at the time of paying, any maintenance fee due after the date on which status as a small entity is no longer appropriate.

Respectfully submitted,

Date: May 2, 2014

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51167
(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	13412243			
Filing Date:	05-Mar-2012			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.045C1			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 1 month with \$0 paid	2251	1	100	100

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				100

Electronic Acknowledgement Receipt

EFS ID:	18934451
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C1
Receipt Date:	02-MAY-2014
Filing Date:	05-MAR-2012
Time Stamp:	23:52:01
Application Type:	Utility under 35 USC 111(a)

Payment information:

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Payment Type	Credit Card
Payment was successfully received in RAM	\$ 100
RAM confirmation Number	7201
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1		2014-05-02_Amendment_NETL045C1.pdf	131239 5b3fb8095ea1e7c07a2e35c53322c074d52c6929	yes	9
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Req. Reconsideration-After Non-Final Reject		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	9	
Warnings:					
Information:					
2	Assertion of entitlement to small entity status	2014-05-02_Small_Entity_Status_NETL045C1.pdf	58297 60aa3bc7d0a86da72f883ffc249ed95f47d277ab	no	1
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30012 33ab4c94eb1b789dd8c82a52222ee18d513b1655	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			219548		
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 13/412,243	Filing Date 03/05/2012	☐ To be Mailed
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 ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED – PART I

(Column 1)

(Column 2)

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	05/02/2014	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	X \$80 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	X \$420 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0

(Column 1)

(Column 2)

(Column 3)

AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	MINUS	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
						TOTAL ADD'L FEE

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

 LIE
 /JASON EADDY/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	1	of	11	Attorney Docket Number	NETL.045C1

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
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	3.	4,571,676	2/18/1986	Mantellina et al.	
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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
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	72.	Reexam Appl. No. 95/000,577 for U.S. Pat. No. 7,289,386	10/20/2010	Netlist, Inc.	Entire File History
	73.	Reexam Appl. No. 95/000,578 for U.S. Pat. No. 7,619,912	10/20/2010	Netlist, Inc.	Entire File History
	74.	Reexam Appl. No. 95/000,579 for U.S. Pat. No. 7,619,912	10/21/2010	Netlist, Inc.	Entire File History
	75.	Reexam App. No. 95/001,337 for U.S. Pat. No. 7,636,274	06/04/2010	Netlist, Inc.	Entire File History
	76.	Reexam Appl. No. 95/001,338; for U.S. Pat. No. 7,532,537	04/19/2010	Netlist, Inc.	Entire File History
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	81.	Reexam Appl. No. 95/002,399 for U.S. Pat. No. 8,250,295	09/15/2010	Smart Modular Technologies, Inc.	Entire File History
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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C1
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US PATENT APPLICATIONS					
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	96.	12/629,827	12/2/2009	Netlist, Inc.	Entire Prosecution History
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	103.	13/032,470	2/22/2011	Netlist, Inc.	Entire Prosecution History
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	108.	13/288,850	11/3/2011	Netlist, Inc.	Entire Prosecution History
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	110.	13/412,243	3/5/2012	Netlist, Inc.	Entire Prosecution History
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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C1
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	165.	Letter from G. Hopkins Guy III, Orrick, Herrington & Sutcliffe LLP, to R. Scott Oliver, Morrison & Foerster, (Apr. 14, 2009).	
	166.	Luthra et al. "Interface Synthesis Using Memory Mapping for an FPGA Platform," Proceedings of the 21st International Conference on Computer Design, 2003.	
	167.	MetaRAM, Inc. v. Netlist, Inc. No. 3:09-cv-01309-VRW, MetaRam's Reply to Netlist's Counterclaims, (N.D. Ca. Filed Jun. 3, 2009).	
	168.	MetaRam, Inc. v. Netlist, Inc., No. 3:09-cv-01309-VRW, Netlist's Answer to Complaint and Counterclaims, (N.D. Ca, filed May 11, 2009). .	
	169.	MetaRAM, Inc. v. Netlist, Inc., No. C09 01309, Complaint for Patent Infringement, (N.D. Ca. Filed Mar. 25, 2009).	
	170.	Micron "DDR2 SDRAM Registered DIMM (RDIMM)," 2003 Micron Technology, Inc. 18 pages.	
	171.	Micron "Synchronous DRAM Module MT18LSDT472," 1998, Micron Technology, Inc., 17 pages.	
	172.	Micron Technical Note,"Decoupling Capacitor Calculations for a DDR Memory Channel," 2004, 3 pages.	
	173.	Miles J. Murdocca et al., "Principles of Computer Architecture", Prentice Hall, Inc., Upper Saddle River, NJ, 2000, pp. 243-252.	
	174.	Murdocca et al., "Principles of Computer Architecture," Prentice Hall, 2000, pp. 249-251.	
	175.	Netlist, Inc. v. MetaRam, Inc., No. 09-165-GMS, MetaRAM, Inc.'s Answer and Affirmative Defenses to Plaintiff's Complaint, dated Apr. 20, 2009.	
	176.	Netlist, Inc. v. MetaRAM, Inc., No. 1:09-ccv-00165-GMS, Complaint for Patent Infringement, (D. Del. Filed Mar. 12, 2009).	
	177.	Non-Final Action Closing Prosecution mailed September 1, 2010, for Control No. 95/001,339, filed April 10, 2010, 17 pages.	

EXAMINER

DATE CONSIDERED

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C1
Sheet	8	of	11		

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²
	178.	Non-Final Action Closing Prosecution mailed June 21, 2011, for Control No. 95/001,381, filed June 9, 2010, 34 pages.	
	179.	Non-Final Action Closing Prosecution mailed March 12, 2012, for Control No. 95/001,337, filed April 19, 2010, 33 pages.	
	180.	Non-Final Action mailed August 27, 2010, for Control No. 95/000,546, filed May 11, 2010, 16 pages.	
	181.	Non-Final Action mailed September 8, 2010, for Control No. 95/001,381, filed June 9, 2010, 17 pages.	
	182.	Non-Final Action mailed April 4, 2011, for Control No. 95/001,339, filed April 20, 2010, 61 pages. (merged with 95/000,578 and 95/000,579)	
	183.	Non-Final Action mailed June 15, 2011, for Control No. 95/001,381, filed June 9, 2010, 33 pages.	
	184.	Non-Final Action mailed September 27, 2011, for Control No. 95/001,337, filed April 19, 2010, 19 pages.	
	185.	Non-Final Action mailed October 4, 2011, for Control No. 95/001,339, filed April 20, 2010, 77 pages. (merged with 95/000,578 and 95/000,579)	
	186.	Non-Final Action mailed October 14, 2011, for Control No. 95/001,339, filed April 30, 2010, 99 pages. (merged with 95/000,578 and 95/000,579)	
	187.	Non-Final Office Action mailed November 16, 2011, for Application No. 95/001,758 filed September 14, 2011, 25 pages.	
	188.	Order Granting Request for Inter Partes Reexamination mailed November 16, 2011, for Application No. 95/001,758, filed September 14, 2011, 13 pages.	
	189.	Order Granting Request for Inter Partes Reexamination mailed August 9, 2010, for Control No. 95/000,546, filed May 11, 2010, 22 pages.	
	190.	Order Granting Request for Inter Partes Reexamination mailed August 27, 2010, for Control No. 95/001,337, filed April 19, 2010, 21 pages.	
	191.	Order Granting Request for Inter Partes Reexamination mailed September 1, 2010, for Control No. 95/001,339, filed April 20, 2010, 14 pages.	
	192.	Order Granting Request for Inter Partes Reexamination mailed September 8, 2010, for Control No. 95/000,381, filed June 9, 2010, 21 pages.	
	193.	Order Granting Request for Inter Partes Reexamination mailed January 14, 2011, for Control No. 95/000,579, filed Oct. 21, 2010, 12 pages.	
	194.	Order Granting Request for Inter Partes Reexamination mailed January 18, 2011, for Control No. 95/000,577, filed Oct. 20, 2010, 17 pages.	
	195.	Right of appeal Notice mailed Feb. 7, 2012, for Control No. 95/001,381, 33 pages.	
	196.	Right of Appeal Notice mailed June 22, 2012, for Control No. 95/001,337, filed June 4 2010, 34 pages.	

EXAMINER

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				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C1
Sheet	9	of	11		

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
	197.	PC133 SDRAM Registered DIMM Design Specification, Revision 1.1, Aug. 1999, 62 pages.		
	198.	Texas Instruments, "TM2SR72EPN 2097152 by 72-Bit, TM4SRT2EPN 4194304 by 72-Bit, Synchronous Dynamic RAM Modules," 1997, 15 pages.		
	199.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. Inphi Corporation, Complaint For Patent Infringement, filed Sept. 22, 2009 in 10 pages.		
	200.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Defendant Inphi Corporation's Answer to Plaintiff's Complaint for Patent Infringement, filed Nov. 12, 2009 in 6 pages.		
	201.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Defendant Inphi Corporation's Answer to Plaintiff's First Amended Complaint for Patent Infringement, filed Feb. 11, 2010 in 9 pages.		
	202.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Defendant Inphi Corporation's Notice of Motion and Motion for Stay Pending Reexaminations and Interference Proceeding Regarding the Patents-In-Suit; Memorandum of Points and Authorities in Support Thereof, filed Apr. 21, 2010 in 28 pages.		
	203.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Plaintiff Netlist Inc's Opposition to Defendant Inphi Corporation's Motion for Stay Pending Reexaminations and Interference Proceedings Regarding the Patents-In-Suit, filed May 3, 2010 in 23 pages.		
	204.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. Inphi Corporation, Plaintiff Netlist, Inc's First Amended Complaint For Patent Infringement, filed Dec. 23, 2009 in 8 pages.		
	205.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Defendant Netlist, Inc.'s Claim Construction Reply Brief, filed Sept. 22, 2009 in 19 pages.		
	206.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Defendant Netlist, Inc.'s Opening Claim Construction Brief, filed July 28, 2009 in 21 pages.		
	207.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Defendant Netlist, Inc.'s Opposition to Google Inc's Motion for Summary Judgment of Invalidity, filed July 6, 2010 in 13 pages.		
	208.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Exhibit A To Joint Claim Construction And Prehearing Statement, filed June 12, 2009 in 2 pages.		
	209.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Exhibit B to Joint Claim Construction and Prehearing Statement, filed June 12, 2009 in 36 pages.		

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				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C1
Sheet	10	of	11		

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
	210.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Joint Claim Construction and Prehearing Statement, filed June 12, 2009 in 5 pages.		
	211.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Netlist, Inc.'s Answer to Complaint and Counterclaims, filed Nov. 18, 2008 in 9 pages.		
	212.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Order Re Claim Construction, filed Nov. 16, 2009 in 1 page.		
	213.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Plaintiff Google's Reply to Counterclaims, filed Dec. 8, 2008 in 4 pages.		
	214.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Stipulation Re: Additional Agreed-Upon Claim Constructions, filed Oct. 28, 2009 in 3 pages.		
	215.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., [Redacted] Google Inc.'s Responsive Claim Construction Brief, filed Aug. 25, 2009 in 30 pages.		
	216.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Amended Exhibit A To Joint Claim Construction And Prehearing Statement, filed Oct. 28, 2009 in 1 page.		
	217.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Appendix 1 To Google's Responsive Claim Construction Brief, filed Nov. 12, 2009 in 4 pages.		
	218.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Attachment 1 to Exhibit B To Joint Claim Construction And Prehearing Statement, filed June 12, 2009 in 7 pages.		
	219.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Attachment 2 To Exhibit B To Joint Claim Construction And Prehearing Statement, filed June 12, 2009 in 12 pages.		
	220.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Complaint for Declaratory Relief, filed Aug. 29, 2008 in 49 pages.		
	221.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Complaint For Patent Infringement, filed Dec. 4, 2009 in 47 pages.		
	222.	U.S. District Court Northern District of California, Case No. CV09 05718, NETLIST, INC. vs. GOOGLE, INC., Defendant Google Inc's Responsive Claim Construction Brief, filed Aug. 4, 2010 in 27 pages.		

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				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
				Attorney Docket Number	NETL.045C1
Sheet	11	of	11		

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	223.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Exhibit A to Joint Claim Construction and Prehearing Statement under Patent L.R. 4-3, filed June 25, 2010 in 2 pages.	
	224.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Exhibit B to Joint Claim Construction and Prehearing Statement under Patent L.R. 4-3, filed Jun. 25, 2010 in 23 pages.	
	225.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Google's Answer to Plaintiff's Complaint for Patent Infringement; and Assertion of Counterclaims, filed Feb. 12, 2010 in 13 pages.	
	226.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Joint Claim Construction And Prehearing Statement Under Patent Local Rule 4-3, filed June 25, 2010 in 5 pages.	
	227.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Plaintiff Netlist, Inc.'s Reply Claim Construction Brief, filed Aug. 16, 2010 in 17 pages.	
	228.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Plaintiff Netlist, Inc.'s Reply to Defendant Google Inc.'s Counterclaim, filed March 8, 2010 in 11 pages.	
	229.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Plaintiff Netlist, Inc.'s Opening Claim Construction Brief, filed Jul. 16, 2010 in 29 pages.	
	230.	US District Court Civil Docket; Google Inc. v. Netlist Inc.; 4:08cv04144; filed Aug. 29, 2008.	
	231.	US District Court Civil Docket; Netlist Inc. v. Google Inc.; 4:09cv5718, filed Dec. 4, 2009 in 10 pages.	
	232.	US District Court Civil Docket; Netlist Inc. v. Inphi Corporation; 2:09cv6900; Date filed Sep. 22, 2009.	
	233.	Vogt, Pete, "Fully Buffered DIMM (FB-DIMM) Server Memory Architecture: Capacity, Performance, Reliability, and Longevity," Intel, Feb. 18, 2004, 33 pages.	

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Electronic Patent Application Fee Transmittal

Application Number:	13412243			
Filing Date:	05-Mar-2012			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.045C1			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Submission- Information Disclosure Stmt	2806	1	90	90
Total in USD (\$)				90

Electronic Acknowledgement Receipt

EFS ID:	18983833
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C1
Receipt Date:	08-MAY-2014
Filing Date:	05-MAR-2012
Time Stamp:	19:19:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$90
RAM confirmation Number	4799
Deposit Account	
Authorized User	

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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1	Transmittal Letter	2014-05_08_IDS_Cover_NETL045C1.pdf	75244 588d1cef8607de213ef366a0f9c5f8671a210d0d	no	2
Warnings:					
Information:					
2	Information Disclosure Statement (IDS) Form (SB08)	2014-05-08_IDS_NETL045C1.pdf	319737 7abb231e2c2c83eedd7221c04ea0bdcf51bde3be	no	11
Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
3	Fee Worksheet (SB06)	fee-info.pdf	30329 9745d40317566960e60e17ecf896196449ea41b9	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			425310		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					

INFORMATION DISCLOSURE STATEMENT

Applicant	: Lee, et al.
App. No.	: 13/412,243
Filed	: March 5, 2012
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: BANSAL, GURTEJ
Art Unit	: 2189
Conf. No.	: 2878
Docket No.	: NETL.045C1

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a Form PTO/SB/08a/b Equivalent listing 233 references. Copies of foreign documents and/or their translation and non-patent literature (NPL) are submitted together with information disclosure statements (IDS) filed in related patent applications, which are accessible by PAIR. The table below serves as a guide in locating the NPL's.

The applicants would appreciate the Examiner initialing and returning the Form PTO/SB/08a/b Equivalent indicating that the information has been considered and made of record herein.

The information contained in this Information Disclosure Statement is not to be construed as a representation that: (i) a complete search has been made; (ii) additional information material to this application does not exist; (iii) the information, protocols, results and the like reported by third parties are accurate or enabling; or (iv) the above information constitutes prior art to the subject invention.

Application No.: 13/412,243
Filing Date: March 5, 2012

	The Present Application	Related Applications				
Appl. No.	13/412,243	12/504,131	12/761,179	13/970,606	13/970,606	13/287,042
IDS Date	5/8/2014	1/17/2013	6/27/2013	4/22/2014	4/23/2014	4/7/2014
Cite No.	117		8			
	118-119			30-31		
	120-125				65-70	
	126-146			32-52		
	147-149	4-6				
	150					52
	151					65
	152		9			
	153-158					2-7
	159-233				71-145	

The Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to
Deposit Account No. 50-5963.

Respectfully submitted,

Dated: May 8, 2014

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141
(650) 273-6008

51167
(Reg. No.)



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Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 05/29/2014
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

EXAMINER

BANSAL, GURTEJ

ART UNIT

PAPER NUMBER

2189

DATE MAILED: 05/29/2014

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/412,243

03/05/2012

Hyun Lee

NETL.045C1

2878

TITLE OF INVENTION: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	08/29/2014

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 05/29/2014
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/412,243	03/05/2012	Hyun Lee	NETL.045C1	2878

TITLE OF INVENTION: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	SMALL	\$480	\$0	\$0	\$480	08/29/2014

EXAMINER	ART UNIT	CLASS-SUBCLASS
BANSAL, GURTEJ	2189	711-149000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363). ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached. ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. Use of a Customer Number is required.	2. For printing on the patent front page, list (1) The names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____ (2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____ 3 _____
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3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted: ☐ Issue Fee ☐ Publication Fee (No small entity discount permitted) ☐ Advance Order - # of Copies _____	4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above) ☐ A check is enclosed. ☐ Payment by credit card. Form PTO-2038 is attached. ☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credits any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).
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5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/412,243	03/05/2012	Hyun Lee	NETL.045C1	2878
79141	7590	05/29/2014	EXAMINER	
The Law Office of Jamie Zheng, Ph.D Esq. P.O. Box 60573 Palo Alto, CA 94306			BANSAL, GURTEJ	
			ART UNIT	PAPER NUMBER
			2189	
DATE MAILED: 05/29/2014				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.** Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.

Notice of Allowability	Application No. 13/412,243	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2189	AIA (First Inventor to File) Status No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to Examiner Initiated Interview and Examiner's Amendment dated May 23, 2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has **THREE MONTHS FROM THE "MAILING DATE"** of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|--|---|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☒ Examiner's Amendment/Comment |
| 2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date <u>05/08/2014</u> | 6. ☐ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 7. ☐ Other _____. |
| 4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date <u>20140523</u> . | |

/GURTEJ BANSAL/
Primary Examiner, Art Unit 2189

EXAMINER'S AMENDMENT

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given in a telephone interview with JAMIE ZHENG (Reg. No. 51,167) on May 23, 2014.

The application has been amended as follows:

1. (Previously Presented) A memory module operable to communicate signals with a system memory controller, comprising:

- a printed circuit board comprising at least one connector configured to be operatively coupled to the system memory controller;

- a plurality of memory devices on the printed circuit board;

- a circuit, comprising:

- a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and

- a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

- a switching circuit configured to selectively and operatively couple one or more

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ports of the second set of ports to one or more selected ports of the first set of ports, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, wherein the switching circuit is further configured to reshape signals between the system memory controller and the one or more first memory devices so as to ameliorate signal deterioration due to transmission over lossy transmission lines having discontinuity regions between the system memory controller and the one or more first memory devices.

2. (Previously Presented) The memory module of claim 1, wherein the switching circuit is further configured to and to selectively isolate one or more un-driven ports of the first set of ports from the second set of ports provide a known static value to the un-driven ports, thereby reducing dynamic power consumption associated with floating or toggling electrical nodes.

3. (Previously Presented) The memory module of claim 1, wherein the switching circuit includes at least one router circuit and at least one signal correction circuit for each of a plurality of signal paths between the system memory controller and the plurality of memory devices, the at least one signal correction circuit including a repeater/amplifier element that can be enabled to generate a respective output signal for a respective input signal, the respective output signal having an earlier transition trigger point and a faster slew rate than the respective input signal.

4. (Previously Presented) The memory module of claim 3, wherein the at least one router circuit can be enabled to operatively couple a corresponding one of the first set of ports to a corresponding one of the second set of ports or disabled to isolate the corresponding one of the first set of ports from the corresponding one of the second set of ports.
5. (Previously Presented) The memory module of claim 3, wherein the at least one signal correction circuit further includes a filter, a programmable impedance matching circuit and a self-adjusting damper circuit.
6. (Previously Presented) The memory module of claim 5, wherein the programmable impedance matching circuit includes a network of programmable resistors that are dynamically controllable to match a characteristic impedance of a corresponding signal path.
7. (Previously Presented) The memory module of claim 5, wherein the self-adjusting damper circuit includes an input terminal adapted to receive an input signal comprising an input noise component and an input signal component, a first sub-circuit adapted to generate a first signal responsive to at least a portion of the noise component, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of at least a portion of the input signal, the self-adjusting damper circuit combining the first signal with the second signal to generate an output signal.

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8. (Previously Presented) The memory module of claim 1, wherein the switching circuit comprises a plurality of circuits each corresponding to a subset of the plurality of devices and each having a data width that is a fraction of a data width of the memory module, the fraction being selected to optimize interconnection characteristics selected from the group consisting of timing, power consumption and signal integrity between the switching circuit and the plurality of memory devices.
9. (Previously Presented) The memory module of claim 1, further comprising a local memory controller to control the selective and operative coupling and the selective isolating in the switching circuit based on control signals from the system memory controller.
10. (Previously Presented) The memory module of claim 1, wherein the switching circuit establishes a one-to-one signal path for transferring data from each port of the second set of ports to a selected corresponding port of the first set of ports.
11. (Previously Presented) The memory module of claim 1, wherein the switching circuit establishes a one-to-many connection for broadcasting data from each port of the second set of ports to selected corresponding ports of the first set of ports.
12. (Previously Presented) A method of operating a memory module to communicate signals with a system memory controller via a plurality of signal transmission lines, the memory module including a plurality of memory devices on a module board, the module board including at least one connector configured to be operatively coupled to the plurality of signal transmission lines, the method comprising:

selectively and operatively coupling one or more ports of a second set of ports to one or more selected ports of a first set of ports, the first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, the second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

reshaping signals between the memory controller and the one or more first memory devices so as to ameliorate signal deterioration due to transmission over lossy transmission lines having discontinuity regions between the system memory controller and the one or more first memory devices.

13. (Previously Presented) The method of claim 12, further comprising:

selectively isolating one or more un-driven ports of the first set of ports from the second set of ports, and the one or more un-driven ports being operatively coupled to one or more second memory devices that are not being accessed by the system memory controller; and

providing a known static value to the un-driven ports, thereby reducing dynamic power consumption associated with floating or toggling electrical nodes.

14. (Previously Presented) The method of claim 12, wherein reshaping the input signals to generate reshaped signals comprises generating a respective output signal

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for a respective input signal, the respective output signal having an earlier transition trigger point and a faster slew rate than the respective input signal.

15. (Previously Presented) The method of claim 12, further comprising dynamically controlling a network of programmable resistors associated with each active signal path between the system memory controller and the plurality of memory devices so as to match a characteristic impedance of the each active signal path.

16. (Previously Presented) The method of claim 12, wherein each input signal comprises an input noise component and an input signal component, the method further comprising, for each input signal:

generating a first signal responsive to at least a portion of the noise component and a second signal corresponding to a delayed version of at least a portion of the input signal; and

combining the first signal with the second signal to generate an output signal.

17. (Previously Presented) The method of claim 12, further comprising controlling the selective and operative coupling and the selective isolating in the switching circuit based on control signals from the system memory controller.

18. (Currently Amended) A memory module operable to communicate signals with a system memory controller via a memory bus ~~plurality of signal transmission lines~~, comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to the memory bus ~~plurality of signal transmission lines~~;

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a plurality of memory devices on the printed circuit board;

a circuit, comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices;

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

a switching circuit configured to selectively and operatively couple one or more ports of the second set of ports to one or more selected ports of the first set of ports and to selectively isolate one or more un-driven ports of the first set of ports from the second set of ports, the one or more selected ports being operatively coupled to one or more first memory devices that are being accessed by the system memory controller, and the one or more un-driven ports being operatively coupled to one or more second memory devices that are not being accessed by the system memory controller, wherein the switching circuit is further configured to reshape signals communicated between the system memory controller and the one or more first memory devices via the selected ports and to provide a known static value to the un-driven ports, thereby ameliorating signal deterioration due to transmission over lossy transmission lines having discontinuity regions and reducing dynamic power consumption associated with floating or toggling electrical nodes.

19. (Currently Amended) The memory module of claim 18, wherein the switching circuit provides one or more signal paths between the one or more ports of the second

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set of ports and the one or more selected ports of the first set of ports, wherein the
switching circuit includes ~~at least one router circuit and~~ at least one signal correction
circuit for each of the one or more ~~a plurality of~~ signal paths ~~between the system~~
~~memory controller and the plurality of memory devices~~, the at least one signal correction
circuit including a repeater/amplifier element that can be enabled to generate a
respective output signal for a respective input signal, the respective output signal having
an earlier transition trigger point and a faster slew rate than the respective input signal.

20. (Previously Presented) The memory module of claim 18, wherein the switching
circuit includes at least one programmable impedance matching circuit for each of a
plurality of signal paths between the system memory controller and the plurality of
memory devices, the programmable impedance matching circuit having a network of
programmable resistors that are dynamically controllable to match a characteristic
impedance of a corresponding signal path.

Conclusion

Any inquiry concerning this communication or earlier communications from the
examiner should be directed to GURTEJ BANSAL whose telephone number is
(571)270-5588. The examiner can normally be reached on Monday - Friday, 7:30 a.m. -
5:00 p.m., EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's
supervisor, Reginald Bragdon can be reached on (571)272-4204. The fax phone

Art Unit: 2189

number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

GURTEJ BANSAL
Primary Examiner
Art Unit 2189

/GURTEJ BANSAL/
Primary Examiner, Art Unit 2189

<i>Applicant-Initiated Interview Summary</i>	Application No. 13/412,243	Applicant(s) LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2189	

All participants (applicant, applicant's representative, PTO personnel):

(1) GURTEJ BANSAL. (3) ____.

(2) JAMIE ZHENG. (4) ____.

Date of Interview: 23 May 2014.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☒102 ☐103 ☐Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 1 and 18.

Identification of prior art discussed: N/A.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Examiner suggested amending claim 18 to include the newly added limitation recited in claim 1. Applicant agreed and authorized the Examiner to make the necessary amendments to the claims.

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/GURTEJ BANSAL/ Primary Examiner, Art Unit 2189	
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Summary of Record of Interview Requirements

Manual of Patent Examining Procedure (MPEP), Section 713.04, Substance of Interview Must be Made of Record

A complete written statement as to the substance of any face-to-face, video conference, or telephone interview with regard to an application must be made of record in the application whether or not an agreement with the examiner was reached at the interview.

Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews

Paragraph (b)

In every instance where reconsideration is requested in view of an interview with an examiner, a complete written statement of the reasons presented at the interview as warranting favorable action must be filed by the applicant. An interview does not remove the necessity for reply to Office action as specified in §§ 1.111, 1.135. (35 U.S.C. 132)

37 CFR §1.2 Business to be transacted in writing.

All business with the Patent or Trademark Office should be transacted in writing. The personal attendance of applicants or their attorneys or agents at the Patent and Trademark Office is unnecessary. The action of the Patent and Trademark Office will be based exclusively on the written record in the Office. No attention will be paid to any alleged oral promise, stipulation, or understanding in relation to which there is disagreement or doubt.

The action of the Patent and Trademark Office cannot be based exclusively on the written record in the Office if that record is itself incomplete through the failure to record the substance of interviews.

It is the responsibility of the applicant or the attorney or agent to make the substance of an interview of record in the application file, unless the examiner indicates he or she will do so. It is the examiner's responsibility to see that such a record is made and to correct material inaccuracies which bear directly on the question of patentability.

Examiners must complete an Interview Summary Form for each interview held where a matter of substance has been discussed during the interview by checking the appropriate boxes and filling in the blanks. Discussions regarding only procedural matters, directed solely to restriction requirements for which interview recordation is otherwise provided for in Section 812.01 of the Manual of Patent Examining Procedure, or pointing out typographical errors or unreadable script in Office actions or the like, are excluded from the interview recordation procedures below. Where the substance of an interview is completely recorded in an Examiners Amendment, no separate Interview Summary Record is required.

The Interview Summary Form shall be given an appropriate Paper No., placed in the right hand portion of the file, and listed on the "Contents" section of the file wrapper. In a personal interview, a duplicate of the Form is given to the applicant (or attorney or agent) at the conclusion of the interview. In the case of a telephone or video-conference interview, the copy is mailed to the applicant's correspondence address either with or prior to the next official communication. If additional correspondence from the examiner is not likely before an allowance or if other circumstances dictate, the Form should be mailed promptly after the interview rather than with the next official communication.

The Form provides for recordation of the following information:

- Application Number (Series Code and Serial Number)
- Name of applicant
- Name of examiner
- Date of interview
- Type of interview (telephonic, video-conference, or personal)
- Name of participant(s) (applicant, attorney or agent, examiner, other PTO personnel, etc.)
- An indication whether or not an exhibit was shown or a demonstration conducted
- An identification of the specific prior art discussed
- An indication whether an agreement was reached and if so, a description of the general nature of the agreement (may be by attachment of a copy of amendments or claims agreed as being allowable). Note: Agreement as to allowability is tentative and does not restrict further action by the examiner to the contrary.
- The signature of the examiner who conducted the interview (if Form is not an attachment to a signed Office action)

It is desirable that the examiner orally remind the applicant of his or her obligation to record the substance of the interview of each case. It should be noted, however, that the Interview Summary Form will not normally be considered a complete and proper recordation of the interview unless it includes, or is supplemented by the applicant or the examiner to include, all of the applicable items required below concerning the substance of the interview.

A complete and proper recordation of the substance of any interview should include at least the following applicable items:

- 1) A brief description of the nature of any exhibit shown or any demonstration conducted,
- 2) an identification of the claims discussed,
- 3) an identification of the specific prior art discussed,
- 4) an identification of the principal proposed amendments of a substantive nature discussed, unless these are already described on the Interview Summary Form completed by the Examiner,
- 5) a brief identification of the general thrust of the principal arguments presented to the examiner,
(The identification of arguments need not be lengthy or elaborate. A verbatim or highly detailed description of the arguments is not required. The identification of the arguments is sufficient if the general nature or thrust of the principal arguments made to the examiner can be understood in the context of the application file. Of course, the applicant may desire to emphasize and fully describe those arguments which he or she feels were or might be persuasive to the examiner.)
- 6) a general indication of any other pertinent matters discussed, and
- 7) if appropriate, the general results or outcome of the interview unless already described in the Interview Summary Form completed by the examiner.

Examiners are expected to carefully review the applicant's record of the substance of an interview. If the record is not complete and accurate, the examiner will give the applicant an extendable one month time period to correct the record.

Examiner to Check for Accuracy

If the claims are allowable for other reasons of record, the examiner should send a letter setting forth the examiner's version of the statement attributed to him or her. If the record is complete and accurate, the examiner should place the indication, "Interview Record OK" on the paper recording the substance of the interview along with the date and the examiner's initials.



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[Analysis of lossy transmission lines with arbitrary nonlinear terminal networks](#)AR Djordjevic, JK Sarkar... - Microwave Theory and ..., 1986 - [ieeexplore.ieee.org](#)

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... line **ports** to obtain a time-stepping solution for the **port** voltages and currents. ... On the other hand, the analysis of **lossy transmission** lines (as well as lines with frequency-dependent parameters) can be performed only in frequency domain. ...

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[\[CITATION\] The generalized method of characteristics for waveform relaxation analysis of lossy coupled transmission lines](#)FY Chang - ... Theory and Techniques, IEEE Transactions on, 1989 - [ieeexplore.ieee.org](#)... cal discrete-time simulation, a phenomenal two order re- duction in computer simulation time and one order savings in computer **memory** requirements have ... In Section 111, the classical MC is extended for deriving the equivalent 2-**port** network of a **lossy transmission** line. ... Cited by 139 Related articles All 5 versions Cite Save

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☒ include patents☒ include citations[Memory and apparatus for input based control of discards in a lossy packet network](#)ASJ Chapman, HT Kung - US Patent 6,304,552, 2001 - [Google Patents](#)... a **lossy** switch fabric capable of selectively establishing a plurality of logical pathways between said ... from knowledge of the switch input and output **ports** involved in the data packet **transmission**. ... The links are unidirectional; that is, data packets are **transmitted** in one direction only ... Cited by 66 Related articles All 2 versions Cite Save

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[Networks on chips: a new SoC paradigm](#)L Benini, G De Micheli - Computer, 2002 - [ieeexplore.ieee.org](#)... Thus, **transmitting** digital values on wires will be inherently unreliable and nondeterministic. ... As a result, future global wires will function as **lossy transmission** lines,¹ as opposed to today's lumped or ... In this architecture, all communication devices share the **transmission** medium. ... Cited by 3096 Related articles All 44 versions Cite Save[\[PDF\] from nsysu.edu.tw](#)[Waveform relaxation analysis of nonuniform lossy transmission lines characterized with frequency-dependent parameters](#)FY Chang - Circuits and Systems, IEEE Transactions on, 1991 - [ieeexplore.ieee.org](#)... $u_2(t)$). By simultaneously solving the terminal voltages at both **ports**, we have ... can be implemented on any existing circuit simulator since the two-**port** asymmetric characteristic ... Three **lossy transmission**-line structures arranged in the following order of increasing complexity are ... Cited by 57 Related articles All 3 versions Cite Save[Digital signal processing system](#)J Seo - US Patent 5,446,456, 1996 - [Google Patents](#)... to the error flag signal being output from third frame **memory** 48, to thereby **transmit** the respective ... Jul 6, 1998, Jul 17, 2001, Sony Corporation, Apparatus and method for partial buffering **transmitted** data to provide robust error recovery in a **lossy transmission** environment. ... Cited by 42 Related articles All 2 versions Cite Save[Scattering parameter transient analysis of transmission lines loaded with nonlinear terminations](#)JE Schuit-Aine, R Mittra - Microwave Theory and Techniques, ..., 1995 - [ieeexplore.ieee.org](#)... With the design of fast devices having switching times in the picosecond range, **transmitting** data at high ... linear one-**port** model at time $t_n + l = (n + l)h$ (see Fig. ... Simulations of pulse propagation through **lossy transmission** lines terminated with active devices can be very useful in ... Cited by 159 Related articles All 6 versions Cite Save More[Selection of lumped element models for coupled lossy transmission lines](#)I Jhansara, R Da Zutter - Computer-Aided Design of Integrated ..., 1992 - [ieeexplore.ieee.org](#)... of high- speed systems is mainly determined and limited by the ability to **transmit** signals undisturbed ... Relative Error on Propagation Factor and Characteristic Impedance: Each two-**port** is characterized ... DHAENE AND DE ZUTTER: COUPLED LOSSY TRANSMISSION LINES ... Cited by 113 Related articles All 11 versions Cite Save[\[PDF\] from ugent.be](#)[Simulation of arbitrary transmission line networks with nonlinear terminations](#)D Winklerstein, MB Stangl... - Circuits and Systems, ..., 1991 - [ieeexplore.ieee.org](#)... external **ports** in the **transmission** line system, and 1 At $g_{ij}(t) = -F^{-1}[G_{ij}(w)]$ (4 ... yr, is in turn defined by (7). The circuit simulator calls this user-defined routing once for each **port** and each ... [5] AR Djordjevic, T. K. Sarkar, and R. F. Harrington, "Analysis of **lossy transmission** lines with ... Cited by 71 Related articles All 3 versions Cite Save[Transient simulation of nonuniform coupled lossy transmission lines characterized with frequency-dependent parameters. I. Waveform relaxation analysis](#)FY Chang - Circuits and Systems I: Fundamental Theory and ..., 1992 - [ieeexplore.ieee.org](#)... In Section IT, the generalized characteristic model of an n-conductor nonuniform **lossy** coupled **transmission**-line system is derived in terms of the ABCD matrices of the conductor system. The characteristic model consists of two disjoint n-**port** networks connected with voltage ... Cited by 54 Related articles All 4 versions Cite Save

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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	13	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:39
L2	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:41
L3	34	signal with discontinuit\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:43
L4	3171	365/191.ccls. 365/196.ccls. 711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2014/05/23 18:48
S1	7	"5559970".pn. "6700823".pn. "6812869".pn. "6981089".pn. "7136958".pn. "7184916".pn. "20070013429".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 15:58
S2	496	bidirectional with port with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S3	89	bidirectional with port with (crossbar switch) same memory	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:02
S4	366	bidirectional with ports with (crossbar switch)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:05
S5	48	bidirectional with ports with (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:15
S6	2546	bi\$1directional near2 ports	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:20
S7	666	bi\$1directional near2 ports same (crossbar switch\$3)	US-PGPUB; USPAT	OR	OFF	2013/12/28 16:40
S8	1	"7130308".pn.	US-PGPUB; USPAT	OR	OFF	2013/12/28 17:14
S9	786	711/149.ccls.	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:03
S10	78	port with noise with reduc\$4 and PCB	US-PGPUB; USPAT	OR	OFF	2013/12/28 22:16
S11	20	signal with deteriorat\$3 with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 17:59
S12	1	signal with deteriorat\$3 with switch same lossy	US-PGPUB;	OR	OFF	2014/05/04 18:01

			USPAT			
S13	16	signal with loss with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:01
S14	19	signal with (loss deteriorat\$3 distort\$3) with switch same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S15	1227	signal with (loss deteriorat\$3 distort\$3) same lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:16
S16	14	signal with (loss deteriorat\$3 distort\$3) with lossy same (correct\$3 ameliorat\$3)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:17
S17	2	signal with (loss deteriorat\$3 distort\$3) with lossy same (switch\$3 with circuit)	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:18
S18	529	signal with (loss deteriorat\$3 distort\$3) with lossy	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S19	11	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	OFF	2014/05/04 18:20
S20	19	signal with (loss deteriorat\$3 distort\$3) with lossy same port	US-PGPUB; USPAT	OR	ON	2014/05/04 18:20
S21	116	signal with (loss deteriorat\$3 distort\$3) with lossy same circuit	US-PGPUB; USPAT	OR	ON	2014/05/04 18:22
S22	8	signal with (loss deteriorat\$3 distort\$3) with lossy same discontin\$6	US-PGPUB; USPAT	OR	ON	2014/05/04 18:23
S23	66	signal with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:36
S24	47	transmission with (loss deteriorat\$3 distort\$3) with lossy same (repeater amplifier)	US-PGPUB; USPAT	OR	ON	2014/05/04 18:39

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L5	2421	365/191.ccls. 365/196.ccls. 711/149.ccls.	USPAT; UPAD	OR	OFF	2014/05/23 18:51
L6	1498	lossy.clm.	USPAT; UPAD	OR	OFF	2014/05/23 18:52
L7	0	L5 and L6	USPAT; UPAD	OR	OFF	2014/05/23 18:52

5/ 23/ 2014 6:52:32 PM
C:\ Users\ gbansal\ Documents\ EAST\ Workspaces\ 13412243.wsp

Search Notes 	Application/Control No. 13412243	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2189

CPC- SEARCHED		
Symbol	Date	Examiner

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner
711	149	12/28/2013	GB
365	191; 196	05/23/2014	GB

SEARCH NOTES		
Search Notes	Date	Examiner
Inventor Search in eDAN	12/28/2013	GB
Text Search in EAST (See Attached)	12/28/2013	GB
NPL Search in Google Scholar (See Attached)	12/28/2013	GB
Updated Text Search in EAST (See Attached)	05/23/2014	GB
Updated NPL Search in Google Scholar (See Attached)	05/23/2014	GB

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner
711	149	05/23/2014	GB
365	191; 196	05/23/2014	GB

/GURTEJ BANSAL/ Examiner.Art Unit 2189	
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Issue Classification 	Application/Control No. 13412243	Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner GURTEJ BANSAL	Art Unit 2189	

CPC						
Symbol					Type	Version
H05K		2201	/	10053	A	2013-01-01
H05K		2201	/	10189	A	2013-01-01
H05K		1	/	0243	F	2013-01-01
H05K		2201	/	10159	A	2013-01-01
H05K		1	/	0216	A	2013-01-01
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CPC Combination Sets				
Symbol	Type	Set	Ranking	Version

NONE		Total Claims Allowed:	
(Assistant Examiner)		20	
(Date)			
/GURTEJ BANSAL/ Primary Examiner.Art Unit 2189	05/23/2014	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	10

Issue Classification 	Application/Control No. 13412243	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner GURTEJ BANSAL	Art Unit 2189

☐ Claims renumbered in the same order as presented by applicant															
☐ CPA															
☐ T.D.															
☐ R.1.47															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	1	17	17												
2	2	18	18												
7	3	19	19												
8	4	20	20												
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12	12														
13	13														
14	14														
15	15														
16	16														

NONE	Total Claims Allowed: 20	
(Assistant Examiner)	(Date)	
/GURTEJ BANSAL/ Primary Examiner.Art Unit 2189	05/23/2014	O.G. Print Claim(s) 1
(Primary Examiner)	(Date)	O.G. Print Figure 10

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	1	of	11	Attorney Docket Number	NETL.045C1

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	4,249,253	2/3/1981	Gentili et al.	
	2.	4,368,515	1/11/1983	Nielsen	
	3.	4,571,676	2/18/1986	Mantellina et al.	
	4.	4,592,011	5/27/1986	Mantellina et al.	
	5.	5,060,188	10/22/1991	Zulian et al.	
	6.	5,345,412	9/6/1994	Shiratsuchi	
	7.	5,463,590	10/31/1995	Watanabe	
	8.	5,513,135	4/30/1996	Dell et al.	
	9.	5,532,954	7/2/1996	Bechtolsheim et al.	
	10.	5,537,584	7/16/1996	Miyai et al.	
	11.	5,541,448	7/30/1996	Carpenter	
	12.	5,572,691	11/5/1996	Koudmani	
	13.	5,581,498	12/3/1996	Ludwig	
	14.	5,590,071	12/31/1996	Kolor	
	15.	5,617,559	4/1/1997	Le et al.	
	16.	5,649,159	7/15/1997	Le et al.	
	17.	5,717,851	2/10/1998	Yishay et al.	
	18.	5,724,604	3/3/1998	Moyer	
	19.	5,729,716	5/17/1998	Lee et al.	
	20.	5,745,914	4/28/1998	Connolly	
	21.	5,784,705	7/21/1998	Leung, Wingyu	
	22.	5,802,395	9/1/1998	Connolly	
	23.	5,802,541	9/1/1998	Reed	
	24.	5,926,827	7/20/1999	Dell	
	25.	5,953,280	9/14/1999	Watsui	
	26.	5,973,392	10/26/1999	Senba et al.	
	27.	6,173,357	1/9/2011	Ju	
	28.	6,188,641	2/13/2001	Uchida	
	29.	6,205,516	3/20/2001	Usami	
	30.	6,209,074	3/27/2001	Dell	
	31.	6,233,650	5/15/2001	Johnson	
	32.	6,260,127	7/10/2001	Olarig et al.	
	33.	6,414,868	7/2/2002	Wong	
	34.	6,438,062	8/20/2002	Curtis	
	35.	6,446,158	9/3/2002	Karabatsos	
	36.	6,446,184	9/3/2002	Dell	
	37.	6,480,439	11/12/2002	Tokutome et al.	

EXAMINER

/Gurtej Bansal/ (05/23/2014)

DATE CONSIDERED

05/23/2014

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	2	of	11	Attorney Docket Number	NETL.045C1

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	38.	6,553,449	4/22/2003	Dodd et al.	
	39.	6,618,791	9/9/2003	Dodd et al.	
	40.	6,717,855	4/6/2004	Underwood	
	41.	6,788,592	9/7/2004	Nakata et al.	
	42.	6,948,084	9/20/2005	Manapat et al	
	43.	7,130,308	10/31/2006	Haddock et al.	
	44.	7,133,960	11/7/2006	Thompson et al.	
	45.	7,379,361	5/27/2008	Co et al.	
	46.	7,532,537	5/12/2009	Solomon et al.	
	47.	7,881,150	2/1/2011	Solomon et al.	
	48.	7,916,574	3/29/2011	Solomon et al.	
	49.	8,001,434	8/16/2011	Lee et al.	
	50.	8,130,560	3/6/2012	Rajan	
	51.	8,250,295	8/21/2012	Amidi et al.	
	52.	8,516,188	8/20/2013	Solomon et al.	
	53.	US RE 36229	06-15-1999	Cady	
	54.	20020048195	4/25/2002	Klein, Dean A.	
	55.	20030070052	4/10/2003	Lai, Jiin	
	56.	20040098528	5/20/2004	Janzen, Jefferey W.	
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				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	3	of	11	Attorney Docket Number	NETL.045C1

US PATENT APPLICATIONS					
Examiner Initials *	Cite No. ¹	Application Number	Filing Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
	71.	Reexam Appl. No. 95/000,546 for U.S. Pat. No. 7,289,386	05/11/2010	Netlist, Inc.	Entire File History
	72.	Reexam Appl. No. 95/000,577 for U.S. Pat. No. 7,289,386	10/20/2010	Netlist, Inc.	Entire File History
	73.	Reexam Appl. No. 95/000,578 for U.S. Pat. No. 7,619,912	10/20/2010	Netlist, Inc.	Entire File History
	74.	Reexam Appl. No. 95/000,579 for U.S. Pat. No. 7,619,912	10/21/2010	Netlist, Inc.	Entire File History
	75.	Reexam App. No. 95/001,337 for U.S. Pat. No. 7,636,274	06/04/2010	Netlist, Inc.	Entire File History
	76.	Reexam Appl. No. 95/001,338; for U.S. Pat. No. 7,532,537	04/19/2010	Netlist, Inc.	Entire File History
	77.	Reexam Appl. No. 95/001,339 for U.S. Pat. No. 7,619,912	06/08/2010	Netlist, Inc.	Entire File History
	78.	Reexam Appl. No. 95/001,381 for U.S. Pat. No. 7,532,537	06/09/2010	Netlist, Inc.	Entire File History
	79.				
	80.	Reexam Appl. No. 95/001,758 for U.S. Pat. No. 7,864,627	09/15/2011	Netlist, Inc.	Entire File History
	81.	Reexam Appl. No. 95/002,399 for U.S. Pat. No. 8,250,295	09/15/2010	Smart Modular Technologies, Inc.	Entire File History
	82.	95/000,578	10/20/2010	Netlist, Inc.	
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	84.	95/001,338	Not available	Netlist, Inc.	
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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
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	95.	12/577,682	10/12/2009	Netlist, Inc.	Entire Prosecution History
	96.	12/629,827	12/2/2009	Netlist, Inc.	Entire Prosecution History
	97.	12/761,179	4/15/2010	Netlist, Inc.	Entire Prosecution History
	98.	12/902,073	10/11/2010	Netlist, Inc.	
	99.	12/912,623	10/26/2010	Netlist, Inc.	Entire Prosecution History
	100.	12/954,492	11/24/2010	Netlist, Inc.	Entire Prosecution History
	101.	12/955,711	11/29/2010	Netlist, Inc.	Entire Prosecution History
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				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	5	of	11	Attorney Docket Number	NETL.045C1

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	129.	Non-final office action, U.S. Patent Application No. 13/288,850, October 11, 2013, 24 pages	
	130.	Non-final office action, U.S. Patent Application No. 13/411,413, December 31, 2013, 28 pages	
	131.	Non-final office action, U.S. Patent Application No. 13/473,413, November 17, 2011, 46 pages	
	132.	Response to non-final office action dated October 14, 2013 for U.S. Patent Application No. 13/288,850, filed January 13, 2014, 15 pages	
	133.	Response to non-final office action dated 12/31/2013 for U.S. Patent Application No. 13/411,344, filed March 31, 2014, 12 pages	
	134.	Patent Owner's Response to Office Action mailed November 13, 2012 for Reexamination Control Nos.95/000,578; 95/000,579, and 95/001,339, filed January 14, 2013, 96 pages.	
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				Examiner Name	BANSAL, GURTEJ
Sheet	6	of	11	Attorney Docket Number	NETL.045C1

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	140.	Patent Trial and Appeal Board Decision on Appeal for Reexamination Control No.95/001/381, mailed January 16, 2014, 24 pages	
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	150.	Response to non-final office action dated September 13, 2012 for U.S. Patent Application No. 12/761,179, filed March 13, 2013, 16 pages	
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NON PATENT LITERATURE DOCUMENTS			
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	192.	Order Granting Request for Inter Partes Reexamination mailed September 8, 2010, for Control No. 95/000,381, filed June 9, 2010, 21 pages.	
	193.	Order Granting Request for Inter Partes Reexamination mailed January 14, 2011, for Control No. 95/000,579, filed Oct. 21, 2010, 12 pages.	
	194.	Order Granting Request for Inter Partes Reexamination mailed January 18, 2011, for Control No. 95/000,577, filed Oct. 20, 2010, 17 pages.	
	195.	Right of appeal Notice mailed Feb. 7, 2012, for Control No. 95/001,381, 33 pages.	
	196.	Right of Appeal Notice mailed June 22, 2012, for Control No. 95/001,337, filed June 4 2010, 34 pages.	

EXAMINER /Gurtej Bansal/ (05/23/2014)

DATE CONSIDERED 05/23/2014

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.¹ Applicant's unique citation designation number (optional).² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	9	of	11	Attorney Docket Number	NETL.045C1

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
	197.	PC133 SDRAM Registered DIMM Design Specification, Revision 1.1, Aug. 1999, 62 pages.		
	198.	Texas Instruments, "TM2SR72EPN 2097152 by 72-Bit, TM4SRT2EPN 4194304 by 72-Bit, Synchronous Dynamic RAM Modules," 1997, 15 pages.		
	199.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. Inphi Corporation, Complaint For Patent Infringement, filed Sept. 22, 2009 in 10 pages.		
	200.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Defendant Inphi Corporation's Answer to Plaintiff's Complaint for Patent Infringement, filed Nov. 12, 2009 in 6 pages.		
	201.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Defendant Inphi Corporation's Answer to Plaintiff's First Amended Complaint for Patent Infringement, filed Feb. 11, 2010 in 9 pages.		
	202.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Defendant Inphi Corporation's Notice of Motion and Motion for Stay Pending Reexaminations and Interference Proceeding Regarding the Patents-In-Suit; Memorandum of Points and Authorities in Support Thereof, filed Apr. 21, 2010 in 28 pages.		
	203.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. INPHI Corporation, Plaintiff Netlist Inc's Opposition to Defendant Inphi Corporation's Motion for Stay Pending Reexaminations and Interference Proceedings Regarding the Patents-In-Suit, filed May 3, 2010 in 23 pages.		
	204.	U.S. District Court Central District of California, Case No. CV09 06900, Netlist, Inc. vs. Inphi Corporation, Plaintiff Netlist, Inc's First Amended Complaint For Patent Infringement, filed Dec. 23, 2009 in 8 pages.		
	205.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Defendant Netlist, Inc.'s Claim Construction Reply Brief, filed Sept. 22, 2009 in 19 pages.		
	206.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Defendant Netlist, Inc.'s Opening Claim Construction Brief, filed July 28, 2009 in 21 pages.		
	207.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Defendant Netlist, Inc.'s Opposition to Google Inc's Motion for Summary Judgment of Invalidity, filed July 6, 2010 in 13 pages.		
	208.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Exhibit A To Joint Claim Construction And Prehearing Statement, filed June 12, 2009 in 2 pages.		
	209.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Exhibit B to Joint Claim Construction and Prehearing Statement, filed June 12, 2009 in 36 pages.		

EXAMINER

/Gurtej Bansal/ (05/23/2014)

DATE CONSIDERED

05/23/2014

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				Application Number	13/412,243
				Filing Date	March 5, 2012
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				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	10	of	11	Attorney Docket Number	NETL.045C1

NON PATENT LITERATURE DOCUMENTS				
Examiner Initials*	Cite No. ¹	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ²	
	210.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Joint Claim Construction and Prehearing Statement, filed June 12, 2009 in 5 pages.		
	211.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Netlist, Inc.'s Answer to Complaint and Counterclaims, filed Nov. 18, 2008 in 9 pages.		
	212.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Order Re Claim Construction, filed Nov. 16, 2009 in 1 page.		
	213.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Plaintiff Google's Reply to Counterclaims, filed Dec. 8, 2008 in 4 pages.		
	214.	U.S. District Court Northern District of California, Case No. CV08 04144, Google Inc. v. Netlist, Inc., Stipulation Re: Additional Agreed-Upon Claim Constructions, filed Oct. 28, 2009 in 3 pages.		
	215.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., [Redacted] Google Inc.'s Responsive Claim Construction Brief, filed Aug. 25, 2009 in 30 pages.		
	216.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Amended Exhibit A To Joint Claim Construction And Prehearing Statement, filed Oct. 28, 2009 in 1 page.		
	217.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Appendix 1 To Google's Responsive Claim Construction Brief, filed Nov. 12, 2009 in 4 pages.		
	218.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Attachment 1 to Exhibit B To Joint Claim Construction And Prehearing Statement, filed June 12, 2009 in 7 pages.		
	219.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Attachment 2 To Exhibit B To Joint Claim Construction And Prehearing Statement, filed June 12, 2009 in 12 pages.		
	220.	U.S. District Court Northern District of California, Case No. CV08 04144, GOOGLE INC. vs. NETLIST, INC., Complaint for Declaratory Relief, filed Aug. 29, 2008 in 49 pages.		
	221.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Complaint For Patent Infringement, filed Dec. 4, 2009 in 47 pages.		
	222.	U.S. District Court Northern District of California, Case No. CV09 05718, NETLIST, INC. vs. GOOGLE, INC., Defendant Google Inc's Responsive Claim Construction Brief, filed Aug. 4, 2010 in 27 pages.		

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	223.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Exhibit A to Joint Claim Construction and Prehearing Statement under Patent L.R. 4-3, filed June 25, 2010 in 2 pages.		
	224.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Exhibit B to Joint Claim Construction and Prehearing Statement under Patent L.R. 4-3, filed Jun. 25, 2010 in 23 pages.		
	225.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Google's Answer to Plaintiff's Complaint for Patent Infringement; and Assertion of Counterclaims, filed Feb. 12, 2010 in 13 pages.		
	226.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Joint Claim Construction And Prehearing Statement Under Patent Local Rule 4-3, filed June 25, 2010 in 5 pages.		
	227.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Plaintiff Netlist, Inc.'s Reply Claim Construction Brief, filed Aug. 16, 2010 in 17 pages.		
	228.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Plaintiff Netlist, Inc.'s Reply to Defendant Google Inc.'s Counterclaim, filed March 8, 2010 in 11 pages.		
	229.	U.S. District Court Northern District of California, Case No. CV09 05718, Netlist, Inc. vs. Google, Inc., Plaintiff Netlist, Inc.'s Opening Claim Construction Brief, filed Jul. 16, 2010 in 29 pages.		
	230.	US District Court Civil Docket; Google Inc. v. Netlist Inc.; 4:08cv04144; filed Aug. 29, 2008.		
	231.	US District Court Civil Docket; Netlist Inc. v. Google Inc.; 4:09cv5718, filed Dec. 4, 2009 in 10 pages.		
	232.	US District Court Civil Docket; Netlist Inc. v. Inphi Corporation; 2:09cv6900; Date filed Sep. 22, 2009.		
	233.	Vogt, Pete, "Fully Buffered DIMM (FB-DIMM) Server Memory Architecture: Capacity, Performance, Reliability, and Longevity," Intel, Feb. 18, 2004, 33 pages.		

EXAMINER

/Gurtej Bansal/ (05/23/2014)

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The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306



**Courtesy Reminder for
Application Serial No: 13/412,243**

Attorney Docket No: NETL.045C1

Customer Number: 79141

Date of Electronic Notification: 05/29/2014

This is a courtesy reminder that new correspondence is available for this application. If you have not done so already, please review the correspondence. The official date of notification of the outgoing correspondence will be indicated on the form PTOL-90 accompanying the correspondence.

An email notification regarding the correspondence was sent to the following email address(es) associated with your customer number:

jz@jzpatent.com

To view your correspondence online or update your email addresses, please visit us anytime at <https://portal.uspto.gov/secure/myportal/privatepair>. If you have any questions, please email the Electronic Business Center (EBC) at EBC@uspto.gov or call 1-866-217-9197.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
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INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Customer No. 79141

The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

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I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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13/412,243	03/05/2012	Hyun Lee	NETL.045C1	2878
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TITLE OF INVENTION:

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
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nonprovisional	SMALL	\$480	\$0	\$0	\$480	08/29/2014
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EXAMINER	ART UNIT	CLASS-SUBCLASS
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1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 JAMIE J. ZHENG, Ph.D, Esq.

2

3

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

☒ Issue Fee

☐ Publication Fee (No small entity discount permitted)

☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

☐ A check is enclosed.

☒ Payment by credit card. Form PTO-2038 is attached.

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 50-5963 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

☒ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27.

☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature /Jamie J. Zheng/

Date June 9, 2014

Typed or printed name Jamie J. Zheng

Registration No. 51,167

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
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8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	13412243			
Filing Date:	05-Mar-2012			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NETL.045C1			
Filed as Small Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl Issue Fee	2501	1	480	480
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				480

Electronic Acknowledgement Receipt

EFS ID:	19253424
Application Number:	13412243
International Application Number:	
Confirmation Number:	2878
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045C1
Receipt Date:	09-JUN-2014
Filing Date:	05-MAR-2012
Time Stamp:	18:47:47
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$480
RAM confirmation Number	5635
Deposit Account	
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Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
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Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	29972 16b155ba96b687a5aaa96fd71d3ba8b9213b6a44	no	2
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Receipt date: 05/08/2014

13412243 - GAU: 2189

PTO/SB/08a/b(07-09)

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Change(s) applied

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to document,

L.A.S./

6/10/2014

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

(Use as many sheets as necessary)

Complete if Known

Application Number	13/412,243
Filing Date	March 5, 2012
First Named Inventor	Hyun Lee
Art Unit	2189
Examiner Name	BANSAL, GURTEJ
Attorney Docket Number	NETL.045C1

Sheet 3 of 11

US PATENT APPLICATIONS

Examiner Initials *	Cite No. ¹	Application Number	Filing Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
	71.	Reexam Appl. No. 95/000,546 for U.S. Pat. No. 7,289,386	05/11/2010 10-30-2007	Netlist, Inc. Bhakta et al.	Entire File History
	72.	Reexam Appl. No. 95/000,577 for U.S. Pat. No. 7,289,386	10/20/2010 10-30-2007	Netlist, Inc. Bhakta et al.	Entire File History
	73.	Reexam Appl. No. 95/000,578 for U.S. Pat. No. 7,619,912	10/20/2010 11-17-2009	Netlist, Inc. Bhakta et al.	Entire File History
	74.	Reexam Appl. No. 95/000,579 for U.S. Pat. No. 7,619,912	10/21/2010 11-17-2009	Netlist, Inc. Bhakta et al.	Entire File History
	75.	Reexam App. No. 95/001,337 for U.S. Pat. No. 7,636,274	06/04/2010 12-22-2009	Netlist, Inc. Solomon et al.	Entire File History
	76.	Reexam Appl. No. 95/001,338; for U.S. Pat. No. 7,532,537	04/19/2010 5-12-2009	Netlist, Inc. Solomon et al.	Entire File History
	77.	Reexam Appl. No. 95/001,339 for U.S. Pat. No. 7,619,912	06/08/2010 11-17-2009	Netlist, Inc. Bhakta et al.	Entire File History
	78.	Reexam Appl. No. 95/001,381 for U.S. Pat. No. 7,532,537	06/09/2010 5-12-2009	Netlist, Inc. Solomon et al.	Entire File History
	79.				
	80.	Reexam Appl. No. 95/001,758 for U.S. Pat. No. 7,864,627	09/15/2011 1-4-2011	Netlist, Inc. Bhakta et al.	Entire File History
	81.	Reexam Appl. No. 95/002,399 for U.S. Pat. No. 8,250,295	09/15/2010 8-21-2012	Smart Modular Technologies, Inc.	Entire File History Amidi et al.
	82.	95/000,578	10/20/2010	Netlist, Inc.	
	83.	95/000,579	10/21/2010	Netlist, Inc.	
	84.	95/001,338	Not available	Netlist, Inc.	
	85.	95/001,758	9/14/2011	Netlist, Inc.	
	86.	11/075,395	3/7/2005	Netlist, Inc.	Entire Prosecution History
	87.	11/142,989	12/07/2006	Netlist, Inc.	
	88.	11/173,175	7/1/2005	Netlist, Inc.	Entire Prosecution History
	89.	11/335,875	1/19/2006	Netlist, Inc.	Entire Prosecution History
	90.	11/862,931	9/27/2007	Netlist, Inc.	Entire Prosecution History
	91.	12/408,652	3/20/2009	Netlist, Inc.	Entire Prosecution History
	92.	12/422,853	4/13/2009	Netlist, Inc.	Entire Prosecution History
	93.	12/422,925	4/13/2009	Netlist, Inc.	Entire Prosecution History
	94.	12/504,131	7/16/2009	Netlist, Inc.	Entire Prosecution History

EXAMINER

/Gurtej Bansal/ (05/23/2014)

DATE CONSIDERED

05/23/2014

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

Receipt date: 05/08/2014

13412243 - GAU: 2189

PTO/SB/08a/b(07-09)

Approved for use through 7/31/2012. OMB 0651-0031

US Patent & Trademark Office; U.S. DEPARTMENT OF COMMERCE

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	2	of	11	Attorney Docket Number	NETL.045C1

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	38.	6,553,449	4/22/2003	Dodd et al.	
	39.	6,618,791	9/9/2003	Dodd et al.	
	40.	6,717,855	4/6/2004	Underwood	
	41.	6,788,592	9/7/2004	Nakata et al.	
	42.	6,948,084	9/20/2005	Manapat et al	
	43.	7,130,308	10/31/2006	Haddock et al.	
	44.	7,133,960	11/7/2006	Thompson et al.	
	45.	7,379,361	5/27/2008	Co et al.	
	46.	7,532,537	5/12/2009	Solomon et al.	
	47.	7,881,150	2/1/2011	Solomon et al.	
	48.	7,916,574	3/29/2011	Solomon et al.	
	49.	8,001,434	8/16/2011	Lee et al.	
	50.	8,130,560	3/6/2012	Rajan	
	51.	8,250,295	8/21/2012	Amidi et al.	
	52.	8,516,188	8/20/2013	Solomon et al.	
	53.	US RE 36229	06-15-1999	Cady	
	54.	20020048195	4/25/2002	Klein, Dean A.	
	55.	20030070052	4/10/2003	Lai, Jiin	
	56.	20040098528	5/20/2004	Janzen, Jefferey W.	
	57.	20050010737	1/1/2005	Ware et al.	
	58.	20060117152	6/1/2006	Amidi	
	59.	20060233012	10/19/2006	Tomonori et al.	Sekiguchi et al.
	60.	20060262586	11/23/2006	Solomon, et al	
	61.	20070058409	3/15/2007	Ruckerbauer et al.	
	62.	20070293094	12/1/2007	Aekins, Robert A.	
	63.	20080025137	1/31/2008	Natarajan et al.	Rajan et al.
	64.	20080046631	2/21/2008	Takaku et al.	
	65.	20080162790	7/3/2008	Im, Jeon-Taek	
	66.	20090103387	4/23/2009	Shau, Jeng-Jye	
	67.	20100070690	3/18/2010	Amer et al.	
	68.	20100091540	4/15/2010	Bhakta et al.	
	69.	20110090749	4/21/2011	Bhakta et al.	
	70.	20110125966	5/26/2011	Amidi et al.	

Change(s) applied
to document,
/G.R.P./
6/16/2014

EXAMINER

/Gurtej Bansal/ (05/23/2014)

DATE CONSIDERED

05/23/2014

* EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant. ¹ Applicant's unique citation designation number (optional). ² See Kinds Codes of USPTO Patents Documents at www.uspto.gov or MPEP 901.04. ³ Enter Office that issued the document, by the two-letter code (WIPO Standard ST. 3). ⁴ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁵ Kind of document by the appropriate symbols as indicated on the document and WIPO Standard ST. 16 if possible. ⁶ Applicant is to place a check mark here if English language Translation is attached.

ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /G.B./

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Substitute for form 1449/PTO INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Use as many sheets as necessary)</i>				Complete if Known	
				Application Number	13/412,243
				Filing Date	March 5, 2012
				First Named Inventor	Hyun Lee
				Art Unit	2189
				Examiner Name	BANSAL, GURTEJ
Sheet	1	of	11	Attorney Docket Number	NETL.045C1

US PATENT DOCUMENTS					
Examiner Initials *	Cite No. ¹	Document Number	Publication Date (MM-DD-YYYY)	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number-Kind Code ² (if known)			
	1.	4,249,253	2/3/1981	Gentili et al.	
	2.	4,368,515	1/11/1983	Nielsen	
	3.	4,571,676	2/18/1986	Mantellina et al.	
	4.	4,592,011	5/27/1986	Mantellina et al.	
	5.	5,060,188	10/22/1991	Zulian et al.	
	6.	5,345,412	9/6/1994	Shiratsuchi	
	7.	5,463,590	10/31/1995	Watanabe	
	8.	5,513,135	4/30/1996	Dell et al.	
	9.	5,532,954	7/2/1996	Bechtolsheim et al.	
	10.	5,537,584	7/16/1996	Miyai et al.	
	11.	5,541,448	7/30/1996	Carpenter	
	12.	5,572,691	11/5/1996	Koudmani	
	13.	5,581,498	12/3/1996	Ludwig	
	14.	5,590,071	12/31/1996	Kolor	
	15.	5,617,559	4/1/1997	Le et al.	
	16.	5,649,159	7/15/1997	Le et al.	
	17.	5,717,851	2/10/1998	Yishay et al.	
	18.	5,724,604	3/3/1998	Moyer	
	19.	5,729,716	03/17/1998	Lee et al.	
	20.	5,745,914	4/28/1998	Connolly	
	21.	5,784,705	7/21/1998	Leung, Wingyu	
	22.	5,802,395	9/1/1998	Connolly	
	23.	5,802,541	9/1/1998	Reed	
	24.	5,926,827	7/20/1999	Dell	
	25.	5,953,280	9/14/1999	Watsui	
	26.	5,973,392	10/26/1999	Senba et al.	
	27.	6,173,357	4/9/2001	Ju	01/09/2001
	28.	6,188,641	2/13/2001	Uchida	
	29.	6,205,516	3/20/2001	Usami	
	30.	6,209,074	3/27/2001	Dell	
	31.	6,233,650	5/15/2001	Johnson	
	32.	6,260,127	7/10/2001	Olarig et al.	
	33.	6,414,868	7/2/2002	Wong	
	34.	6,438,062	8/20/2002	Curtis	
	35.	6,446,158	9/3/2002	Karabatsos	
	36.	6,446,184	9/3/2002	Dell	
	37.	6,480,439	11/12/2002	Tokutome et al.	

EXAMINER

/Gurtej Bansal/ (05/23/2014)

DATE CONSIDERED

05/23/2014

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/412,243	07/15/2014	8782350	NETL.045C1	2878

79141 7590 06/25/2014
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 206 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY.DOCKET NO./TITLE	REQUEST ID
13/412,243	03/05/2012	Hyun Lee	129980-5042US	67074

Acknowledgement of Loss of Entitlement to Entity Status Discount

The entity status change request below filed through Private PAIR on 07/06/2018 has been accepted.

CERTIFICATIONS:

Change of Entity Status:

✕ Applicant changing to regular undiscounted fee status.

NOTE: Checking this box will be taken to be notification of loss of entitlement to small or micro entity status, as applicable.

This portion must be completed by the signatory or signatories making the entity status change in accordance with 37 CFR 1.4(d)(4).

Signature:	/Jamie J. Zheng/
Name:	Jamie J. Zheng
Registration Number:	51167



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PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
13/412,243	03/05/2012	411121

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine , CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, Virginia 22313-1450
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FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
13/412,243	03/05/2012	411121

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167

UTILITY APPLICATION	Attorney Docket No.: NETL.045A
	First Named Inventor: Hyun Lee et al.
	Title: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Direct all correspondence to Customer No.: 20995	
Date: April 13, 2009 Page 1	

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

The following enclosures are transmitted herewith to be filed in the patent application of:

Inventor(s):

- | | |
|-----------------------|-------------------------|
| 1. Hyun Lee | 2. Jayesh R. Bhakta |
| 3. Jeffrey C. Solomon | 4. Mario Jesus Martinez |
| 5. Chi-She Chen | |

APPLICATION:

- (X) Specification in 28 pages.
(X) Drawings in 9 sheets.

CONTINUITY INFORMATION:

Application	Relationship	Parent App. No.	Filing Date	Status
This Application	Non-Provisional of	61/044839	04/14/2008	Pending
This Application	Non-Provisional of	61/044825	04/14/2008	Pending
This Application	Non-Provisional of	61/044801	04/14/2008	Pending

Reference to prior domestic applications is made in the:

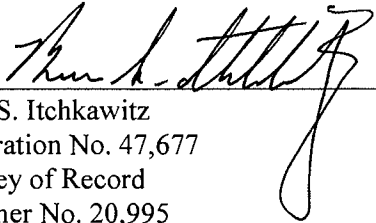
- (X) Specification.
(X) Enclosed Application Data Sheet (ADS).

FILING FEES:

FEE CALCULATION				
FEE TYPE		FEE CODE	CALCULATION	TOTAL
Basic Utility	1.16(a)(1)	1011 (\$330)		\$330
Search Fee	1.16(k)	1111 (\$540)		\$540
Examination Fee	1.16(o)	1311 (\$220)		\$220
Excess Claims > 20	23 - 20 = 3	1202 (\$52)	3 x 52 =	\$156
			TOTAL FEE DUE	\$1246

UTILITY APPLICATION	Attorney Docket No.: NETL.045A
	First Named Inventor: Hyun Lee et al.
	Title: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Direct all correspondence to Customer No.: 20995	
Date: April 13, 2009 Page 2	

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment to Account No. 11-1410.

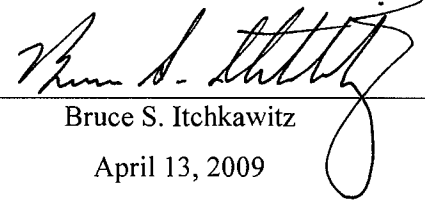


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

**NONPUBLICATION REQUEST
UNDER
35 U.S.C. § 122(b)(2)(B)(i)**

I hereby certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral agreement, that requires publication at eighteen months after filing.

I hereby request that the attached application not be published under 35 USC 122(b).



Bruce S. Itchkawitz
April 13, 2009

This request is being signed in compliance with 37 CFR 1.33(b) and submitted with the application **upon filing**.

Applicant understands that this nonpublication request may be rescinded at any time. If applicant rescinds a request that an application not be published under 35 USC 122(b), the application will be scheduled for publication at eighteen months from the earliest claimed filing date for which a benefit is claimed.

Applicant also understands that if applicant subsequently files an application directed to the invention disclosed in the attached application in another country, or under a multilateral international agreement, that requires publication of applications eighteen months after filing, the applicant **must** notify the United States Patent and Trademark Office of such filing within forty-five (45) days after the date of the filing of such foreign or international application. **Failure to do so will result in abandonment of this application (35 USC 122(b)(2)(B)(iii)).**

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045A
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence	US
Citizenship under 37 CFR 1.41(b)		US			
Mailing Address of Applicant:					
Address 1	21 Thalia				
Address 2					
City	Ladera Ranch	State/Province	CA		
Postal Code	92694	Country	US		
Applicant 2					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jayesh	R.	Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cerritos	State/Province	CA	Country of Residence	US
Citizenship under 37 CFR 1.41(b)		US			
Mailing Address of Applicant:					
Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country	US		
Applicant 3					
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
				☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jeffrey	C.	Solomon		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Irvine	State/Province	CA	Country of Residence	US

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045A
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

Citizenship under 37 CFR 1.41(b)		US	
Mailing Address of Applicant:			
Address 1	16 Silver Fir		
Address 2			
City	Irvine	State/Province	CA
Postal Code	92604	Country	US
Applicant 4			
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
		☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name
	Mario	Jesus	Martinez
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service			
City	Laguna Niguel	State/Province	CA
		Country of Residence	US
Citizenship under 37 CFR 1.41(b)		US	
Mailing Address of Applicant:			
Address 1	19 Lindall Street		
Address 2			
City	Laguna Niguel	State/Province	CA
Postal Code	92677	Country	US
Applicant 5			
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117	
		☐ Party of Interest under 35 U.S.C. 118	
Prefix	Given Name	Middle Name	Family Name
	Chi-She		Chen
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service			
City	Walnut	State/Province	CA
		Country of Residence	US
Citizenship under 37 CFR 1.41(b)		TW	
Mailing Address of Applicant:			
Address 1	944 Crystal Water Lane		
Address 2			
City	Walnut	State/Province	CA
Postal Code	91789	Country	US
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.			

Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	20995

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045A
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
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Application Information:

Title of the Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
Attorney Docket Number	NETL.045A	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)		Suggested Figure for Publication (if any)	

Publication Information:

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Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61/044839	2008-04-14
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61/044825	2008-04-14
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045A
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)
	non provisional of	61/044801	2008-04-14
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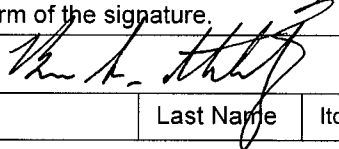
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Signature			Date (YYYY-MM-DD)	2009-04-13	
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045A
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

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CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application claims the benefit of priority from U.S. Provisional Application No. 61/044,839, filed April 14, 2008, from U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and from U.S. Provisional Application No. 61,044,801, filed April 14, 2008, each of which are incorporated in their entirety by reference herein. This application is related to U.S. Patent Application No. _____, filed on even date herewith and entitled “Self-Adjusting Damper”, and to U.S. Application No. _____, filed on even date herewith and entitled “Memory Board with Self-Testing Capability”, both of which are incorporated in their entirety by reference herein.

BACKGROUNDField

[0002] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0003] As the demand on the performance of electronic systems increase, the systems include higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0004] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system 10 which incorporates a multiplexer-demultiplexer 12 to isolate unused subsystems of a plurality of subsystems 16, 18. This arrangement can reduce the load on each driver 14 when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver 14, the receiver 20 and the multiple destinations. However, the multiplexer and/or demultiplexer 12 can present transmission line discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with

electrical lengths longer than 1/4 of the wavelength of the operational frequency or the signal transition rate). In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

[0005] In memory applications, some systems control “chip select” or device select signals, along with the memory address and control signals, based on the device select signals, in order to disable memory devices when they are not being accessed. Figure 2 schematically illustrates an example of such a system 20 which isolates unused subsystems of a plurality of subsystems 16, 18 using device select signals 22, 24 (e.g., chip select signals). The devices are disabled by not activating (asserting) the address, control, and/or device select signals, thus logically isolating the memory devices which are not accessed. This arrangement generally requires a device select signal for each physical memory rank and the address and device select signals are generally controllable based on whether the corresponding memory device or group of memory devices is being accessed. Generating controllable address and device select signals can introduce timing delays on the device select, address, and control paths. In addition, this arrangement does not address the power dissipation and may degrade performance degradation by, for example, requiring that secondary control signals be generated for the memory devices.

SUMMARY

[0006] Certain embodiments described herein include a memory module comprising a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module comprises a plurality of memory devices on the printed circuit board and a circuit. The circuit comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The memory module further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector. The memory module of certain embodiments comprises a switching circuit

configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0007] A method of using a memory module with a computer system is described in certain embodiments. The method includes providing a memory module comprising a printed circuit board having at least one connector configured to be operatively coupled to a memory controller of a computer system and to a plurality of memory devices. The memory module further including a circuit which comprises a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices. The circuit further comprises a second set of ports comprising one or more bi-directional ports. Each port of the second set of ports is operatively coupled to the at least one connector. The circuit is configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports in some embodiments. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit. The correction circuit reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports. The method includes activating the circuit to selectively operatively couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports. The method also includes activating the circuit to reduce noise in the one or more signals.

[0008] Certain embodiments are described herein provide a circuit comprising a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports may be configured to be operatively coupled to at least one memory device of a plurality of memory devices on a printed circuit board. The circuit further comprises a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one connector on the printed circuit board. The at least one connector is configured to be operatively coupled to a memory controller of a

computer system. The circuit further comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduce noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0009] In certain embodiments, a circuit comprises a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports can be configured to be operatively coupled to at least one subsystem of a plurality of subsystems. The circuit may comprise a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one subsystem of one or more subsystems. In certain embodiments, the circuit comprises a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each comprise a correction circuit in certain embodiments. The correction circuit can reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Figure 1 schematically illustrates an example system which isolates unused subsystems and incorporates a multiplexer-demultiplexer.

[0011] Figure 2 schematically illustrates an example system which isolates unused subsystems using device select signals.

[0012] Figure 3 schematically illustrates an example memory system incorporating a memory module in accordance with certain embodiments described herein.

[0013] Figure 4 schematically illustrates an example correction circuit in accordance with certain embodiments described herein.

[0014] Figure 5 schematically illustrates an example programmable impedance matching circuit in accordance with certain embodiments described herein.

[0015] Figure 6 schematically illustrates an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0016] Figure 7 schematically illustrates example configuration including an example filter element connected to an example repeater/amplifier in accordance with certain embodiments described herein.

[0017] Figure 8 schematically illustrates an example system having an example circuit connecting the system with one or more other systems in accordance with certain embodiments described herein.

[0018] Figure 9 schematically illustrates an example system including an example circuit in accordance with certain embodiments described herein.

[0019] Figure 10 is a flowchart of an example method 400 of using a memory module with a computer system in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0020] Electronic systems incorporating devices and methods in accordance with embodiments described herein may include memory subsystems that contain multiple memory boards or modules that are driven by a single memory controller which synchronizes memory access and manages memory coherency. However, having a single memory controller may gradually degrade system performance and increase power consumption as the physical and the logical size of the memory subsystem increases. Increases in memory speed may further increase the power consumption. Additionally, impedance mismatches between the system board, the memory boards, and the memory devices may also substantially degrade the performance and power efficiency of traditional memory systems. Such memory systems may be used with a power supply which helps to stabilize the voltage level, and/or a thermo-mechanical component to moderate memory subsystem temperature, and/or software for throttling the processor to lower the memory access rate.

[0021] Embodiments described herein address these speed and power issues in memory systems. For example, a circuit of certain embodiments can comprise an isolation switch and can logically isolate inactive subsystems and reduce signal reflection and/or other noise at, for example, the interface between the system board and the memory board and at the interface between the memory board and the memory devices.

[0022] Certain embodiments described herein present a circuit for isolating inactive sub-systems and maintaining and improving signal integrity in a variety of systems. Certain embodiments described herein are used to increase the performance of electronic systems without increasing the power dissipation or the memory access time. In certain embodiments, a plurality of circuits are configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits and a plurality of corresponding devices (e.g., memory devices) of a system or subsystem (e.g., a memory module of a computer system). For example, the data width of each of the plurality of circuits may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example.

[0023] Certain embodiments described herein describe a circuit for use in high performance electronic systems which may contain transmission lines and relatively heavy fan-outs. A circuit is described herein which, in certain embodiments, includes at least some of the following components: a signal router, a programmable coarse impedance matching circuit, a self-adjusting damper circuit, a filter and/or a repeater/amplifier.

[0024] Figure 3 schematically illustrates an example system incorporating a memory module 100 in accordance with certain embodiments described herein. The memory module 100 comprises a printed circuit board (PCB) 102. The PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108. The memory module 100 includes a plurality of memory devices 110 on the PCB 102.

[0025] The memory module 100 of certain embodiments further includes a circuit 112 comprising a first set of ports 122 comprising a plurality of bi-directional ports 124, each port 124 of the first set of ports 122 operatively coupled to at least one memory device 110 of the plurality of memory devices 110. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports 128, each port 128 of the second set of ports 126 operatively coupled to the at least one connector 104. The circuit 112 further comprises a switching sub-circuit 140 configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one

or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises a correction circuit 150 (see, e.g., Figure 4) which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0026] In certain embodiments, the memory module 100 has a memory capacity of 512-MB, 1-GB, 2-GB, 4-GB, 8-GB, or higher or lower. Other memory capacities are also compatible with certain embodiments described herein. In addition, memory modules 100 having widths of 4 bits, 8 bits, 16 bits, 32 bits, as well as other widths (in bytes or in bits), are compatible with embodiments described herein. In certain embodiments, the PCB 102 has an industry-standard form factor. For example, the PCB 102 can have a low profile (LP) form factor with a height of 30 millimeters and a width of 133.35 millimeters. In certain other embodiments, the PCB 102 has a very high profile (VHP) form factor with a height of 50 millimeters or more. In certain other embodiments, the PCB 102 has a very low profile (VLP) form factor with a height of 18.3 millimeters. Other form factors including, but not limited to, small-outline (SO-DIMM), unbuffered (UDIMM), registered (RDIMM), fully-buffered (FBDIMM), mini-DIMM, mini-RDIMM, VLP mini-DIMM, micro-DIMM, and SRAM DIMM are also compatible with certain embodiments described herein. For example, in other embodiments, certain non-DIMM form factors are possible such as, for example, single in-line memory module (SIMM), multi-media card (MMC), and small computer system interface (SCSI).

[0027] In certain embodiments, the plurality of memory devices 110 of the memory module 100 may be arranged as one or more ranks, each rank of memory generally having a bit width. In certain embodiments, each rank may comprise an independent set of memory devices 111 of the plurality of memory devices 110 that can be accessed by the memory controller 106 to access the full bit-width of the memory bus of the memory module 100. For example, a memory module 100 in which each rank of the memory module is 64 bits wide is described as having an “x 64” organization. Similarly, a memory module 100 having 72-bit-wide ranks is described as having an “x 72” organization. The number of memory devices 111 of a memory module 100 can be increased by increasing the number of memory devices 111 per rank or by increasing the number of ranks of the memory module

100. For example, a memory module with four ranks with each rank having N 512-MB memory devices 111 has double the memory capacity of a memory module with two ranks with each rank having N 512-MB memory devices 111 and four times the memory capacity of a memory module with one rank with each rank having N 512-MB memory devices.

[0028] During operation, one or more ranks of a memory module 100 may be selected or activated from the plurality of ranks by control signals that are received from a component of the system (e.g., a system memory controller 106 or a local memory controller of the memory module 100). Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. In certain other embodiments, the memory module 100 comprises only one rank of memory devices 111.

[0029] As discussed, the PCB 102 includes at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108, such as a host computer system 108. For example, the connector 104 is electrically coupled, logically coupled, or both, with the memory controller 106 in some other manner. In certain embodiments, the connector 104 comprises a wireless radio frequency (RF) interface or a portion thereof and is wirelessly connected (e.g., via one or more RF antennas) to the memory controller 106. In some embodiments, some other connection mechanism is used besides the connector 104. Examples of host computer systems 108 include, but are not limited to, blade servers, 1U servers, personal computers (PCs), data storage systems and other applications in which space is constrained or limited. The memory controller 106 may comprise a disk controller of the computer system 108, for example. The memory controller 106 of certain embodiments may be mounted on a system board 109 of the host computer 108. The connector 104 can comprise a plurality of edge connections distributed along one or more edges of the PCB 102 which fit into a corresponding slot connector of the host system. The connector 104 of certain embodiments provides a conduit for power voltage as well as data, address, and control signals between the memory module 100 and the host system 108. For example, the connector 104 can comprise a standard 240-pin DDR2 edge connector. Additionally, in certain embodiments, more than one memory module 100 is coupled to the host system 108.

[0030] The plurality of memory devices 110 on the PCB 102 may include one or more volatile memory components. For example, the plurality of memory devices 110 of certain embodiments comprises two or more dynamic random-access memory (DRAM) elements 111. For example, in the block diagram of Figure 3, the plurality of memory devices 110 comprises eight DRAM devices 111. Types of DRAM devices 111 compatible with certain embodiments described herein include, but are not limited to, DDR, DDR2, DDR3, and synchronous DRAM (SDRAM). The memory devices 111 may comprise other types of memory elements such as static random-access memory (SRAM). In addition, volatile memory devices 111 having bit widths of 4, 8, 16, 32, as well as other bit widths, are compatible with certain embodiments described herein. Memory devices 111 compatible with certain embodiments described herein have packaging which include, but are not limited to, thin small-outline package (TSOP), ball-grid-array (BGA), fine-pitch BGA (FBGA), micro-BGA (μ BGA), mini-BGA (mBGA), and chip-scale packaging (CSP).

[0031] The first set of ports 122 of the circuit 112 comprises a plurality of bi-directional ports 124 where each port of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. In addition, the second set of ports 126 of the connector 104 comprises one or more bi-directional ports 128 where each port of the second set of ports 126 is operatively coupled to the at least one connector 104. The switching sub-circuit 140 is configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The operative coupling of the first set of ports 122 to the memory devices 110, the second set of ports 126 to the connector 104, and the selective operative coupling between the first and second set of ports 122, 126 may comprise electrical coupling, logical coupling, or both. In other embodiments, optical coupling or some other manner of coupling may be employed.

[0032] In certain embodiments, each bi-directional port 128 of the second set of ports 126 may correspond to a signal of a memory bus 130 which operatively couples the memory controller 106 and the memory module 100 via the connector 104. For example, the memory controller 106 may generally read data from and/or write data to the plurality of memory devices 110 on the memory module 100 through the memory bus 130 and through

the bi-directional ports 124, 128. In certain embodiments, the number of bi-directional ports 128 of the second set of ports 126 may correspond to the number of data bits in the memory bus 130 (i.e., the bit-width of the memory bus 130).

[0033] In certain embodiments, each of the bi-directional ports 124 in the first set of ports 122 may correspond to and be operatively coupled to a signal port of one of the memory devices 111 of the plurality of memory devices 110. The bi-directional ports 124 correspond to the data signal ports, for example, of the plurality of memory devices 110. Because each memory device 111 includes associated data signal ports, the number of bi-directional ports 124 in the first set of ports of certain embodiments may be larger than the number of bi-directional ports 128 in the second set of ports 122 (e.g., the number of bits in the memory bus 130). In general, the overall number of ports of the plurality of memory devices 110 and corresponding bi-directional ports 124 in the first set of ports 126 may depend on the organization of the memory module. For example, the number of ports may depend on the number of memory devices and/or the number of ranks in the memory module. In certain embodiments, the memory module 100 includes a plurality of circuits 112. The circuits 112 may be configured so as to improve the interconnection (e.g., timing, power consumption, and/or signal integrity) between each of the plurality of circuits 112 and a plurality of corresponding memory devices 111 of memory module 100. For example, the data width of each of the plurality of circuits 112 may be selected to optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In various embodiments, the data width of each of the plurality of devices may be 4, 8, or 16 bits, for example. Other data widths are possible in other embodiments. In some embodiments, the data width of each circuit 112 corresponds to the data width of each of the memory devices 111 such that each circuit 112 is operatively coupled to one of the plurality of memory devices 110. In other embodiments, the data width of each circuit 112 corresponds to a multiple of the data width of each memory device 111 such that each circuit 112 may be operatively coupled to the data bits of more than one memory device 111. In various embodiments, the data width of each circuit 112 corresponds to a segment of the data width of each memory device 111 or to segments of multiple memory devices 111. As such, in some embodiments, each circuit 112 may be operatively coupled to a segment of the data bits

of one memory device 111 or to segments of more than one memory device 111. Other configurations are possible which optimize the interconnection (e.g., timing, power consumption, and/or signal integrity). In addition, in certain embodiments, the location of each of the plurality of circuits 112 on the PCB 102 is selected to optimize the interconnection, timing, power consumption, and signal integrity.

[0034] In other embodiments, the bi-directional ports 124 correspond to one or more other bits of the memory bus 130 instead of, or in addition to, the data bits. For example, one or more of the ports 124 may correspond to address, chip select, control (e.g., read/write) signals. Moreover, in some embodiments, one or more ports of the first set of ports 122 is not bi-directional. For example, in one embodiment, signals corresponding to address, chip select and/or control signals from the first set of ports 122 to the plurality of memory devices 110 and not vice versa. For example, one or more of the address, chip select, and control signals may be used to control the routing of the bi-directional ports 124 which correspond to data bits.

[0035] As discussed, the switching sub-circuit 140 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the switching sub-circuit 112 may be capable of interconnecting any number of the second set of ports 126 to any number of the first set of ports 122 simultaneously. In certain embodiments, the switching sub-circuit 140 can pass signals (e.g., information such as the data signals described above) from one port 128 of the second set of ports 126 to an operatively coupled port 124 of the first set of ports 122 and vice versa. Additionally, the switching sub-circuit 140 of some embodiments can pass signals from one port 128 of the second set of ports 126 to multiple operatively coupled ports 124 of the first set of ports 122 and vice versa. In certain embodiments, the switching sub-circuit 140 isolates and provides a known static value to all un-driven ports, thereby reducing any dynamic power consumption associated with floating or toggling electrical nodes.

[0036] The switching sub-circuit 140 of certain embodiments can function as an isolation switch which isolates and/or inactivates individual subsystems not actively transferring information. For example, in an example embodiment, the plurality of memory

devices 110 comprises multiple independently accessible ranks. When the memory controller accesses one of the ranks of the memory module 100, the switching sub-circuit 140 isolates and/or inactivates at least the one or more other, un-accessed ranks. For example, the switching sub-circuit 140 does not operatively couple ports from the second of set of ports 126 to the first set of ports 122 which are coupled to the memory devices 111 of the un-accessed one or more ranks of memory devices 110. In certain embodiments, the switching sub-circuit 140 may function as or be characterized as a scalable isolation switch router which routes the information from one subsystem to another (e.g., host computer 108 to the memory module 100 via the memory controller 106), establishing a one-to-one connection for transferring data from point-to-point, and/or establishing a one-to-many connection for broadcasting data. The switching sub-circuit 140 comprises or is characterized as a bi-directional multiplexer-demultiplexer in certain embodiments. In other embodiments, the switching sub-circuit 140 comprises or is characterized as a router.

[0037] In certain embodiments, the memory module 100 further includes circuitry (not shown) for controlling the operation of the circuit 112 and the components therein (e.g., the switching sub-circuit 140 and the correction circuit 150, as described more fully below). For example, in one embodiment, the memory module 100 comprises a local memory controller which is operatively coupled to one or more control inputs (not shown) of the circuit 112 to control the selective coupling of the second set of ports 126 to the first set of ports 122. In certain other embodiments, some other component of the memory module 100 controls the operation of the circuit 112 or the circuit 112 comprises internal control circuitry. In various embodiments, the circuitry for controlling the circuit 112 includes discrete logic, one or more microprocessors, a field-programmable gate array (FPGA), or a computer-programmable logic device (CPLD). In yet other configurations, an external component controls the operation of the circuit 112.

[0038] The circuit 112 of certain embodiments comprises one physically integral component (e.g., one integrated circuit package mounted on the memory module 100). Various alternative configurations for the physical organization of the circuit 112 are possible, however. In certain other embodiments, the circuit 112 comprises multiple, physically separate components. For example, in one embodiment, the circuit 112 comprises

one physically separate component per memory device 111 (e.g., integrate circuit packages mounted on the memory module 100). In another embodiment, the circuit 112 comprises one physically separate component per memory rank. In some embodiments, the circuit 112 comprises one physically separate component corresponding to multiple memory devices 111.

[0039] Figure 4 schematically illustrates an example correction circuit 150 in accordance with certain embodiments described herein. As shown, in certain embodiments, the correction circuit 150 comprises one or more correction elements (e.g., at least one coarse correction element 152 to provide coarse correction of the noise and at least one fine correction element 154 to provide fine correction of the noise). The correction circuit 150 of certain embodiments further comprises at least one filter element 156. In certain embodiments, the correction circuit 150 further comprises at least one repeater/amplifier element 158 which improves the signal quality of the one or more signals.

[0040] Referring to Figures 3 and 4, the correction circuit 150 reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. For example, the correction circuit 150 may reduce noise due to impedance mismatches (e.g., due to discontinuity regions) at nodes between the memory controller 106 and the plurality of memory devices 110. In certain embodiments, the noise may comprise signal reflections. For example, in certain embodiments, the signal paths between the host computer 108 and the plurality of memory devices 110 or portions thereof may be characterized as transmission lines if they are greater than or about equal to a particular fraction of the wavelength of the signal, such as, for example, greater than or equal to about 1/4 of the wavelength of the signal. In other configurations, the signal paths may be characterized as transmission lines when the signal path is greater than or equal to some other fraction of the wavelength of the signal such as, for example, 1/2, 1/8, or 1/10 of the wavelength. In certain embodiments, the system (e.g., memory module 100) incorporating the circuit 112 has signal delays which are related to signal characteristics such as the signal transition rate and/or the rise/fall time of the signals. For example, the system incorporating the circuit 112 may include signal delays which are more than some fraction (e.g., 1/2) of the signal transition rate and/or the rise/fall time of the

signals. In other embodiments, the signal delays are related in some other way to the signal transition rate and/or the rise/fall time of the signals (e.g., are more than 1/4, 1/8 of those values) or are related to some other signal property.

[0041] The reflections may be caused by one or more discontinuity regions along the signal paths. For example, discontinuity regions comprise points or regions at which the characteristic impedance of the corresponding signal path changes. For example, the interface between the system board 109 of the host computer 108 and the memory module 100 (e.g., the connector 104) may comprise a discontinuity region along each of the signal paths. The interfaces between the connector 104 of the memory module 100 and each of the memory devices 111 mounted on the PCB 102 may comprise one or more discontinuity regions along each of the signal paths.

[0042] The noise may comprise cross-coupling between the signal paths in certain embodiments. For example, a cross-coupling impedance may be present between one or more of the signal paths or portions thereof. In certain embodiments, the cross-coupling impedance may include cross-coupling due to cross-coupling capacitance, inductance (e.g., mutual inductance), transconductance, or a combination thereof. Additionally, a variety of other noise are possible such as, for example, electromagnetic interference or other types of noise.

[0043] Referring to Figure 4, in certain embodiments, the correction circuit 150 includes one or more elements to provide for coarse correction of signal noise due to impedance mismatches (e.g., discontinuity regions) in the system. The at least one coarse correction element 152 comprises at least one programmable impedance matching circuit in certain embodiments. Figure 5 schematically illustrates an example programmable impedance matching circuit 160 in accordance with certain embodiments described herein. For example, the programmable impedance matching circuit 160 may comprises at least one programmable resistor 162. The resistance of the programmable resistor 162 may be controllable by a control signal 164. In certain embodiments, the coarse correction element 152 may be integrated with the switching sub-circuit 140, the fine correction element 154, the filter element 156, and/or the repeater/amplifier element 158. Alternatively, in other embodiments, the coarse correction element 152 may be implemented with discrete components.

[0044] In other embodiments, the coarse correction element 152 comprises a network of resistors 162 which can be configured to provide programmable impedance matching. In some embodiments, the coarse correction element 152 may comprise a set of programmable coarse impedance matching circuits 160 that are connected to each of the ports 124, 128 of the first set of ports 122 and the ports 128 of the second set of ports 126. Advantageously, the impedance values of the coarse correction element 152 may be dynamically controllable, configurable, and/or programmable, so as to match the characteristic impedances of the transmission lines (e.g., signal paths) in the system. The values of these impedances may differ (e.g., may be programmed differently) based on which particular signal path is activated (e.g., by the switching sub-circuit 140). Where the coarse correction element 152 comprises at least one impedance matching circuit, the granularities of the configurable impedance values may be configured based on the application of the particular system. In some embodiments, the impedance matching circuit of the coarse correction element 152 may include a real time controllable impedance control input for pre-programming and/or dynamically controlling the impedance matching circuit. In this manner, the impedance matching circuit 152 may, in certain embodiments, provide dynamically controllable variable impedance which can be pre-set for matching the impedance of the subsystem that drives the circuit 112.

[0045] In certain embodiments, the at least one fine correction element 154 of the correction circuit 150 further reduces signal reflections due to impedance mismatches and/or reduces other noise. In certain embodiments, for example, the coarse correction element 152 may be characterized as a first refining stage and the fine correction element 154 may be referred to as a second refining stage. The fine correction element 154 may smooth out the signal after it has undergone coarse correction (e.g., impedance matching) and reduces or removes transmission line characteristics from the received signal. Referring again to Figure 4, the fine correction element 154 can be connected, for example, in parallel with the programmable coarse correction element 152. In some embodiments, the fine correction element 154 operates to reduce the impedance mismatches by introducing a current source that damps to the slew rate of the input signal, reducing signal reflection. The fine correction element 154 actively counteracts the noise in certain embodiments. In addition, the fine

correction element 154 of certain embodiments is dynamically responsive to one or more changes in the noise, such as, for example, changes in the frequency of the noise component.

[0046] The fine correction element 154 may include at least one damper circuit. The damper circuit includes at least one self-adjusting damper circuit 170 in certain embodiments, such as the example self-adjusting damper circuit 170 schematically illustrated by Figure 6. In certain embodiments, the self-adjusting damper circuit 170 comprises an input terminal 172 adapted to receive an input signal 174 comprising an input noise component and an input signal component. The damper circuit 170 further comprises an output terminal 176. In certain embodiments, the damper circuit 170 includes a first sub-circuit 184 adapted to generate a first signal 187 responsive to at least a portion of the noise component. For example, the first sub-circuit 184 may generate the first signal 187 in response to the portion of the noise component not reduced by the coarse correction element 152 in certain embodiments. The damper circuit 170 of certain embodiments further comprises a second sub-circuit 188 adapted to generate a second signal 190 corresponding to a delayed version of at least a portion of the input signal. The damper circuit 170 combines the first signal 187 with the second signal 190 to generate an output signal 108 on the output terminal 176 wherein the output signal comprises an output noise component smaller than the input noise component. For example, the first signal 187 and the second signal 190 may be combined at the node 182 of the damper circuit 170. The output signal 176 of certain embodiments is substantially representative of the signal component of the input signal 174. The first sub-circuit 184 of certain embodiments comprises one or more current sources 185, 186. For example, the current sources 185, 186 may each comprise a controlled current source such as a voltage-controlled current source 185, 186 in certain embodiments. In some embodiments, the first sub-circuit 184 may comprise or be characterized as a negative feedback current source. In various embodiments, the self-adjusting damper circuit 170 includes an enable control input (not shown) for enabling the self-adjusting damper circuit 170. Examples of self-adjusting damper circuits 170 in accordance with certain embodiments described herein are described in U.S. Provisional Application No. No. 61/044,825, filed April 14, 2008, and co-pending U.S. Application No. _____, filed by the assignee of the present

application on the same day as the present application, and entitled "Self-Adjusting Damper." Both applications are incorporated in their entirety by reference herein.

[0047] In certain embodiments, the at least one filter element 156 and the at least one repeater/amplifier element 158 may also reduce noise from other sources such as, for example, signal coupling (e.g., cross-coupling). The filter element 156 may reduce noise from the one or more signals where the noise includes a predetermined range of frequencies. For example, the filter element 156 may include at least one band-pass filter and may reduce any low and/or high frequency noise components. In other embodiments, the filter element 156 comprises at least one other type of a filter such as a notch filter, a low-pass filter, or a high-pass filter.

[0048] One or more of the filter element 156 and the repeater/amplifier element 158 of certain embodiments enhance signal slew rate to reduce the transient power dissipation at target devices or components (e.g., one or more of the memory devices 110). Figure 7 schematically illustrates example configuration including an example filter element 156 connected to an example repeater/amplifier element 158 in accordance with certain embodiments described herein. The repeater/amplifier element 158 includes an output enable control input 159 for controlling the bi-directional signal path. In various embodiments one or more of the filter element 156 and repeater/amplifier element 158 are programmable and/or dynamically controllable. The repeater/amplifier element may further include a reference input 157. In certain embodiments, the repeater/amplifier element 158 may include one or more operational amplifiers, for example. Referring again to Figure 4, in certain embodiments, only filter element 156, only the repeater, only the amplifier, or any combination of the three elements may be included in the circuit.

[0049] The filter element 156 and/or the repeater/amplifier element 158 of certain embodiments provide wave reshaping functions and generally reduces signal noise. For example, the filter element 156 and/or the repeater/amplifier element 158 may reduce signal ringing due to reflection (e.g., reflections at one or more interfaces along the signal paths such as interfaces between the circuit 112, the connector 104, the memory devices 110 and/or the system board 109 of Figure 3). In certain embodiments, the repeater/amplifier element 158 reshapes the signal which may be deteriorated due to transmission over lossy

transmission lines having discontinuity regions. The repeater/amplifier element 158 may be configured to repeat the input signal, generating an output signal with earlier transition trigger points and faster slew rates in order to regain speed and to reduce the transient power.

[0050] In certain embodiments, the circuit 112 or portions thereof (e.g., the switching sub-circuit 140, the correction circuit 150 or elements thereof, etc.) is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD). For example, in certain embodiments, the circuit 112 or portions of thereof is implemented in one or more of a PLD, ASIC, FPGA, or CPLD. In other embodiments, the circuit 112 or portions thereof is implemented using some other technology, such as, for example, using discrete components.

[0051] In various embodiments, the elements of the circuit 112 such as the switching sub-circuit 140, the coarse correction element 152, the fine correction element 154, the filter element 156, and the repeater/amplifier element 158, comprise or may be characterized as independent, fully controllable functional blocks. Additionally, in certain embodiments, any subset of one or more of these functional blocks may be implemented in order to enhance performance, reduce power dissipation, or provide other advantages.

[0052] Figure 8 schematically illustrates an example system 200 having an example circuit 112 connecting a system 200 with one or more other systems 220 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 8, for example. The system 200 of Figure 8 includes a driver 202, a receiver 201, and two subsystems 203, 204. For illustration purposes, the system 200 of Figure 8 is described as an open system that communicates with other systems 220 via the driver 202 and the receiver 201. Furthermore, in some embodiments, a connector (not shown) connects the system 200 with other systems by means of transmission lines. For example, the system 200 may be a memory system and the receiver 201 and the driver 202 of Figure 8 may represent memory signals (e.g., data, address and/or control signals) transmitted between the system 220 (e.g., a memory controller and the subsystems 203, 204 (e.g., two sets of independently addressable memory devices such as DRAM devices)). In certain embodiments, for example, the circuit

112 may be disposed on a memory board that is populated with sets of memory devices 203, 204.

[0053] In various embodiments, the receiver 201 may or may not contain any active circuits. For example, the receiver 201 may be a connector such as a socket or DIMM interface with a system board (e.g., the connector 104 described above in regards to Figure 3). The output signal from the receiver 201 may exhibit the characteristics of the signal on a transmission line such as, for example, wave dispersion, attenuation, and reflection. In certain embodiments, the output signal from the receiver 201 goes through a number of refining, filtering and/or amplification stages in circuit 112 as described herein to improve the performance of the overall system. For example, the signal quality is enhanced by introducing at least one circuit 112 including at least one correction circuit 150 comprising one or more coarse correction elements 152 (e.g., impedance matching circuits), fine correction elements 154 (e.g., self-adjusting dampers), filter elements 156 (e.g., bandpass filters) and/or repeater/amplifier elements 158 as described herein on the signal paths (e.g., from receiver 101 to subsystem 103). For example, the circuit 112 may reduce power consumption by selectively isolating the inactive (or un-accessed) subsystems and may enhance the access rate of a targeted subsystem. For example, in one embodiment, the system 200 comprises a memory module 100 such as one of the memory modules described herein, the subsystems 203 and 204 comprise memory devices such as the memory devices 111 described herein, and the other system 220 comprises a host computer 108 including a memory controller 106 as described herein.

[0054] In certain embodiments, the performance is enhanced and power dissipation is reduced by removing the load associated with other (e.g., unused or inactive) subsystems (e.g., the subsystem 204) and reshaping the signal waveform using a filter element 156 and/or the repeater/amplifier element 158. In some embodiments, there are two refining stages for reducing signal reflections due to impedance mismatches. In some embodiments, for example, the first refining stage is the coarse correction element 152 (e.g., an impedance matching circuit) that matches the impedance of the transmission line (e.g., wire) prior to a receiver 101 with the characteristic impedance of the transmission line (e.g., wire) in the system 100. The second stage may be the fine correction element 154 (e.g., a self-

adjusting damper for further reducing signal reflections due to impedance mismatches and/or reducing other noise.

[0055] Figure 9 schematically illustrates an example system 300 including an example circuit 112 in accordance with certain embodiments described herein. The circuit 112 may be any of the circuits 112 discussed herein, such as the circuit 112 discussed above with respect to Figure 3, for example. The circuit 112 comprises a first set of ports 122 comprising a plurality of bi-directional ports, each port of the first set of ports 122 configured to be operatively coupled to at least one subsystem 311 of a plurality of subsystems 310. The circuit 112 further comprises a second set of ports 126 comprising one or more bi-directional ports, each port of the second set of ports 126 configured to be operatively coupled to at least one subsystem 321 of one or more subsystems 320. In certain embodiments, the circuit 112 is configured to selectively operatively couple one or more ports of the second set of ports 126 to one or more ports of the first set of ports 122. In certain embodiments, the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126 each comprises a correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports 122 and the one or more ports of the second set of ports 126. The correction circuit 150 of certain embodiments may comprise one of the correction circuits 150 described herein such as the correction circuits 150 described above with respect to Figures 4-7 and may include at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158, or any combination or sub-combination thereof. In other embodiments, the fine correction element 154 may be placed after the switching sub-circuit 140 in order to achieve better results. For example, the fine correction element 154 may be connected after the filter element 156 (e.g., between the filter 156 and the repeater/amplifier element 158) in certain embodiments.

[0056] In certain embodiments, the system 300 includes a first set of interfaces 330 positioned between (e.g., logically and/or physically between) and operatively coupled to the first set of ports 122 and to the plurality of subsystems 310. In addition, the system 300 of certain embodiments may include a second set of interfaces 340 positioned between (e.g., logically and/or physically between) and operatively coupled to the second set of ports 126

and the one or more subsystems 320. For example, the first and/or second sets of interfaces 330, 340 may comprise port connections.

[0057] In an example configuration, the plurality of subsystems 310 include a plurality of memory devices 311 (e.g., DRAM devices) mounted on a memory module and the first set of interfaces 330 comprise interfaces between the circuit 112 and the plurality of memory devices 311. For example, each of the first set of ports 122 may correspond to a memory signal (e.g., a data, address, and/or control signal) and each of the first set of interfaces 330 is operatively coupled to a corresponding memory signal port of a corresponding memory device 311. The one or more subsystems 320 of the example configuration comprise a host computer system and the second set of interfaces 340 comprises memory board (e.g., a PCB of a memory module) interfaces to the host system. For example, the second set of interfaces 340 form part of a connector. Each of the second set of ports 126 and corresponding interfaces of the second set of interfaces 340 corresponds to a memory signal (e.g., a data, address, and/or control signal) of at least one memory bus between the host system and the memory module.

[0058] The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 126 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the system board and the memory board. The coarse correction elements 152 (e.g., impedance matching circuits) and the fine correction elements (e.g., self-adjusting dampers) of the second set of ports 122 of the example configuration are configured to reduce the signal reflection due to the impedance mismatching between the circuit 112, the memory board and/or the plurality of memory devices. The filter elements 156 (e.g., a bandpass filter) reduce or filter out cross-coupling between signals, for example. For example, the filter elements 158 reduce or filter out cross-coupling between data, address, and/or control signals (e.g., wires).

[0059] Figure 10 is a flowchart of an example method 400 of using a memory module 100 with a computer system in accordance with certain embodiments described herein. While the method 400 is described herein by reference to the memory module 100 schematically illustrated by Figure 3, other memory modules, electronic systems or

subsystems, and/or circuits are also compatible with the embodiments described herein. In certain embodiments, at operational block 402, the method 400 comprises providing a memory module 100 comprising a printed circuit board 102 having at least one connector 104 configured to be operatively coupled to a memory controller 106 of a computer system 108 and to a plurality of memory devices 110. The memory module 100 further comprises a circuit 112 including a first set of ports 122 comprising a plurality of bi-directional ports 124. In certain embodiments, each port 124 of the first set of ports 122 is operatively coupled to at least one memory device 111 of the plurality of memory devices 110. The circuit of certain embodiments further includes a second set of ports 126 comprising one or more bi-directional ports 128 where each port 128 of the second set of ports 126 is operatively coupled to the at least one connector 104.

[0060] The circuit 112 of certain embodiments is configured to selectively operatively couple one or more ports 128 of the second set of ports 126 to one or more ports 124 of the first set of ports 122. The one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126 each comprises at least one correction circuit 150 which reduces noise in one or more signals transmitted between the one or more ports 124 of the first set of ports 122 and the one or more ports 128 of the second set of ports 126.

[0061] The method 400 of certain embodiments further includes activating the circuit 112 at operational block 404 to selectively operatively couple at least one of the one or more ports 128 of the second set of ports 126 to at least one of the one or more ports 124 of the first set of ports 122. For example, in certain embodiments, activating the circuit 112 to perform the selective operative coupling may include sending the appropriate control signal or combination of control signals to a switching sub-circuit 140 of the circuit 112. In certain embodiments, the method 400 further includes activating the circuit 112 to reduce noise in the one or more signals at operational block 406. For example, in one embodiment, activating the circuit 112 to reduce noise comprises activating one or more of the elements of the correction circuit 150 (e.g., at least one coarse correction element 152, at least one fine correction element 154, at least one filter element 156, and at least one repeater/amplifier element 158) of ports 124, 128 of the first and second sets of ports 122, 126 corresponding to

active signal paths between the memory controller 106 and the memory devices 110. Activating one or more of the elements 152, 154, 156, 158 of the correction circuit 150 may include sending control signals (e.g., enable signals) to the appropriate elements 152, 154, 156, 158 and/or receiving the control signals (e.g., enable signals) at the appropriate elements 152, 154, 156, 158.

[0062] The operational blocks 402, 404, 406 of the method 400 may be performed in a different order than provided or one or more of the operational blocks may be performed substantially at the same time or during overlapping time periods. For example, in certain embodiments, at least the operational block 404 and the operational block 406 are performed substantially simultaneously. In addition, in some embodiments, one or more of the operational blocks 402, 404, 406 may not be included, an additional operational block may be included, or both. The devices and methods described herein may be incorporated in any system (e.g. a board level system) or device that contains one or more connectors or wires.

[0063] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory module comprising:
 - a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board;
 - a circuit, comprising:
 - a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and
 - a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and
 - a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.
2. The memory module of Claim 1, wherein the noise comprises signal reflections.
3. The memory module of Claim 1, wherein the correction circuit comprises at least one coarse correction element to provide coarse correction of the noise.
4. The memory module of Claim 3, wherein the coarse correction element comprises at least one programmable impedance matching circuit.
5. The memory module of Claim 3, wherein the correction circuit further comprises at least one fine correction element to provide fine correction of the noise.
6. The memory module of Claim 5, wherein the fine correction element comprises at least one damper circuit.
7. The memory module of Claim 6, wherein the damper circuit comprises at least one self-adjusting damper circuit.

8. The circuit of Claim 5, wherein the fine correction element actively counteracts the noise.

9. The circuit of Claim 5, wherein the fine correction circuit is dynamically responsive to one or more changes in the noise.

10. The circuit of Claim 9, wherein the fine correction circuit is dynamically responsive to changes in the frequency of the noise component.

11. The memory module of Claim 6, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one filter element which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

12. The memory module of Claim 11, wherein the filter element comprises at least one band-pass filter.

13. The memory module of Claim 11, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one repeater/amplifier element which improves the signal quality of the one or more signals.

14. The memory module of Claim 1, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

15. The memory module of Claim 1, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

16. A method of using a memory module with a computer system, the method comprising:

providing a memory module comprising a printed circuit board having at least one connector configured to be operatively coupled to a memory controller of a computer system and to a plurality of memory devices, the memory module further comprising a circuit, the circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

wherein the circuit is configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports; and

activating the circuit to selectively operatively couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports; and

activating the circuit to reduce noise in the one or more signals.

17. The method of Claim 16, wherein the noise comprises signal reflections.

18. The method of Claim 16, wherein the correction circuit comprises at least one coarse correction element to provide coarse correction of the noise.

19. The method of Claim 18, wherein the correction circuit further comprises at least one fine correction element to provide fine correction of the noise.

20. The circuit of Claim 16, wherein the fine correction element is dynamically responsive to one or more changes in the noise.

21. The method of Claim 16, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one filter element which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

22. The method of Claim 21, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one repeater/amplifier element which improves the signal quality of the one or more signals.

23. A circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports configured to be operatively coupled to at least one subsystem of a plurality of subsystems; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one subsystem of one or more subsystems;

a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

ABSTRACT

Certain embodiments described herein include a memory module having a printed circuit board including at least one connector configured to be operatively coupled to a memory controller of a computer system. The memory module further includes a plurality of memory devices on the printed circuit board and a circuit including a first set of ports operatively coupled to at least one memory device. The circuit further includes a second set of ports operatively coupled to the at least one connector. The circuit includes a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports. Each port of the first set and the second set comprises a correction circuit which reduces noise in one or more signals transmitted between the first set of ports and the second set of ports.

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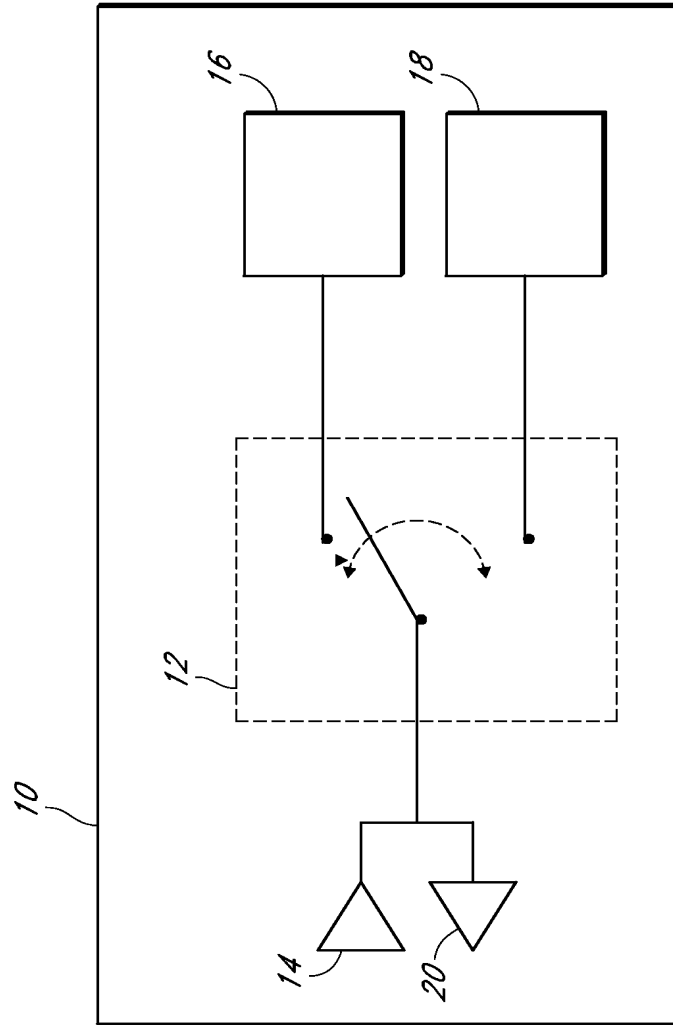


FIG. 1

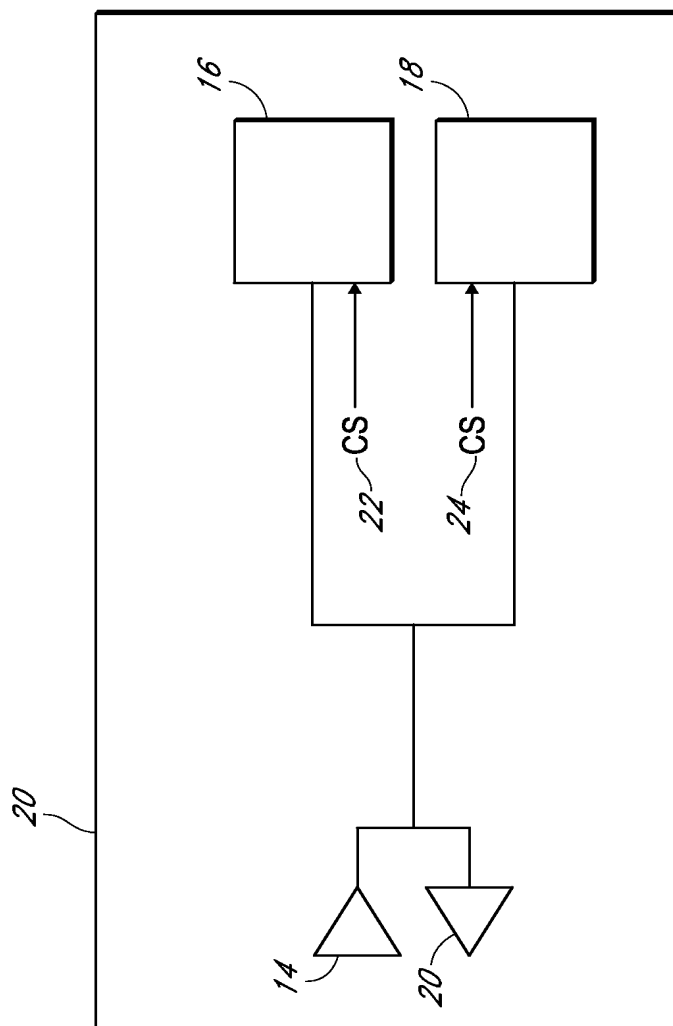


FIG. 2

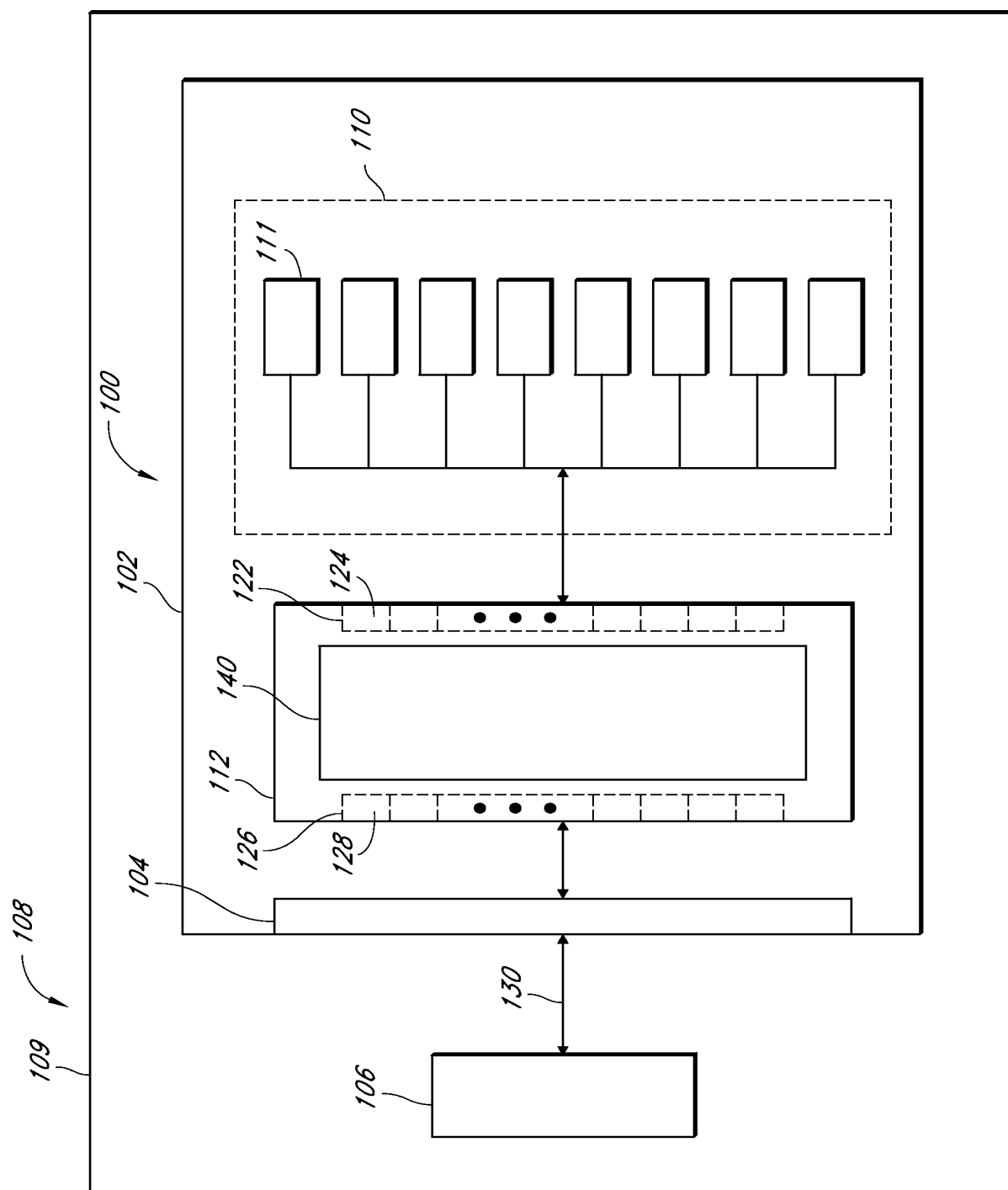


FIG. 3

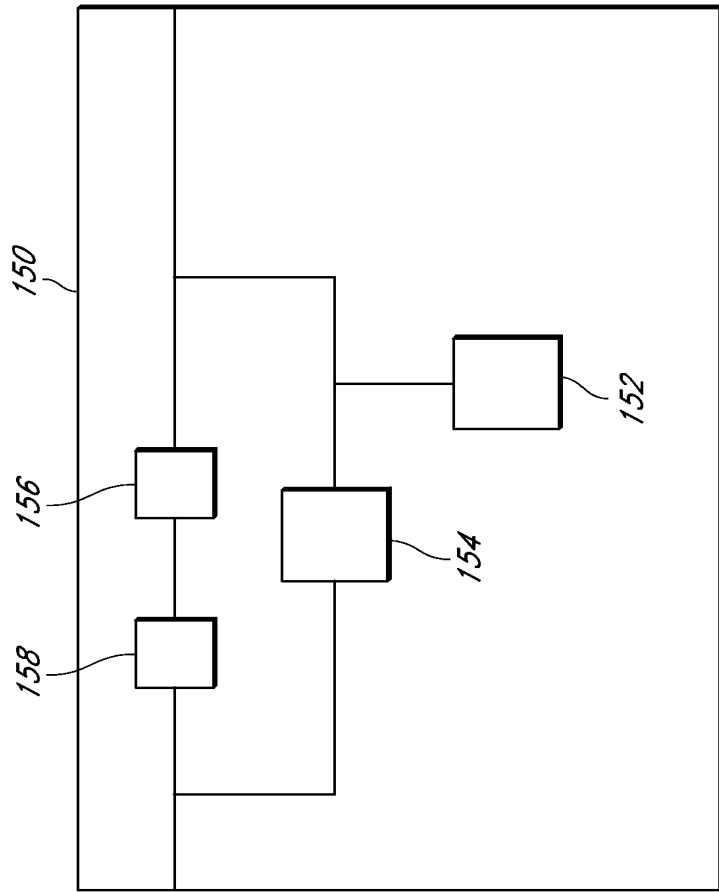


FIG. 4

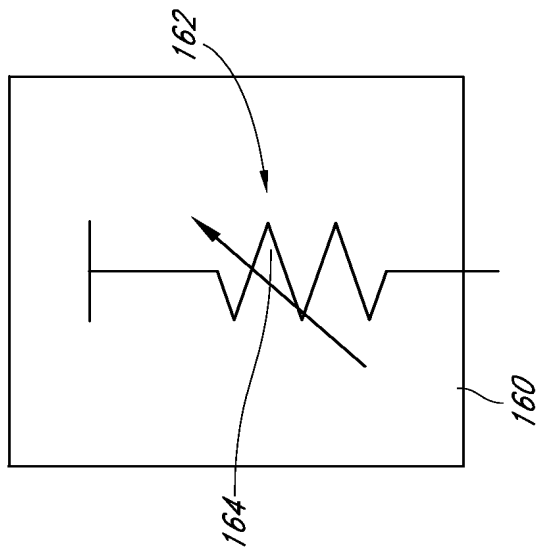


FIG. 5

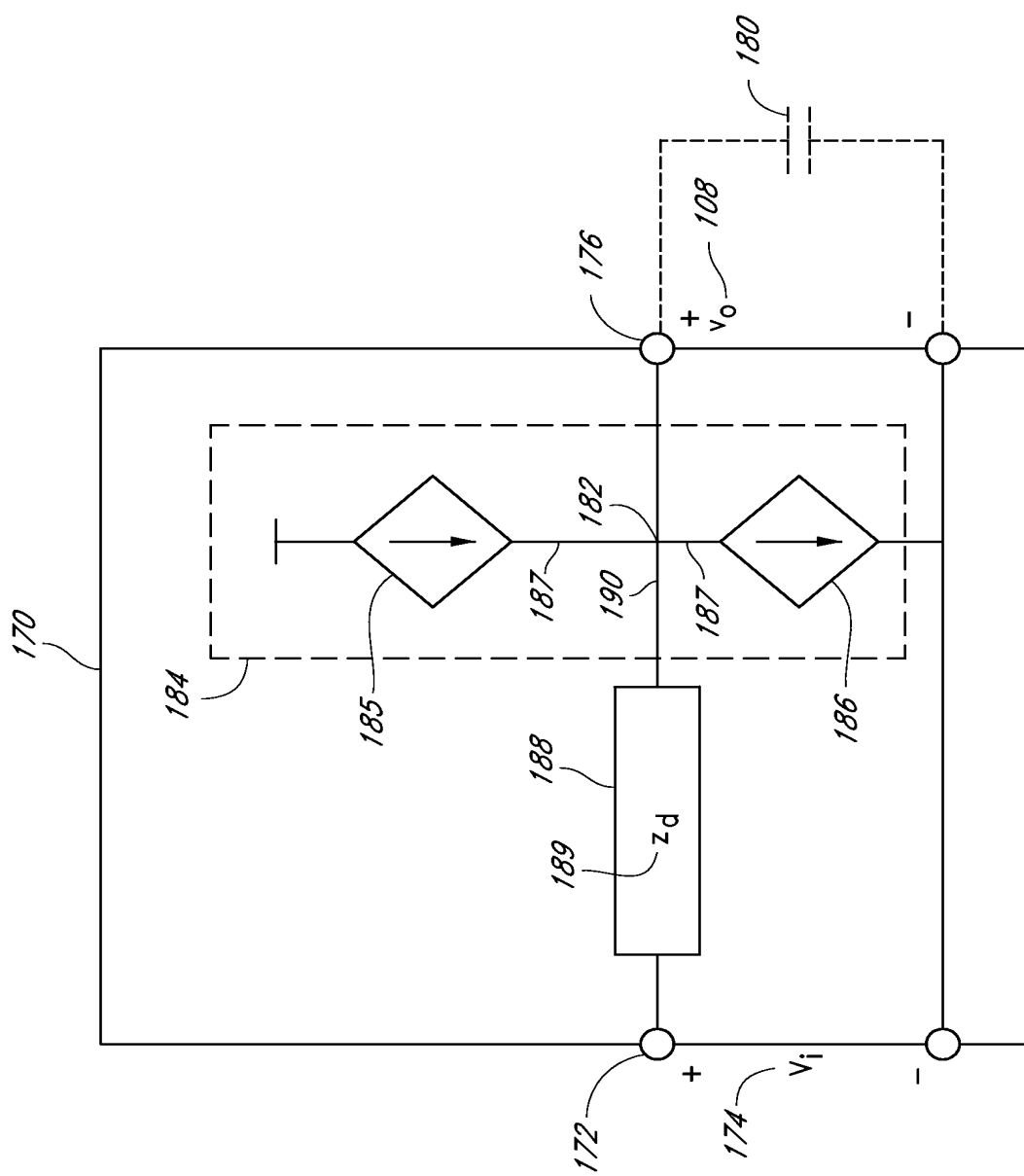


FIG. 6

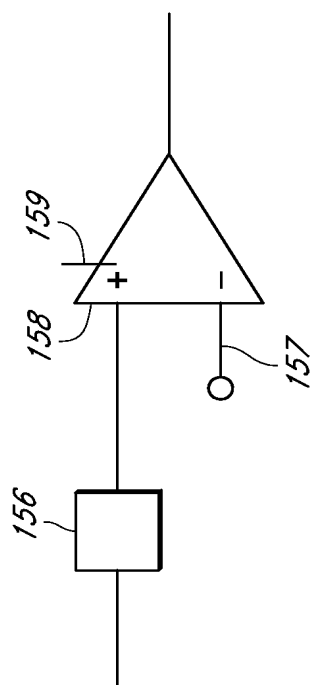


FIG. 7

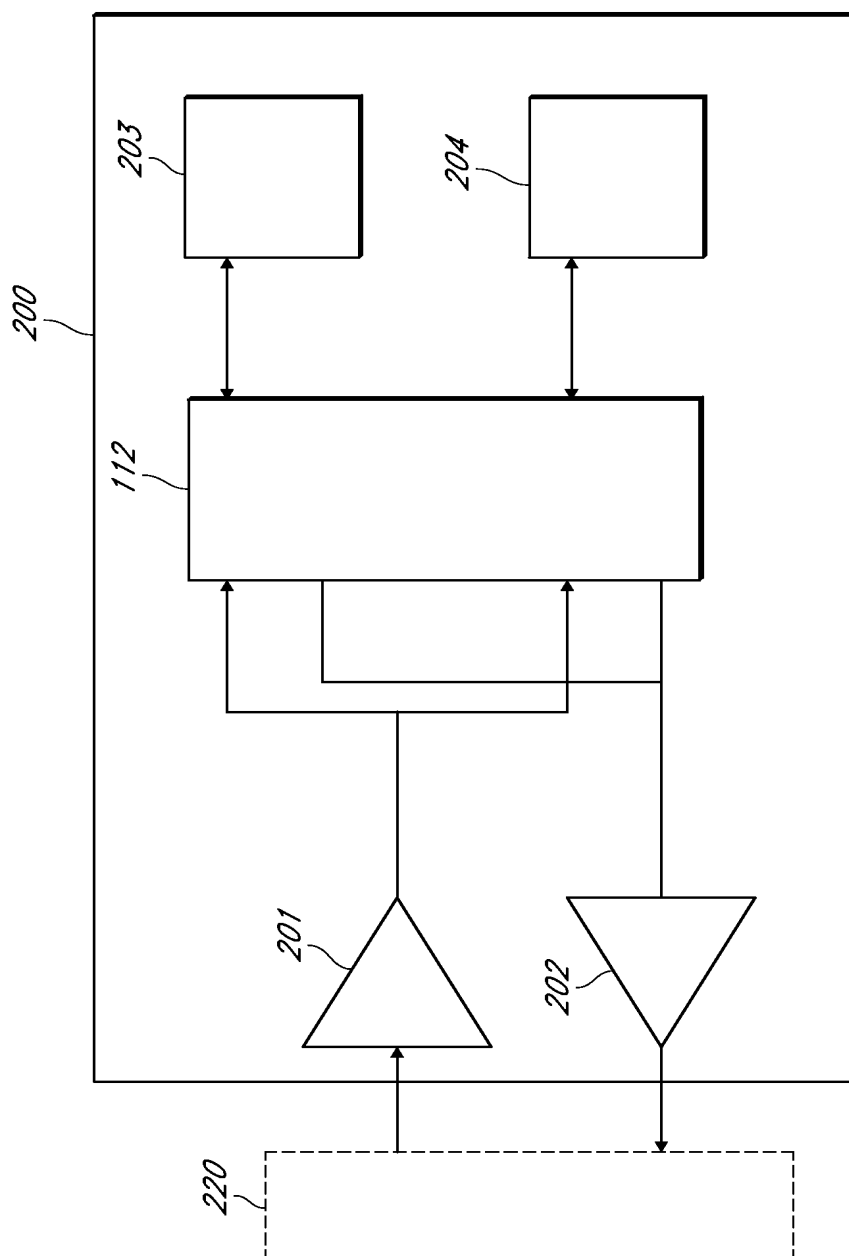


FIG. 8

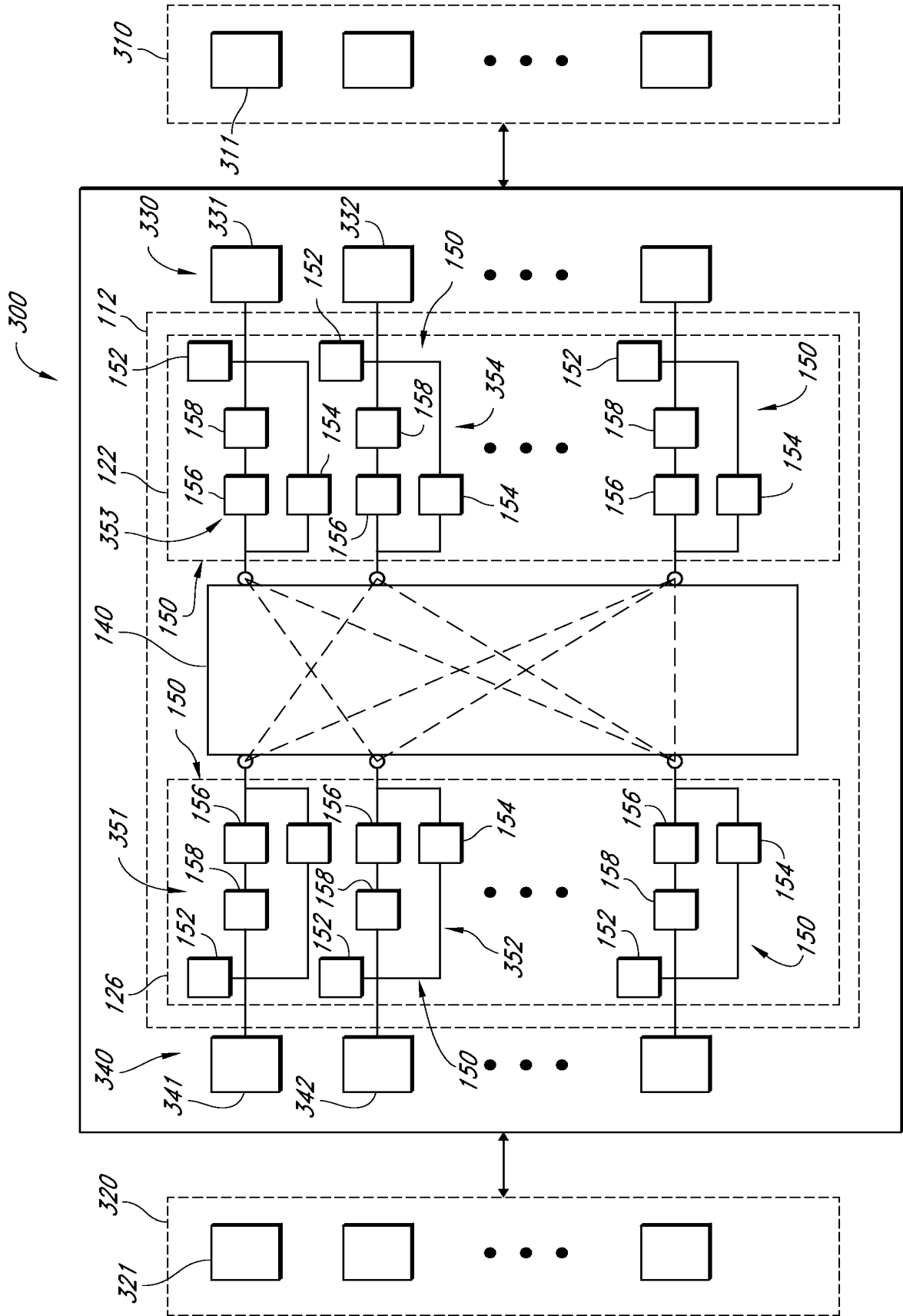


FIG. 9

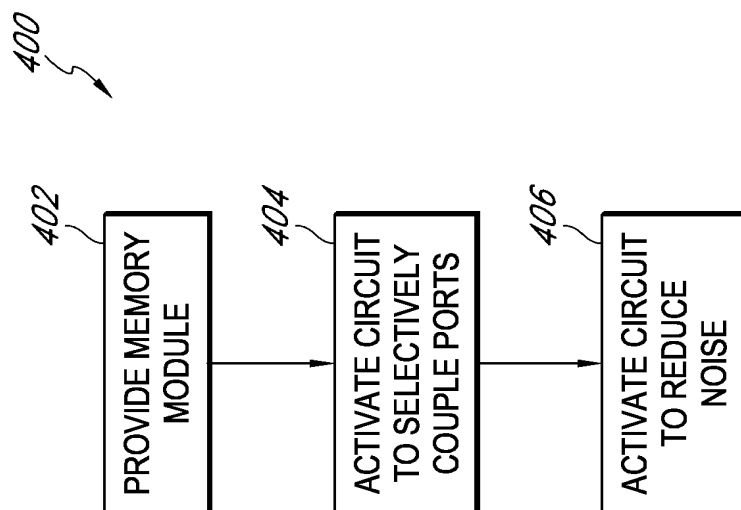


FIG. 10

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

First Named Inventor/Applicant Name:

Hyun Lee

Filer:

Bruce S. Itchkawitz/Jodi Marasia

Attorney Docket Number:

NETL.045A

Filed as Large Entity

Utility under 35 USC 111(a) Filing Fees

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Utility application filing	1011	1	330	330
Utility Search Fee	1111	1	540	540
Utility Examination Fee	1311	1	220	220

Pages:

Claims:

Claims in excess of 20	1202	3	52	156
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Miscellaneous-Filing:

Petition:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1246

Electronic Acknowledgement Receipt

EFS ID:	5142926
Application Number:	12422853
International Application Number:	
Confirmation Number:	7481
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Lori Larson
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045A
Receipt Date:	13-APR-2009
Filing Date:	
Time Stamp:	19:11:24
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1246
RAM confirmation Number	4719
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	Transmittal-NETL045A.pdf	95920 14d8e040ce3bf3b1c3b471f027d661d718a96311	no	2
Warnings:					
Information:					
2	Application Data Sheet	ADS-NETL045A.pdf	357945 9ce0d4780c1090f3feeb03a5cb6724ef0dbcc786a	no	6
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
3		Specification-NETL045A.pdf	1723925 dc362ce7bdeb687bce45dce17df42f80b5b0c12e	yes	28
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	23	
	Claims		24	27	
	Abstract		28	28	
Warnings:					
Information:					
4	Drawings-only black and white line drawings	Drawings-NETL045A.pdf	112756 1d8af39f910ce8d3831a8cf652f285190f232b2f	no	9
Warnings:					
Information:					
5	Fee Worksheet (PTO-06)	fee-info.pdf	36689 182e216c4614bfb250dc905048fef54b56a30e4	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			2327235		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

UTILITY APPLICATION	Attorney Docket No.: NETL.045A
	First Named Inventor: Hyun Lee et al.
	Title: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Direct all correspondence to Customer No.: 20995	
Date: April 13, 2009 Page 1	

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

The following enclosures are transmitted herewith to be filed in the patent application of:

Inventor(s):

- | | |
|-----------------------|-------------------------|
| 1. Hyun Lee | 2. Jayesh R. Bhakta |
| 3. Jeffrey C. Solomon | 4. Mario Jesus Martinez |
| 5. Chi-She Chen | |

APPLICATION:

- (X) Specification in 28 pages.
(X) Drawings in 9 sheets.

CONTINUITY INFORMATION:

Application	Relationship	Parent App. No.	Filing Date	Status
This Application	Non-Provisional of	61/044839	04/14/2008	Pending
This Application	Non-Provisional of	61/044825	04/14/2008	Pending
This Application	Non-Provisional of	61/044801	04/14/2008	Pending

Reference to prior domestic applications is made in the:

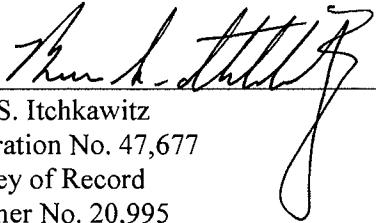
- (X) Specification.
(X) Enclosed Application Data Sheet (ADS).

FILING FEES:

FEE CALCULATION				
FEE TYPE		FEE CODE	CALCULATION	TOTAL
Basic Utility	1.16(a)(1)	1011 (\$330)		\$330
Search Fee	1.16(k)	1111 (\$540)		\$540
Examination Fee	1.16(o)	1311 (\$220)		\$220
Excess Claims > 20	23 - 20 = 3	1202 (\$52)	3 x 52 =	\$156
			TOTAL FEE DUE	\$1246

UTILITY APPLICATION	Attorney Docket No.: NETL.045A
	First Named Inventor: Hyun Lee et al. Title: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Direct all correspondence to Customer No.: 20995	
Date: April 13, 2009 Page 2	

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment to Account No. 11-1410.

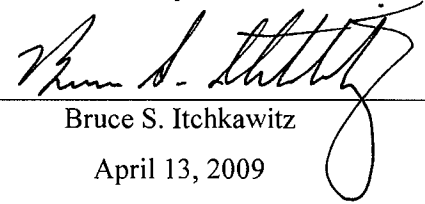


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

**NONPUBLICATION REQUEST
UNDER
35 U.S.C. § 122(b)(2)(B)(i)**

I hereby certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral agreement, that requires publication at eighteen months after filing.

I hereby request that the attached application not be published under 35 USC 122(b).



Bruce S. Itchkawitz
April 13, 2009

This request is being signed in compliance with 37 CFR 1.33(b) and submitted with the application **upon filing**.

Applicant understands that this nonpublication request may be rescinded at any time. If applicant rescinds a request that an application not be published under 35 USC 122(b), the application will be scheduled for publication at eighteen months from the earliest claimed filing date for which a benefit is claimed.

Applicant also understands that if applicant subsequently files an application directed to the invention disclosed in the attached application in another country, or under a multilateral international agreement, that requires publication of applications eighteen months after filing, the applicant **must** notify the United States Patent and Trademark Office of such filing within forty-five (45) days after the date of the filing of such foreign or international application. **Failure to do so will result in abandonment of this application (35 USC 122(b)(2)(B)(iii)).**

Filing Date: 04/13/09

Approved for use through 7/31/2006. OMB 0651-0032

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/422,853	
APPLICATION AS FILED – PART I						
(Column 1)		(Column 2)		SMALL ENTITY		OR
OTHER THAN SMALL ENTITY						
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	RATE (\$)	FEE (\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A		N/A	330
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A		N/A	540
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A		N/A	220
TOTAL CLAIMS (37 CFR 1.16(j))	23	minus 20 = 3	x\$26		x\$52	156
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3	minus 3 = *	x\$110		x\$220	
APPLICATION SIZE FEE (37 CFR 1.16(s))			If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$270 (\$135 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR			
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))			195		390	
			TOTAL		TOTAL	1246
* If the difference in column 1 is less than zero, enter "0" in column 2.						
APPLICATION AS AMENDED – PART II						
(Column 1)		(Column 2)		(Column 3)		
SMALL ENTITY		OTHER THAN SMALL ENTITY				
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)
Total (37 CFR 1.16(i))	*	Minus **	=	X =		X =
Independent (37 CFR 1.16(h))	*	Minus ***	=	X =		X =
Application Size Fee (37 CFR 1.16(s))						
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				N/A		N/A
				TOTAL ADD'T FEE		TOTAL ADD'T FEE
(Column 1)		(Column 2)		(Column 3)		
SMALL ENTITY		OTHER THAN SMALL ENTITY				
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	RATE (\$)
Total (37 CFR 1.16(i))	*	Minus **	=	X =		X =
Independent (37 CFR 1.16(h))	*	Minus ***	=	X =		X =
Application Size Fee (37 CFR 1.16(s))						
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				N/A		N/A
				TOTAL ADD'T FEE		TOTAL ADD'T FEE
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/422,853	04/13/2009	Hyun Lee	NETL.045A

CONFIRMATION NO. 7481

FORMALITIES LETTER



OC000000035647574

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/29/2009

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The oath or declaration is missing.

A properly signed oath or declaration in compliance with 37 CFR 1.63, identifying the application by the above Application Number and Filing Date, is required.

Note: If a petition under 37 CFR 1.47 is being filed, an oath or declaration in compliance with 37 CFR 1.63 signed by all available joint inventors, or if no inventor is available by a party with sufficient proprietary interest, is required.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee, search fee, examination fee or oath or declaration) as set forth in 37 CFR 1.16(f) of \$130 for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is \$130 for a non-small entity

- \$130 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/bnguyen/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/422,853	04/13/2009	2827	1246	NETL.045A	23	3

CONFIRMATION NO. 7481

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



OC000000035647573

Date Mailed: 04/29/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Assignment For Published Patent Application

NETLIST, INC, Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications

If Required, Foreign Filing License Granted: 04/23/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/422,853**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

365

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 1 OF 4	Attorney Docket No.	NETL.045A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
	2	3,757,235	09-04-1973	McCormick et al.	
	3	4,305,091	12-08-1981	Cooper	
	4	4,586,168	04-29-1986	Adlhoch, et al.	
	5	4,701,845	10-20-1987	Andreasen et al.	
	6	4,752,741	06-21-1988	Kim et al.	
	7	4,782,487	11-01-1988	Smelser	
	8	4,837,743	06-06-1989	Chiu et al.	
	9	4,885,799	12-05-1989	Van Horn	
	10	4,903,266	02-20-1990	Hack	
	11	4,910,597	03-20-1990	Harada et al.	
	12	4,942,556	07-17-1990	Sasaki et el.	
	13	4,987,321	01-22-1991	Toohy	
	14	5,138,619	08-11-1992	Fasang et al.	
	15	5,173,906	12-22-1992	Dreibelbis et al.	
	16	5,222,066	06-22-1993	Grula et al.	
	17	5,241,503	08-31-1993	Cheng	
	18	5,304,856	04-19-1994	Rainal	
	19	5,337,254	08-09-1994	Knee et al.	
	20	5,359,235	10-25-1994	Coyle et al.	
	21	5,394,037	02-28-1995	Josephson et al.	
	22	5,430,335	07-04-1995	Tanoi	
	23	5,525,917	06-11-1996	Wong et al.	
	24	5,841,296	11-24-1998	Churcher et al.	
	25	5,914,543	06-22-1999	Scherpenberg, et al.	
	26	6,000,048	12-07-1999	Krishna et al.	
	27	6,044,481	03-28-2000	Kornachuk et al.	
	28	6,070,217	05-30-2000	Connolly et al.	
	29	6,169,696	01-02-2001	Bissey	

Examiner Signature

Date Considered

***Examiner:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 2 OF 4	Attorney Docket No.	NETL.045A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	30	6,194,959	02-27-2001	Kamoshida et al.	
	31	6,216,240	04-10-2001	Won et al.	
	32	6,467,056	10-15-2002	Satou et al.	
	33	6,560,740	05-06-2003	Zuraski, Jr. et al.	
	34	6,614,702	09-02-2003	Lee	
	35	6,721,150	04-13-2004	Guerrero, Jr.	
	36	6,812,869	11-02-2004	Rahman et al.	
	37	6,829,728	12-07-2004	Cheng et al.	
	38	6,918,072	07-12-2005	Cowels et al.	
	39	6,928,024	08-09-2005	Pfeiffer et al.	
	40	6,928,593	08-09-2005	Halbert et al.	
	41	6,930,509	08-16-2005	Banik	
	42	6,934,900	08-23-2005	Cheng et al.	
	43	7,036,064	04-25-2006	Kebichi et al.	
	44	7,053,470	05-30-2006	Sellers et al.	
	45	7,062,696	06-13-2006	Barry et al.	
	46	7,184,915	02-27-2007	Hansquine et al.	
	47	7,190,210	03-13-2007	Azrai et al.	
	48	7,203,873	04-10-2007	Adams	
	49	7,253,652	08-07-2007	Azimi et al.	
	50	7,284,166	10-16-2007	Zappa et al.	
	51	2001/0048342	12-06-2001	Yoshida et al.	
	52	2002/0000847	01-03-2002	Taguchi	
	53	2002/0131535	09-19-2002	Huber	
	54	2002/0140523	10-03-2002	Park et al.	
	55	2003/0098742	05-29-2003	Nakagawa et al.	
	56	2003/0126346	07-03-2003	Kuo	
	57	2003/0197797	10-23-2003	Segura	
	58	2003/0218491	11-27-2003	Nagasue	

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 3 OF 4	Attorney Docket No.	NETL.045A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	59	2004/0155702	08-12-2004	Danielsson	
	60	2004/0199843	10-07-2004	Hansquaine et al.	
	61	2005/0093620	05-05-2005	Ho et al.	
	62	2005/0127989	06-16-2005	Miyagi	
	63	2005/0289423	12-29-2005	Yabuta	
	64	2006/0082383	04-20-2006	Choi	
	65	2006/0107156	05-18-2006	Lee et al.	
	66	2006/0140015	06-26-2006	Kasamsetty	
	67	2006/0144015	07-06-2006	Cash et al.	
	68	2006/0147217	07-06-2006	Hahin et al.	
	69	2006/0192653	08-31-2006	Atkinson et al.	
	70	2006/0242458	10-26-2006	Feldman et al.	
	71	2006/0262586	11-23-2006	Solomon et al.	
	72	2006/0271748	11-30-2006	Jain et al.	
	73	2007/0030814	02-08-2007	Shin et al.	
	74	2007/0058471	03-15-2007	Rajan et al.	
	75	2007/0079199	04-05-2007	Chorn et al.	
	76	2007/0109707	05-17-2007	Honda	
	77	2007/0152743	07-05-2007	Keeth et al.	
	78	2007/0204075	08-30-2007	Rajan et al.	
	79	2007/0223296	09-27-2007	Miller et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹

Examiner Signature	Date Considered
--------------------	-----------------

***Examiner:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

T¹ - Place a check mark in this area when an English language Translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
(Multiple sheets used when necessary)	Examiner	Unknown
SHEET 4 OF 4	Attorney Docket No.	NETL.045A

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
	80	DER-CHANG ET AL. "A parallel built-in self-diagnostic method for embedded memory arrays", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems, April 2002, Volume 21, Issue 4, Pages 449-465	
	81	BAYARD "On the LTI properties of adaptive feed forward systems with tap delay-line regressors", IEEE Transactions on Signal Processing, May 1999, Volume 47, Issue 5, pages 1288-1296	
	82	MUTOH et al. "EMI Noise controlling methods suitable for electric vehicle drive systems", Industrial Electronics Society, 2004, IECON 2004, November 2-6, 2004, Volume 1, pages 963-968	
	83	SEKIGUCHI et al "Low-noise, high-speed data transmission using a ringing-canceling output buffer", IEEE Journal of Solid-State Circuits, Dec. 1995, Volume 30, Issue 12, pages 1569-1574	

7333908
061909

Examiner Signature	Date Considered
*Examiner: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.	

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Electronic Acknowledgement Receipt

EFS ID:	5671085
Application Number:	12422853
International Application Number:	
Confirmation Number:	7481
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Chelsea Pearsall
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045A
Receipt Date:	09-JUL-2009
Filing Date:	13-APR-2009
Time Stamp:	14:41:48
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		IDS_NETL045A.pdf	284001 14cc28964bfe9939745c1d3671f44b02504a510a	yes	5

	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Transmittal Letter		1	1	
	Information Disclosure Statement (IDS) Filed (SB/08)		2	5	
Warnings:					
Information:					
2	NPL Documents	Der-Chang_2002.PDF	351911	no	17
			9d99ac67a7fedfd832c8a72c41db496210e7045a		
Warnings:					
Information:					
3	NPL Documents	Bayard_1999.PDF	426723	no	9
			2378ecec191f2e9ec715ed2b1ed73de94916c07c		
Warnings:					
Information:					
4	NPL Documents	Mutoh_2004.pdf	529276	no	6
			65af400f07b5c6424cab138fa7ad7fb2b39f946e		
Warnings:					
Information:					
5	NPL Documents	Sekiguchi_1995.pdf	620932	no	6
			cea3ee0409d761ddb879a98ffb38bd9d344b9163		
Warnings:					
Information:					
Total Files Size (in bytes):			2212843		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT

Applicant	: Hyun Lee et al.
App. No.	: 12/422,853
Filed	: April 13, 2009
For	: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
Examiner	: Unknown
Art Unit	: 2827
Conf No.	: 7481

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Enclosed for filing in the above-identified application is a PTO/SB/08 Equivalent listing 83 references to be considered by the Examiner. Also enclosed are 4 non-patent literature as listed on the Information Disclosure Statement.

This Information Disclosure Statement is being filed before the receipt of a first Office Action on the merits, and presumably no fee is required. If a first Office Action on the merits was mailed before the mailing date of this Statement, the Commissioner is authorized to charge the fee set forth in 37 C.F.R. § 1.17(p) to Deposit Account No. 11-1410.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: _____

7/9/09

By: _____

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20995
(949) 760-0404

Please Direct All Correspondence to Customer Number 20995

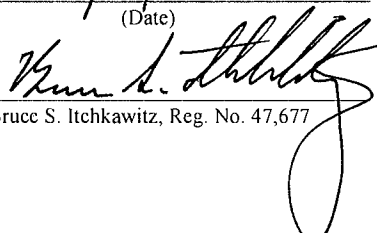
RESPONSE TO FORMALITIES NOTICE

Applicant : Hyun Lee et al.
App. No : 12/422,853
Filed : April 13, 2009
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Art Unit : Unknown

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

7/29/09
(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

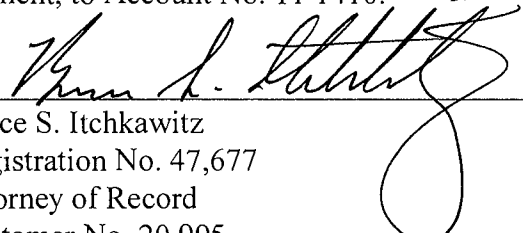
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without a Declaration. Enclosed in compliance with 37 CFR 1.53(f) are the following.

- (X) A Declaration in 6 pages.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404

7072966/Jlm
043009

Doc Code: OATH

Document Description: Oath or declaration filed

Approved for use through 06/30/2010. OMB 0651-0032
U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

DECLARATION FOR UTILITY OR DESIGN PATENT APPLICATION (37 CFR 1.63)	Attorney Docket Number	NETL.045A		
	First Named Inventor	Hyun Lee		
	COMPLETE IF KNOWN			
	Application Number	12/422,853		
	Filing Date	April 13, 2009		
	Art Unit	Unknown		
☐ Declaration Submitted With Initial Filing	OR	☒ Declaration Submitted after Initial Filing (surcharge (37 CFR 1.16 (f)) required)	Examiner Name	Unknown

I hereby declare that: (1) Each inventor's residence, mailing address, and citizenship are as stated below next to their name; and (2) I believe the inventor(s) named below to be the original and first inventor(s) of the subject matter which is claimed and for which a patent is sought on the invention entitled:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

(Title of the Invention)

the application of which

☐ is attached hereto

OR



was filed on (MM/DD/YYYY)

04/13/2009

as United States Application Number or PCT International

Application Number

12/422,853

and was amended on (MM/DD/YYYY)

(if applicable).

I hereby state that I have reviewed and understand the contents of the above identified application, including the claims, as amended by any amendment specifically referred to above.

I acknowledge the duty to disclose information which is material to patentability as defined in 37 CFR 1.56, including for continuation-in-part applications, material information which became available between the filing date of the prior application and the national or PCT international filing date of the continuation-in-part application.

Authorization To Permit Access To Application by Participating Offices

☐ If checked, the undersigned hereby grants the USPTO authority to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), and any other intellectual property offices in which a foreign application claiming priority to the above-identified application is filed access to the above-identified patent application. See 37 CFR 1.14(c) and (h). This box should not be checked if the applicant does not wish the EPO, JPO, KIPO, or other intellectual property office in which a foreign application claiming priority to the above-identified application is filed to have access to the application.

In accordance with 37 CFR 1.14(h)(3), access will be provided to a copy of the application-as-filed with respect to: 1) the above-identified application, 2) any foreign application to which the above-identified application claims priority under 35 USC 119(a)-(d) if a copy of the foreign application that satisfies the certified copy requirement of 37 CFR 1.55 has been filed in the above-identified US application, and 3) any U.S. application from which benefit is sought in the above-identified application.

In accordance with 37 CFR 1.14(c), access may be provided to information concerning the date of filing the Authorization to Permit Access to Application by Participating Offices.

[Page 1 of 3]

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

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DECLARATION — Utility or Design Patent Application

Claim of Foreign Priority Benefits

I hereby claim foreign priority benefits under 35 U.S.C. 119(a)-(d) or (f), or 365(b) of any foreign application(s) for patent, inventor's or plant breeder's rights certificate(s), or 365(a) of any PCT international application which designated at least one country other than the United States of America, listed below and have also identified below, by checking the box, any foreign application for patent, inventor's or plant breeder's rights certificate(s), or any PCT international application having a filing date before that of the application on which priority is claimed.

Prior Foreign Application Number(s)	Country	Foreign Filing Date (MM/DD/YYYY)	Priority Not Claimed	Certified Copy Attached?	
				YES	NO
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐
			☐	☐	☐

☐ Additional foreign application numbers are listed on a supplemental priority data sheet PTO/SB/02B attached hereto.

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DECLARATION — Utility or Design Patent Application

Direct all correspondence to:	☒ The address associated with Customer Number:	20995	OR	☐ Correspondence address below
Name				
Address				
City State			ZIP	
Country T	elephone	Email		
WARNING: Petitioner/applicant is cautioned to avoid submitting personal information in documents filed in a patent application that may contribute to identity theft. Personal information such as social security numbers, bank account numbers, or credit card numbers (other than a check or credit card authorization form PTO-2038 submitted for payment purposes) is never required by the USPTO to support a petition or an application. If this type of personal information is included in documents submitted to the USPTO, petitioners/applicants should consider redacting such personal information from the documents before submitting them to the USPTO. Petitioner/applicant is advised that the record of a patent application is available to the public after publication of the application (unless a non-publication request in compliance with 37 CFR 1.213(a) is made in the application) or issuance of a patent. Furthermore, the record from an abandoned application may also be available to the public if the application is referenced in a published application or an issued patent (see 37 CFR 1.14). Checks and credit card authorization forms PTO-2038 submitted for payment purposes are not retained in the application file and therefore are not publicly available. Petitioner/applicant is advised that documents which form the record of a patent application (such as the PTO/SB/01) are placed into the Privacy Act system of records DEPARTMENT OF COMMERCE, COMMERCE-PAT-7, System name: <i>Patent Application Files</i> . Documents not retained in an application file (such as the PTO-2038) are placed into the Privacy Act system of COMMERCE/PAT-TM-10, System name: <i>Deposit Accounts and Electronic Funds Transfer Profiles</i> .				
I hereby declare that all statements made herein of my own knowledge are true and that all statements made on information and belief are believed to be true; and further that these statements were made with the knowledge that willful false statements and the like so made are punishable by fine or imprisonment, or both, under 18 U.S.C. 1001 and that such willful false statements may jeopardize the validity of the application or any patent issued thereon.				
NAME OF SOLE OR FIRST INVENTOR:		☐ A petition has been filed for this unsigned inventor		
Given Name (first and middle [if any]) Hyun		Family Name or Surname Lee		
Inventor's Signature 		Date 7/8/09		
Residence: City Ladera Ranch	State CA	Country US	Citizenship US	
Mailing Address 21 Thalia				
City State Ladera Ranch	CA	Zip 92694	Country US	
☒ Additional inventors or a legal representative are being named on the 2 supplemental sheet(s) PTO/SB/02A or 02LR attached hereto.				

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DECLARATION**ADDITIONAL INVENTOR(S)**
Supplemental SheetPage 1 of 2

Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Jayesh R.		Bhakta	
Inventor's Signature <i>Jayesh Bhakta</i>		Date <i>7-7-09</i>	
Residence: City	Cerritos	State	CA
Country	US	Citizenship	US
Mailing Address 12220 Rose Street			
City	Cerritos	State	CA
Zip	90703	Country	US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Jeffrey C. <i>Jeffrey C. Solomon</i>		Solomon	
Inventor's Signature		Date	
Residence: City	Irvine	State	CA
Country	US	Citizenship	US
Mailing Address 16 Silver Fir			
City	Irvine	State	CA
Zip	92604	Country	US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Mario Jesus		Martinez	
Inventor's Signature <i>Mario Jesus Martinez</i>		Date <i>7/9/09</i>	
Residence: City	Laguna Niguel	State	CA
Country	US	Citizenship	US
Mailing Address 19 Lindall Street			
City	Laguna Niguel	State	CA
Zip	92677	Country	US

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

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DECLARATION**ADDITIONAL INVENTOR(S)**
Supplemental SheetPage 2 of 2

Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Chi-She		Chen	
Inventor's Signature 		Date 07-07-09	
Residence: City	Walnut	State	CA
		Country	US
		Citizenship	TW
Mailing Address 944 Crystal Water Lane			
City	Walnut	State	CA
		Zip	91789
		Country	US
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Inventor's Signature		Date	
Residence: City		State	
		Country	
		Citizenship	
Mailing Address			
City		State	
		Zip	
		Country	
Name of Additional Joint Inventor, if any:		☐ A petition has been filed for this unsigned inventor	
Given Name (first and middle (if any))		Family Name or Surname	
Inventor's Signature		Date	
Residence: City		State	
		Country	
		Citizenship	
Mailing Address			
City		State	
		Zip	
		Country	

This collection of information is required by 35 U.S.C. 115 and 37 CFR 1.63. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 21 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 (1-800-786-9199) and select option 2.

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Patent Application Fee Transmittal

Application Number:	12422853			
Filing Date:	13-Apr-2009			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Bruce S. Itchkawitz/Jodi Marasia			
Attorney Docket Number:	NETL.045A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late filing fee for oath or declaration	1051	1	130	130
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	130	130
Miscellaneous:				
Total in USD (\$)				260

Electronic Acknowledgement Receipt

EFS ID:	5789128
Application Number:	12422853
International Application Number:	
Confirmation Number:	7481
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Valerie Jones
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045A
Receipt Date:	29-JUL-2009
Filing Date:	13-APR-2009
Time Stamp:	14:59:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 260
RAM confirmation Number	1237
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		DECLARATION_NETL045A.pdf	434901 c1ea7ed9b34f96462423dd8d6e940cd542cf388c	yes	7
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Applicant Response to Pre-Exam Formalities Notice		1	1	
	Oath or Declaration filed		2	7	
Warnings:					
Information:					
2	Fee Worksheet (PTO-875)	fee-info.pdf	32482 9ec17117f183c8a4f515afa03acb2411991d821d	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			467383		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
12/422,853	04/13/2009	2827	1376	NETL.045A	23	3

CONFIRMATION NO. 7481

UPDATED FILING RECEIPT



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 08/07/2009

Receipt is acknowledged of this non-provisional patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Assignment For Published Patent Application

NETLIST, INC, Irvine, CA

Power of Attorney: None

Domestic Priority data as claimed by applicant

This appln claims benefit of 61/044,839 04/14/2008
and claims benefit of 61/044,825 04/14/2008
and claims benefit of 61/044,801 04/14/2008

Foreign Applications

If Required, Foreign Filing License Granted: 04/23/2009

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 12/422,853**

Projected Publication Date: Request for Non-Publication Acknowledged

Non-Publication Request: Yes

Early Publication Request: No

Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

Preliminary Class

365

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicant	:	Lee et al.	Group Art Unit: 2827
Appl. No.	:	12/422,853	
Filed	:	April 13, 2009	
For	:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION	
Examiner	:	Alexander Sofocleous	
Conf. No.	:	7481	

PRELIMINARY AMENDMENT**Mail Stop Amendment**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

Prior to examination on the merits, please amend the above-captioned application as described herein.

Amendments to the Claims: Begin on page 2 of this Preliminary Amendment.

Remarks: Begin on page 7 of this Preliminary Amendment.

Appl. No. : 12/422,853
Filed : April 13, 2009

AMENDMENTS TO THE CLAIMS

Please amend Claims 1, 4, 6, 8-10, 14, 16, 20, and 23, as indicated below.

Please cancel Claims 3, 5, 18, and 19 without prejudice, as indicated below.

Please add Claims 24-35, as indicated below.

A complete listing of all claims is presented below with insertions underlined (e.g., insertion), and deletions struckthrough or in double brackets (e.g., ~~deletion~~ or [[deletion]]):

1. (Currently Amended) A memory module comprising:

a printed circuit board comprising at least one connector configured to be operatively coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board;

a circuit, comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

a switching circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports, wherein the correction circuit comprises at least one coarse correction element to provide coarse correction of the noise and wherein the correction circuit further comprises at least one fine correction element to provide fine correction of the noise.

2. (Original) The memory module of Claim 1, wherein the noise comprises signal reflections.

3. (Canceled)

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Filed : April 13, 2009

4. (Currently Amended) The memory module of Claim [[3]]1, wherein the coarse correction element comprises at least one programmable impedance matching circuit.

5. (Canceled)

6. (Currently Amended) The memory module of Claim [[5]]1, wherein the fine correction element comprises at least one damper circuit.

7. (Original) The memory module of Claim 6, wherein the damper circuit comprises at least one self-adjusting damper circuit.

8. (Currently Amended) The memory module~~element~~ of Claim [[5]]1, wherein the fine correction element actively counteracts the noise.

9. (Currently Amended) The memory module~~element~~ of Claim [[5]]1, wherein the fine correction element~~is~~ is dynamically responsive to one or more changes in the noise.

10. (Currently Amended) The memory module~~element~~ of Claim 9, wherein the fine correction element~~is~~ is dynamically responsive to changes in the frequency of the noise component.

11. (Original) The memory module of Claim 6, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one filter element which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

12. (Original) The memory module of Claim 11, wherein the filter element comprises at least one band-pass filter.

13. (Original) The memory module of Claim 11, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one repeater/amplifier element which improves the signal quality of the one or more signals.

14. (Currently Amended) The memory module of Claim 1, wherein the correction circuits reduce impedance mismatches at nodes between the memory ~~board~~-controller and the plurality of memory devices.

15. (Original) The memory module of Claim 1, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

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Filed : April 13, 2009

16. (Currently Amended) A method of using a memory module with a computer system, the method comprising:

providing a memory module comprising a printed circuit board having at least one connector configured to be operatively coupled to a memory controller of a computer system and to a plurality of memory devices, the memory module further comprising a circuit, the circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports operatively coupled to at least one memory device of the plurality of memory devices; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports operatively coupled to the at least one connector; and

wherein the circuit is configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports, wherein the correction circuit comprises at least one coarse correction element to provide coarse correction of the noise, wherein the correction circuit further comprises at least one fine correction element to provide fine correction of the noise; and

activating the circuit to selectively operatively couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports; and

activating the circuit to reduce noise in the one or more signals.

17. (Original) The method of Claim 16, wherein the noise comprises signal reflections.

18. (Canceled)

19. (Canceled)

20. (Currently Amended) The ~~method~~circuit of Claim 16, wherein the fine correction element is dynamically responsive to one or more changes in the noise.

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Filed : April 13, 2009

21. (Original) The method of Claim 16, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one filter element which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

22. (Original) The method of Claim 21, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises at least one repeater/amplifier element which improves the signal quality of the one or more signals.

23. (Currently Amended) A circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports configured to be operatively coupled to at least one memory subsystem of a plurality of memory subsystems; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be operatively coupled to at least one connector configured to be operatively coupled to a memory controller of a computer systems~~subsystem of one or more subsystems~~;

a switching sub-circuit configured to selectively operatively couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports, wherein the correction circuit comprises at least one coarse correction element to provide coarse correction of the noise, wherein the correction circuit further comprises at least one fine correction element to provide fine correction of the noise.

24. (New) The circuit of Claim 23, wherein the noise comprises signal reflections.

25. (New) The circuit of Claim 23, wherein the coarse correction element comprises at least one programmable impedance matching circuit.

26. (New) The circuit of Claim 23, wherein the fine correction element comprises at least one damper circuit.

27. (New) The circuit of Claim 26, wherein the damper circuit comprises at least one self-adjusting damper circuit.

Appl. No. : 12/422,853
Filed : April 13, 2009

28. (New) The circuit of Claim 23, wherein the fine correction element actively counteracts the noise.

29. (New) The circuit of Claim 23, wherein the fine correction element is dynamically responsive to one or more changes in the noise.

30. (New) The circuit of Claim 29, wherein the fine correction element is dynamically responsive to changes in the frequency of the noise component.

31. (New) The circuit of Claim 23, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprise at least one filter element which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

32. (New) The circuit of Claim 31, wherein the filter element comprises at least one band-pass filter.

33. (New) The circuit of Claim 31, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprise at least one repeater/amplifier element which improves the signal quality of the one or more signals.

34. (New) The circuit of Claim 23, wherein the correction circuit reduces impedance mismatches at nodes between the memory controller and the plurality of memory subsystems.

35. (New) The circuit of Claim 23, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

Appl. No. : 12/422,853
Filed : April 13, 2009

REMARKS

Claims 1, 4, 6, 8-10, 14, 16, 20, and 23 have been amended, Claims 3, 5, 18, and 19 have been canceled without prejudice, and new Claims 24-35 have been added. Applicants believe that none of the foregoing amendments have added new matter to the application.

Summary of Interview

Applicants thank Examiner Alexander Sofocleous for extending the courtesy of conducting a telephonic interview with Applicants' representative, Bruce S. Itchkawitz, on November 29, 2011. During the interview, Claims 5 and 19 were discussed, and agreement was reached that these claims are allowable. Applicants agreed to consider whether to file a preliminary amendment to incorporate the features of Claim 5 into independent Claim 1, to incorporate the features of Claim 19 into independent Claim 16, and to amend independent Claim 23 in a similar manner. The amendments described herein are in accordance with this discussion.

Comments on Amendments

Besides the amendments to Claims 1, 16, and 23, Applicants have amended Claims 4, 6, 8-10, 14, and 20 to provide proper dependency and proper antecedent basis.

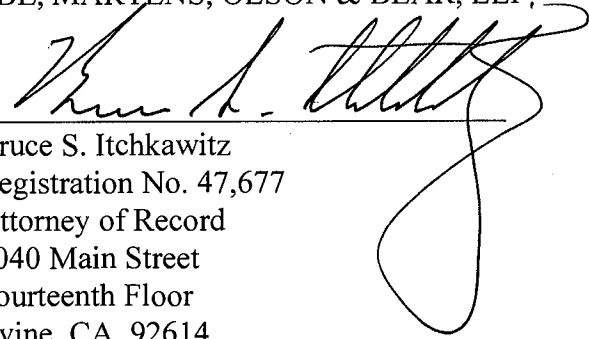
Conclusion

All of the claims currently pending in the present application are believed to be allowable over the prior art, and such action is earnestly solicited. If, however, any matters remain which could be resolved by Examiner's Amendment, the Examiner is cordially invited to contact the undersigned by telephone so that any such matters can be promptly resolved.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP,

Dated: 12/2/11

By: 

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
2040 Main Street
Fourteenth Floor
Irvine, CA 92614
(949) 760-0404

Electronic Patent Application Fee Transmittal

Application Number:	12422853			
Filing Date:	13-Apr-2009			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Bruce S. Itchkawitz			
Attorney Docket Number:	NETL.045A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Claims in excess of 20	1202	8	60	480
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				480

Electronic Acknowledgement Receipt

EFS ID:	11536257
Application Number:	12422853
International Application Number:	
Confirmation Number:	7481
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Garett Gomez
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045A
Receipt Date:	02-DEC-2011
Filing Date:	13-APR-2009
Time Stamp:	20:19:23
Application Type:	Utility under 35 USC 111(a)

Payment information:

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Payment Type	Credit Card
Payment was successfully received in RAM	\$480
RAM confirmation Number	6042
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		amendment.pdf	310777 969489cd558a87e6793a9e6e7751e21246ce00a8	yes	7
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Preliminary Amendment		1	1	
	Claims		2	6	
	Applicant Arguments/Remarks Made in an Amendment		7	7	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	30616 25d940f649152b95ef1efa8e7939646a8bc3be3d	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			341393		
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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875					Application or Docket Number 12/422,853		Filing Date 04/13/2009		☐ To be Mailed	
APPLICATION AS FILED – PART I										
(Column 1)			(Column 2)			SMALL ENTITY ☐ OR		OTHER THAN SMALL ENTITY		
FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)	OR	RATE (\$)	FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A				
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$	=		X \$	=			
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$	=		X \$	=			
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$250 (\$125 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))										
* If the difference in column 1 is less than zero, enter "0" in column 2.										
APPLICATION AS AMENDED – PART II										
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	12/02/2011	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 31	Minus	** 23	= 8	X \$	=	OR	X \$60=	480
	Independent (37 CFR 1.16(h))	* 3	Minus	***3	= 0	X \$	=	OR	X \$250=	0
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	480
(Column 1)			(Column 2)			SMALL ENTITY OR		OTHER THAN SMALL ENTITY		
AMENDMENT	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	OR	RATE (\$)	ADDITIONAL FEE (\$)		
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$	=	OR	X \$	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$	=	OR	X \$	=
	☐ Application Size Fee (37 CFR 1.16(s))									
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))									
						TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3. ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20". *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3". The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.										

Legal Instrument Examiner:
/ANITA JOHNSON/

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/422,853

04/13/2009

Hyun Lee

NETL.045A

7481

20995

7590

12/05/2011

Knobbe Martens Olson & Bear LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

EXAMINER

SOFOCLEOUS, ALEXANDER

ART UNIT

PAPER NUMBER

2824

NOTIFICATION DATE

DELIVERY MODE

12/05/2011

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

jcartee@kmob.com
efiling@kmob.com
eOAPilot@kmob.com

<i>Examiner-Initiated Interview Summary</i>	Application No. 12/422,853	Applicant(s) LEE ET AL.	
	Examiner ALEXANDER SOFOCLEOUS	Art Unit 2824	

All participants (applicant, applicant's representative, PTO personnel):

(1) Alex Sofocleous (PTO personnel). (3) ____.

(2) Bruce Itchkawitz (app's rep). (4) ____.

Date of Interview: 29 November 2011.

Type: ☒ Telephonic ☐ Video Conference
 ☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
 If Yes, brief description: _____.

Issues Discussed ☐101 ☐112 ☐102 ☐103 ☒Others
 (For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 5 and 19.

Identification of prior art discussed: n/a.

Substance of Interview
 (For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

On November 29, 2011, claims 5 and 19 were identified as allowable if incorporated into their independent claims. Also, if similar amendments were made to claim 23, then the application would be in condition for allowance. To expedite application to allowance, it was proposed to make such amendment. At the momen, it was preferred (if such proposed amendment is acceptable) to file a preliminary amendment. It was agreed to consider the amendment and follow-up later.

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

/ALEXANDER SOFOCLEOUS/ Primary Examiner, Art Unit 2824	
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Alexandria, Virginia 22313-1450
www.uspto.gov

NOTICE OF ALLOWANCE AND FEE(S) DUE

20995 7590 12/14/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

EXAMINER

SOFOCLEOUS, ALEXANDER

ART UNIT

PAPER NUMBER

2824

DATE MAILED: 12/14/2011

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/422,853

04/13/2009

Hyun Lee

NETL.045A

7481

TITLE OF INVENTION: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$0	\$0	\$1740	03/14/2012

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the SMALL ENTITY status shown above.

If the SMALL ENTITY is shown as YES, verify your current SMALL ENTITY status:

A. If the status is the same, pay the TOTAL FEE(S) DUE shown above.

B. If the status above is to be removed, check box 5b on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and twice the amount of the ISSUE FEE shown above, or

If the SMALL ENTITY is shown as NO:

A. Pay TOTAL FEE(S) DUE shown above, or

B. If applicant claimed SMALL ENTITY status before, or is now claiming SMALL ENTITY status, check box 5a on Part B - Fee(s) Transmittal and pay the PUBLICATION FEE (if required) and 1/2 the ISSUE FEE shown above.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Utility patents issuing on applications filed on or after Dec. 12, 1980 may require payment of maintenance fees. It is patentee's responsibility to ensure timely payment of maintenance fees when due.

PART B - FEE(S) TRANSMITTAL

**Complete and send this form, together with applicable fee(s), to: Mail Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
or Fax (571)-273-2885**

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

20995 7590 12/14/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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12/422,853 04/13/2009 Hyun Lee NETL.045A 7481

TITLE OF INVENTION: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$0	\$0	\$1740	03/14/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
SOFOCLEOUS, ALEXANDER	2824	365-051000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively, 1 _____
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed. 2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE (B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☐ Issue Fee
☐ Publication Fee (No small entity discount permitted)
☐ Advance Order - # of Copies _____

4b. Payment of Fee(s); (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
☐ Payment by credit card. Form PTO-2038 is attached.
☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number _____ (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____ Date _____

Typed or printed name _____ Registration No. _____

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/422,853	04/13/2009	Hyun Lee	NETL.045A	7481

20995 7590 12/14/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

EXAMINER

SOFOCLEOUS, ALEXANDER

ART UNIT	PAPER NUMBER
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2824

DATE MAILED: 12/14/2011

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment to date is 549 day(s). If the issue fee is paid on the date that is three months after the mailing date of this notice and the patent issues on the Tuesday before the date that is 28 weeks (six and a half months) after the mailing date of this notice, the Patent Term Adjustment will be 549 day(s).

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No.	Applicant(s)	
	12/422,853	LEE ET AL.	
	Examiner	Art Unit	
	ALEXANDER SOFOCLEOUS	2824	

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the Preliminary Amendment filed December 2, 2011.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 1,2,4,6-17 and 20-35.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).
 - a) ☐ All b) ☐ Some* c) ☐ None of the:
 1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. ____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: ____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ A SUBSTITUTE OATH OR DECLARATION must be submitted. Note the attached EXAMINER'S AMENDMENT or NOTICE OF INFORMAL PATENT APPLICATION (PTO-152) which gives reason(s) why the oath or declaration is deficient.
 6. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
 - (a) ☐ including changes required by the Notice of Draftsperson's Patent Drawing Review (PTO-948) attached
 - 1) ☐ hereto or 2) ☐ to Paper No./Mail Date ____.
 - (b) ☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date ____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
7. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|---|
| 1. ☒ Notice of References Cited (PTO-892) | 5. ☐ Notice of Informal Patent Application |
| 2. ☐ Notice of Draftsperson's Patent Drawing Review (PTO-948) | 6. ☐ Interview Summary (PTO-413),
Paper No./Mail Date ____. |
| 3. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date ____ | 7. ☐ Examiner's Amendment/Comment |
| 4. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material | 8. ☒ Examiner's Statement of Reasons for Allowance |
| | 9. ☐ Other ____. |

/ALEXANDER SOFOCLEOUS/
Primary Examiner, Art Unit 2824

DETAILED ACTION

1. This action is responsive to the following communication: Application filed on April 13, 2009, the Information Disclosure Statement filed on July 9, 2009, and the Preliminary Amendment filed December 2, 2011.
2. Claims 1, 2, 4, 6-17, and 20-35 are pending. Claims 3, 5, 18, and 19 are cancelled. Claims 1, 4, 6, 8, 9, 10, 14, 16, 20, and 23 are currently amended. Claims 24-35 are new. Claims 1, 16, and 23 are independent.

Information Disclosure Statement

3. Acknowledgment is made of Applicant's Information Disclosure Statement (IDS) Form PTO-1449 filed on July 9, 2009. This IDS has been considered.

Allowable Subject Matter

4. **Claims 1, 2, 4, 6-17, and 20-35 are allowed.**
5. The following is an examiner's statement of reasons for allowance:

With respect to independent claim 1, there is no teaching, suggestion, or motivation for combination in the prior art to a memory module having a circuit including a first set of bi-directional ports operatively coupled to at least one memory device (of the plurality of memory devices on the printed circuit board of the memory module), a second set of one or more bi-directional ports operatively coupled to the at least one connector, and a switching circuit configured to selectively operatively couple one or

Art Unit: 2824

more of the second set of ports to one or more of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit (which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports) that includes at least one coarse correction element to provide coarse correction of the noise and at least one fine correction element to provide fine correction of the noise, in combination with the other limitations.

With respect to independent claim 16, there is no teaching, suggestion, or motivation for combination in the prior art to a method of using a memory module that includes a connector to connect to a memory controller and memory devices, and a circuit that includes a first set of bi-directional ports operatively coupled to at least one memory device (of the plurality of memory devices), a second set of one or more hi-directional ports operatively coupled to the at least one connector, wherein the circuit is configured to selectively operatively couple one or more of the second set of ports to one or more of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit (which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports) that includes at least one coarse correction element to provide coarse correction of the noise and at least one fine correction element to provide fine correction of the noise, in combination with the other limitations.

With respect to independent claim 23, there is no teaching, suggestion, or

Art Unit: 2824

motivation for combination in the prior art to a circuit including a first set of bi-directional ports configured to be operatively coupled to at least one memory subsystem, a second set of one or more bi-directional ports configured to be operatively coupled to at least one connector configured to be operatively coupled to a memory controller of a computer system, and a switching sub-circuit configured to selectively operatively couple one or more of the second set of ports to one or more of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit (which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports) that includes at least one coarse correction element to provide coarse correction of the noise and at least one fine correction element to provide fine correction of the noise, in combination with the other limitations.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

CONCLUSION

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure: Sharma (U.S. Patent 5,559,970), Rahman et al. '823 (U.S. Patent

Art Unit: 2824

6,700,823), Rahman et al. '869 (U.S. Patent 6,812,869), Dodd et al. (U.S. Patent 6,981,089), Jeddelloh (U.S. Patent 7,136,958), Resnick (U.S. Patent 7,184,916), and Katoch et al. (U.S. Patent Application Publication 2007/0013429).

Sharma teaches a crossbar switch for coupling different memory modules to different processors.

Rahman et al. '823 teach line drivers with termination resistors to reduce signal reflections due to system impedance mismatches.

Rahman et al. '869 teach employing, into I/O circuits, pass transistors that function as lowpass filters to reduce noise and improve signal integrity.

Dodd et al. teach a memory system in which each memory rank includes its own termination circuit for its port that connects to the memory controller.

Jeddelloh teaches a multiprocessor system that includes a plurality of memory modules, wherein one memory module has a first set of memory access ports to connect to the processor, that one memory module has a second set of memory access ports to couple to another memory module, and that one memory module includes a memory hub that has a crossbar switch to couple the ports.

Resnick teaches a memory daughter card that includes a crossbar switch to link SERDES interfaces to L3 Cache that connects to a DDR2 memory controller which connects to a memory module.

Katoch et al. teach a damper circuit to reduce peak crosstalk noise over a wire.

None of the prior art, alone or in combination, meet all of the following limitations:

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a memory module having a circuit including a first set of bi-directional ports operatively coupled to at least one memory device (of the plurality of memory devices on the printed circuit board of the memory module), a second set of one or more bi-directional ports operatively coupled to the at least one connector, and a switching circuit configured to selectively operatively couple one or more of the second set of ports to one or more of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit (which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports) that includes at least one coarse correction element to provide coarse correction of the noise and at least one fine correction element to provide fine correction of the noise, **as recited in claim 1**; a method of using a memory module that includes a connector to connect to a memory controller and memory devices, and a circuit that includes a first set of bi-directional ports operatively coupled to at least one memory device (of the plurality of memory devices), a second set of one or more hi-directional ports operatively coupled to the at least one connector, wherein the circuit is configured to selectively operatively couple one or more of the second set of ports to one or more of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit (which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports) that includes at least one coarse correction element to provide coarse correction of the noise and at

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least one fine correction element to provide fine correction of the noise, **as recited in claim 16**; or a first set of bi-directional ports configured to be operatively coupled to at least one memory subsystem, a second set of one or more bi-directional ports configured to be operatively coupled to at least one connector configured to be operatively coupled to a memory controller of a computer system, and a switching sub-circuit configured to selectively operatively couple one or more of the second set of ports to one or more of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit (which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports) that includes at least one coarse correction element to provide coarse correction of the noise and at least one fine correction element to provide fine correction of the noise, **as recited in claim 23**.

When responding to this office action, applicants are advised to provide the examiner with the line numbers and page numbers in the application and/or references cited to assist the examiner in locating appropriate paragraphs.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Alexander Sofocleous whose telephone number is 571-272-0635. The examiner can normally be reached on 7:00am - 4pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Richard Elms can be reached on 571-272-1869. The fax phone number for

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the organization where this application or proceeding is assigned is (571) 273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free).

/ALEXANDER SOFOCLEOUS/
Primary Examiner, Art Unit 2824

Notice of References Cited	Application/Control No. 12/422,853		Applicant(s)/Patent Under Reexamination LEE ET AL.	
	Examiner ALEXANDER SOFOCLEOUS		Art Unit 2824	Page 1 of 1

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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	Classification
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*	B	US-6,700,823	03-2004	Rahman et al.	365/189.05
*	C	US-6,812,869	11-2004	Rahman et al.	341/78
*	D	US-6,981,089	12-2005	Dodd et al.	710/308
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*	F	US-7,184,916	02-2007	Resnick et al.	702/118
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	H	US-			
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*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	Classification
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*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
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*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i> SHEET 1 OF 4	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
	Examiner	Unknown
	Attorney Docket No.	NETL.045A

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Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
	1	3,660,675	05-02-1972	Andrews, Jr.	
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Examiner Signature	/Alexander Sofocleous/	Date Considered	12/05/2011
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***Examiner:** Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

T¹ - Place a check mark in this area when an English language translation is attached. **ALL REFERENCES CONSIDERED EXCEPT WHERE LINED THROUGH. /AGS/**

INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
	Examiner	Unknown
SHEET 2 OF 4	Attorney Docket No.	NETL.045A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
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Examiner Signature	/Alexander Sofocleous/	Date Considered	12/05/2011
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i> SHEET 3 OF 4	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
	Examiner	Unknown
	Attorney Docket No.	NETL.045A

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number Number - Kind Code (if known) Example: 1,234,567 B1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear
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Examiner Initials	Cite No.	Foreign Patent Document Country Code-Number-Kind Code Example: JP 1234567 A1	Publication Date MM-DD-YYYY	Name of Patentee or Applicant	Pages, Columns, Lines Where Relevant Passages or Relevant Figures Appear	T ¹

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT <i>(Multiple sheets used when necessary)</i>	Application No.	12/422,853
	Filing Date	April 13, 2009
	First Named Inventor	Hyun Lee et al.
	Art Unit	2827
	Examiner	Unknown
SHEET 4 OF 4	Attorney Docket No.	NETL.045A

NON PATENT LITERATURE DOCUMENTS			
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published.	T ¹
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<i>Index of Claims</i> 	Application/Control No. 12422853	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ALEXANDER SOFOCLEOUS	Art Unit 2824

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
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☐ Claims renumbered in the same order as presented by applicant			☐ CPA			☐ T.D.			☐ R.1.47		
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Search Notes 	Application/Control No. 12422853	Applicant(s)/Patent Under Reexamination LEE ET AL.
	Examiner ALEXANDER SOFOCLEOUS	Art Unit 2824

SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
711/105, 115, 149 (text search only - see EAST search history printout)	11/25/2011	AGS
326/30 (text search only - see EAST search history printout)	12/5/2011	AGS
365/51, 191, 198, 230.05 (text search only - see EAST search history printout)	12/5/2011	AGS
see EAST search history	12/5/2011	AGS

INTERFERENCE SEARCH			
Class	Subclass	Date	Examiner
326	30	12/5/2011	AGS
365	51, 191, 198, 230.05	12/5/2011	AGS

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EAST Search History

EAST Search History (Prior Art)

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"6240048"	"6243282"	"6252807"
"6260097"	"6260154"	"6262938"
"6266285"	"6274395"	"6279069"
"6295572"	"6298426"	"6304511"
"6307769"	"6314051"	"6317352"
"6317381"	"6324120"	"6326810"
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"6338113"	"6341347"	"6353561"
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"6363031"	"6378020"	"6381188"
"6381668"	"6392304"	"6414868"
"6418034"	"6426916"	"6429029"
"6430103"	"6437600"	"6438057"
"6442698"	"6452826"	"6452863"
"6453400"	"6453402"	"6453434"
"6457095"	"6459651"	"6473831"
"6480929"	"6487102"	"6493789"
"6496440"	"6496897"	"6498766"
"6510097"	"6510503"	"6512392"
"6526471"	"6526484"	"6545895"
"6546446"	"6553450"	"6560158"
"6563337"	"6563759"	"6564281"
"6564285"	"6574150"	"6584037"
"6587912"	"6590822"	"6594770"
"6597617"	"6614700"	"6618791"
"6621760"	"6631086"	"6639820"
"6646939"	"6650594"	"6657918"
"6657919"	"6658530"	"6665224"
"6665227"	"6683372"	"6697295"
"6701446"	"6705877"	"6708144"
"6714891"	"6724684"	"6731527"
"6744687"	"6747887"	"6751113"
"6751696"	"6754129"	"6754132"
"6757751"	"6762948"	"6765812"
"6771526"	"6772359"	"6779097"
"6785767"	"6791877")	PN. OR
("6795899"	"6799241"	"6801989"
"6807598"	"6807655"	"6816991"
"6819602"	"6819617"	"6820163"
"6820169"	"6826104"	"6839290"

		"6845055" "6847582" "6850449" "6862202" "6862249" "6862653" "6873534" "6894933" "6898683" "6908314" "6912778" "6917219" "6922371" "6930900" "6930903" "6938119" "6961281" "6968416" "6968419" "6970968" "6980021" "6986118" "6992501" "6992950" "7000062" "7003618" "7003639" "7007175" "7010642" "7010736" "7024518" "7028234" "7033861" "7035150" "7046538" "7058776" "7058863" "7061784" "7061823" "7075175" "7079441" "7079446" "7085152" "7085941" "7089438" "7093101" "7103730" "7120727" "7126399" "7133960" "7136978" "7149145" "7149824" "7173863" "7200021" "7205789" "7224595" "7228264" "7233541" "7245541" "7266639" "7269042" "7269708" "7289386" "7296754" "7299330" "7307863" "7363422" "7379316" "7386656" "7392338" "7408393" "7409492" "7464225" "7472220" "7496777" "7532537" "RE36839").PN. OR ("7609567").URPN.				
S3	1	("7609567").PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 12:51
S5	29	((("3660675") or ("3757235") or ("4305091") or ("4586168") or ("4701845") or ("4752741") or ("4782487") or ("4837743") or ("4885799") or ("4903266") or ("4910597") or ("4942556") or ("4987321") or ("5138619") or ("5173906") or ("5222066") or ("5241503") or ("5304856") or ("5337254") or ("5359235") or ("5394037") or ("5430335") or ("5525917") or ("5841296") or ("5914543") or ("6000048") or ("6044481") or ("6070217") or ("6169696")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 13:17
S6	29	((("6194959") or ("6216240") or ("6467056") or ("6560740") or ("6614702") or ("6721150") or ("6812869") or ("6829728") or ("6918072") or ("6928024") or ("6928593") or ("6930509") or ("6934900") or ("7036064") or ("7053470") or ("7062696") or ("7184915") or ("7190210") or ("7203873") or ("7253652") or ("7284166") or ("20010048342") or ("20020000847") or ("20020131535") or ("20020140523") or ("20030098742") or ("20030126346") or ("20030197797") or ("20030218491")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 13:31
S9	40	(port with (impedance near match\$3)) (memory adj module)	US-PGPUB; USPAT; USOCR;	AND	ON	2011/11/25 13:34

			FPRS; EPO; JPO; DERWENT; IBM_TDB			
S8	2775	(port with (impedance near match\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 13:34
S7	3127	(port with (impedance near2 match\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 13:34
S10	1277	(port with (reflect\$3 or noise)) same (memory)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 13:39
S11	682	(port with (reflect\$3 or noise) with memory)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 13:40
S12	21	((("20040155702") or ("20040199843") or ("20050093620") or ("20050127989") or ("20050289423") or ("20060082383") or ("20060107156") or ("20060140015") or ("20060144015") or ("20060147217") or ("20060192653") or ("20060242458") or ("20060262586") or ("20060271748") or ("20070030814") or ("20070058471") or ("20070079199") or ("20070109707") or ("20070152743") or ("20070204075") or ("20070223296")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 13:44
S13	372	(port with (reflect\$3 or noise) with memory) (terminat\$3 or match\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:05
S15	212	(port with (reflect\$3 or noise) with memory) (terminat\$3)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:06
S14	274	(port with (reflect\$3 or noise) with memory) (terminat\$3 or impedance)	US-PGPUB; USPAT; USOCR; FPRS;	AND	ON	2011/11/25 14:06

			EPO; JPO; DERWENT; IBM_TDB			
S16	57	(port with ((reduc\$4 or minimiz\$3 or eliminat\$3 or compensat\$3 or correct\$3) near2 (reflect\$3 or noise)) with memory)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:07
S20	38	(memory adj (circuit or chip)) same (cross\$1bar with port)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:11
S19	1276	memory same (cross\$1bar with port)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:11
S18	48	(memory adj module) same (cross\$1bar with port)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:11
S17	305	(memory adj module) (cross\$1bar with port)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:11
S23	49	S21 S22	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:27
S22	40622	(port with (reflect\$3 or noise))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:27
S21	728	(memory with cross\$1bar with port)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:27
S25	175	S21 S24	US-PGPUB; USPAT;	AND	ON	2011/11/25 14:29

			USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			
S24	111339	(port with (reflect\$3 or noise or match\$3 or terminat\$3))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:29
S27	18	S21 S26	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:30
S26	16304	(port with signal with (reflect\$3 or noise or match\$3 or terminat\$3 or integrity))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:30
S28	2514	(326/30).CCLS.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 14:33
S29	1534	(365/189.19,230.05).CCLS.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 14:40
S31	4153	(711/105,115,149).CCLS.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 14:43
S30	528	(710/317).CCLS.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 14:43
S33	43	(S28 or S29 or S30 or S31) S26 (memory adj (chip or circuit or module))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:44
S32	209	(S28 or S29 or S30 or S31) S26	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 14:44
S34	5	dampner with noise with i\$1o	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/25 15:03
S35	41	(impedance adj match\$3) with (coarse	US-PGPUB;	AND	ON	2011/11/25

		with fine)	USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB			15:09
S36	4	((12/863831) or (12/846053) or (12/836201) or (13/181957) or (12/016801) or (12/691819) or (13/168241)).APP.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/25 15:42
S42	140	(cross\$1bar with memory adj module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 11:07
S43	1	(10/283765).APP.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/29 11:29
S44	0	(cross\$1bar with memory adj module) (port with (noise or termination or integrity))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 11:50
S46	4	((("5559970") or ("6812869") or ("7184916") or ("6981089")).PN.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/29 11:51
S45	1	(cross\$1bar with memory adj module) (i\$1o with (noise or termination or integrity))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 11:51
S47	97	(cross\$1bar with memory adj module) (port)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 11:54
S49	2	damper adj circuit with reflection	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 11:59
S48	8	damper adj circuit with noise	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 11:59
S50	5	("20040008054" "6326821" "6456124").PN. OR ("7495469").URPN.	US-PGPUB; USPAT;	AND	ON	2011/11/29 16:02

			USOCR			
S51	2517	(326/30).CCLS.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/11/29 16:16
S52	20537	(coarse or fine) with reflect\$3	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 16:59
S53	67	(coarse or fine) with reflect\$3 with port	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:00
S55	2	(coarse or fine) with adjust\$3 with port (memory adj module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:02
S54	449	(coarse or fine) with adjust\$3 with port	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:02
S59	22	(coarse or fine) with (correct\$4 or adjust\$4) port with (memory adj module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:03
S58	22	(coarse or fine) with (correct\$3 or adjust\$4) port with (memory adj module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:03
S57	7	(coarse or fine) with correct\$3 port with (memory adj module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:03
S56	2	(coarse or fine) with correct\$3 with port (memory adj module)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/11/29 17:03

L3	4	1 2	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/12/05 15:19
L2	389	(memory adj module) port (coarse fine)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	AND	ON	2011/12/05 15:19
L1	7913	((326/30) or (365/51,191,198,230.05)).CCLS.	US-PGPUB; USPAT; USOCR	OR	OFF	2011/12/05 15:19

EAST Search History (I nterference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S2	1	"Term Removed"	US-PGPUB; USPAT; UPAD	OR	OFF	2011/09/13 18:19
S37	127	((memory adj module) (port near5 (select\$3 or multi\$1plex\$3))).clm.	US-PGPUB; USPAT; UPAD	AND	ON	2011/11/25 12:27
S40	59	damper with noise with circuit	US-PGPUB; USPAT; UPAD	AND	ON	2011/11/25 15:04
S39	2754	damper with noise	US-PGPUB; USPAT; UPAD	AND	ON	2011/11/25 15:04
S38	0	damper with noise with i\$1o	US-PGPUB; USPAT; UPAD	AND	ON	2011/11/25 15:04
S41	4	"Term Removed"	US-PGPUB; USPAT; UPAD	OR	OFF	2011/11/25 15:45
L6	5	4 5	US-PGPUB; USPAT; UPAD	AND	ON	2011/12/05 15:21
L5	398	(memory adj module) port (coarse fine)	US-PGPUB; USPAT; UPAD	AND	ON	2011/12/05 15:21
L4	7652	((326/30) or (365/51,191,198,230.05)).CCLS.	US-PGPUB; USPAT; UPAD	OR	OFF	2011/12/05 15:21

12/ 5/ 2011 3:22:08 PM

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BIB DATA SHEET

CONFIRMATION NO. 7481

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
12/422,853	04/13/2009	365	2824	NETL.045A	
RULE					
APPLICANTS Hyun Lee, Ladera Ranch, CA; Jayesh R. Bhakta, Cerritos, CA; Jeffrey C. Solomon, Irvine, CA; Mario Jesus Martinez, Laguna Niguel, CA; Chi-She Chen, Walnut, CA;					
** CONTINUING DATA ***** This appln claims benefit of 61/044,839 04/14/2008 and claims benefit of 61/044,825 04/14/2008 and claims benefit of 61/044,801 04/14/2008					
** FOREIGN APPLICATIONS *****					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 04/23/2009					
Foreign Priority claimed ☐ Yes ☒ No 35 USC 119(a-d) conditions met ☐ Yes ☒ No Verified and /ALEXANDER SOFOCLEOUS/ Acknowledged Examiner's Signature	☐ Met after Allowance Initials	STATE OR COUNTRY CA	SHEETS DRAWINGS 9	TOTAL CLAIMS 23	INDEPENDENT CLAIMS 3
ADDRESS KNOBBE MARTENS OLSON & BEAR LLP 2040 MAIN STREET FOURTEENTH FLOOR IRVINE, CA 92614 UNITED STATES					
TITLE CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION					
FILING FEE RECEIVED 1856	FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT No. _____ for following:		☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time) ☐ 1.18 Fees (Issue) ☐ Other _____ ☐ Credit		

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), to: **Mail** **Mail Stop ISSUE FEE**
Commissioner for Patents
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or Fax (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

20995 7590 12/14/2011
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

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I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/422,853	04/13/2009	Hyun Lee	NETL.045A	7481

TITLE OF INVENTION: CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

APPLN. TYPE	SMALL ENTITY	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	NO	\$1740	\$0	\$0	\$1740	03/14/2012

EXAMINER	ART UNIT	CLASS-SUBCLASS
SOFOCLEOUS, ALEXANDER	2824	365-051000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

- ☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
- ☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

- (1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
- (2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1. Knobbe, Martens,
Olson & Bear LLP

2. _____
 3. _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

NETLIST, INC.

IRVINE, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:

- ☒ Issue Fee
- ☒ Publication Fee (No small entity discount permitted)
- ☐ Advance Order - # of Copies _____

4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above)

- ☐ A check is enclosed.
- ☐ Payment by credit card. Form PTO-2038 is attached.
- ☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment, to Deposit Account Number 11-1410 (enclose an extra copy of this form).

5. Change in Entity Status (from status indicated above)

- ☐ a. Applicant claims SMALL ENTITY status. See 37 CFR 1.27. ☐ b. Applicant is no longer claiming SMALL ENTITY status. See 37 CFR 1.27(g)(2).

NOTE: The Issue Fee and Publication Fee (if required) will not be accepted from anyone other than the applicant; a registered attorney or agent; or the assignee or other party in interest as shown by the records of the United States Patent and Trademark Office.

Authorized Signature _____

Date _____

Typed or printed name Bruce Itchkawitz

Registration No. 47,677

This collection of information is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450.

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicants : Lee et al.

Group Art Unit 2827

Appl. No. : 12/422,853

Filed : April 13, 2009

For : CIRCUIT PROVIDING LOAD ISOLATION
AND NOISE REDUCTION

Examiner : Alexander Sofocleous

Conf. No. : 7481

COMMENTS ON EXAMINER'S STATEMENT OF REASONS FOR ALLOWANCE

Mail Stop- ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Arlington, VA 22313-1450

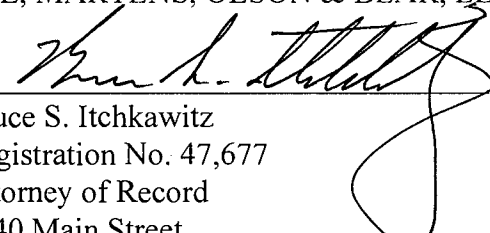
Dear Sir:

Applicants respectfully disagree with the Examiner's Statement of Reasons for Allowance to the extent that the features recited by the Examiner are not present in all of the claims.

To the extent that there is any implication that the patentability of the claims rests on the recitation of a single feature, Applicants respectfully disagree with the Examiner's Statement because it is the combination of features that makes the claims patentable. Accordingly, Applicants submit that the claims of the present application are allowable because each of the claims recites a combination of features that are not anticipated or rendered obvious by the prior art.

Respectfully submitted,

KNOBBE, MARTENS, OLSON & BEAR, LLP

Dated: 3/5/12By: 

Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
2040 Main Street
Fourteenth Floor

Appl. No. : 12/422,853
Filed : April 13, 2009

Irvine, CA 92614
(949) 760-0404

12869640
030212

Electronic Patent Application Fee Transmittal

Application Number:	12422853			
Filing Date:	13-Apr-2009			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Bruce S. Itchkawitz/Nicole Jevsevar			
Attorney Docket Number:	NETL.045A			
Filed as Large Entity				
Utility under 35 USC 111(a) Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Utility Appl issue fee	1501	1	1740	1740
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				1740

Electronic Acknowledgement Receipt

EFS ID:	12224822
Application Number:	12422853
International Application Number:	
Confirmation Number:	7481
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/ThuyQuyen Nguyen
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045A
Receipt Date:	05-MAR-2012
Filing Date:	13-APR-2009
Time Stamp:	16:26:57
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$ 1740
RAM confirmation Number	3211
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	issue_fee.pdf	101072	no	1
			c7cb347c146b2caf136407d4b2cc9a5341b eb104		
Warnings:					
Information:					
2	Miscellaneous Incoming Letter	comments.pdf	44151	no	2
			b4df73ba6a15e7fb23955844e670dc0c6a6 0ad7e		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	30394	no	2
			68cc2ad6f3d10d11f44748155b399ce7dec5 054a		
Warnings:					
Information:					
Total Files Size (in bytes):			175617		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

correction element 154 of certain embodiments is dynamically responsive to one or more changes in the noise, such as, for example, changes in the frequency of the noise component.

[0046] The fine correction element 154 may include at least one damper circuit. The damper circuit includes at least one self-adjusting damper circuit 170 in certain embodiments, such as the example self-adjusting damper circuit 170 schematically illustrated by Figure 6. In certain embodiments, the self-adjusting damper circuit 170 comprises an input terminal 172 adapted to receive an input signal 174 comprising an input noise component and an input signal component. The damper circuit 170 further comprises an output terminal 176. In certain embodiments, the damper circuit 170 includes a first sub-circuit 184 adapted to generate a first signal 187 responsive to at least a portion of the noise component. For example, the first sub-circuit 184 may generate the first signal 187 in response to the portion of the noise component not reduced by the coarse correction element 152 in certain embodiments. The damper circuit 170 of certain embodiments further comprises a second sub-circuit 188 adapted to generate a second signal 190 corresponding to a delayed version of at least a portion of the input signal. The damper circuit 170 combines the first signal 187 with the second signal 190 to generate an output signal 108 on the output terminal 176 wherein the output signal comprises an output noise component smaller than the input noise component. For example, the first signal 187 and the second signal 190 may be combined at the node 182 of the damper circuit 170. The output signal 176 of certain embodiments is substantially representative of the signal component of the input signal 174. The first sub-circuit 184 of certain embodiments comprises one or more current sources 185, 186. For example, the current sources 185, 186 may each comprise a controlled current source such as a voltage-controlled current source 185, 186 in certain embodiments. In some embodiments, the first sub-circuit 184 may comprise or be characterized as a negative feedback current source. In various embodiments, the self-adjusting damper circuit 170 includes an enable control input (not shown) for enabling the self-adjusting damper circuit 170. Examples of self-adjusting damper circuits 170 in accordance with certain embodiments described herein are described in U.S. Provisional Application No. No. 61/044,825, filed April 14, 2008, and co-pending U.S. Application No. 12/422,912, filed by the assignee of the present

Change(s)
applied
to document,
/J.L.B./
12/23/2011

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION**CROSS-REFERENCE TO RELATED APPLICATIONS**

[0001] The present application claims the benefit of priority from U.S. Provisional Application No. 61/044,839, filed April 14, 2008, from U.S. Provisional Application No. 61/044,825, filed April 14, 2008, and from U.S. Provisional Application No. 61,044,801, filed April 14, 2008, each of which are incorporated in their entirety by reference herein. This application is related to U.S. Patent Application No. 12/422,912, filed on even date herewith and entitled "Self-Adjusting Damper", and to U.S. Application No. 12/422,925, filed on even date herewith and entitled "Memory Board with Self-Testing Capability", both of which are incorporated in their entirety by reference herein.

Change(s)
applied
to document,
/J.L.B./
12/23/2011

BACKGROUNDField

[0002] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0003] As the demand on the performance of electronic systems increase, the systems include higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0004] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system 10 which incorporates a multiplexer-demultiplexer 12 to isolate unused subsystems of a plurality of subsystems 16, 18. This arrangement can reduce the load on each driver 14 when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver 14, the receiver 20 and the multiple destinations. However, the multiplexer and/or demultiplexer 12 can present transmission line discontinuity regions in electronic systems (e.g., in electronic systems with signal paths with



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
12/422,853	04/10/2012	8154901	NETL.045A	7481

20995 7590 03/21/2012
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 549 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

**REVOCATION
AND
GENERAL POWER OF ATTORNEY**

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Dear Sir:

The undersigned is an empowered representative of the Assignee and hereby appoints the registrant, Jamie J. Zheng, Customer No. 79141, as attorney to represent the Assignee before the United States Patent and Trademark Office (USPTO) in connection with the patent assigned to the Assignee according to the USPTO assignment records or assignment documents supplied with the accompanying Statement Under 37 CFR § 3.73(b). This appointment is to be to the exclusion of the inventor(s) and his attorney(s) in accordance with the provisions of 37 CFR § 3.71.

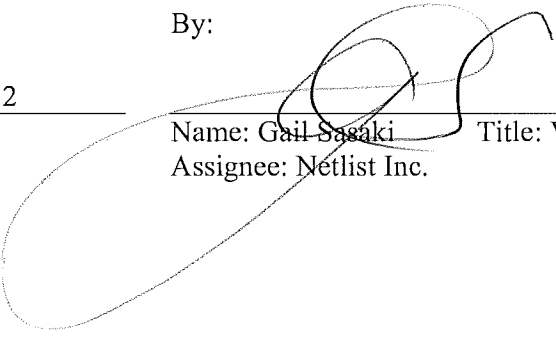
Submission of this paper in connection with any matter of the below named assignee, together with a statement under 37 CFR 3.73(b), shall serve to revoke any previous powers of attorneys in that matter.

A Statement under 37 CFR § 3.73(b) is attached setting forth a full chain of title for the subject patent owned by the Assignee named below.

Please recognize or change the correspondence address for the patent identified in the attached Statement to Customer No. 79141.

By:

Date: October 1, 2012


Name: Gail Sasaki Title: Vice President and CFO
Assignee: Netlist Inc.

STATEMENT UNDER 37 CFR 3.73(b)Applicant/Patent Owner: Netlist, Inc.Application No./Patent No.: 8,154,901 Filed/Issue Date: 10-Apr-2012

Titled:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTIONNetlist, Inc. , a Corporation

(Name of Assignee)

(Type of Assignee, e.g., corporation, partnership, university, government agency, etc.)

states that it is:

1. ☒ the assignee of the entire right, title, and interest in;
2. ☐ an assignee of less than the entire right, title, and interest in
(The extent (by percentage) of its ownership interest is _____ %); or
3. ☐ the assignee of an undivided interest in the entirety of (a complete assignment from one of the joint inventors was made)
the patent application/patent identified above, by virtue of either:

A. ☒ An assignment from the inventor(s) of the patent application/patent identified above. The assignment was recorded in the United States Patent and Trademark Office at Reel 025301, Frame 0961, or for which a copy therefore is attached.

OR

B. ☐ A chain of title from the inventor(s), of the patent application/patent identified above, to the current assignee as follows:

1. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

2. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

3. From: _____ To: _____

The document was recorded in the United States Patent and Trademark Office at

Reel _____, Frame _____, or for which a copy thereof is attached.

☐ Additional documents in the chain of title are listed on a supplemental sheet(s).

☒ As required by 37 CFR 3.73(b)(1)(i), the documentary evidence of the chain of title from the original owner to the assignee was, or concurrently is being, submitted for recordation pursuant to 37 CFR 3.11.

[NOTE: A separate copy (i.e., a true copy of the original assignment document(s)) must be submitted to Assignment Division in accordance with 37 CFR Part 3, to record the assignment in the records of the USPTO. See MPEP 302.08]

The undersigned (whose title is supplied below) is authorized to act on behalf of the assignee.

Signature

Gail Sasaki

Printed or Typed Name

October 1, 2012

Date

VP and CFO

Title

This collection of information is required by 37 CFR 3.73(b). The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

Electronic Acknowledgement Receipt

EFS ID:	13925555
Application Number:	12422853
International Application Number:	
Confirmation Number:	7481
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	20995
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NETL.045A
Receipt Date:	05-OCT-2012
Filing Date:	13-APR-2009
Time Stamp:	18:55:55
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Power of Attorney	poa8154901.pdf	233357 fc4a5033c318c0816c849a937e55a51c690621de	no	2

Warnings:

The page size in the PDF is too large. The pages should be 8.5 x 11 or A4. If this PDF is submitted, the pages will be resized upon entry into the Image File Wrapper and may affect subsequent processing

Information:

Total Files Size (in bytes):

233357

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
12/422,853	04/13/2009	Hyun Lee	NETL.045A

79141
The Law Office of Jamie Zheng, Ph.D Esq.
P.O. Box 60573
Palo Alto, CA 94306

CONFIRMATION NO. 7481
POA ACCEPTANCE LETTER



Date Mailed: 10/25/2012

NOTICE OF ACCEPTANCE OF POWER OF ATTORNEY

This is in response to the Power of Attorney filed 10/05/2012.

The Power of Attorney in this application is accepted. Correspondence in this application will be mailed to the above address as provided by 37 CFR 1.33.

/zabraha/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101



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CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
12/422,853	04/13/2009	411127

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine , CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



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Alexandria, Virginia 22313-1450
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FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
12/422,853	04/13/2009	411127

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045PR
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
		☐ Party of Interest under 35 U.S.C. 118		
Prefix	Given Name	Middle Name	Family Name	Suffix
	To be provided later			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residence i	
Citizenship under 37 CFR 1.41(b) i				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Country i		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				
Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).	
☐ An Address is being provided for the correspondence Information of this application.	
Customer Number	20995
Email Address	efiling@kmob.com
Add Email Remove Email	

Application Information:

Title of the Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		
Attorney Docket Number	NETL.045PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	6	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045PR
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

Publication Information:
☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.
Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

			Remove
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

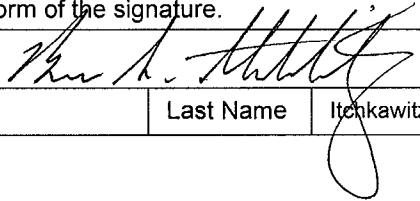
Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

Assignee 1

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.045PR
		Application Number	
Title of Invention	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION		

If the Assignee is an Organization check here. ☒			
Organization Name	NETLIST, INC.		
Mailing Address Information:			
Address 1	51 Discovery, #150		
Address 2			
City	Irvine	State/Province	CA
Country	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	949-435-0031
Email Address	hlee@netlist.com		
Additional Assignee Data may be generated within this form by selecting the Add button.			

Signature:

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
Signature			Date (YYYY-MM-DD)	2008-04-14	
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION**BACKGROUND**Field

[0001] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0002] As the demand on the performance of electronic systems increase, the systems require higher speed processors and higher memory densities with relatively heavy fan-outs. However, higher speed processors and higher memory densities translate to higher power dissipation and increased memory access time, impairing the performance of the system.

[0003] Traditional systems sometimes incorporate multiplexers and/or demultiplexers to help quiet inactive signal paths, minimizing the overall dynamic power dissipation, and isolating unused subsystems. Figure 1 schematically illustrates an example of such a system which incorporates a multiplexer-demultiplexer to isolate unused subsystems. This arrangement can reduce the load on each driver when the original signal path includes multiple destinations by inserting a multiplexer on each signal path between the driver and the multiple destinations. However, the multiplexer and/or demultiplexer can present a transmission line discontinuity point in electronic systems with signal paths with electrical lengths longer than $\frac{1}{4}$ of the wavelength of the operational frequency or the signal transition rate. In such systems, signal reflections may occur, changing wave characteristics and degrading system performance. These signal reflections may cause signal distortions and signal integrity issues which can contribute to reduced system performance and eventual failure of the system, thereby limiting the effectiveness of the multiplexer and/or demultiplexer in addressing the signal integrity and power dissipation issues.

[0004] In memory applications, some systems control "chip select" or device select signals, along with the memory address and control signals, based on the bank device select signals, in order to disable memory devices when they are not being accessed. Figure 2

schematically illustrates an example of such a system which isolates unused subsystems using device select signals. The devices are disabled by not activating (asserting) the address, control, and/or device select signals, thus logically isolating the memory devices which are not accessed. This arrangement generally requires a bank device select signal for each physical memory rank and the address and device select signals are generally controllable based on whether the corresponding memory device or group of memory devices is being accessed. Generating controllable address and device select signals can introduce timing delays on the bank device select, address, and control paths. In addition, this arrangement does not address the power dissipation and may degrade performance degradation by, for example, requiring that secondary control signals be generated for the memory devices.

SUMMARY

[0005] In certain embodiments, a memory module is provided which includes a printed circuit board having at least one connector configured to be electrically coupled to a memory controller of a computer system. The memory module also includes a plurality of memory devices on the printed circuit board and a circuit. The circuit includes a first set of ports having a plurality of bi-directional ports. Each port of the first set of ports is electrically coupled to at least one memory device of the plurality of memory devices. The circuit also includes a second set of ports having one or more bi-directional ports. Each port of the second set of ports is electrically coupled to the at least one connector. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0006] In certain embodiments, a method of using a memory module with a computer system is provided which involves providing a memory module including a printed circuit board having at least one connector configured to be electrically coupled to a memory controller of a computer system and to a plurality of memory devices. The memory module further includes a circuit having a first set of ports comprising a plurality of bi-directional ports. Each port of the first set of ports electrically coupled to at least one memory device of

the plurality of memory devices. The circuit also includes a second set of ports comprising one or more bi-directional ports. Each port of the second set of ports electrically coupled to the at least one connector. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports. The method further includes activating the circuit to selectively electrically couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports. The method also involves activating the circuit to reduce noise in the one or more signals.

[0007] In certain embodiments, a circuit is provided which includes a first set of ports having a plurality of bi-directional ports. Each port of the first set of ports is configured to be electrically coupled to at least one memory device of a plurality of memory devices on a printed circuit board. The circuit also includes a second set of ports having one or more bi-directional ports. Each port of the second set of ports is configured to be electrically coupled to at least one connector on the printed circuit board. The at least one connector is configured to be electrically coupled to a memory controller of a computer system. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

[0008] In certain embodiments, a circuit is provided which includes a first set of ports having a plurality of bi-directional ports. Each port of the first set of ports is configured to be electrically coupled to at least one subsystem of a plurality of subsystems. The circuit also includes a second set of ports having one or more bi-directional ports. Each port of the second set of ports is configured to be electrically coupled to at least one subsystem of one or more subsystems. The circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports. The one or more ports

of the first set of ports and the one or more ports of the second set of ports each includes a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Figure 1 schematically illustrates an example system which isolates unused subsystems and incorporates a multiplexer-demultiplexer.

[0010] Figure 2 schematically illustrates an example system which isolates unused subsystems using device select signals.

[0011] Figure 3 schematically illustrates an example system having an example circuit connecting two subsystems in accordance with certain embodiments described herein.

[0012] Figure 4 schematically illustrates an example circuit in accordance with certain embodiments described herein.

[0013] Figure 5 schematically illustrates an example programmable impedance matching circuit in accordance with certain embodiments described herein.

[0014] Figure 6 schematically illustrates an example filter/repeater/amplifier circuit in accordance with certain embodiments described herein.

BRIEF DESCRIPTION OF THE APPENDIX

[0015] Appendix: "Self-Adjusting Damper," incorporated in its entirety by reference herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0016] Certain embodiments described herein present a circuit for isolating inactive sub-systems and maintaining and improving signal integrity in a variety of systems. Certain embodiments described herein are implemented in a memory system of a computer system. Certain embodiments described herein are used to increase the performance of electronic systems without increasing the power dissipation or the memory access time.

[0017] Certain embodiments described herein describe a circuit for use in high performance electronic systems which may contain transmission lines and relatively heavy fan-outs. A circuit is described herein which, in certain embodiments, includes at least some of the following components: a signal router, a programmable coarse impedance matching circuit, a self-adjusting damper circuit, and an amplifier/repeater/filter.

[0018] In certain embodiments, the circuit includes multiple bi-directional ports. The signal router may comprise a multiplexer-demultiplexer. In certain embodiments, the signal router may be capable of interconnecting any number of the ports simultaneously. In certain embodiments, the circuit can pass signals (information) from one port to another port or from one port to multiple ports via multiplexer-demultiplexer. In certain embodiments, the multiplexer-demultiplexer isolates and provides a known static value to all un-driven ports, reducing any dynamic power consumption associated with floating or toggling electrical nodes. The multiplexer-demultiplexer can function as an isolation switch which isolates and inactivates individual subsystems not actively transferring information. In certain embodiments, the multiplexer-demultiplexer may function as a scalable isolation switch router which routes the information from one subsystem to another, establishing a one-to-one connection for transferring data from point-to-point and establishing a one-to-many connection for broadcasting data.

[0019] In certain embodiments, one or more of the ports in the circuit advantageously include a correction circuit for reducing noise. In certain embodiments, the correction circuit includes one or more elements to provide for coarse correction of signal noise due to impedance mismatches in the system. In some embodiments, the coarse correction elements may be a set of programmable coarse impedance matching circuits that are connected to the bi-directional ports. The impedance values of the impedance matching circuits may be dynamically controllable, configurable, and programmable, so as to match the characteristic impedances of the transmission lines in the system. The values of these impedances may differ when, for example, a particular signal path is connected or disconnected by the multiplexer-demultiplexer.

[0020] Advantageously, in certain embodiments, the correction circuit includes one or more fine correction elements which further reduce signal reflections due to impedance mismatches and/or reduce other noise. In certain embodiments the fine correction element is a damper circuit. For example, a self-adjusting damper circuit may be connected to each port of the multiplexer-demultiplexer. The damper circuit may be connected, for example, in parallel with the programmable coarse impedance matching circuit. In some embodiments, the damper circuit operates to reduce the impedance mismatches by introducing a current source

that damps to the slew rate of the input signal, reducing signal reflection at the interface of each port. In some embodiments, the current source is a negative feedback current source. In various embodiments, the self-adjusting damper includes an enable control input for enabling the self-adjusting damper circuit. Examples of the self-adjusting damper circuit in accordance with certain embodiments described herein are described in the Appendix which is incorporated in its entirety by reference herein.

[0021] In some embodiments, the circuit includes a filter/repeater/amplifier which provides wave reshaping functions and generally reduces signal noise. For example, the filter/repeater/amplifier may reduce signal ringing due to reflection at the multiplexer-demultiplexer. In certain embodiments, only the filter, only the repeater, only the amplifier, or any combination of the three elements may be included in the circuit. In certain embodiments, the filter/repeater/amplifier may also reduce noise from other sources such as, for example, signal coupling. In some embodiments, the filter/repeater/amplifier may include a band-pass filter and may reduce any low and/or high frequency noise components. The filter/repeater/amplifier may also, in certain embodiments enhance the signal slew rate to reduce the transient power dissipation at the target devices or components.

[0022] Figure 3 schematically illustrates an example system having an example circuit 200 connecting two subsystems in accordance with certain embodiments described herein. In some embodiments, the circuit 200 includes several independent, fully controllable functional blocks. For example, the circuit 200 may include a signal router, a programmable coarse impedance matching circuit, a self-adjusting damper circuit, and an amplifier/repeater/filter. In certain embodiments, any subset of one or more of these functional blocks may be implemented in order to enhance performance and reduce power dissipation. Figure 4 schematically illustrates a circuit 200 comprising a router 210 and which connects a subsystem 103 to another subsystem 104, or a subsystem 103 to multiple subsystems. The router 210 may include a router control input for controlling the connection paths of the router 210. The devices and methods described herein may be incorporated in any system (e.g. a board level system) or device that contains one or more connectors or wires. In certain embodiments the system has a signal delay which is more than $\frac{1}{2}$ the signal transition rate or the rise/fall time of the signal.

[0023] The system 100 of Figure 3 includes a driver 102, a receiver 101, and two subsystems 103, 104. For illustration purposes, the system 100 of Figure 3 is described as an open system that communicates with other systems via the driver 102 and the receiver 101. Furthermore, in some embodiments, a connector connects the system 100 with other systems by means of transmission lines. In some embodiments, a transmission line may be characterized as any wire or connector with an electrical length longer than the $\frac{1}{2}$ of the rise/fall time of the signal that is being transmitted. In various embodiments, the receiver 101 may or may not contain any active circuits. The receiver 101 may be a simple connector such as a socket or DIMM interface with a system board. The output signal from the receiver 101 may exhibit the characteristics of the signal on a transmission line such as, for example, wave dispersion, attenuation, and reflection. In certain embodiments, the output signal from the receiver goes through a number of refining stages in circuit 200 to improve the performance of the overall system. For example, the circuit 200 may reduce power consumption by selectively isolating the inactive (or un-accessed) subsystems and may enhance the access rate of a targeted subsystem.

[0024] Figure 4 schematically illustrates an example circuit 200 in accordance with certain embodiments described herein. In certain embodiments, the signal quality is enhanced by introducing the impedance matching circuit 230, the self-adjusting damper 240, and the filter/repeater/amplifier 250 on a signal path (e.g., from receiver 101 to subsystem 103). In certain embodiments, the performance is enhanced and power dissipation is reduced by removing the load associated with other subsystems (e.g., subsystem 104) and reshaping the signal waveform using the filter/repeater/amplifier 250. In some embodiments, there are two refining stages for reducing signal reflections due to impedance mismatches. In some embodiments, for example, the first refining stage is the impedance matching circuit 230 that matches the impedance of the transmission wire prior to the receiver 101 with the characteristic impedance of the wire in the system 100. The granularities of the configurable impedance values may be configured based on the application of the particular system. Advantageously, in some embodiments, the impedance matching circuit 230 is programmable. In some embodiments, the impedance matching circuit 230 may include a real time controllable impedance control input for pre-programming and/or dynamically controlling the

impedance matching circuit 230. In this manner, the impedance matching circuit 230 may, in certain embodiments, provide a dynamically controllable variable impedance which can be pre-set for matching the impedance of the subsystem that drives the isolation switch.

[0025] In the illustrated embodiment, the second refining stage comprises a self-adjusting damper circuit 240 which smoothes out the impedance-matched signal and removes the transmission line characteristics from the received signals. In other embodiments, the self-adjusting damper circuit 240 may be placed after the router 210 in order to achieve better results.

[0026] In certain embodiments, the second refining stage further comprises a filter/repeater/amplifier 250. As schematically shown in Figure 6, the filter/repeater/amplifier circuit 250 of certain embodiments includes a filter 253 which reduces high frequency noise that is induced by cross-coupling between wires or transmission lines. In certain embodiments, the filter 253 may reduce low frequency noise in addition to or instead of high frequency noise, and/or may reduce noise having a particular range of frequencies. In addition, in some embodiments, the filter/repeater/amplifier enable signal 251 puts the filter/repeater/amplifier 250 in a high impedance mode when the impedance matching circuit 230 is enabled or the self-adjusting damper 240 is active. In certain embodiments, for example, when the circuit 200 receives a signal (e.g., a signal traveling from receiver 101 to subsystem 103 in Figure 3, or from a first subsystem 220 to a second subsystem 420 in Figure 4), the impedance matching circuit 230, the damper circuit 240, and the filter/repeater/amplifier circuit 250, along with the dotted path in the router 210 are active and all other elements may be inactive, providing one-to-one connection between the first subsystem 220 and the second subsystem 420.

[0027] Figure 5 schematically illustrates an example programmable impedance matching circuit 230 in accordance with certain embodiments described herein. In certain embodiments, the impedance matching circuit 230 may be integrated with the router 210 and with the filter/repeater/amplifier 250. Alternatively, in other embodiments, the impedance matching circuit 230 may be implemented with discrete components.

[0028] Figure 6 schematically illustrates an example filter/repeater/amplifier circuit 250 in accordance with certain embodiments described herein. In certain embodiments, the

filter/repeater/amplifier 250 can be integrated with the router 210 and impedance matching circuit 230. In other embodiments, the filter/repeater/amplifier 250 may be implemented with discrete components. In certain embodiments, the filter/repeater/amplifier 250 comprises a filter 253 and a repeater/amplifier 252 with an output enable control input 251 for controlling the bi-directional signal path. In certain embodiments, the filter/repeater/amplifier 250 is programmable and dynamically controllable.

[0029] In some embodiments, the circuit 200 may be implemented in a memory subsystem of an electronic system (e.g., a personal computer, server computer, or data storage system). In general, such electronic systems include memory subsystems that contain multiple memory boards or modules that are driven by a single memory controller which synchronizes memory access and manages memory coherency. However, having a single memory controller may gradually degrade system performance and increase power consumption as the physical and the logical size of the memory subsystem increases. The increase in the memory speed may further impair the performance and increase the power consumption. The memory system may advantageously be used with a power supply which stabilizes the voltage level, and/or a thermo-mechanical component to moderate memory subsystem temperature, and/or software for throttling the processor to lower the memory access rate. Additionally, impedance mismatches between the system board, the memory boards, and the memory devices may also substantially degrade the performance and power efficiency of traditional memory systems.

[0030] In certain embodiments, the circuit 200 described herein addresses these speed and power issues in memory systems. For example, the circuit 200 of certain embodiments can comprise an isolation switch and can logically isolate inactive subsystems and reduce signal reflection and/or other noise at, for example, the interface between the system board and the memory board and at the interface between the memory board and the memory devices. In an embodiment in which the circuit 200 is incorporated in a memory system, the receiver 101 and the driver 102 of Figure 3 would represent data and address signals between the memory controller and the subsystems 103, 104 which may represent two sets of independently addressable memory devices. In certain embodiments, the circuit 200 may be disposed on the same memory board that is populated with memory devices 103, 104.

[0031] Referring to Figure 4, and with respect to an example memory application, the subsystem port connections 220, 320 represent a memory board interface to a system board while the subsystem port connections 420, 520 represent the isolation circuit 200 interface to the memory devices. In some embodiments, the impedance matching circuits 230, 330 and the self-adjusting dampers 240, 340 are employed to reduce the signal reflection due to the impedance mismatching between the system board and the memory board. In certain embodiments, the filter/repeater/amplifier 250 filters out, for example, cross-coupling between signals. For example, the filter 253 of the filter/repeater/amplifier 250 of Figure 6 may filter out cross-coupling between data, address, and/or control wires in a memory application. In certain embodiments, the repeater/amplifier 252 of the filter/repeater/amplifier of Figure 6 reshapes the signal which may be deteriorated due to transmission over lossy transmission lines having discontinuity points. In certain embodiments, the repeater/amplifier 252 may be configured to repeat the input signal, generating an output signal with earlier transition trigger points and faster slew rates in order to regain speed and to reduce the transient power.

[0032] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A memory module comprising:
 - a printed circuit board comprising at least one connector configured to be electrically coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board;
 - a circuit, comprising:
 - a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports electrically coupled to at least one memory device of the plurality of memory devices; and
 - a second set of ports comprising one or more bi-directional ports, each port of the second set of ports electrically coupled to the at least one connector;
 - wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.
2. The memory module of Claim 1, wherein the noise comprises signal reflections.
3. The memory module of Claim 1, wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.
4. The memory module of Claim 3, wherein the coarse correction element comprises a programmable impedance matching circuit.
5. The memory module of Claim 3, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.
6. The memory module of Claim 5, wherein the fine correction element comprises a damper circuit.
7. The memory module of Claim 6, wherein the damper circuit is a self-adjusting damper circuit.

8. The memory module of Claim 6, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

9. The memory module of Claim 8, wherein the filter circuit comprises a band-pass filter.

10. The memory module of Claim 8, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

11. The memory module of Claim 1, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

12. The memory module of Claim 1, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

13. A method of using a memory module with a computer system, the method comprising:

providing a memory module comprising a printed circuit board having at least one connector configured to be electrically coupled to a memory controller of a computer system and to a plurality of memory devices, the memory module further comprising a circuit, the circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports electrically coupled to at least one memory device of the plurality of memory devices; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports electrically coupled to the at least one connector; and

wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of

ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports; and

activating the circuit to selectively electrically couple at least one of the one or more ports of the second set of ports to at least one of the one or more ports of the first set of ports; and

activating the circuit to reduce noise in the one or more signals.

14. The method of Claim 13, wherein the noise comprises signal reflections.

15. The method of Claim 13, wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.

16. The method of Claim 15, wherein the coarse correction element comprises a programmable impedance matching circuit.

17. The method of Claim 15, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.

18. The method of Claim 17, wherein the fine correction element comprises a damper circuit.

19. The method of Claim 18, wherein the damper circuit is a self-adjusting damper circuit.

20. The method of Claim 18, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

21. The method of Claim 20, wherein the filter circuit comprises a band-pass filter.

22. The method of Claim 20, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

23. The method of Claim 13, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

24. A circuit comprising:

a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports configured to be electrically coupled to at least one memory device of a plurality of memory devices on a printed circuit board; and

a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be electrically coupled to at least one connector on the printed circuit board, the at least one connector configured to be electrically coupled to a memory controller of a computer system;

wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

25. The circuit of Claim 24, wherein the noise comprises signal reflections.

26. The circuit of Claim 24, wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.

27. The circuit of Claim 26, wherein the coarse correction element comprises a programmable impedance matching circuit.

28. The circuit of Claim 26, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.

29. The circuit of Claim 28, wherein the fine correction element comprises a damper circuit.

30. The circuit of Claim 29, wherein the damper circuit is a self-adjusting damper circuit.

31. The circuit of Claim 29, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

32. The circuit of Claim 31, wherein the filter circuit comprises a band-pass filter.

33. The circuit of Claim 31, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

34. The circuit of Claim 24, wherein the correction circuits reduce impedance mismatches at nodes between the memory board controller and the memory devices.

35. The circuit of Claim 24, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

36. A circuit comprising:

- a first set of ports comprising a plurality of bi-directional ports, each port of the first set of ports configured to be electrically coupled to at least one subsystem of a plurality of subsystems; and

- a second set of ports comprising one or more bi-directional ports, each port of the second set of ports configured to be electrically coupled to at least one subsystem of one or more subsystems;

- wherein the circuit is configured to selectively electrically couple one or more ports of the second set of ports to one or more ports of the first set of ports, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports each comprises a correction circuit which reduces noise in one or more signals transmitted between the one or more ports of the first set of ports and the one or more ports of the second set of ports.

37. The circuit of Claim 36, wherein the noise comprises signal reflections.

38. The circuit of Claim 36 wherein the correction circuit comprises a coarse correction element to provide coarse correction of the noise.

39. The circuit of Claim 38, wherein the coarse correction element comprises a programmable impedance matching circuit.

40. The circuit of Claim 38, wherein the correction circuit further comprises a fine correction element to provide fine correction of the noise.

41. The circuit of Claim 40, wherein the fine correction element comprises a damper circuit.

42. The circuit of Claim 41, wherein the damper circuit is a self-adjusting damper circuit.

43. The circuit of Claim 41, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a filter circuit which reduces noise from the one or more signals, the noise having a predetermined range of frequencies.

44. The circuit of Claim 43, wherein the filter circuit comprises a band-pass filter.

45. The circuit of Claim 43, wherein the one or more ports of the first set of ports and the one or more ports of the second set of ports comprises a repeater/amplifier circuit which improves the signal quality of the one or more signals.

46. The circuit of Claim 36, wherein the correction circuits reduce impedance mismatches at nodes between the one or more subsystems and the plurality of subsystems.

47. The circuit of Claim 36, wherein the circuit is selected from a group consisting of: a programmable-logic device (PLD), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a custom-designed semiconductor device, and a complex programmable-logic device (CPLD).

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

ABSTRACT

A memory module is provided which includes a printed circuit board having memory devices and a connector which can connect to a memory controller. The memory module also includes a circuit including a first set of ports comprising bi-directional ports. Each port of the first set of ports couples to at least one memory device. The circuit also includes a second set of ports comprising bi-directional ports. Each port of the second set of ports couples to the connector. The circuit is configured to selectively couple ports of the second set of ports to ports of the first set of ports. The ports of the first set of ports and the ports of the second set of ports include a correction circuit which reduces noise in signals transmitted between the ports of the first set of ports and the ports of the second set of ports.

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APPENDIX

SELF-ADJUSTING DAMPER**BACKGROUND**Field

[0001] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0002] Complex electronic systems often contain multiple sources of electrical noise. Such sources may include noise from signal cross coupling, transmission line reflections, ground bounce noise, noise due to an unstable power supply voltage, noise due to external EM waves, etc. In some cases these noise sources may be generally negligible depending on the noise tolerance of the system. However, many modern electronic systems incorporate relatively low supply voltages to accommodate state-of-art semiconductor processing technology. In such systems, the noise tolerance of the system is generally decreased. This phenomenon is clearly demonstrated in dynamic random-access memory (DRAM) systems where the operating voltage is sometimes substantially reduced in order to reduce the leakage current and optimize the performance.

[0003] The low power supply voltage can reduce the noise tolerance and can be compensated for, at least in part, by additional electronic components to ensure proper operation of the DRAM modules. For example, terminating resistors may be added in some cases (e.g., for DDR2 and DDR3 memory modules) at transmission line discontinuity points. Although terminating resistors may reduce the signal noise by lowering the amplitude of reflected waves, there may still remain substantial signal ringing since, in most cases, values of the terminating resistor on the memory module may not completely match the system board impedance. Furthermore, because each electrical component in a system exhibits its own unique electrical characteristics including the output driving capabilities, the signals generated from each electrical component produce their own unique reflected wave patterns, which cannot be completely matched with a set of impedance values.

[0004] Circuits often used to remove noises (e.g., to counteract ringing signals and/or reflected signals with small amplitudes) from the received signals include band-pass

filters, impedance matching circuits, echo cancellers, and error-correction circuits that reconstruct the transmitted signal. Each of these circuits exhibits specific characteristics and are used in well-defined application areas. In general, band-pass filters are analog circuits which generally pass frequencies within a certain range and reject frequencies outside that range. Impedance matching circuits are generally characterized by a fixed impedance value that exhibits impedance characteristics similar to those of the transmission line to which it is interfacing so as to reduce signal reflections. Echo cancellers and the error-correction circuits are typically digital circuits that interact with a micro-processor.

SUMMARY

[0005] In certain embodiments, a self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is, in certain embodiments, substantially representative of the signal component of the input signal.

[0006] In certain embodiments, a method of removing a noise component of an electrical signal is provided. The method involves receiving an input signal including a noise component and a signal component. The method further includes generating a first signal corresponding to the noise component of the input signal and generating a second signal corresponding to a delayed version of the input signal. The method also involves adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal which, in certain embodiments, is substantially representative of the signal component of the input signal.

[0007] In certain embodiments, a self-adjusting damper circuit is provided which includes a means for receiving an input signal comprising a noise component and a signal component, a first generation means for generating a first signal corresponding to the noise component of the input signal, and a second generation means for generating a second signal

corresponding to a delayed version of the input signal. The self-adjusting damper circuit also includes a means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal. In certain embodiments, the output signal is substantially representative of the signal component of the input signal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure 1 is a block diagram illustrating an example system incorporating self-adjusting damper circuits in accordance with certain embodiments described herein.

[0009] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0010] Figure 3 is a schematic diagram illustrating another example self-adjusting damper circuit in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0011] In many electronic systems, while multiple noise sources contribute to the distortion of the transmitted signal, for conventional systems there are usually one or two major noise sources while other noise sources make minor contributions. Thus, system designers have been able to implement one or two noise removing circuits in conventional electronic systems to remove the major noise sources and to ignore the minor noise sources. However, in some high performance systems such as memory subsystems, even relatively minor noises may contribute significant signal distortions due to, for example, the lower supply voltages. For example, a major noise contribution can be the reflected signals at the transmission line discontinuity points, and a minor noise contribution can be due to signal cross-coupling. Such systems may benefit from a noise filtering scheme that may include various types of noise removing circuits but may also potentially increase cost, complexity, and/or the time for system development.

[0012] Certain embodiments described herein present a self-adjusting electrical damper applicable to a wide frequency range. Certain embodiments described herein are implemented in an application-specific integrated circuit (ASIC) or in a System-on-Chip (SoC) device.

[0013] Certain embodiments described herein provide a self-adjusting damper configured to remove signal noise contributed by, for example signal cross-coupling and the

signal reflections. In certain embodiments, the self-adjusting damper is a component of a system comprising signal buses and transmission lines. In certain embodiments, the self-adjusting damper employs band-pass filtering and echo cancellation without any external control inputs.

[0014] In certain embodiments, the self-adjusting damper can be implemented either with discrete components or integrated into an ASIC device. Advantageously, the damper of certain embodiments is capable of reducing noise due to, for example, signal coupling between wires or signal reflection caused by mismatched terminations of transmission lines. In certain embodiments, the self-adjusting damper includes a delay component, a negative voltage controlled current source, and a positive voltage controlled current source. In certain embodiments, the signal is delayed through at least one of the current sources. In certain embodiments, the signal delay through the delay component is substantially similar to the signal delay through the current sources.

[0015] In certain embodiments, the voltage controlled current sources react to a change of the input voltage and have corresponding current source multiplication factors g_1 and g_2 . These multiplication factors are variables that are functions of the slew rate of the input signal such that a faster input transition produces higher transconductance on the current sources, smoothing out or reducing the signal ringing on the output. Advantageously, the damper circuit of certain embodiments actively counteracts signal noise, thereby allowing noise reduction over a broad range of noise frequencies rather than over, for example, a frequency window. Moreover, in certain embodiments, the damper circuit actively counteracts signal noise regardless of the periodicity of the input signal and/or of the noise component of the signal. In certain embodiments, the damper is implemented without analog circuit technology (e.g., within an ASIC without invoking analog circuit technology), thereby enabling easy design migration to various process technologies.

[0016] Figure 1 is a block diagram illustrating an example system incorporating example self-adjusting damper circuits 141, 142 in accordance with certain embodiments described herein. A plurality of drivers 101, 102 transmit signals via transmission lines or wires with a cross coupling impedance of Z_c 120. Z_c 120 includes cross-coupling capacitance and (mutual) inductance as well as any transconductance between the wires. In certain

embodiments, the impedance matching components 131, 132 may comprise a fixed termination resistor for the respective transmission line 111, 112. The self-adjusting damper circuits 141, 142 of certain embodiments advantageously reduce cross coupling noise due to Z_c and/or the remaining reflected wave after impedance matching is performed by the impedance matching components 131, 132.

[0017] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein. According to certain embodiments, the mathematical representation of the output 202 with load capacitance C_L can be expressed by the following equations:

$$V_o(t) = \int \left[\left(I(t) - \frac{g1 \delta V_i(t - \Delta \tau)}{dt} - \frac{g2 \delta V_i(t - \Delta \tau)}{dt} \right) C_L \right] dt$$

$$= \left[\int I(t) dt - (g1 \delta V_i(t - \Delta \tau) + g2 \delta V_i(t - \Delta \tau)) \right] C_L$$

Where,

$$I(t) = \frac{(V_o(t) - V_i(t))}{Z_d}$$

Thus,

$$Z_d * dV_o(t) / dt = V_i(t) - V_o(t) - (g1 + g2) * C_L * d(\delta V_i(t - \Delta \tau)) / dt$$

And,

$$V_o(t) = V_i(t) - (g1 + g2) * C_L * d(\delta V_i(t - \Delta \tau)) / dt - Z_d * dV_o(t) / dt$$

[0018] The above equations show that the output voltage of certain embodiments is a function of the change rate of the input signal transition rate and the change rate of the output itself. Because the change rate of the input transition rate and the output change rate have a negative effect on the output voltage, the damper of certain embodiments reduces noise having a slew rate that is faster than the slew rate of the non-noise component of the signal. Thus, certain embodiments provide a self-adjusting damper that effectively removes or reduces noises such as cross-coupling noises and reflected signals in complex electrical and electronic systems.

[0019] Figure 3 is a schematic diagram of another example of a self-adjusting damper circuit in accordance with certain embodiments described herein. The illustrated

embodiment comprises two voltage controlled current sources 340, 350 which are included in a first sub-circuit. The current source 340 of the illustrated embodiment is an AC-coupled, negative-gain pmos voltage-controlled current source for increasing the current flow from the power supply when the input sees a negative noise component, thus reducing the negative noise component on the output. The current source 340 includes two pmos transistors 341, 342. The pmos transistor 341 is a current-limiting transistor that ensures that the current supplied to the output has a relatively smaller magnitude than does the current to/from the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other pmos transistor 342 provides the negative-voltage-controlled current source function by AC coupling with the input signal V_i 301, on an input terminal, via a capacitor 311. Thus, the transconductance of this transistor 342 changes inversely proportionally to the slew rate of the input.

[0020] The current source 350 of the illustrated embodiment is an AC-coupled, positive-gain nmos current source for increasing the current flow to ground when the input sees a positive noise component, thus reducing the positive noise component on the output. The current source 350 includes two nmos transistors 351, 352. The nmos transistor 351 is a current-limiting transistor that ensures the current taken away from the output has a relatively smaller magnitude than does the current from/to the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other nmos transistor 352 provides the positive-voltage-controlled current source function by AC coupling with the input signal V_i 301 via a capacitor 311. Thus, the transconductance of this transistor 342 changes proportionally with the slew rate of the input.

[0021] In the illustrated embodiment, the resistor R_d 310 is included in a second sub-circuit. The resistor R_d 310 is a delay element which introduces a signal delay compatible with the current source delay from the first sub-circuit and also may reduce the reverse feedback of the output signal back to the input. Since the delay time from input V_i 301 on the input terminal to output V_o 302 on the output terminal is the function of the time constant, $R_d * C_L$, the resistance of the resistor R_d 310 can be selected based on the estimated value of C_L 360. Thus, in certain embodiments, the input signal can be delayed to match the reaction time of the voltage-controlled current sources that generate a negative feedback current to remove

signal noise. In certain embodiments the damper 141 can reduce cross coupling noise and reflections having various slew rate ranges.

[0022] In certain embodiments, the damper also includes a current-limiting, self-biasing op bias for biasing the reference voltage so that the operation point can be established in the middle of the linear range.

[0023] Advantageously, in certain embodiments, the damper is configured to be implemented in an application specific integrated circuit (“ASIC”) without requiring analog circuit technologies, enabling easy migration to other process technologies. For example, the damper is configured to be implemented in digital process technologies comprising a component resolution of approximately plus or minus 20 percent. Moreover, in certain embodiments, the damper may be implemented with no external control signals or memory elements.

[0024] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-adjusting damper circuit, comprising:
 - an input terminal adapted to receive an input signal comprising a noise component and a signal component;
 - an output terminal;
 - a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal; and
 - a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal, wherein the circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal, the output signal substantially representative of the signal component of the input signal.
2. The circuit of Claim 1, wherein the second sub-circuit comprises a resistor.
3. The circuit of Claim 1, wherein the first sub-circuit comprises a current source.
4. The circuit of Claim 3, wherein the current source comprises a voltage-controlled current source.
5. The circuit of Claim 4, wherein voltage-controlled current source comprises one or more transistors.
6. The circuit of Claim 1, wherein the first sub-circuit comprises a negative current source and a positive current source.
7. The circuit of Claim 6, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.
8. The circuit of Claim 7, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the trans-conductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.
9. The circuit of Claim 7, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

10. The circuit of Claim 9, wherein the first and second current limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

11. The circuit of Claim 10, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current limiting component, and the second current limiting component comprise one or more transistors.

12. The circuit of Claim 1, wherein the first sub-circuit is adapted to generate the first signal by generating a positive signal when the noise component of the input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

13. The circuit of Claim 12, wherein the first sub-circuit is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

14. The circuit of Claim 1, wherein the first signal comprises negative feedback current.

15. The circuit of Claim 1, wherein the circuit is configured to be operate on a digital semiconductor process technology.

16. The circuit of Claim 1, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution of less than plus or minus 20 percent.

17. The circuit of Claim 1, wherein the circuit is implemented on a memory module.

18. The circuit of Claim 1, wherein the noise-counteracting signal actively counteracts a noise component of the delayed version of the input signal.

19. A method of removing a noise component of an electrical signal, the method comprising:

receiving an input signal comprising a noise component and a signal component;

generating a first signal corresponding to the noise component of the input signal;

generating a second signal corresponding to a delayed version of the input signal; and

adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal substantially representative of the signal component of the input signal.

20. The method of Claim 19, wherein generating the first signal comprises generating a positive signal when the noise component of the input signal is negative and generating a negative signal when the noise component of the input signal is positive.

21. The method of Claim 20, wherein generating a positive signal comprises increasing the current flow from a power supply to the first signal and generating a negative signal comprises increasing the current flow from the first signal to ground.

22. The method of Claim 19, further comprising limiting the magnitude of the first signal in relation to the second signal.

23. The method of Claim 19, wherein adding the noise-counteracting signal to the second signal actively counteracts a noise component of the delayed version of the input signal.

24. A self-adjusting damper circuit, comprising:

means for receiving an input signal comprising a noise component and a signal component;

first generation means for generating a first signal corresponding to the noise component of the input signal; and

second generation means for generating a second signal corresponding to a delayed version of the input signal,

means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal, the output signal substantially representative of the signal component of the input signal.

25. The circuit of Claim 24, wherein the second generation means comprises a resistor.

26. The circuit of Claim 24, wherein the first generation means comprises a current source.

27. The circuit of Claim 26, wherein the current source comprises a voltage-controlled current source.

28. The circuit of Claim 27, wherein voltage-controlled current source comprises one or more transistors.

29. The circuit of Claim 24, wherein the first generation means comprises a negative current source and a positive current source.

30. The circuit of Claim 29, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.

31. The circuit of Claim 30, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the transconductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.

32. The circuit of Claim 31, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

33. The circuit of Claim 32, wherein the first and second current-limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

34. The circuit of Claim 33, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current-limiting component, and the second current-limiting component comprise one or more transistors.

35. The circuit of Claim 24, wherein the first generation means is adapted to generate the first signal by generating a positive signal when the noise component of the input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

36. The circuit of Claim 35, wherein the first generation means is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

37. The circuit of Claim 24, wherein the first signal comprises negative feedback current.

38. The circuit of Claim 24, wherein the circuit is configured to be operate on a digital semiconductor process technology.

39. The circuit of Claim 24, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution less than plus or minus 20 percent.

40. The circuit of Claim 24, wherein the circuit is implemented on a memory module.

41. The circuit of Claim 24, wherein the addition means actively counteracts a noise component of the delayed version of the input signal.

SELF-ADJUSTING DAMPER

ABSTRACT

A self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is substantially representative of the signal component of the input signal.

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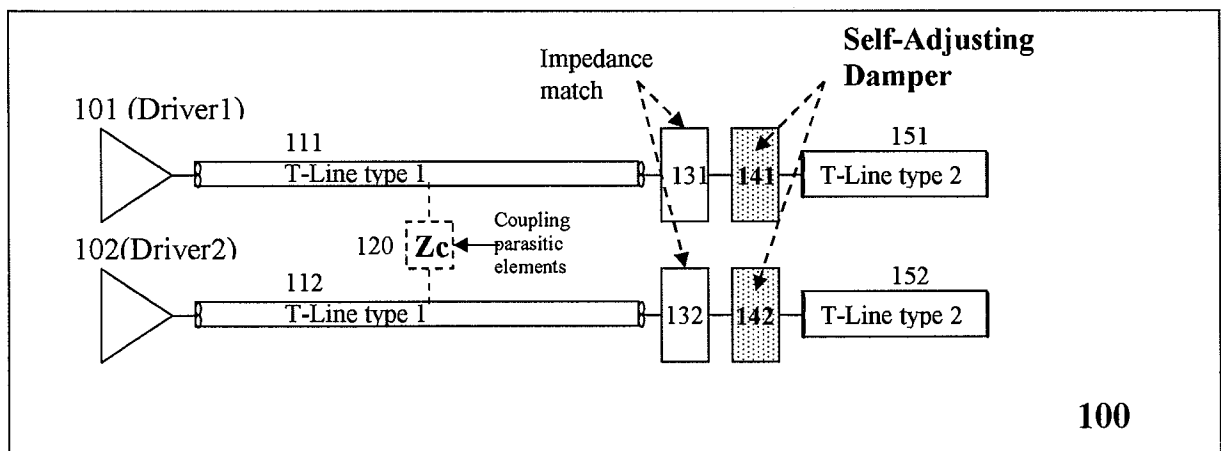


Figure 1

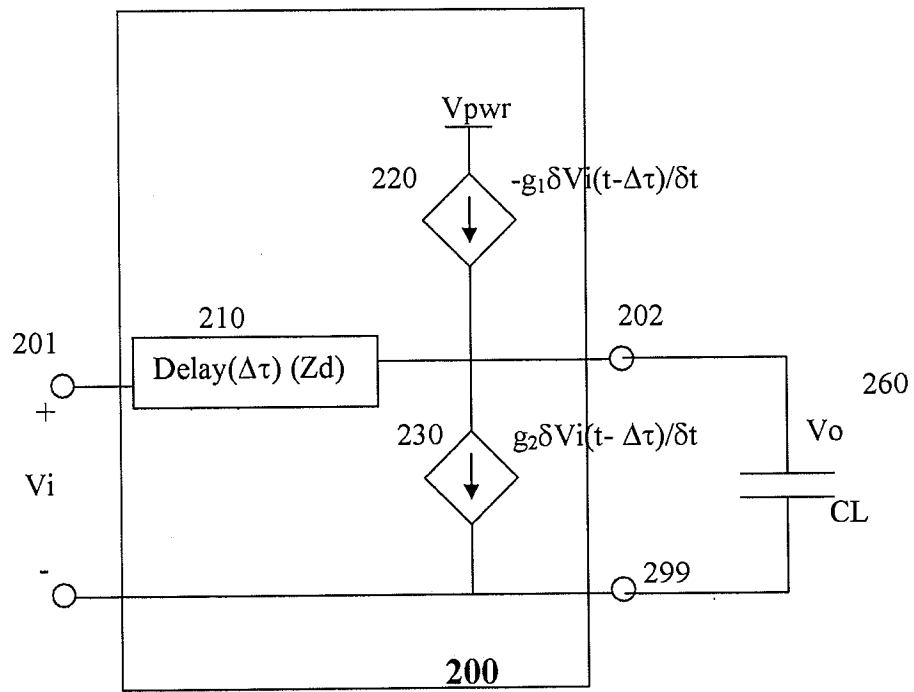


Figure2

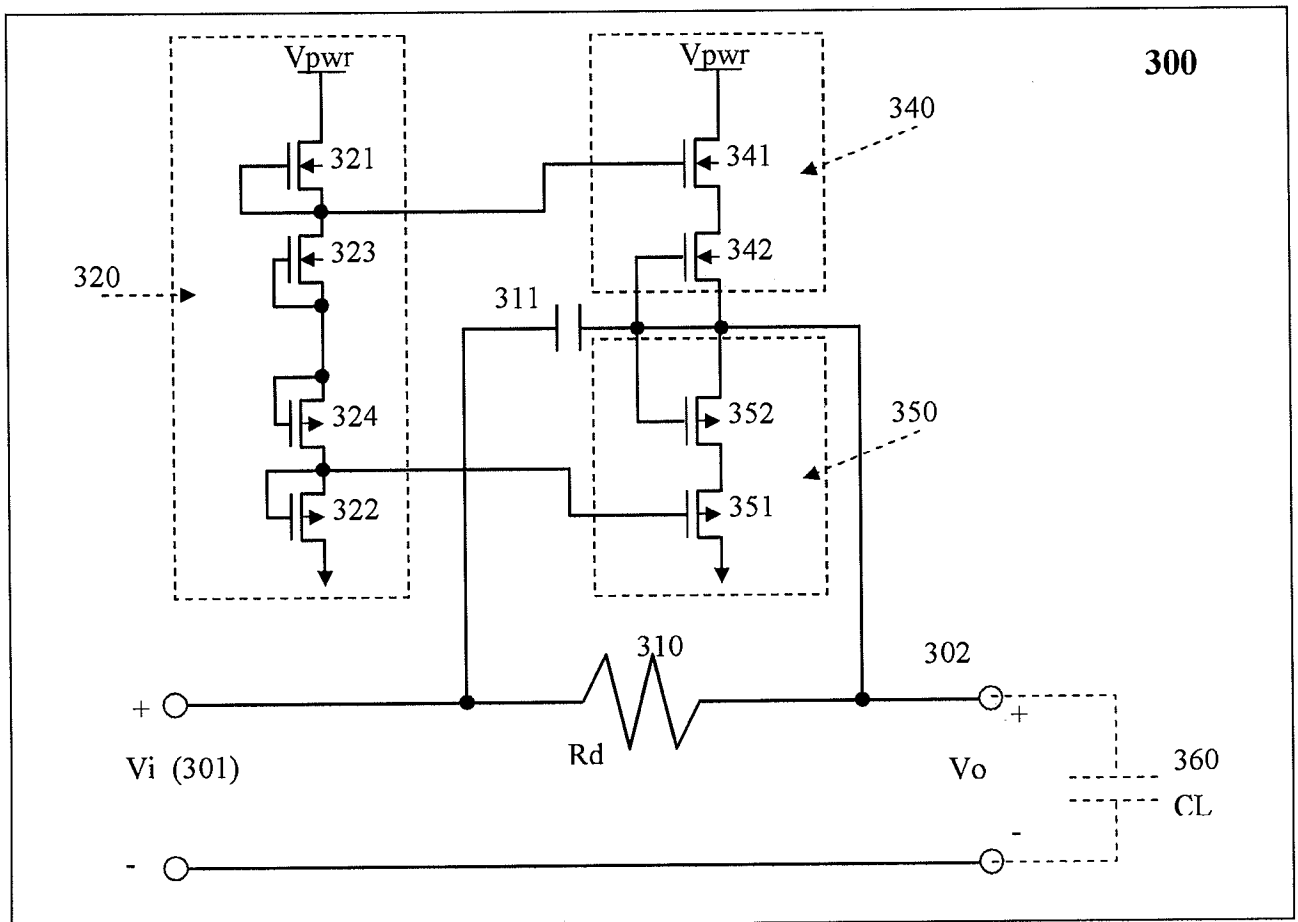


Figure 3

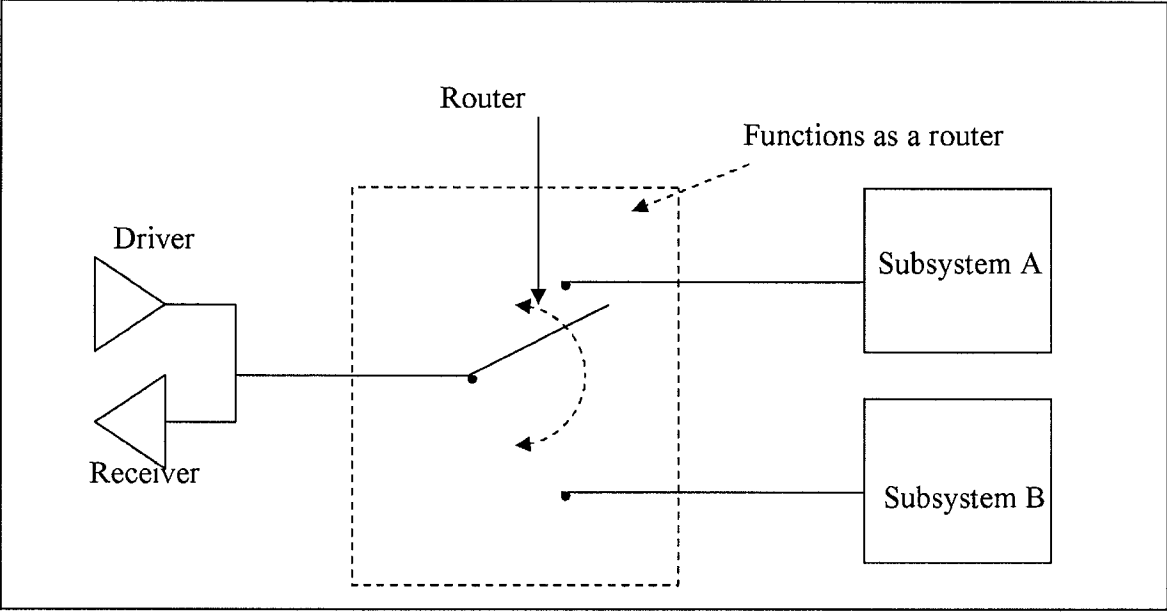


Figure 1

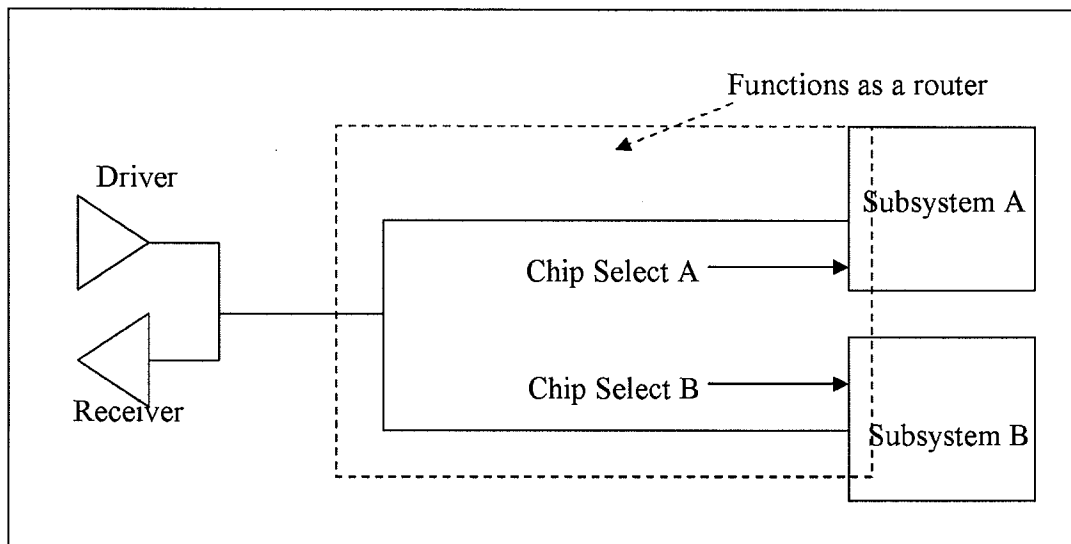


Figure 2

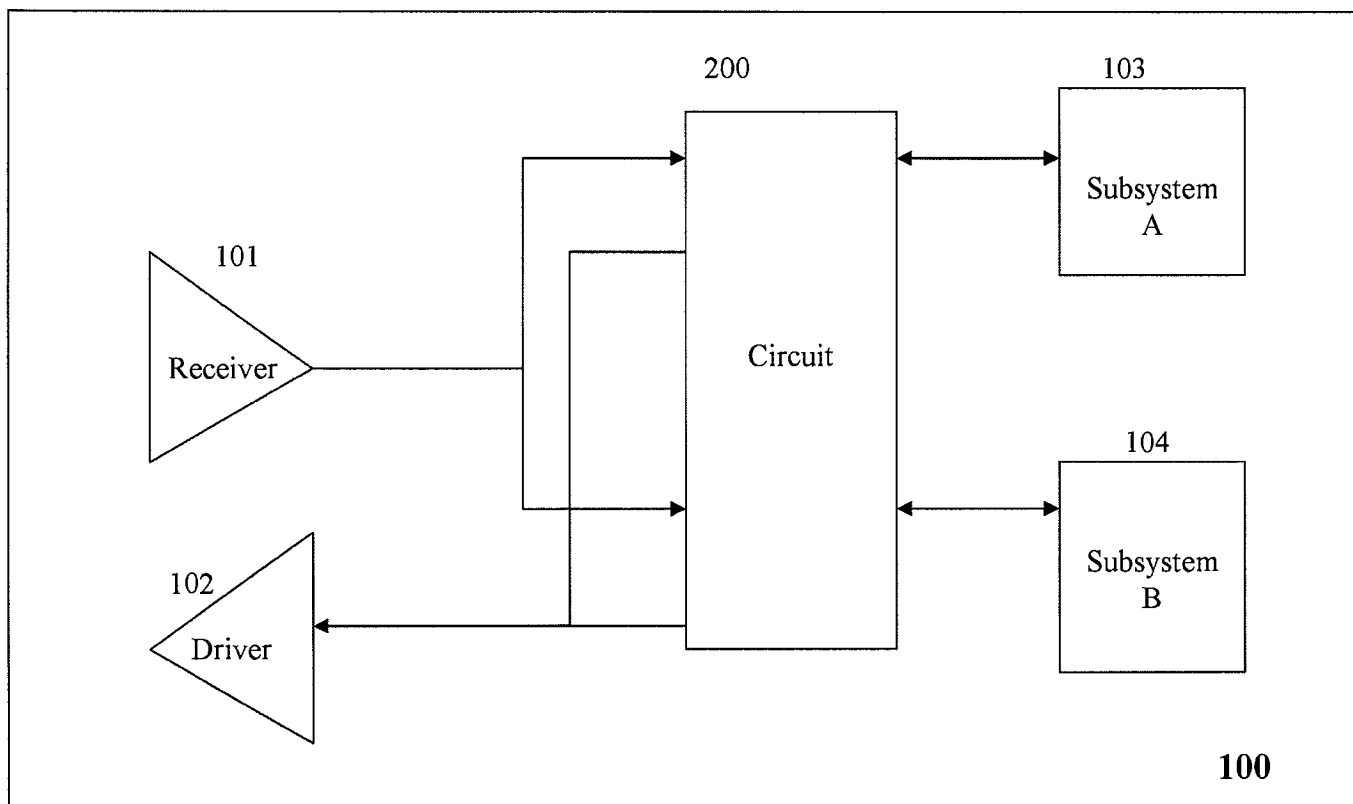


Figure 3

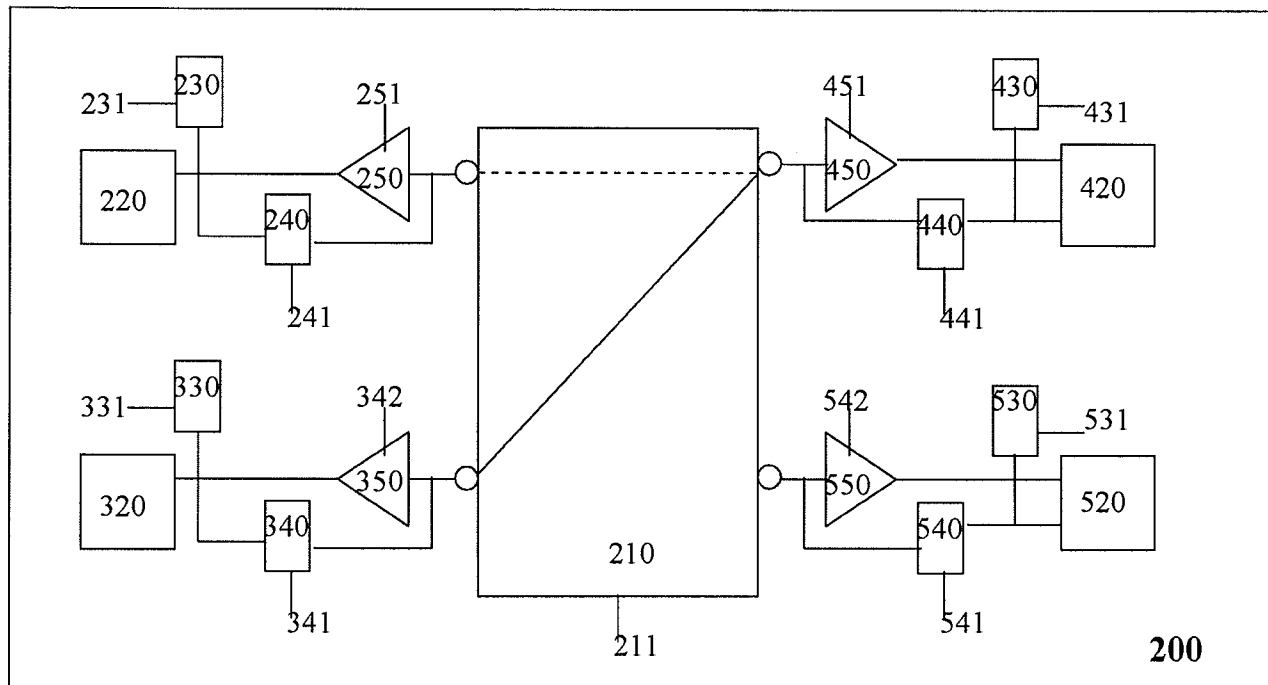


Figure 4

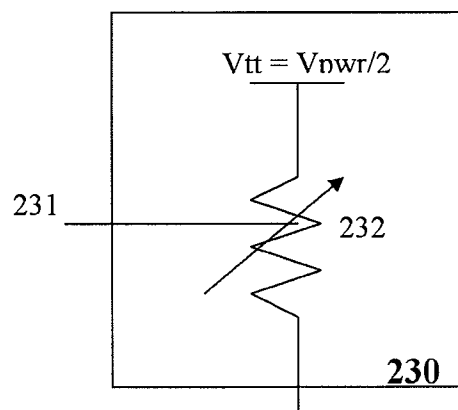
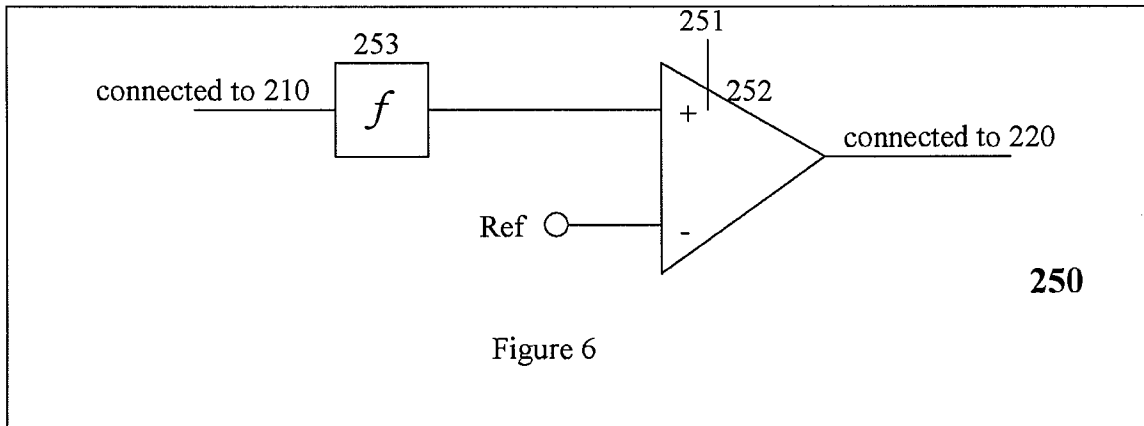


Figure 5



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Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

First Named Inventor/Applicant Name:

To be provided later .

Filer:

Bruce S. Itchkawitz/Dee Manena

Attorney Docket Number:

NETL.045PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

Provisional application filing

1005

1

210

210

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				210

Electronic Acknowledgement Receipt

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International Application Number:	
Confirmation Number:	1764
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First Named Inventor/Applicant Name:	To be provided later .
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Quyen Lieu
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045PR
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Application Type:	Provisional

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NETL045PR_ADS.pdf	243412 6fe62a8c8c1875ea429c67ca1975c8e79e771125	no	4
Warnings:					
Information:					
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2		NETL045PR_APPLICATION.pdf	1484241 ae311f5ba5e9257dbc18cc928d5b60517f1bbeb36	yes	34
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	10	
	Claims		11	16	
	Abstract		17	17	
	Appendix to the Specification		18	34	
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3	Drawings-only black and white line drawings	NETL045PR_DRAWINGS.pdf	42457 70fcb3793d9dc2d11f1e9ff14d311dcf1996cd9	no	6
Warnings:					
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4	Fee Worksheet (PTO-06)	fee-info.pdf	8142 af8709838e65915db8ed3d0e5379b7b16da146dc	no	2
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61/044,839	04/14/2008		210	NETL.045PR		

CONFIRMATION NO. 1764

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FILING RECEIPT



OC000000029445407

Date Mailed: 04/22/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please write to the Office of Initial Patent Examination's Filing Receipt Corrections. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney: The patent practitioners associated with Customer Number 20995

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The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,839**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).



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United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,839	04/14/2008		NETL.045PR

CONFIRMATION NO. 1764

FORMALITIES LETTER



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/22/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of **\$50** for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$50** for a non-small entity

- **\$50** Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/mduong/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Please Direct All Correspondence to Customer Number 20995

RESPONSE TO FORMALITIES NOTICE

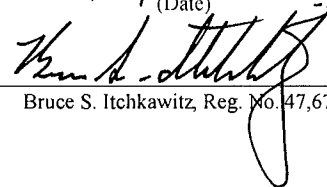
Applicant : Hyun Lee et al.
App. No : 61/044,839
Filed : April 14, 2008
For : CIRCUIT PROVIDING LOAD
ISOLATION AND NOISE
REDUCTION
Art Unit : Unknown

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

6/16/08

(Date)



Bruce S. Itchkawitz, Reg. No. 47,677

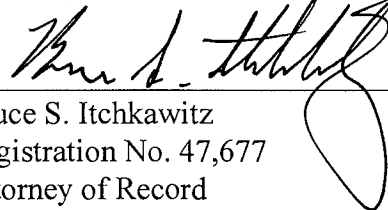
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without the inventors information and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following:

- (X) A Notice to File Missing Parts dated April 22, 2008.
- (X) Supplemental Application Data Sheet.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.



Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,839	04/14/2008		NETL.045PR

CONFIRMATION NO. 1764

FORMALITIES LETTER



OC000000029445408

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/22/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of \$50 for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is \$50 for a non-small entity

- \$50 Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
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Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/mduong/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/044,839
Filing Date:: April 14, 2008
Application Type:: Provisional
Subject Matter:: Utility
CD-ROM or CD-R?:: None
Title:: CIRCUIT PROVIDING LOAD ISOLATION AND
NOISE REDUCTION
Attorney Docket Number:: NETL.045PR
Request for Early Publication?:: No
Request for Non-Publication?:: No
Total Drawing Sheets:: 6
Small Entity?:: No
Petition Included?:: No
Secrecy Order in Parent Appl.?:: No

Inventor Information

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Hyun
Family Name:: Lee
City of Residence:: Ladera Ranch
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 21 Thalia Street

City:: Ladera Ranch
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92694

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Jayesh
Middle Name:: R.
Family Name:: Bhakta
City of Residence:: Cerritos
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 12220 Rose Street
City:: Cerritos
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 90703

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Jeffrey
Middle Name:: C.
Family Name:: Solomon
City of Residence:: Irvine
State or Prov. of Residence:: CA

Country of Residence:: U.S.
Street:: 16 Silver Fir
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92604

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Mario
Middle Name:: Jesus
Family Name:: Martinez
City of Residence:: Laguna Niguel
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 19 Lindall Street
City:: Laguna Niguel
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92677

Applicant Authority Type:: Inventor
Primary Citizenship Country:: Taiwan
Status:: Full Capacity
Given Name:: Chi-She
Family Name:: Chen

City of Residence:: Walnut
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 944 Crystal Water Lane
City:: Walnut
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 91789

Correspondence Information

Correspondence Customer Number:: 20,995
Phone Number:: (949) 760-0404
Fax Number:: (949) 760-9502
E-Mail Address:: efilings@kmob.com

Representative Information

Representative Customer Number:: 20,995

Assignment Information

Assignee Name:: Netlist, Inc.
Street:: 51 Discovery, Ste. #150
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92618

5519318
061208

Electronic Patent Application Fee Transmittal

Application Number:	61044839			
Filing Date:	14-Apr-2008			
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/alexis swan			
Attorney Docket Number:	NETL.045PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	3464635
Application Number:	61044839
International Application Number:	
Confirmation Number:	1764
Title of Invention:	CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Heide Young
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.045PR
Receipt Date:	16-JUN-2008
Filing Date:	14-APR-2008
Time Stamp:	19:07:24
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	3759
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	NETL_045PR.pdf	201021	no	7
			35bb1b32a82863f1b229c06677111108a72ad1af		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	8170	no	2
			ce73b9b98f6cf828e46a4e6ef04d1e54abb57ba3		

Warnings:**Information:**

Total Files Size (in bytes):			209191
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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United States Patent and Trademark Office
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www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,839	04/14/2008		260	NETL.045PR		

CONFIRMATION NO. 1764

UPDATED FILING RECEIPT

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614



OC000000030523670

Date Mailed: 06/23/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please submit a written request for a Filing Receipt Correction. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Jeffrey C. Solomon, Irvine, CA;
Mario Jesus Martinez, Laguna Niguel, CA;
Chi-She Chen, Walnut, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/16/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,839**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

CIRCUIT PROVIDING LOAD ISOLATION AND NOISE REDUCTION

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing

of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

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For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of

State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.044PR
		Application Number	
Title of Invention	SELF-ADJUSTING DAMPER		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority		☒ Inventor		☐ Legal Representative under 35 U.S.C. 117
		☐ Party of Interest under 35 U.S.C. 118		
Prefix	Given Name	Middle Name	Family Name	Suffix
	To be provided later			
Residence Information (Select One)				
☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residenceⁱ	
Citizenship under 37 CFR 1.41(b)ⁱ				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Countryⁱ		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.				
				Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	SELF-ADJUSTING DAMPER		
Attorney Docket Number	NETL.044PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	3	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.044PR
		Application Number	
Title of Invention	SELF-ADJUSTING DAMPER		

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	20995		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, or 365(c) or indicate National Stage entry from a PCT application. Providing this information in the application data sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78(a)(2) or CFR 1.78(a)(4), and need not otherwise be made part of the specification.

Prior Application Status		Remove	
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Foreign Priority Information:

This section allows for the applicant to claim benefit of foreign priority and to identify any prior foreign application for which priority is not claimed. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55(a).

Remove			
Application Number	Country ⁱ	Parent Filing Date (YYYY-MM-DD)	Priority Claimed
			☐ Yes ☒ No

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Assignee Information:

Providing this information in the application data sheet does not substitute for compliance with any requirement of part 3 of Title 37 of the CFR to have an assignment recorded in the Office.

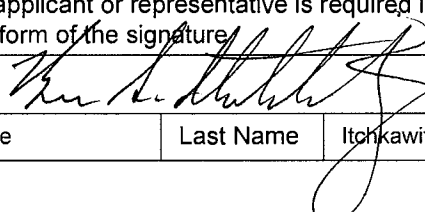
Assignee 1

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.044PR
		Application Number	
Title of Invention	SELF-ADJUSTING DAMPER		

If the Assignee is an Organization check here. ☒			
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A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 10.18. Please see 37 CFR 1.4(d) for the form of the signature.					
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SELF-ADJUSTING DAMPER**BACKGROUND**Field

[0001] The present application relates generally to the field of electrical signal integrity and more specifically to circuits for improving electrical signal integrity.

Description of the Related Art

[0002] Complex electronic systems often contain multiple sources of electrical noise. Such sources may include noise from signal cross coupling, transmission line reflections, ground bounce noise, noise due to an unstable power supply voltage, noise due to external EM waves, etc. In some cases these noise sources may be generally negligible depending on the noise tolerance of the system. However, many modern electronic systems incorporate relatively low supply voltages to accommodate state-of-art semiconductor processing technology. In such systems, the noise tolerance of the system is generally decreased. This phenomenon is clearly demonstrated in dynamic random-access memory (DRAM) systems where the operating voltage is sometimes substantially reduced in order to reduce the leakage current and optimize the performance.

[0003] The low power supply voltage can reduce the noise tolerance and can be compensated for, at least in part, by additional electronic components to ensure proper operation of the DRAM modules. For example, terminating resistors may be added in some cases (e.g., for DDR2 and DDR3 memory modules) at transmission line discontinuity points. Although terminating resistors may reduce the signal noise by lowering the amplitude of reflected waves, there may still remain substantial signal ringing since, in most cases, values of the terminating resistor on the memory module may not completely match the system board impedance. Furthermore, because each electrical component in a system exhibits its own unique electrical characteristics including the output driving capabilities, the signals generated from each electrical component produce their own unique reflected wave patterns, which cannot be completely matched with a set of impedance values.

[0004] Circuits often used to remove noises (e.g., to counteract ringing signals and/or reflected signals with small amplitudes) from the received signals include band-pass

filters, impedance matching circuits, echo cancellers, and error-correction circuits that reconstruct the transmitted signal. Each of these circuits exhibits specific characteristics and are used in well-defined application areas. In general, band-pass filters are analog circuits which generally pass frequencies within a certain range and reject frequencies outside that range. Impedance matching circuits are generally characterized by a fixed impedance value that exhibits impedance characteristics similar to those of the transmission line to which it is interfacing so as to reduce signal reflections. Echo cancellers and the error-correction circuits are typically digital circuits that interact with a micro-processor.

SUMMARY

[0005] In certain embodiments, a self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is, in certain embodiments, substantially representative of the signal component of the input signal.

[0006] In certain embodiments, a method of removing a noise component of an electrical signal is provided. The method involves receiving an input signal including a noise component and a signal component. The method further includes generating a first signal corresponding to the noise component of the input signal and generating a second signal corresponding to a delayed version of the input signal. The method also involves adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal which, in certain embodiments, is substantially representative of the signal component of the input signal.

[0007] In certain embodiments, a self-adjusting damper circuit is provided which includes a means for receiving an input signal comprising a noise component and a signal component, a first generation means for generating a first signal corresponding to the noise component of the input signal, and a second generation means for generating a second signal

corresponding to a delayed version of the input signal. The self-adjusting damper circuit also includes a means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal. In certain embodiments, the output signal is substantially representative of the signal component of the input signal.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure 1 is a block diagram illustrating an example system incorporating self-adjusting damper circuits in accordance with certain embodiments described herein.

[0009] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein.

[0010] Figure 3 is a schematic diagram illustrating another example self-adjusting damper circuit in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0011] In many electronic systems, while multiple noise sources contribute to the distortion of the transmitted signal, for conventional systems there are usually one or two major noise sources while other noise sources make minor contributions. Thus, system designers have been able to implement one or two noise removing circuits in conventional electronic systems to remove the major noise sources and to ignore the minor noise sources. However, in some high performance systems such as memory subsystems, even relatively minor noises may contribute significant signal distortions due to, for example, the lower supply voltages. For example, a major noise contribution can be the reflected signals at the transmission line discontinuity points, and a minor noise contribution can be due to signal cross-coupling. Such systems may benefit from a noise filtering scheme that may include various types of noise removing circuits but may also potentially increase cost, complexity, and/or the time for system development.

[0012] Certain embodiments described herein present a self-adjusting electrical damper applicable to a wide frequency range. Certain embodiments described herein are implemented in an application-specific integrated circuit (ASIC) or in a System-on-Chip (SoC) device.

[0013] Certain embodiments described herein provide a self-adjusting damper configured to remove signal noise contributed by, for example signal cross-coupling and the

signal reflections. In certain embodiments, the self-adjusting damper is a component of a system comprising signal buses and transmission lines. In certain embodiments, the self-adjusting damper employs band-pass filtering and echo cancellation without any external control inputs.

[0014] In certain embodiments, the self-adjusting damper can be implemented either with discrete components or integrated into an ASIC device. Advantageously, the damper of certain embodiments is capable of reducing noise due to, for example, signal coupling between wires or signal reflection caused by mismatched terminations of transmission lines. In certain embodiments, the self-adjusting damper includes a delay component, a negative voltage controlled current source, and a positive voltage controlled current source. In certain embodiments, the signal is delayed through at least one of the current sources. In certain embodiments, the signal delay through the delay component is substantially similar to the signal delay through the current sources.

[0015] In certain embodiments, the voltage controlled current sources react to a change of the input voltage and have corresponding current source multiplication factors g_1 and g_2 . These multiplication factors are variables that are functions of the slew rate of the input signal such that a faster input transition produces higher transconductance on the current sources, smoothing out or reducing the signal ringing on the output. Advantageously, the damper circuit of certain embodiments actively counteracts signal noise, thereby allowing noise reduction over a broad range of noise frequencies rather than over, for example, a frequency window. Moreover, in certain embodiments, the damper circuit actively counteracts signal noise regardless of the periodicity of the input signal and/or of the noise component of the signal. In certain embodiments, the damper is implemented without analog circuit technology (e.g., within an ASIC without invoking analog circuit technology), thereby enabling easy design migration to various process technologies.

[0016] Figure 1 is a block diagram illustrating an example system incorporating example self-adjusting damper circuits 141, 142 in accordance with certain embodiments described herein. A plurality of drivers 101, 102 transmit signals via transmission lines or wires with a cross coupling impedance of Z_c 120. Z_c 120 includes cross-coupling capacitance and (mutual) inductance as well as any transconductance between the wires. In

certain embodiments, the impedance matching components 131, 132 may comprise a fixed termination resistor for the respective transmission line 111, 112. The self-adjusting damper circuits 141, 142 of certain embodiments advantageously reduce cross coupling noise due to Z_c and/or the remaining reflected wave after impedance matching is performed by the impedance matching components 131, 132.

[0017] Figure 2 is a block diagram illustrating an example self-adjusting damper circuit in accordance with certain embodiments described herein. According to certain embodiments, the mathematical representation of the output 202 with load capacitance C_L can be expressed by the following equations:

$$V_o(t) = \int \left[\left(I(t) - \frac{g_1 \delta V_i(t - \Delta \tau)}{dt} - \frac{g_2 \delta V_i(t - \Delta \tau)}{dt} \right) C_L \right] dt$$

$$= \left[\int I(t) dt - (g_1 \delta V_i(t - \Delta \tau) + g_2 \delta V_i(t - \Delta \tau)) \right] C_L$$

Where,

$$I(t) = \frac{(V_o(t) - V_i(t))}{Z_d}$$

Thus,

$$Z_d * dV_o(t) / dt = V_i(t) - V_o(t) - (g_1 + g_2) * C_L * d(\delta V_i(t - \Delta \tau)) / dt$$

And,

$$V_o(t) = V_i(t) - (g_1 + g_2) * C_L * d(\delta V_i(t - \Delta \tau)) / dt - Z_d * dV_o(t) / dt$$

[0018] The above equations show that the output voltage of certain embodiments is a function of the change rate of the input signal transition rate and the change rate of the output itself. Because the change rate of the input transition rate and the output change rate have a negative effect on the output voltage, the damper of certain embodiments reduces noise having a slew rate that is faster than the slew rate of the non-noise component of the signal. Thus, certain embodiments provide a self-adjusting damper that effectively removes or reduces noises such as cross-coupling noises and reflected signals in complex electrical and electronic systems.

[0019] Figure 3 is a schematic diagram of another example of a self-adjusting damper circuit in accordance with certain embodiments described herein. The illustrated

embodiment comprises two voltage controlled current sources 340, 350 which are included in a first sub-circuit. The current source 340 of the illustrated embodiment is an AC-coupled, negative-gain pmos voltage-controlled current source for increasing the current flow from the power supply when the input sees a negative noise component, thus reducing the negative noise component on the output. The current source 340 includes two pmos transistors 341, 342. The pmos transistor 341 is a current-limiting transistor that ensures that the current supplied to the output has a relatively smaller magnitude than does the current to/from the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other pmos transistor 342 provides the negative-voltage-controlled current source function by AC coupling with the input signal V_i 301, on an input terminal, via a capacitor 311. Thus, the transconductance of this transistor 342 changes inversely proportionally to the slew rate of the input.

[0020] The current source 350 of the illustrated embodiment is an AC-coupled, positive-gain nmos current source for increasing the current flow to ground when the input sees a positive noise component, thus reducing the positive noise component on the output. The current source 350 includes two nmos transistors 351, 352. The nmos transistor 351 is a current-limiting transistor that ensures the current taken away from the output has a relatively smaller magnitude than does the current from/to the input V_i 301 such that the damper circuit has a limited effect on the non-noise component of the signal. The other nmos transistor 352 provides the positive-voltage-controlled current source function by AC coupling with the input signal V_i 301 via a capacitor 311. Thus, the transconductance of this transistor 342 changes proportionally with the slew rate of the input.

[0021] In the illustrated embodiment, the resistor R_d 310 is included in a second sub-circuit. The resistor R_d 310 is a delay element which introduces a signal delay compatible with the current source delay from the first sub-circuit and also may reduce the reverse feedback of the output signal back to the input. Since the delay time from input V_i 301 on the input terminal to output V_o 302 on the output terminal is the function of the time constant, $R_d \cdot C_L$, the resistance of the resistor R_d 310 can be selected based on the estimated value of C_L 360. Thus, in certain embodiments, the input signal can be delayed to match the reaction time of the voltage-controlled current sources that generate a negative feedback

current to remove signal noise. In certain embodiments the damper 141 can reduce cross coupling noise and reflections having various slew rate ranges.

[0022] In certain embodiments, the damper also includes a current-limiting, self-biasing op bias for biasing the reference voltage so that the operation point can be established in the middle of the linear range.

[0023] Advantageously, in certain embodiments, the damper is configured to be implemented in an application specific integrated circuit (“ASIC”) without requiring analog circuit technologies, enabling easy migration to other process technologies. For example, the damper is configured to be implemented in digital process technologies comprising a component resolution of approximately plus or minus 20 percent. Moreover, in certain embodiments, the damper may be implemented with no external control signals or memory elements.

[0024] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-adjusting damper circuit, comprising:
 - an input terminal adapted to receive an input signal comprising a noise component and a signal component;
 - an output terminal;
 - a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal; and
 - a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal, wherein the circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal, the output signal substantially representative of the signal component of the input signal.
2. The circuit of Claim 1, wherein the second sub-circuit comprises a resistor.
3. The circuit of Claim 1, wherein the first sub-circuit comprises a current source.
4. The circuit of Claim 3, wherein the current source comprises a voltage-controlled current source.
5. The circuit of Claim 4, wherein voltage-controlled current source comprises one or more transistors.
6. The circuit of Claim 1, wherein the first sub-circuit comprises a negative current source and a positive current source.
7. The circuit of Claim 6, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.
8. The circuit of Claim 7, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the trans-conductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.

9. The circuit of Claim 7, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

10. The circuit of Claim 9, wherein the first and second current limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

11. The circuit of Claim 10, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current limiting component, and the second current limiting component comprise one or more transistors.

12. The circuit of Claim 1, wherein the first sub-circuit is adapted to generate the first signal by generating a positive signal when the noise component of the input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

13. The circuit of Claim 12, wherein the first sub-circuit is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

14. The circuit of Claim 1, wherein the first signal comprises negative feedback current.

15. The circuit of Claim 1, wherein the circuit is configured to be operate on a digital semiconductor process technology.

16. The circuit of Claim 1, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution of less than plus or minus 20 percent.

17. The circuit of Claim 1, wherein the circuit is implemented on a memory module.

18. The circuit of Claim 1, wherein the noise-counteracting signal actively counteracts a noise component of the delayed version of the input signal.

19. A method of removing a noise component of an electrical signal, the method comprising:

receiving an input signal comprising a noise component and a signal component;

generating a first signal corresponding to the noise component of the input signal;

generating a second signal corresponding to a delayed version of the input signal; and

adding a noise counteracting signal to the second signal in response to the first signal to generate an output signal substantially representative of the signal component of the input signal.

20. The method of Claim 19, wherein generating the first signal comprises generating a positive signal when the noise component of the input signal is negative and generating a negative signal when the noise component of the input signal is positive.

21. The method of Claim 20, wherein generating a positive signal comprises increasing the current flow from a power supply to the first signal and generating a negative signal comprises increasing the current flow from the first signal to ground.

22. The method of Claim 19, further comprising limiting the magnitude of the first signal in relation to the second signal.

23. The method of Claim 19, wherein adding the noise-counteracting signal to the second signal actively counteracts a noise component of the delayed version of the input signal.

24. A self-adjusting damper circuit, comprising:

means for receiving an input signal comprising a noise component and a signal component;

first generation means for generating a first signal corresponding to the noise component of the input signal; and

second generation means for generating a second signal corresponding to a delayed version of the input signal,

means for adding a noise counteracting signal to the second signal in response to the first signal to the second signal to generate an output signal, the output signal substantially representative of the signal component of the input signal.

25. The circuit of Claim 24, wherein the second generation means comprises a resistor.

26. The circuit of Claim 24, wherein the first generation means comprises a current source.

27. The circuit of Claim 26, wherein the current source comprises a voltage-controlled current source.

28. The circuit of Claim 27, wherein voltage-controlled current source comprises one or more transistors.

29. The circuit of Claim 24, wherein the first generation means comprises a negative current source and a positive current source.

30. The circuit of Claim 29, wherein the negative current source comprises a negative voltage-controlled current source component and the positive current source comprises a positive voltage-controlled current source component.

31. The circuit of Claim 30, wherein the transconductance of the negative voltage-controlled current source component is inversely proportional to the slew rate of the input signal and the transconductance of the positive voltage-controlled current source component is proportional to the slew rate of the input signal.

32. The circuit of Claim 31, wherein the negative current source further comprises a first current-limiting component and the positive current source further comprises a second current-limiting component.

33. The circuit of Claim 32, wherein the first and second current-limiting components are adapted to limit the magnitude of the first signal in relation to the second signal.

34. The circuit of Claim 33, wherein each of the positive voltage-controlled current source, the negative voltage-controlled current source component, the first current-limiting component, and the second current-limiting component comprise one or more transistors.

35. The circuit of Claim 24, wherein the first generation means is adapted to generate the first signal by generating a positive signal when the noise component of the

input signal is negative and by generating a negative signal when the noise component of the input signal is positive.

36. The circuit of Claim 35, wherein the first generation means is adapted to generate the positive signal by increasing the current flow from a power supply to the first signal and to generate the negative signal by increasing the current flow from the first signal to ground.

37. The circuit of Claim 24, wherein the first signal comprises negative feedback current.

38. The circuit of Claim 24, wherein the circuit is configured to be operate on a digital semiconductor process technology.

39. The circuit of Claim 24, wherein the circuit is configured to be operate on a semiconductor process technology having a device resolution less than plus or minus 20 percent.

40. The circuit of Claim 24, wherein the circuit is implemented on a memory module.

41. The circuit of Claim 24, wherein the addition means actively counteracts a noise component of the delayed version of the input signal.

SELF-ADJUSTING DAMPER

ABSTRACT

A self-adjusting damper circuit is provided which includes an input terminal adapted to receive an input signal. The input signal includes a noise component and a signal component. The self-adjusting damper also includes an output terminal, a first sub-circuit adapted to generate a first signal corresponding to the noise component of the input signal, and a second sub-circuit adapted to generate a second signal corresponding to a delayed version of the input signal. The circuit is responsive to the first signal to add a noise counteracting signal to the second signal to generate an output signal on the output terminal. The output signal is substantially representative of the signal component of the input signal.

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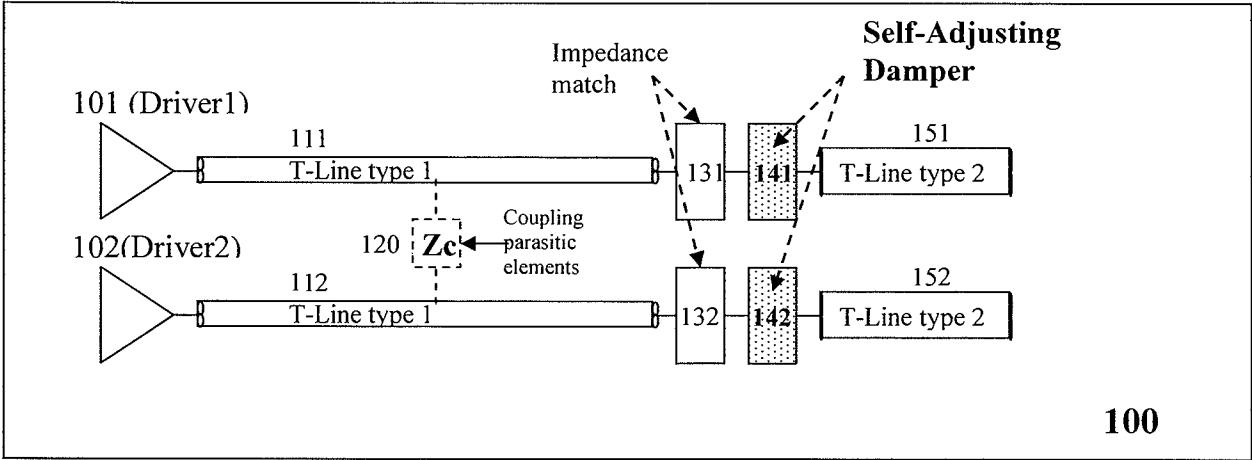


Figure 1

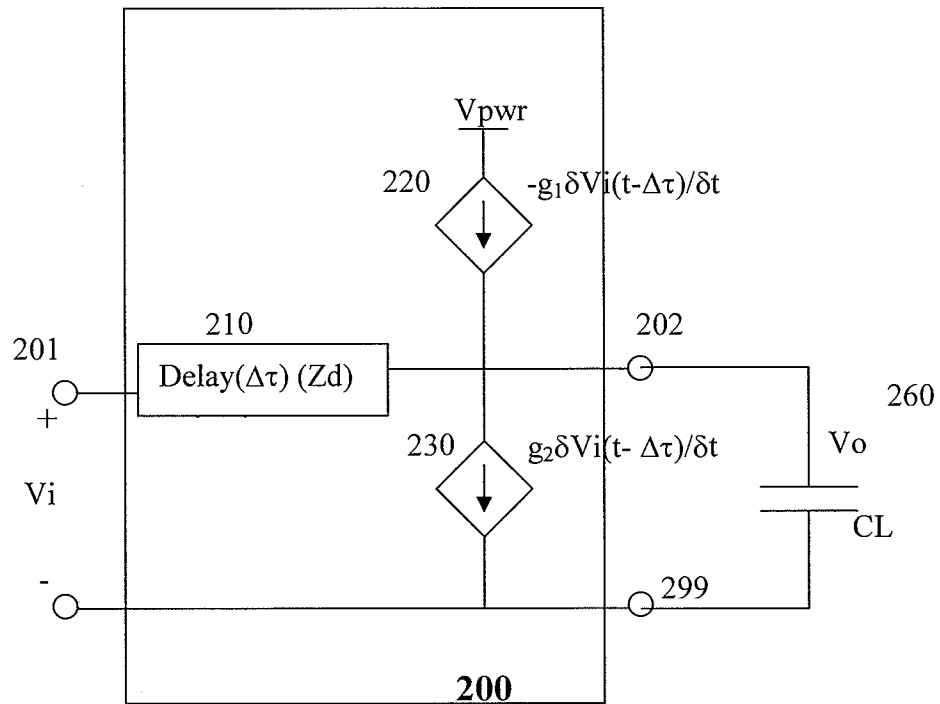


Figure2

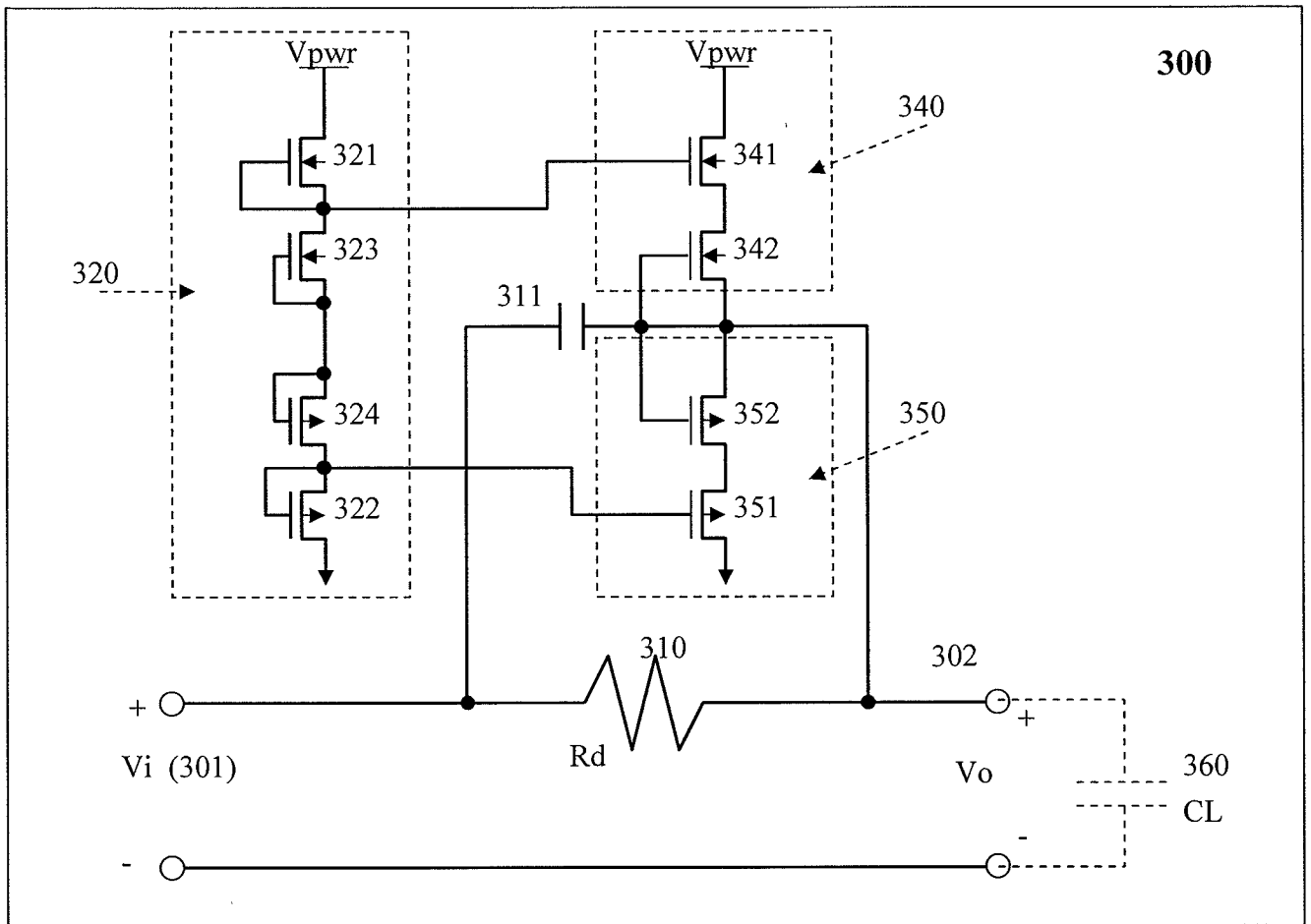


Figure 3

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

SELF-ADJUSTING DAMPER

First Named Inventor/Applicant Name:

To be provided later .

Filer:

Bruce S. Itchkawitz/Dee Manena

Attorney Docket Number:

NETL.044PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

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Basic Filing:

Provisional application filing

1005

1

210

210

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

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Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				210

Electronic Acknowledgement Receipt

EFS ID:	3150288
Application Number:	61044825
International Application Number:	
Confirmation Number:	1720
Title of Invention:	SELF-ADJUSTING DAMPER
First Named Inventor/Applicant Name:	To be provided later .
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Nicole Lauer
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.044PR
Receipt Date:	14-APR-2008
Filing Date:	
Time Stamp:	19:09:54
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$210
RAM confirmation Number	3984
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Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NETL044PR_ADS.pdf	238840 d8c233ab171200d8daae81b1a3fc40b7a5df15607	no	4
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2		NETL044PR_APPLICATION.pdf	612132 72bb8ada9928f27e6e6b1eb33298c875a35411226	yes	13
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	7	
	Claims		8	12	
	Abstract		13	13	
Warnings:					
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3	Drawings-only black and white line drawings	NETL044PR_DRAWINGS.pdf	26554 1a63cf466dd06eb6b89db8757150145d5d600aa1	no	3
Warnings:					
Information:					
4	Fee Worksheet (PTO-06)	fee-info.pdf	8110 69319f0fd4ac76f78647331ad62842de550b993a	no	2
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,825	04/14/2008		210	NETL.044PR		

CONFIRMATION NO. 1720

FILING RECEIPT

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614



Date Mailed: 04/18/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please write to the Office of Initial Patent Examination's Filing Receipt Corrections. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/17/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,825**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

SELF-ADJUSTING DAMPER

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,825	04/14/2008		NETL.044PR

CONFIRMATION NO. 1720

FORMALITIES LETTER



20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/18/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of **\$50** for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is **\$50** for a non-small entity

- **\$50** Surcharge.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web.
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at **1-866-217-9197** or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/aabranys/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Please Direct All Correspondence to Customer Number 20995

RESPONSE TO FORMALITIES NOTICE

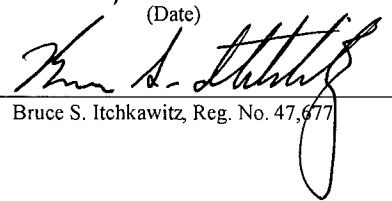
Applicant : Hyun Lee et al.
App. No : 61/044,825
Filed : April 14, 2008
For : SELF-ADJUSTING DAMPER
Art Unit : Unknown

**CERTIFICATE OF EFS WEB
TRANSMISSION**

I hereby certify that this correspondence, and any other attachment noted on the automated Acknowledgement Receipt, is being transmitted from within the Pacific Time zone to the Commissioner for Patents via the EFS Web server on:

6/16/08

(Date)


Bruce S. Itchkawitz, Reg. No. 47,677

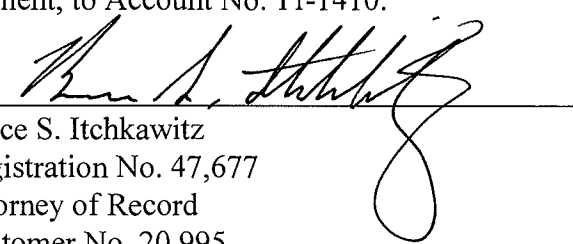
**Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450**

Dear Sir:

The above-captioned application was filed without the inventors information and/or filing fees. Enclosed in compliance with 37 CFR 1.53(f) are the following:

- (X) A Notice to File Missing Parts dated April 18, 2008.
- (X) Supplemental Application Data Sheet
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

The Commissioner is hereby authorized to charge any additional fees which may be required, now or in the future, or credit any overpayment, to Account No. 11-1410.


Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404



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UNITED STATES DEPARTMENT OF COMMERCE
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,825	04/14/2008		NETL.044PR

CONFIRMATION NO. 1720

FORMALITIES LETTER



0000000029444808

20995

KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/18/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

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- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of \$50 for a non-small entity, must be submitted with the missing items identified in this notice.

SUMMARY OF FEES DUE:

Total additional fee(s) required for this application is \$50 for a non-small entity

- \$50 Surcharge.

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Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

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<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

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If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

/aabranys/

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/044,825
Filing Date:: April 14, 2008
Application Type:: Provisional
Subject Matter:: Utility
CD-ROM or CD-R?:: None
Title:: SELF-ADJUSTING DAMPER
Attorney Docket Number:: NETL.044PR
Request for Early Publication?:: No
Request for Non-Publication?:: No
Total Drawing Sheets:: 3
Small Entity?:: No
Petition Included?:: No
Secrecy Order in Parent Appl.?:: No

Inventor Information

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Hyun
Family Name:: Lee
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State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 21 Thalia Street
City:: Ladera Ranch

State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92694

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Jayesh
Middle Name:: R.
Family Name:: Bhakta
City of Residence:: Cerritos
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 12220 Rose Street
City:: Cerritos
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 90703

Applicant Authority Type:: Inventor
Primary Citizenship Country:: Taiwan
Status:: Full Capacity
Given Name:: Chi-She
Family Name:: Chen
City of Residence:: Walnut
State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 944 Crystal Water Lane

City:: Walnut
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 91789

Correspondence Information

Correspondence Customer Number:: 20,995
Phone Number:: (949) 760-0404
Fax Number:: (949) 760-9502
E-Mail Address:: efiling@kmob.com

Representative Information

Representative Customer Number:: 20,995

Assignment Information

Assignee Name:: Netlist, Inc.
Street:: 51 Discovery, Ste. #150
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92618

5519247
061208

Electronic Patent Application Fee Transmittal

Application Number:	61044825			
Filing Date:	14-Apr-2008			
Title of Invention:	SELF-ADJUSTING DAMPER			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/alexis swan			
Attorney Docket Number:	NETL.044PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	3464591
Application Number:	61044825
International Application Number:	
Confirmation Number:	1720
Title of Invention:	SELF-ADJUSTING DAMPER
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Lori Larson
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.044PR
Receipt Date:	16-JUN-2008
Filing Date:	14-APR-2008
Time Stamp:	19:54:53
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	4115
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

Charge any Additional Fees required under 37 C.F.R. Section 1.16 (National application filing, search, and examination fees)

Charge any Additional Fees required under 37 C.F.R. Section 1.17 (Patent application and reexamination processing fees)

File Listing:

Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	NETL_044PR.pdf	175330	no	6
			039d72533893d17831f3da11d1cf751b5fb01700		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	8138	no	2
			c96373900d3d8156a46644ab00c67617abbf1d56		

Warnings:**Information:**

Total Files Size (in bytes):			183468
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

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If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,825	04/14/2008		260	NETL.044PR		

CONFIRMATION NO. 1720

UPDATED FILING RECEIPT



OC000000030527948

20995
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FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/19/2008

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Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Chi-She Chen, Walnut, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

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Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046PR
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2

- ☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Applicant Information:

Applicant 1				
Applicant Authority ☒ Inventor		☐ Legal Representative under 35 U.S.C. 117		☐ Party of Interest under 35 U.S.C. 118
Prefix	Given Name	Middle Name	Family Name	Suffix
	To be provided later			
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service				
City		State/Province	Country of Residence i	
Citizenship under 37 CFR 1.41(b) i				
Mailing Address of Applicant:				
Address 1				
Address 2				
City		State/Province		
Postal Code		Countryi		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	20995		
Email Address	efiling@kmob.com	Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		
Attorney Docket Number	NETL.046PR	Small Entity Status Claimed ☐	
Application Type	Provisional		
Subject Matter	Utility		
Suggested Class (if any)		Sub Class (if any)	
Suggested Technology Center (if any)			
Total Number of Drawing Sheets (if any)	4	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046PR
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under 35 U.S.

☐ C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Enter either Customer Number or complete the Representative Name section below. If both sections are completed the Customer Number will be used for the Representative Information during processing.

Please Select One: ☒ Customer Number ☐ US Patent Practitioner ☐ Limited Recognition (37 CFR 11.9)

Customer Number 20995

Domestic Benefit/National Stage Information:

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Prior Application Status		Remove	
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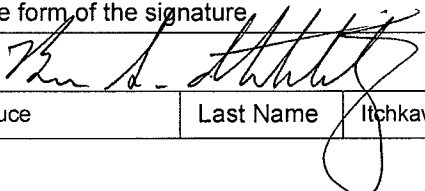
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NETL.046PR
		Application Number	
Title of Invention	MEMORY BOARD WITH SELF-TESTING CAPABILITY		

If the Assignee is an Organization check here. ☒			
Organization Name	NETLIST, INC.		
Mailing Address Information:			
Address 1	51 Discovery, #150		
Address 2			
City	Irvine	State/Province	CA
Country ⁱ	US	Postal Code	92618
Phone Number	949-435-0025	Fax Number	949-435-0031
Email Address	hlee@netlist.com		
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Signature			Date (YYYY-MM-DD)	2008-04-14	
First Name	Bruce	Last Name	Itchkawitz	Registration Number	47677

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MEMORY BOARD WITH SELF-TESTING CAPABILITY**BACKGROUND**Field

[0001] The present invention relates to self-testing electronic modules and, more particularly, to self-testing electronic memory modules.

Description of the Related Art

[0002] The failure of memory components in an electronic system may result in the loss of valid data. Therefore, it is important to ensure proper memory operation in an electronic system. Memory integrated circuits ("memory chips") often go through a series of tests at various stages of system manufacture. Once memory chips are deployed in a system, they also generally go through a system level memory test each time the system is booted. In addition, memory chips may undergo a parity checking process during normal system operation.

[0003] There are typically at least three test phases which memories undergo during system manufacture. Each phase generally tests for memory defects and for the correct operation of the input/output interface. The first test phase is typically conducted by the memory chip manufacturer and generally involves checking for bit failures, correct memory access speed, etc. The second test phase is typically done by memory module manufacturers and generally involves testing the signal quality, the noise susceptibility, and the operational speed of the memory module as a single unit. The second test phase may also include checking for bit failures in individual memory chips. The third phase is usually carried out by the system manufacturer. During the third phase, the interaction of the memory subsystem with other components in the system is tested. During the third phase, the individual memory module operation is also tested again and the memory array is checked for defects. Because of the significant amount of testing that memories undergo during the manufacturing process, there is generally substantial test cost and test time associated with ensuring the proper memory operation. This test cost and test time translate into an increase in system cost and a decrease in system performance.

[0004] There are a number of memory test methodologies that employ either external test hardware, embedded self-test logic ("MBIST"), or both. However, the usefulness of these test methodologies is limited due to the high cost and other limitations associated with them. For instance, external test hardware such as automatic test equipment ("ATE") is very expensive. Moreover, the development time and cost associated with implementing MBIST is relatively high. These costs and limitations are especially significant when testing dynamic random access memory ("DRAM"). For example, technological developments, such as increases in DRAM speed, may require manufacturers to upgrade ATE machines relatively frequently. In addition, MBIST in DRAM chips generally cannot be fully utilized for system level testing of memory boards.

[0005] Because of the increasing cost, complexity, and time involved with fully testing DRAM chips, DRAM manufacturers often provide "effectively tested" ("ETT") DRAM chips to memory module manufacturers at a lower price rather than providing fully tested DRAM chips. Memory module manufacturers often prefer the ETT DRAM chips mainly due to their greater availability. Memory module manufacturers who receive ETT DRAM chips then have to assume a part of the responsibility of validating the DRAM chips, adding to the complexity of the memory module test process.

SUMMARY

[0006] In certain embodiments, a self-testing memory module is provided which includes a printed circuit board configured to be electrically coupled to a memory controller of a computer system. The memory module also includes a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices includes data, address, and control ports. The memory module also includes a test control module configured to generate address and control signals for testing the memory devices. The test control module is further configured to selectively input to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module. The memory module also includes a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices. The test data handler module is further configured to write data to the memory devices by selectively inputting to the data ports of the memory

devices either data signals from the system memory controller or data signals from the test data handler module.

[0007] In certain embodiments, a method of self-testing a memory module is provided which involves providing a memory module which includes a printed circuit board configured to be electrically coupled to a memory controller of a computer system. The memory module also includes a plurality of memory devices on the printed circuit board. Each memory device of the plurality of memory devices includes data, address, and control ports. The memory module further includes a test control module configured to generate address and control signals for testing the memory devices, and a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices. The method further includes selectively inputting to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module. The method also involves writing data to the memory devices by selectively inputting to the data ports of the memory devices either data signals from the system memory controller or data signals from the test data handler module.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein.

[0009] Figure 2 is a block diagram of example control circuitry of an example self-testing memory module in accordance with certain embodiments described herein.

[0010] Figure 3 is a flow diagram of an embodiment of a method for testing a self-testing memory module in accordance with certain embodiments described herein.

[0011] Figure 4 is a flow diagram of an embodiment of a method for executing a test mode of a self-testing memory module in accordance with certain embodiments described herein.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0012] Certain embodiments described herein present a self-testing electronic system, such as for example, a self-testing electronic memory module. Some embodiments described herein present a self-testing memory module that is populated with ETT DRAM

chips. Certain embodiments described herein present a self testing registered dual in-line memory module (“RDIMM”). Some embodiments described herein present a self-testing RDIMM that does not require any additional pins other than the pins on the standard JEDEC RDIMM connector. Certain embodiments described herein present a self-testing memory module that includes a control block and a data block which can generate memory addresses and data according to the JEDEC standard memory protocol.

[0013] Some embodiments described herein presents a self-testing memory module that can be configured through an I²C interface and that allows test results to be read out through the I²C interface. Certain embodiments described herein present a self-testing memory module that allows a test function to be configured, controlled, and executed without a system memory controller. Some embodiments present a self-testing memory module that can be tested without any external test equipment. For example, certain embodiments present a self-testing memory module that can be tested without any system driven test procedure. Various embodiments described herein present a self-testing memory module that can be self-tested at a target system speed. One embodiment described herein present a self-testing memory module that can generate data (DQ) and data-strobe (DQS) signals with wave characteristics that resemble the wave characteristics of DQ and DQS signals from a system memory controller.

[0014] MBIST is commonly used to test memories embedded in application specific integrated circuits (“ASICs”) or system on chip integrated circuits (“SoCs”) such as, for example, advanced memory buffers (“AMBs”). The MBIST implementation usually includes three distinct functional blocks: the address/control generator, the data generator/checker (sometimes referred to as a signature analyzer), and the test interface controller/register (“JTAG”). The test instructions and test patterns are generally loaded to the address/control generator and the data generator/checker through the JTAG interface. The test results are generally read out through the JTAG interface.

[0015] In memory module applications, there are obstacles associated with using MBIST. One of these obstacles is the large number of signals required to interface the three MBIST functional blocks. The inter-block timing requirements can present another obstacle. In addition, the ability to control the test is limited. For example, synchronizing the address/or

control signals with the data signal can be difficult. The relatively large amount of information needed to be gathered and stored while testing the memory presents yet another obstacle.

[0016] Unlike on an ASIC, in which the three MBIST functional blocks are in a single chip, the MBIST functional blocks on a memory module generally would be segregated into multiple chips on the memory module due to physical and electrical limitations and requirements. This makes implementing MBIST on a memory module difficult because, while there is virtually no limit on the number of available interface signals among the three MBIST functional blocks in an ASIC, the memory module can support only a limited amount of interface signals between memory chips. In addition, because the MBIST functional blocks are spread out to multiple chips on a memory module, the inter-block signal delay is generally much longer on a memory module than on an ASIC. It is also generally not feasible to implement cross-checking logic that operates the three MBIST functional blocks in lock-step. This is due to the limitation on the number of interface signals and to the relatively long inter-block signal delay time on a memory module versus on an ASIC. Finally, in most cases, since an ASIC cannot be repaired, the ASIC MBIST is generally only capable of detecting and reporting the pass/fail status of memory tests. Memory module test results, on the other hand, should generally include both addresses of the memory locations where failures occur and the data patterns that were read back from the failed memory locations. This type of reporting can help to facilitate the repair of the memory module by, for example, allowing for the identification and replacement of failed components.

[0017] In one embodiment, the self-testing memory module comprises logic including two blocks: a control block and a data block. The control block may also be referred to as, for example, a test control module. The data block may be referred to as, for example, a test data handler module. The control block may be implemented in the control register. The data block can be implemented as data path switches in the data path. In certain embodiments, the data block generates test data patterns to write to the memory module and checks the data patterns against corresponding data patterns that are expected to be read back from the memory locations. For example, in one embodiment, the data block generates random patterns to write to the memory module. In some embodiments, the data block also isolates the data path from the system board while the module is not accessed by the system.

For example, the data block may isolate the data path from the system board when the module is in a self-testing mode.

[0018] The control block may include, for example, a dual input register for registering address and control signals coming from either the self-testing logic or from the memory controller on the system board. In some embodiments, during testing, the control block generates address and control signals associated with memory locations to be tested and the data block generates corresponding test data patterns and provides them to the appropriate memory chips. For example, the data block may receive a read/write signal from the control block and provide data to be written to certain locations in the memory chips during a write operation. The data block may then read back the data from those locations and compare the read data with expected reference data. If there is a mismatch between the read data and the reference data, the data block may, for example, store the bit failure information and inform the control block about the failure. The control block may save the address of the memory location where the failure occurred.

[0019] In some embodiments, the control block and the data block produce memory addresses and data according to the JEDEC standard memory protocol. In some embodiments, the address sequences and the data patterns may be programmable either through the I²C interface or they may be defaulted to pre-defined values. In certain embodiments, the test speed, for example, may be defined by the speed of the clock. In some embodiments, for example, the control block and the data block generate the memory interface signals with proper edge relationships based on the JEDEC standard. In one embodiment, the data block is also able to generate data (DQ) and data-strobe (DQS) signals with programmable slew rates and programmable peak values so that the signals generally resemble the DQ and DQS signals generated by the system memory controller.

[0020] In some embodiments, the self-testing memory module does not require any additional DIMM connector pins. For example, the control block and the data block may utilize the address and control signals along with a test signal to enable and execute the self testing function.

[0021] Figure 1 is a block diagram of an example self-testing memory module in accordance with certain embodiments described herein. The self-testing memory module 100

may comprise, for example, a dual in-line memory module (“DIMM”). In one embodiment, the memory module 100 self-testing circuitry includes a control block 120 which includes a control mixer 122. In certain embodiments, the control mixer 122 generally controls the address and the control signals for the self-testing function. The data block 110, includes at least one data handler 111 which generally controls the data signals during self-testing operation. The data block 110 may include a plurality of data handlers 111. The data handlers 111 may operate handle different data segments. In some embodiments, for example, the data handler 111 may handle byte or nibble wise data segments. The control block 120 may further include a register 121 which, in some embodiments, may comprise a JEDEC standard component.

[0022] Figure 2 is a block diagram of example control circuitry 200 of an example self-testing memory module in accordance with certain embodiments described herein. The control circuitry 200 may include functional blocks and other features not shown in Figure 2. The data handler 111 may include a data multiplexer/demultiplexer (“data mux/demux”) 212 and a test data handler 213. The data mux/demux 212 may provide a bi-directional data multiplexer function. In some embodiments, the data mux/demux 212 selects the data to be written to the memories during memory write operations from either a system memory controller during normal operation, or from the test data handler 213 during self-test operation. The data mux/demux 212 may, in some embodiments, send the data read from the memories during memory read operations to both test data handler 213 and the system memory controller.

[0023] In some embodiments, the control block 220 includes a control mixer 122. The control mixer 122 may include a DRAM controller 223 and a test controller 224. The DRAM controller 223 generally pre-processes address and control information before it sends the information to the register 221. In one embodiment, the DRAM controller 223 receives address and control signals from both the system memory controller and the test controller 224. The DRAM controller 223 may send either the signals from the system memory controller or, alternatively, the signals from the test controller 224, to the register 221 depending on whether the memory module is in normal mode or test mode, respectively. In one embodiment, the DRAM controller 223 generates the address and control signals for

DRAM operations. The test controller 224 controls the generation of the address and control signal sequences to be used during self-testing operation and also communicates with the test data handler 213. The test data handler 213, based on control information it receives from the test controller 224, may generate data patterns to be written to the memory when the self-testing memory module performs a memory write test. In addition, the test data handler 213 may also check data patterns that are read back from the memories when the self-testing memory module performs a memory read test. The checking of the data patterns may also be done based on the control information the test data handler 213 receives from the test controller 224. In one embodiment, when memory failures occur, the test data handler 213 stores data patterns associated with failed memory locations and the test controller 224 stores the addresses of the failed memory locations. In certain embodiments, the test controller 224 may be updated through an I²C interface with new data patterns and/or with alternative memory access sequences to conduct AC tests.

[0024] Figure 3 is a flow diagram of an embodiment of a method 300 for testing a self-testing memory module in accordance with certain embodiments described herein. At step 310 the method 300 enters an idle state after power up or, in some embodiments, when the memory board is connected to a mother board or to a test board. In one embodiment, at step 310, the contents of the register (e.g., the register 121 of Figure 1) are undefined, and the controllers are in unknown state. After power up, the method 300 resets the board at step 311. At step 311, although the self-test control logic in the data handler and in the control mixer (e.g., the data handler 111 and the control mixer 122 of Figure 1) are set to a default value, the data mux/demux and the DRAM controller 223 (e.g., the data mux/demux 212 and the DRAM controller 223 of Figure 2) do not go into a self-test mode. The method 300 next configures the test mode at step 312 through, for example, an I²C interface.

[0025] The method 300 next initiates the test mode at step 313. At step 313, for example, the tester readies the data handler and the control mixer (e.g. data handler 111 and control mixer 122 of Figure 1) beginning the self-testing of the memory module. At step 313 the method 300 updates the test data handler (e.g., the test data handler 213 of Figure 2) using, for example, the test controller (e.g., the test controller 224 of Figure 2). The method 300 executes the test mode of the memory module at step 314. At step 314 the method, for

example, in one embodiment, asserts or toggles an input signal on a parity-in pin of the memory module. The method 300 continues executing the test mode at step 314 until the self-test is complete or until the method 300 detects a certain number of memory failures such that a failure count exceeds a preset number. If the failure count exceeds the preset number the method 300 enters step 315 and “errors out” of the test mode. If the self-test is completed at step 314 and the failure count does not exceed the preset number, the method also enters step 315. At step 315, the method 300 sends out a test completion indicator using the test controller. At step 316, the method 300 allows the fail status of the test to be read out through, for example, an I²C interface.

[0026] In some embodiments, the method 300 is generally interruptible. For example, the method 300 may abandon the self-test and return to step 310 if the signal going into the parity-in pin is, for example, de-asserted or removed. In other embodiments the interrupt signal or condition may be different and may not be a parity-in signal. For example, in one embodiment, the method 300 may abandon the self-test and return to step 310 if a timer having a pre-determined count expires. When the self-testing is interrupted, the method 300 will return to step 310 and the bit failure information in the control mixer and the data handler becomes invalid.

[0027] Figure 4 is a flow diagram of an embodiment of a method 400 for executing a test mode of a self-testing memory module in accordance with certain embodiments described herein. For example, in one embodiment, the test mode may be similar to step 314 of method 300. In certain embodiments, the method 400 executes various tasks which configure the self-test mode of the self-testing memory module using an I²C interface. For example, the method 400 may configure the test controller (e.g., test controller 224 of Figure 2) through the I²C interface at step 452 and activate the test controller through the I²C at step 453. Alternatively, the method 400 may activate the test controller through the I²C interface without first updating the test mode at step 451. The method 400 enters the self-test mode at step 454 by, for example, asserting or detecting the assertion of the parity-in signal on a parity-in pin.

[0028] At step 413, the method begins the execution of the self-test sequence by clearing, for example, a set of test status registers. Next, the method 400 performs a burst

write function at step 415. For example, in certain embodiments, the method 400 writes multiple memory locations in the memory module that are to be tested at step 415 in a relatively short amount of time (or in a burst). In certain embodiments, the method 400 can perform multiple write bursts at step 415. Next, at step 416, the method 400 performs a read & compare function. For example, in certain embodiments, the method 400 reads back certain memory locations at step 416 that have been written at step 415 and compares the values with certain expected reference data. In some embodiments, the method 400 reads from multiple memory locations in the memory module in a relatively short amount of time (or in a burst). In certain embodiments, the method 400 can execute multiple read bursts at step 416 as it compares the read data with expected reference data. In some embodiments, the method 400 stores any failed read data and the associated expected reference data at step 417 using, for example, the test data handler (e.g. test data handler 213 of Figure 2).

[0029] The method 400 stores the addresses of the memory locations associated with the failed read data using the test controller at step 417. In other embodiments the failed read data, expected reference data, and the addresses of the memory locations associated with the failed read data may be stored using the same functional block. For example, in one embodiment, the failed read data, expected reference data, and the addresses of the memory locations associated with the failed read data are all stored using the test controller at step 417. In some embodiments, the test controller is generally inaccessible through the I²C interface during self-test operation. For example, the test controller is inaccessible through the I2C interface until a test failure occurs at step 417 or until the test passes at step 419. When the test is complete, the test results can be read out through, for example, the I²C interface.

[0030] Although certain preferred embodiments and examples are discussed above, it is understood that the inventive subject matter extends beyond the specifically disclosed embodiments to other alternative embodiments and/or uses of the invention and obvious modifications and equivalents thereof. It is intended that the scope of the inventions disclosed herein should not be limited by the particular disclosed embodiments. Thus, for example, in any method or process disclosed herein, the acts or operations making up the method/process may be performed in any suitable sequence and are not necessarily limited to

any particular disclosed sequence. Various aspects and advantages of the embodiments have been described where appropriate. It is to be understood that not necessarily all such aspects or advantages may be achieved in accordance with any particular embodiment. Thus, for example, it should be recognized that the various embodiments may be carried out in a manner that achieves or optimizes one advantage or group of advantages as taught herein without necessarily achieving other aspects or advantages as may be taught or suggested herein.

WHAT IS CLAIMED IS:

1. A self-testing memory module, comprising:
 - a printed circuit board configured to be electrically coupled to a memory controller of a computer system;
 - a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;
 - a test control module configured to generate address and control signals for testing the memory devices, the test control module further configured to selectively input to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module; and
 - a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices, the test data handler module further configured to write data to the memory devices by selectively inputting to the data ports of the memory devices either data signals from the system memory controller or data signals from the test data handler module.
2. The memory module of Claim 1, wherein the test data handler module further comprises a plurality of switches which perform the selective electrical coupling of the data ports.
3. The memory module of Claim 3, wherein the switches comprise one or more data multiplexer-demultiplexers.
4. The memory module of Claim 1, wherein the test control module further comprises a DRAM controller which performs the selective electrical coupling of the address and control ports.
5. The memory module of Claim 1, wherein the memory module is configured to check for failures in the operation of the memory devices by comparing data read from the memory devices with data written to the memory devices.
6. The memory module of Claim 5, wherein the memory module is configured to report failures in the operation of the memory devices via an I²C interface.

7. The memory module of Claim 5, wherein the test data handler module is configured to check for failures in the operation of the memory devices.

8. The memory module of Claim 5, wherein data associated with failures in the operation of the memory devices is stored in the test data handler module and wherein memory addresses associated with failures in the operation of the memory devices are stored in the test control module.

9. The memory module of Claim 1, wherein the memory module can be controlled via an I²C interface.

10. The memory module of Claim 1, wherein the memory module is configured to be operated in a test mode in which the test control module selectively inputs to the address and control ports of the memory devices the address and control signals from the test control module and wherein the test data handler module writes data to the memory devices by selectively inputting to the data ports of the memory devices the data signals from the test data handler module.

11. The memory module of Claim 1, wherein the test data handler module is configured to generate data signals with programmable slew rates.

12. The memory module of Claim 1, wherein the test data handler module is configured to generate data signals having variable peak values.

13. A method of self-testing a memory module, comprising:

providing a memory module comprising:

a printed circuit board configured to be electrically coupled to a memory controller of a computer system;

a plurality of memory devices on the printed circuit board, each memory device of the plurality of memory devices comprising data, address, and control ports;

a test control module configured to generate address and control signals for testing the memory devices;

a test data handler module configured to generate and write data to the memory devices and to read data from the memory devices; and

selectively inputting to the address and control ports of the memory devices either the address and control signals from the memory controller or the address and control signals from the test control module; and

writing data to the memory devices by selectively inputting to the data ports of the memory devices either data signals from the system memory controller or data signals from the test data handler module.

14. The method of Claim 13, wherein the test data handler module further comprises a plurality of switches which perform the selective electrical coupling of the data ports.

15. The method of Claim 14, wherein the switches comprise one or more data multiplexer-demultiplexers.

16. The method of Claim 13, wherein the test control module further comprises a DRAM controller which performs the selective electrical coupling of the address and control ports.

17. The method of Claim 13, further comprising checking for failures in the operation of the memory devices by comparing data read from the memory devices with data written to the memory devices.

18. The memory module of Claim 17, further comprising reporting failures in the operation of the memory devices via an I²C interface.

19. The method of Claim 17, wherein the test data handler module checks for errors in the memory devices.

20. The method of Claim 17, further comprising:

storing data associated with the failures in the operation of the memory devices in the data handler module; and

storing memory addresses associated with the failures in the operation of the memory devices in the test control module.

21. The method of Claim 13, further comprising configuring the memory module via an I²C interface.

22. The method of Claim 13, further comprising:

selectively inputting to the address and control ports of the memory devices the address and control signals from the test control module when the memory module is operating in a test mode; and

writing data to the memory devices by selectively inputting to the data ports of the memory devices the data signals from the test data handler module when the memory module is operating in the test mode.

23. The method of Claim 13, wherein the data handler module is configured to generate data signals with programmable slew rates.

24. The method of Claim 13, wherein the data handler module is configured to generate data signals having variable peak value.

MEMORY BOARD WITH SELF-TESTING CAPABILITY

ABSTRACT

A self-testing memory module is provided that includes a printed circuit board and is configured to couple to a memory controller. Memory devices are on the printed circuit board and include data, address, and control ports. The memory module includes a test control module which generates address and control signals to test the memories. The test control module is configured to selectively input to the address and control ports of the memories the address and control signals from the memory controller or the address and control signals from the test control module. The memory module also includes a test data handler module which generates and writes data to, and reads data from, the memories. The test data handler module writes data to the memories by selectively inputting to the data ports of the memories data signals from the system memory controller or data signals from the test data handler module.

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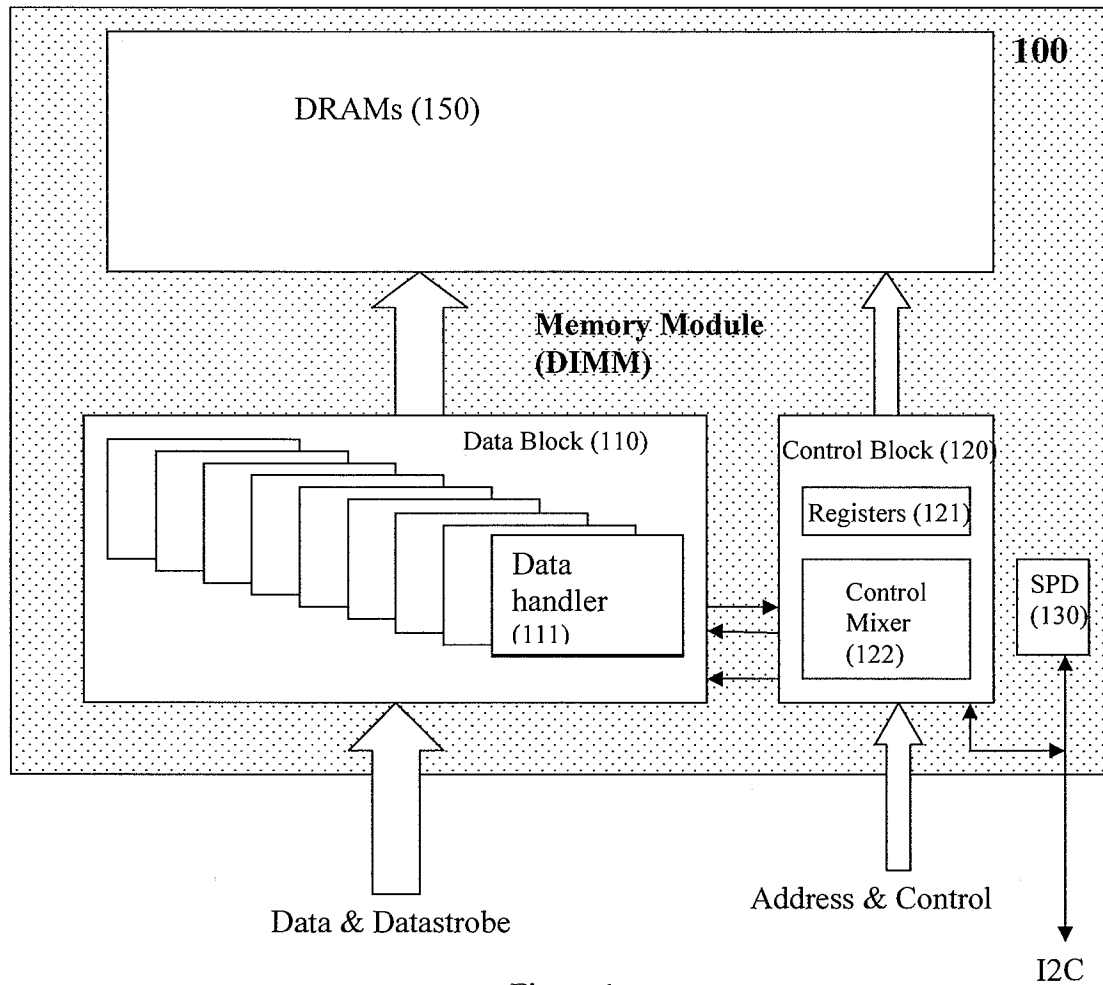


Figure 1

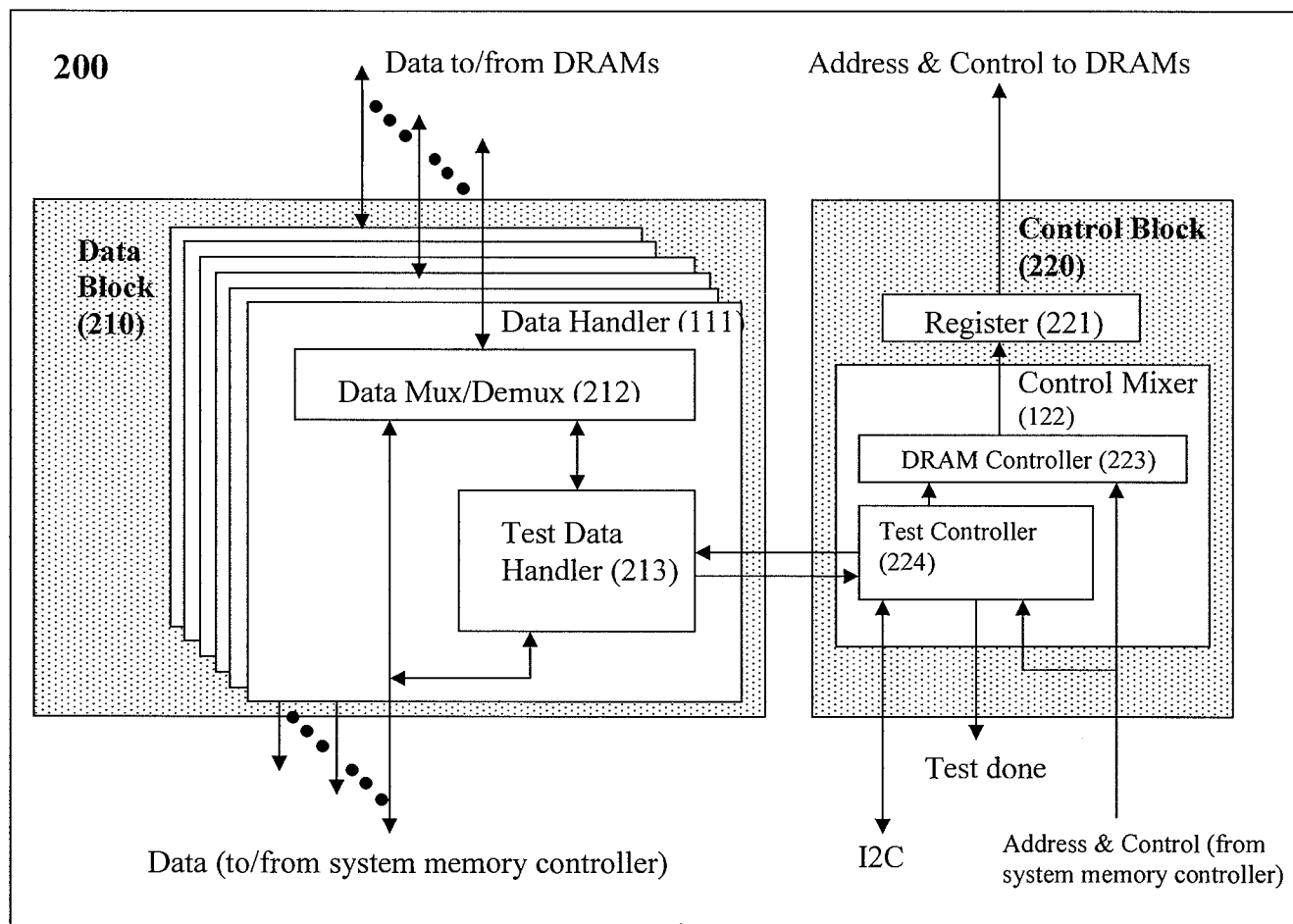


Figure 2

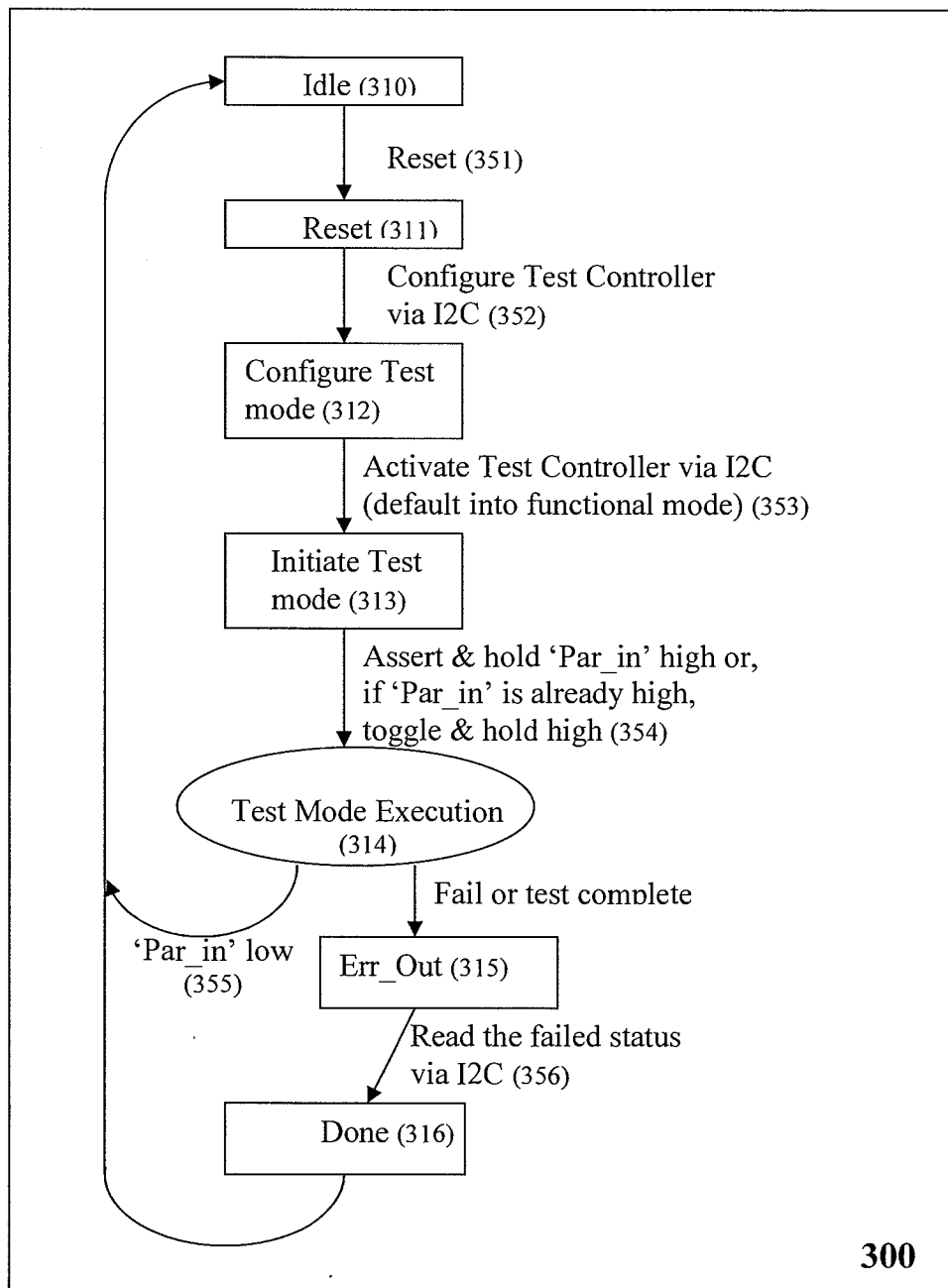


Figure 3: Example of the Test mode State Diagram

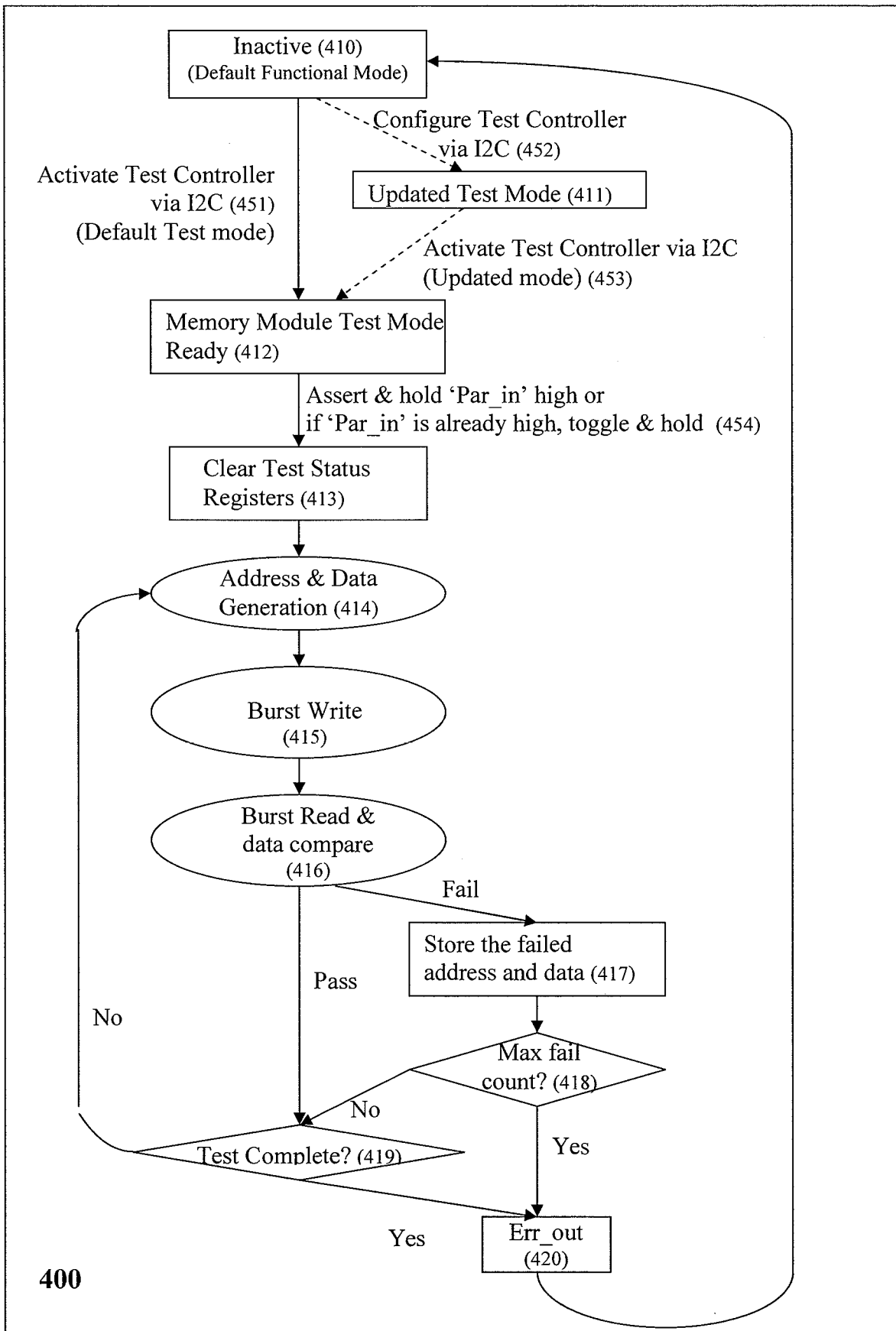


Figure 4

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

MEMORY BOARD WITH SELF-TESTING CAPABILITY

First Named Inventor/Applicant Name:

To be provided at a later date .

Filer:

Bruce S. Itchkawitz

Attorney Docket Number:

NETL.046PR

Filed as Large Entity

Provisional Filing Fees

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

Provisional application filing

1005

1

210

210

Pages:

Claims:

Miscellaneous-Filing:

Petition:

Patent-Appeals-and-Interference:

Post-Allowance-and-Post-Issuance:

Extension-of-Time:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				210

Electronic Acknowledgement Receipt

EFS ID:	3150790
Application Number:	61044801
International Application Number:	
Confirmation Number:	1630
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	To be provided at a later date .
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Segovia Phan
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046PR
Receipt Date:	14-APR-2008
Filing Date:	
Time Stamp:	18:35:25
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$210
RAM confirmation Number	3660
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Application Data Sheet	NETL046PR_ADS.pdf	239940 b5b0f15d382b919d2c2f47e8c158dbdf9ce42f62	no	4
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
2		NETL046PR_APPLICATION.pdf	772497 e337f84847c62bf44922cc6aa248820682bf4bae	yes	16
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Specification		1	11	
	Claims		12	15	
	Abstract		16	16	
Warnings:					
Information:					
3	Drawings-only black and white line drawings	NETL046PR_DRAWINGS.pdf	99372 d70ca11eb8a6d6130f2ffa3a07c40c1b8ca69a6a	no	4
Warnings:					
Information:					
4	Fee Worksheet (PTO-06)	fee-info.pdf	8132 c8d7f32cf135098bfef2c76dae9d3a9d9fbb477	no	2
Warnings:					
Information:					
Total Files Size (in bytes):			1119941		

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National Stage of an International Application under 35 U.S.C. 371

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,801	04/14/2008		NETL.046PR

CONFIRMATION NO. 1630

FORMALITIES LETTER



OC000000029493457

20995
KNOBBE MARTENS OLSON & BEAR LLP
2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 04/23/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

Filing Date Granted

An application number and filing date have been accorded to this provisional application. The items indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items and pay any fees required below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

- To avoid abandonment, a surcharge (for late submission of filing fee or cover sheet) as set forth in 37 CFR 1.16(g) of **\$50** for a non-small entity, must be submitted with the missing items identified in this notice.

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,801	04/14/2008		210	NETL.046PR		

CONFIRMATION NO. 1630

20995
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2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

FILING RECEIPT



Date Mailed: 04/23/2008

Receipt is acknowledged of this provisional patent application. It will not be examined for patentability and will become abandoned not later than twelve months after its filing date. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF APPLICANT, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection. Please verify the accuracy of the data presented on this receipt. **If an error is noted on this Filing Receipt, please write to the Office of Initial Patent Examination's Filing Receipt Corrections. Please provide a copy of this Filing Receipt with the changes noted thereon. If you received a "Notice to File Missing Parts" for this application, please submit any corrections to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections**

Applicant(s)

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/21/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,801**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

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Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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RESPONSE TO FORMALITIES NOTICE

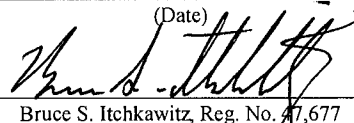
Applicant : Hyun Lee et al.
App. No : 61/044,801
Filed : April 14, 2008
For : MEMORY BOARD WITH SELF-
TESTING CAPABILITY
Art Unit : Unknown

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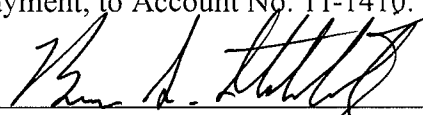
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Dear Sir:

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- (X) A Notice to File Missing Parts dated April 23, 2008.
- (X) Supplemental Application Data Sheet.
- (X) Fees will be paid via EFS Web. Extension of time is requested by payment of any extension fee.

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Bruce S. Itchkawitz
Registration No. 47,677
Attorney of Record
Customer No. 20,995
(949) 760-0404



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
61/044,801	04/14/2008		NETL.046PR

CONFIRMATION NO. 1630

FORMALITIES LETTER



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IRVINE, CA 92614

Date Mailed: 04/23/2008

NOTICE TO FILE MISSING PARTS OF PROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(c)

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- The provisional application cover sheet under 37 CFR 1.51(c)(1), which may be an application data sheet (37 CFR 1.76), is required identifying:
 - the name(s) of the inventor(s).
 - either city and state, or city and foreign country, of the residence of each inventor.

The applicant needs to satisfy supplemental fees problems indicated below.

The required item(s) identified below must be timely submitted to avoid abandonment:

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SUPPLEMENTAL APPLICATION DATA SHEET**Application Information**

Application Number:: 61/044,801
Filing Date:: April 14, 2008
Application Type:: Provisional
Subject Matter:: Utility
CD-ROM or CD-R?:: None
Title:: MEMORY BOARD WITH SELF-TESTING
CAPABILITY
Attorney Docket Number:: NETL.046PR
Request for Early Publication?:: No
Request for Non-Publication?:: No
Total Drawing Sheets:: 4
Small Entity?:: No
Petition Included?:: No
Secrecy Order in Parent Appl.?:: No

Inventor Information

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
Given Name:: Hyun
Family Name:: Lee
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State or Prov. of Residence:: CA
Country of Residence:: U.S.
Street:: 21 Thalia Street

City:: Ladera Ranch
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92694

Applicant Authority Type:: Inventor
Primary Citizenship Country:: U.S.
Status:: Full Capacity
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Middle Name:: R.
Family Name:: Bhakta
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State or Prov. of Residence:: CA
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State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 90703

Applicant Authority Type:: Inventor
Primary Citizenship Country:: Korea
Status:: Full Capacity
Given Name:: Soonju
Family Name:: Choi
City of Residence:: Irvine
State or Prov. of Residence:: CA
Country of Residence:: U.S.

Street:: 195 Woodbury
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92620

Correspondence Information

Correspondence Customer Number:: 20,995
Phone Number:: (949) 760-0404
Fax Number:: (949) 760-9502
E-Mail Address:: efiling@kmob.com

Representative Information

Representative Customer Number:: 20,995

Assignment Information

Assignee Name:: Netlist, Inc.
Street:: 51 Discovery, Ste. #150
City:: Irvine
State or Province:: CA
Country:: U.S.
Postal or Zip Code:: 92618

5519888
061208

Electronic Patent Application Fee Transmittal

Application Number:	61044801			
Filing Date:	14-Apr-2008			
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY			
First Named Inventor/Applicant Name:				
Filer:	Bruce S. Itchkawitz/alexis swan			
Attorney Docket Number:	NETL.046PR			
Filed as Large Entity				
Provisional Filing Fees				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
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Pages:				
Claims:				
Miscellaneous-Filing:				
Late provisional filing fee/cover sheet	1052	1	50	50
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
Total in USD (\$)				50

Electronic Acknowledgement Receipt

EFS ID:	3464654
Application Number:	61044801
International Application Number:	
Confirmation Number:	1630
Title of Invention:	MEMORY BOARD WITH SELF-TESTING CAPABILITY
First Named Inventor/Applicant Name:	
Customer Number:	20995
Filer:	Bruce S. Itchkawitz/Wayne Mahoney
Filer Authorized By:	Bruce S. Itchkawitz
Attorney Docket Number:	NETL.046PR
Receipt Date:	16-JUN-2008
Filing Date:	14-APR-2008
Time Stamp:	19:41:33
Application Type:	Provisional

Payment information:

Submitted with Payment	yes
Payment Type	Credit Card
Payment was successfully received in RAM	\$50
RAM confirmation Number	4026
Deposit Account	111410
Authorized User	KNOBBE MARTENS OLSON AND BEAR

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Document Number	Document Description	File Name	File Size(Bytes) /Message Digest	Multi Part /.zip	Pages (if appl.)
1	Applicant Response to Pre-Exam Formalities Notice	NETL_046PR.pdf	176994	no	6
			f336b1e8594d6ced85182e2b1b8b679b7a1db6b3		

Warnings:**Information:**

2	Fee Worksheet (PTO-06)	fee-info.pdf	8164	no	2
			722b9f0ea5ec0de8cd1f6c163a9461ef9d3910a9		

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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
61/044,801	04/14/2008		260	NETL.046PR		

CONFIRMATION NO. 1630

UPDATED FILING RECEIPT



OC000000030525564

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2040 MAIN STREET
FOURTEENTH FLOOR
IRVINE, CA 92614

Date Mailed: 06/23/2008

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Applicant(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;
Soonju Choi, Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 20995

If Required, Foreign Filing License Granted: 04/21/2008

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 61/044,801**

Projected Publication Date: None, application is not eligible for pre-grant publication

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY BOARD WITH SELF-TESTING CAPABILITY

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international

patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4158).

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Title 37, Code of Federal Regulations, 5.11 & 5.15

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